# The Vertical Jump Bible 2.0

## *Deluxe*

By: Kelly Baggett

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The information in this book is offered for educational purposes only; the reader should be cautioned that there is an inherent risk assumed by the participant with any form of physical activity. With that in mind, those participating in strength and conditioning programs should check with their physician prior to initiating such activities. Anyone participating in these activities should understand that such training initiatives may be dangerous if performed incorrectly. The author assumes no liability for injury; this is purely an educational manual to guide those already proficient with the demands of such programming.

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# *The Vertical Jump Bible 2.0*

Although geared towards the Vertical Jump, this book is really a book for any coach or athlete who wants to better understand the modern science behind increasing athleticism. This obviously includes jumping higher, but also running faster, throwing harder, hitting harder, and virtually any other sport related movement you can think of.

This book was written because I felt it was and is needed. It's been said that information in sports science doubles about every 3 years. That might be a slight exaggeration, but it certainly IS true that there is a lot more information in the field of sports training today than there was 5 years ago, and there will be more next year than today. When I wrote the original Vertical Jump Bible it became FAR more popular than I ever dreamed. I wrote it with the expectation that it might sell a handful of copies, yet 7 years later its popularity obviously exceeded my expectations. I've had countless people tell me it was THE best book they've ever read on training – period. The information has helped thousands and some people give it credit for transforming the entire Vertical Jump information industry.

The VJ Bible was one of the first mainstream products of its kind that promoted correct training principles. Prior to its launch the entire VJ industry was, for the most part, nothing but an assortment of gimmicks and empty promises. Having said that, in my opinion the VJ bible was far from perfect and as I've grown older as a coach I've better learned how to get things accomplished more efficiently and can better explain things in ways that makes the information more applicable for everyone. The shortest distance between 2 points is a straight line and I'm known as a straight line, no bull type of guy. My objective is getting people from point A to point B the quickest and most efficient way possible. Point A is where you are now and point B is where you want to be – jumping out of the building. It's my goal to teach and show you how to do that very efficiently. The original VJ Bible produced terrific results and still does, yet 6 or 7 more years of experience has taught me there are better ways of explaining things, additional information that needs explaining, and quicker ways to progress people. This manual should be just as interesting from an informational standpoint as the original VJ Bible but without any repetition of information. I didn't copy a single paragraph.

My hope is people will find this manual just as valuable and useful as the original VJ Bible. BECAUSE information in sport science expands so quickly, it's important that coaches and aspiring athletes stay on top of things in order to avoid being like the old school nautilus instructor at your local YMCA still preaching the same old 1 set of 10 he was preaching back in the 60's. That's not to say that being "old school" is a bad thing - few if any major principles ever change over time. If you know how to train, or know how to develop an athlete today, you'll know how to get it done 20 years from today. What new information can really do is lead to better understanding and change the *efficiency* of the coaching or training process. Why take twice as long to get something done? I can unequivocally state the information in this book will improve your understanding of the Vertical Jump Development process or help you improve your efficiency as a coach.

As far as the workouts go, I have streamlined the workouts down considerably compared to the original Vertical Jump Bible - they're not as numerous, not as complicated, and are easier to follow. I believe the progressions are also better and more efficient. The workouts in this manual are what I've found to be the "meat and potatoes" with regard to vert development - the workouts, exercises, and combinations that give the best results as efficiently as possible, while also being highly practical and comprehensible. Based on the last 5 years of back-testing I have no doubt the workouts in this manual are even more effective than those in the original Vertical Jump Bible and the material should be just as enjoyable. I hope you enjoy it!

# *Yearning For a Big Jump…*

A sky high vertical jump has long been something coveted by many, yet had by a rare few. Despite it being relatively over-rated when it comes to sporting success (unless you're a high jumper), those possessing it often become the recipients of instant respect and admiration, regardless of whether they can play or not. Simply take a look around at some of the various youtube highlight videos – a lot of the best dunkers can't play a lick, but everyone loves them. Because so many people are interested in improving their vertical jump everyone wants to know what THE secret is – and their typically aren't any shortage of people around to sell you something deemed as the answer. Millions experiment with methods promising magic: Plyometrics, olympic lifting, powerlifting, dynamic mobility, creatine, energizers, flexbands, platform shoes, russian secrets, soft tissue work, the list goes on and on and on. The result is a huge industry, along with what is more often than not a ton of confusion for the VJ-seeking athlete.

With so many things to learn, so many training methods to choose from, and so many systems all promising to be the answer, what are YOU supposed to do to ensure you're on the right path towards attaining your true athletic potential?

Well, fortunately, building the vertical jump is a lot simpler than most marketers would have you believe. I have a reputation for cutting thru confusion and making things understandable and applicable for people, and I don't plan to disappoint here. In this manual I'm gonna give you a step-by-step, no B.S. surefire approach to get you on the right path towards utilizing correct and time tested principles so that you can transform yourself into a Vertical Jump MACHINE and increase your jump to whatever your innate ability allows. To start off let's touch on some common modern myths when it comes to vertical jump:

# *Common Vert Myths…*

**Myth #1: The VJ is mostly genetic**

It is true that most people that jump *extremely* well inherently jump well, but that doesn't mean that people that don't NATURALLY jump well can't do so with proper training. True, not EVERYONE is capable of putting 15 or more inches on their vertical jump - that's just an unfortunate reality. However,

I've yet to meet a person who couldn't, with proper training, put a good 5-6 inches on their vertical jump, and I estimate 90% of the population can do better than that. In the last 6 years I've had the opportunity to work with or consult with many athletes that started with vertical jumps in the low to mid 20's and ultimately hit VJs of 37 inches or higher. These athletes started out as very average yet achieved superior results. That's a minimum of 12 inches improvement and some of them gained 15 inches or more. The fact is EVERYONE is capable of improving their vertical to at least a decent degree. Further on in this manual we'll go into significant detail as far as answering the question, "How much can I gain?"

**Myth #2: The VJ requires a lot of speed**

The fact is, speed of movement and things that go along with it, such as being born with a high percentage of fast twitch fibers, are relatively unimportant for the vertical jump. In the VJ everyone pretty much straightens their legs at the same rate of speed. The rate you develop FORCE is an important factor and highly influences how fast you're going when you leave the ground (which determines how high you jump), but how fast you can inherently move is relatively unimportant. The vertical jump is very similar to a lower body version of throwing a shot-put or hammer. It's a power oriented activity and therefore highly influenced by strength. To quote famous strength coach Tudor Bompa, "I've never seen a powerful athlete that isn't also strong." The VJ is really more about strength & power than it is pure speed.

**Myth #3: Squats (and strength training) make you slow**

Although modern beliefs about strength training have come a long way, there are still a significant number of people who buy into the myth that squats and other strength training methods make you slow. The truth of the matter is unless you push your strength to the limits like a highly competitive powerlifter, squats are more likely to make you faster. A cursory search of squats and rate of force development brings up plenty of research indicating heavy loaded squats actually increase an athlete's ability to apply force quickly, particularly for beginner and intermediate level athletes. Having said that, it is true that the more strength one gains the more likely the additional strength they gain will slow them down.

In other words, taking your squat from 150 lbs to 315 lbs will most likely make you faster and IMPROVE your ability to apply force quickly, yet taking your squat from 400 lbs to 600 lbs may in fact make you slower. Since few VJ seeking athletes are squatting 600 lbs (or even 400), most have little to worry about as far as weights making them slow. The key is to utilize strength training in a correct, vertical jump specific manner, so the strength you gain "supports" your efforts to jump higher and isn't all about just throwing more weight on the bar so you can say you're stronger with no regard to technique or how efficiently that strength transfers to your vert. I will teach you how to do that.

**Myth #4: Plyometrics are the best way to improve vertical jump**

Traditionally, when someone thinks of plyometrics, they think of jumps - and they think of improving jumps. I myself have promoted substantial plyometric training in the past, and it does have a

place. However, in my experience plyometrics come in a relatively modest 3rd place as far as where I rank particular methods for vertical jump development for most athletes. Based on my experience working with and consulting with tons of athletes over a period of many years, I'd rank general strength exercises like squats and loaded ballistic strength exercises like jump squats ahead of plyometrics.

There has long been almost a magical aura surrounding true plyometric exercises like depth jumps ever since the Russians discovered them some 40 + years ago. The magical ability of plyometrics is explained by the notion that they give involuntary muscle recruitment. *Muscle recruitment* refers to how well you use or "turn-on" your existing muscles. The explanation is that great athletes are great because they can utilize, or turn-on, more of their existing muscles in a given task. It was believed that plyometrics eliminate this barrier into muscle recruitment. However, for reasons which I'll go into more detail on later, we now know this is largely a myth. It is true that great athletes get more out of their muscles, but not because of muscle recruitment. Even beginners can recruit all their muscles. The reality is great athletes can really ramp up something called *rate coding* and this is largely due to the inherent excitability of their central nervous systems. I'll thoroughly discuss rate coding in later chapters.

For the most part what plyos really do is enable you to do is *express* your strength and coordination in a high velocity vertical jump specific manner, and they allow you to overload the jump in a manner very specific to the vertical jump. The overload occurs with your bodyweight. Basically, you gain strength so you have more raw horsepower to work with, then you can use plyometrics to express, transfer, and utilize that strength in a jump specific manner. However, that process typically doesn't take that long and MANY people don't have problems expressing strength in the first place. For that reason, if an intense plyometric program is GOING to work for you, you'll generally see gains relatively quickly and the gains tend not to come steady for long unless you're also improving some other strength quality. That doesn't mean plyometric dedicated programs aren't effective, you just have to consider their time and place.

**Myth #5: *All* a person needs to do to develop a big vert is squat (gain strength)**

Just as some are afraid strength training will make them slow, there are others that swing too far to the other side of the fence and think ALL they need to do is strength train. For SOME people squats (or anything else allowing them to gain lower body strength), are all they need to do. Those that inherently express or transfer strength very efficiently can gain additional lower body strength and immediately be able to apply it in sport specific movements of their choice (the vertical jump). These people will be able to base the vast majority of their routines around squats and other lower body strength training movements and will often continuously increase their vert even without any extra work in the way of plyometrics and jumps.

However, a significant percentage of people need to work on their jump efficiency and work on transferring the strength they gain in a vertical jump specific manner. For this reason, most people interested in VJ development should always have some volume of jumps and/or plyometrics in their routine, even if their FOCUS is on strength. It doesn't take much - a few plyometric exercises a couple of times per week in conjunction with actual jumps. But this stabilizes adaptations that allow an athlete to

stay jumping efficient (tendon stiffness, elasticity, coordination) so the strength they gain can be transferred to the VJ.

**Myth #6: A Vertical Jump seeking athlete should always seek to stay as light as possible**

It has long been believed that relative strength, or strength per pound of bodyweight, is one of the most important things for a vertical jump seeking athlete. This IS true from a theoretical and observational perspective - the stronger you are and the lighter you are the better you'll jump, and most people that can inherently jump well are lean and very powerful for their size. However, in my experience many people focus so much on staying light they limit their strength development because they limit their muscle mass gains. A bigger muscle is potentially a stronger muscle, and a stronger muscle can produce more force. Strength gains also come much quicker with at least a slight increase in bodyweight.

A bigger muscle can and usually does generate more than enough additional force to offset any extra bodyweight gains that occur. For THIS reason I believe the typical athlete should be more concerned about BODY-FAT and less concerned about gaining muscle mass. Some muscle mass development of the lower body (and to some degree the upper body), almost universally helps the vertical jump. This is due to the fact that one who gains muscle mass thru strength training methods will typically gain a disproportionate amount of strength. So, if you gain 10 lbs of muscle mass you'll probably gain 30 lbs of strength (at least).

Consider this comparison: Athlete A is 6'6 185 pounds. He wants to jump as high as possible and has read that strength per pound of bodyweight is very important, so he intentionally limits his bodyweight gains. He trains regularly for 3 years and builds his squat from 200 up to 300 pounds, right at around 1.5 x his bodyweight. Although he trained consistently, his strength gains came fairly slow because he was so intent on keeping his bodyweight down. Now let's compare him to athlete B, also 6’6 185 lbs. Athlete B decides he’s gonna get bigger and stronger overall and in the same 3 years takes his bodyweight from 185 to 215 lbs and his squat from 200 to 430 lbs. Now he’s a 215 lb machine at under 10% body-fat. Who's relatively stronger? Who's gonna jump higher? With few exceptions athlete B will be the more explosive athlete and have the higher VJ. If 2 athletes start from the same point and have similar body structures and measurables, usually he who progresses to be relatively stronger will jump the highest, but building relative strength does not mean bodyweight need not increase at times.

If you’re on the thin side 9 times out of 10 muscle mass gains within reason will also dramatically increase your pound per pound strength, and do so at a very rapid rate, as long as those muscle mass gains aren’t also accompanied by a ton of body-fat. Work on getting strong while staying (or getting) relatively lean and you'll be on the right track. This doesn’t mean you should intentionally attempt to gain massive amounts of bodyweight, yet you also need not starve yourself in an effort to stay as light as possible.

**Myth #7: Basketball players have the best vertical jumps and people should train like basketball players**

When people think of a sky high vertical jump they typically think of high flying dunkers, and when they think of high flying dunkers, they inherently think of basketball players. However, basketball players aren't typically at the top of the vert world as a group. Volleyball players, football players, throwers, olympic lifters, and bobsledders all typically have standing vertical jumps ahead of basketball players. Most professional basketball players use training routines that shouldn't be emulated by VJ seeking athletes. In fact, if there's one piece of advice I could give you it's DON'T attempt to emulate the typical training of popular pro basketball players you see. The majority of these athletes usually perform rehab oriented routines to work around injuries and are NOT performance oriented routines.

**Myth #8: More training is better**

There is a long held belief among hard training athletes that more training is always better. After all, *work harder* is a thought that's pounded into our brains from an early age. The typical young athlete believes if he wants to be elite he should adopt the training of elite athletes. What most young athletes fail to realize is that the advanced athlete may have taken 5-10 years to build to that level of training, so they attempt to emulate the routine of their favorite superstar and get destroyed. Another thing people don't often realize is the high flying superstar they seek to emulate may not be getting any gains from his training. Take your favorite NBA high flying superstar and go back and watch college highlight films of him. Chances are he's not jumping any higher now than he was then. To quote my buddy Lyle McDonald, "More training isn't better, *better* training is better". You have to look at what qualities you need in you sport (in this case the VJ), and train accordingly.

The fact is, proper VJ training could be considered a lazy man's activity compared to the weekly volume and hours of training required of most high level sports. A maximal jump is a very short activity that takes less than half a second to complete and it really doesn't take all that much training volume to optimally stimulate. Even at the highest level a jump specialist rarely does an average of much more than an average of an hour of training per day. Compare that to athletes like cyclists, marathon runners, rowers, and professional team sport athletes, who often put in 25 or more hours per week in training. The VJ is a high intensity activity that doesn't really require a ton of volume and time (as far as hours put in per day/week), nor does it tend to respond to a ton of training time for most people. That doesn't mean it's easy, but it is simple. The typical beginner to intermediate athlete will be doing all they can do to fully stimulate VJ gains with about 3-4 hours of TOTAL quality training time per week, not including warm-ups and stretching. That's per WEEK, not per day. Your training volume will progress over time, but the athlete who overloads himself with volume from the get-go usually limits, instead of improving, his gains.

**Myth #9: More variety is better**

There is another long held belief that the more variety you use in your training and the more qualities you can attack, the better results you'll get. Realize the actual vertical jump movement never changes and the motor qualities contributing to it never change. The primary goal of training is increasing your ability to exert tension in movements that syngergize to correspond to the vertical jump.

You should choose exercises that give the best combination of recruitment and optimally hit the quality you're trying to develop – strength, force development, reactivity. An exercise that meets those qualifications today will meet those qualification tomorrow, next year, or 5 years from now. For that reason the best exercises for vertical jump development tend to be a relatively small basic handful of exercises like squats, jump squats, and depth jumps. There are other effective exercises and variety can be important from a psychological perspective - changing things up occasionally can help eliminate boredom, but in my experience those who attempt to overcomplicate their routines by using a smorgasbord of exercises just for the sake of doing so, hurt, rather than help, themselves. Too many people spend too much time learning too many exercises. As a result they end up a jack of all trades and a master of none.

The magic isn't in how many exercises you do, it's how consistently you can progress in your chosen exercises over time, and how your program is arranged over time to deliver a sum greater than the whole. I will show you an example of how to do that using just 3 exercises.

**Myth #10: Gimmicks are the answer to VJ**

Vertical Jump gimmicks have been around for as long as people have been interested in improving it. There are gimmick shoes, gimmick jump training aids, gimmick strength training aids, gimmick vertical jump supplements - you name it and it's out there. There are a few effective gimmicks that serve a specific purpose, and I'll touch on these throughout this manual, but I've YET to see a gimmick, training aid, or training device that is the answer to everything. Without fail the one common theme I've noticed in the 20 years of coaching is that athletes who get the best gains base their routines on basic exercises, not gimmicks. I've yet to see an exception to that.

Throughout this manual I will go into more detail and expand on some of these myths, but those just give you a general idea of our starting point. Now let's move on and talk about some more exciting stuff:

# *Performance Terminology vs Fixin Things..*

This is supposed to be the section where I talk about and define some of the important underlying scientific terms we'll be dealing with throughout this book. I will get around to doing that, but I wanted to talk about something important beforehand:

I try to use terminology my average reader can relate to, so I try not to get overly sciency. In my opinion, sports science terminology is about as important to me as advanced level science is to the average mechanic. No offense to anyone - but that's just not the way I think. To me I look at the body like a machine and I treat it no differently than a mechanic treats your car, at least as far as the thought process goes. I am the diagnostician and the workouts and exercises are just tools.

A good mechanic looks are your car, listens to it, evaluates it, and then determines what's wrong with it (makes a diagnosis). From there he goes to his toolbox, pulls out a tool, or maybe a couple of tools, and makes the necessary repair. Most likely he's NOT thinking about advanced level mechanical engineering equations while he's fixing it (and the guy who knows the mechanical engineering equations probably can't fix your car). Many GOOD coaches are no different. The coach is the diagnostician and exercises and workouts are just tools. It is DEFINITELY possible to know how to fix or influence movement without intimately understanding all the myriad underlying equations behind velocity, force, starting strength, explosive power, etc. You just have to know the general basics of what you're looking at and know what tool does what. I will teach you how to do that.

The major reason I mention this is because I KNOW thru my own experiences that it is easy to become completely overwhelmed and mesmerized by immersing yourself deep into the science of sports training. Back in my 20's I spent 3 years reading as many scientific training texts as I could, then spent the next 5 years trying to figure out what I learned. Throughout much of that 3 year period I was completely flabbergasted. I felt worthless as a coach because I was so confused my head was spinning in so many different directions. What's really interesting is when you learn and memorize different terminology that means the exact same thing. The soviets and Germans use slightly different terms to describe the same thing and that totally threw me off.

You have to remember that educational systems and research typically make any field relatively complicated. So it's not uncommon to see a few months of practical information spread out into 4 or 5 years. I look upon it as my job to make things as practical as possible, so I try and do the opposite: Take 4 or 5 years worth of information and present it in a manner that it can be learned and put into practice in a relatively short time span. That doesn't necessarily mean everything should be simplified, but many times it does. As Einstein once said, "Make things as simple as they need to be - but not any simpler." I would say *practical* is the best word. It's the way I think, it's the way I operate, and it's what I believe in.

Another reason I mention this is I've seen too many people get confused and lost in terminology when what they really need to do is step back and really think about what they're doing from a broader perspective. I normally get about 10 emails per week from aspiring athletes really interested in learning all they can about training and many of them get bogged down in minutia. I once heard an NFL

defensive coordinator say that when his defenses weren't playing well he instinctively knew to cut the playbook in half. Failure to get it done as a defense almost assuredly meant people were thinking too much. It WAS the coaches fault. They knew the game and had spent 30 or 40 years studying the game. They had an entire offseason of 12 hour days to make their playbook as big and complicated as possible and when they weren't getting it done they knew what to do. In my own experiences I often see the same thing with training. If what you're doing isn't getting it done, take a broader perspective of things, simplify your playbook, and get back to basics.

With all that said, let's get on with some of the terminology that will be address throughout this manual:

**Strength:** The ability to generate force

**Explosive strength:** Maximal force in minimal time. For all intents and purposes could be considered the same thing as power. The Vertical Jump is a good example. Explosive strength is formed primarily when starting strength combines with reactive strength.

P**ower:** Work divided by time. Or force x speed. Is often used interchangeably with explosive strength.

**Starting strength:** Force at the very beginning (100 milliseconds) of movement.

**Reactive strength:** How much extra force you get from a countermovement. For example, you can bench press more with a touch and go then you can using a 5 second pause on your chest. The additional weight you can lift with the touch and go style tells you how much additional strength you're getting from reactive strength. You can also jump higher with a countermovement (quick dip down) than you can if you start from a static position. Reactive strength is technically a combination of starting strength and acceleration strength.

**Absolute strength:** Strength irrespective of bodyweight. Example: A 500 pound sumo wrestler is *absolutely* stronger than a 150 lb Greco roman wrestler.

**Relative strength:** Strength per pound of bodyweight. Example: a 150 lb Greco roman wrestler is *relatively* stronger than a 500 lb sumo wrestler.

**Relative power:** Power per pound of bodyweight. F x V/bodyweight

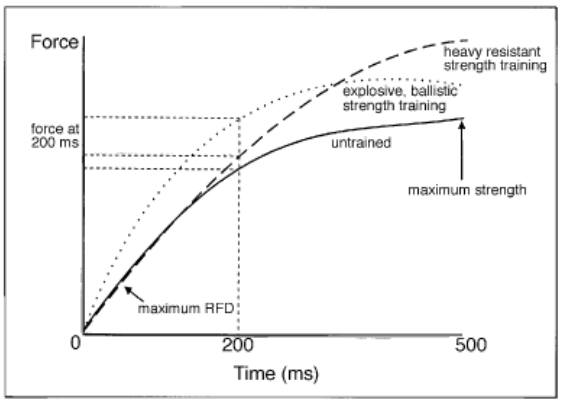
**Strength-speed:** Movement involving a combination of strength and speed in which strength plays more of a primary role. The perfect example of strength speed is an Olympic lift like a clean or snatch. Typically strength-speed movements will involve a loaded exercise (squats, bench presses etc), using a load equivalent to 50% or more of your 1 rep maximum.

**Speed-strength:** Movement involving a combination of strength and speed in which *speed*, or speed of force delivery, plays more of a primary role. The vertical jump is a *display* of speed strength and exercises like plyometrics and jump squats could be considered speed-strength exercises. The cutoff point is the exercise must use less than about 30% 1rm for any given exercise. Thus, a squat with 30% of

your max would be a speed-strength exercise while a squat with 40% of your max would be a strength-speed exercise. Power training and rate of force development training are often used interchangeably with speed strength training.

**Rate of force development:** how rapidly force is developed during any given movement. Technically the slope of the force/time curve.

**Force time curve (F/T curve):** A chart showing levels of force developed and how they relate with time.




**FORCE/TIME CURVE**

**Velocity:** The speed of movement. Technically is distance divided by time or d/t.

**Work:** Force x distance is the equation for it, but honestly all I know is "work" is something I'd rather not do!

I will come back to many of those terms throughout the book but for now I just wanted to introduce them. Now that I have some of the complicated stuff out of the way let's look at the vertical jump from more of the perspective of a mechanic:

# *Vertical Jump – A Mechanics Perspective…*

Although it is often very difficult for the average person to sort through all the often contrasting information in the Vertical Jump development industry, improving the VJ really isn't all that complicated. In fact, when you get down to it, it's really extremely simple. Not necessarily easy – but simple. Methods are many, but principles are few. Any improvement in your vert are really just a matter of increasing two (count them, just two) foundational descriptive qualities. They are:

1. ***Movement efficiency***: How you move. It's the ability to carry out a movement with utmost efficiency. Think of the fluid grace of someone like a Dwayne Wade moving straight ahead like he's shot out of a cannon, stopping on a dime, changing direction, and doing a pirouette like a ballerina, followed by an explosive dunk

2. ***Horsepower***: How much force is behind a movement. The amount of force, power, and speed that is exerted when you move. It's the difference between a ballerina sprinting down the basketball court vs a Lebron James sprinting down the court with ball in hand. They're both fluid, but only one of them has any power behind the fluidity.

All of the aforementioned training methods I mentioned earlier, as well as anything else that improves performance, will affect one or both of these. Any training method, gimmick, recovery aid, diet, or anything else promoted to improve a physical quality like VJ will impact one of those factors. When you get down to it that's the only way the Vertical Jump (or really any other display of athleticism) can improve.

Think about it. What really determines how high you get off the ground when you jump? The same thing that determines how fast or far a baseball flies through the air. How much force (horsepower) that it's hit with, or in the case of jumping, how much force is behind the legs. What determines the ability to jump with perfect speed and fluid mechanics so that you best "leverage" your horsepower recruit the right muscles in the right order, at the right time? The answer is the efficiency of the movement.

If I lift weights, I improve my ability to exert force (horsepower). If I do a lot of stretching and that stretching relaxes my tight muscles, I can then move more efficiently, and that in turn also allows me to exert more force, right? If I drink special pills and lose weight I have less fat mass to carry around and that'll improve my movement efficiency and my pound per pound power. The list goes on and on.

But here is the question: What if, instead of haphazardly throwing all sorts of training methods against the wall to see what sticks (what gives us some semblance of results), we simply worked backwards and found the most direct approach to improving movement efficiency and horsepower? We could ask, "What is the most direct and straight line approach to improve the force I put behind my jump movements??" After we answered that question we'd ask, "Ok, what is the most direct way to increase my movement efficiency?" Hmm…something to think about isn't it?

From a practical perspective, as it relates to the VJ, ***horsepower*** is basically just how hard you push against the ground and how quickly you do it. There are terms that allow us to dig a little deeper, such as relative power and ground reaction force, which I'll cover in detail in a few minutes, but they all influence horsepower.

From a practical perspective, ***movement efficiency*** is simply the ability to carry out a movement with optimum efficiency so as to recruit the right muscles, in the right order, at the right time, so that you generate the greatest amount of power with the least amount of effort. Movement efficiency can be impacted by a ton of things like mobility and muscle balance.

Although rare, it is possible to be fairly powerful and explosive but not be able to jump high and here's why: Every particular movement in sport is characterized by unique movement and recruitment patterns. Movement and recruitment patterns involve the timing of muscular activation for a particular activity. Think of any sporting movement - running, jogging, jumping off both feet, jumping vertically off 1 foot, jumping horizontally, lunging, throwing a baseball, throwing a shotput, throwing a javelin, throwing a hammer, a martial arts kick, kicking a football, kicking a soccer ball, swimming, and the list goes on and on and on. Some of them are related, but none of them are 100% identical. Movement efficiency is simply the optimization of muscular coordination, activation, and timing for a particular movement. Think of it as *skill* for a particular event – in our case jumping. Just like dancers practice several hours per day honing their skill, you need to do enough jumping to optimize the movement and recruitment patterns inherent to your particular activity. The VJ is a LOT simpler than dancing and doesn't take THAT much practice, but the principle is the same.

Every particular movement involves muscles turning on and relaxing at slightly different times. Movements involving similar muscular movements and speed of contraction tend to correlate quite well: For example, sprinting is really just a series of horizontal jumps from one foot to the next down the track. Because it involves mostly the same muscles and the contractions occur at about the same speed as a vertical jump and the external resistance is the same (bodyweight), it tends to correlate fairly well with jumping ability. Someone that can accelerate quickly usually has an above average vertical jump, and vice versa. The main difference with the sprints is they involve significantly more horizontal force application, like a skateboarder, rather than vertical force application. This results in significantly greater hamstring contribution for the sprints. The coordination and activation patterns are a bit different too. Because of this, although rare, it is possible to be very good at sprints and only average at the jumps, and vice versa.

The most dramatic illustration I've seen of this was an athlete who could run a 4.37 forty yard dash but could only vertical jump 24 inches. He simply had never been interested in the jumps and never developed the coordination to jump effectively – even though he definitely had the power to do so. I've also seen guys that could vertical jump 35 inches but would be hard pressed to break 5.0 seconds in the 40. This also explains why horizontal bounding tends to correlate better to sprint times than vertical jumping. From a muscular recruitment and activation standpoint a bound is significantly closer to a sprint than a vertical jump is.

Another example of this involves kicking and throwing. Not all kicks are equal. Back in my younger days I used to take tae-kwon-do. After about 7 years of this my kicks were pretty strong. I could kick thru bricks. Yet I don't think I've ever been able to kick or punt a football or soccer ball more than 15 yards tops (and still can't). I've simply never been interested in kicking a football, and never have practiced enough to learn how to optimally do so.

Movements also vary based on speed and force. The faster a particular movement occurs and the less external resistance is involved, the more the movement is dominated by pure speed and quickness. The slower the movement occurs and the more external resistance involved, the more the movement is dominated by strength and the ability to develop force. For example, throwing a baseball and throwing a football are fairly similar as far as the actual motion. The main difference is the football is significantly heavier and the grip is different. It takes more strength to throw a football. Thus one can be very effective at one and terrible at the other.

Back to my own experience: In Jr. high I went thru a 2 year phase where I was a baseball fanatic. I used to spend all of my free time outside throwing pitches against a brick wall. I'd probably throw 200 pitches per day minimum, and I did this mostly all year around. I was obsessed with throwing. If I wasn't throwing a baseball, I was obsessively throwing rocks, or whatever else I could get my hands on. After a couple of years of that I could throw a baseball about as hard as anyone my age, even though I was only about half as big as a lot of kids my age. However, I'd literally get laughed off the field anytime I tried to throw a football (and still do). My arm was fast, but the football was too heavy and it made my weakness apparent. I am a lot stronger now but still can't throw a football worth a darn, and I doubt if I could throw a baseball any harder than I could as a 12 year old sixth grader. I haven't touched a baseball since 7th grade, and even though I'm probably 3 times as strong now, my movement efficiency for throwing is completely gone. Now, if I were to take a few months and work my way back into throwing, chances are I could throw a lot harder than I used to, but it would take time to develop and hone that movement pattern and coordination.

The same general thing happens to a few people that get so caught up in "training" for the vertical jump they forget to jump. They get so caught up in lifting and bringing up their squat they forget their main activity is jumping, so they don't do enough jumps and lose movement proficiency. Their squats go up but their jumps don't. Or they get so caught up in fancy plyometric drills they get really good at them, but forget to do enough actual jumps. Yes, it is important to focus on foundational training (strength), and getting your squat up is an important part of that, but don't forget you're training to increase your vertical jump, not set a powerlifting record. All the extra foundational training is just a supplement. Training a particular movement skill often (in our case jumping) optimizes a quality called synaptic facilitation, which allows the nervous system to develop more powerful connections between your brain, the nerves, and the muscles they innervate specific to that particular movement. The bottom line is that the mastery of any particular skill demands repetition in that skill.

## Building proper movement efficiency

Many jump programs don't work as effectively as they could because most people's bodies don't work as optimally as they should. They don't recruit the right muscles, to the right degree, at the right order, at the right time. In short, their bodies need a tune-up.

A rather large number of trainees train hard and don't get the results their "training" improvements suggest they should - rather they end up with knee pain, sore backs, sore ankles, and little gains to show for their efforts. These are surefire signs of the body not being in balance. Watch a good jumper jump and most of the time it appears easy, effortless, and quiet. Watch a bad jumper jump and most of the time it's the opposite - loud, rough, and not something that conveys the illusion of artistry. Do you move like a cheetah or like a lumbering bear?

Many people are capable of transmitting a lot more force into the ground than they do, even without gaining additional strength and speed, but they "leak" excessive energy throughout their body. A big reason why some individuals can jump well with no or little training is because they're naturally put together well and inherently transfer energy into the ground very efficiently.

Many athletes with movement efficiency problems develop ***compensation patterns***. A compensation pattern is when an imbalance in a movement or muscle places additional stress on other muscles or joints. For example, say you do squats and you have a problem with the back of your heels coming off the floor. As you descend your body "compensates" by transferring more stress from your feet and onto your knees and back, creating an overload that can result in injury. Most people have some degree of compensation pattern. The mobility and movement assessment that follows later is designed to eliminate these types of problems.

## The Importance of Glute Dominance

One thing you want to emphasize to avoid compensation patterns is a glute dominant hip extension pattern. This simply means that your gluteus (butt) muscles are highly active when you jump. The knee extensors (quadriceps), and ankle extensors (calves), are highly active as well, but the more you can engage your glutes in a jump the better off you'll tend to be both from a performance perspective and from a health perspective. One colleague of mine, Alex Vasquez, took a group of cross country runners and put an average of 5 inches on their vertical jumps in about a month and a half doing NOTHING but supplemental glute work. If you do a ton of jumps and don't feel your glutes pump AT ALL afterwards I'd say you might have problems. If you do a lot of squats and don't feel any soreness at all in your glutes you might have problems. If you take a week and really emphasize glute work you will probably notice the feel of your jump changes, usually for the better. It may not necessarily be higher, but feels smoother and you'll come off the ground easier with less stress on your knees. The key is then maintaining that improved recruitment pattern while strengthening the entire kinetic chain (hips, knees, ankles). More hip involvement means better leverage and less stress on the rest of the kinetic chain.

Another simple observation you can do is stand blindfolded and jump vertically as high as possible, aiming for nothing in particular. See where you land in relationship to where you took off. If you went backward a bit your quads are probably dominant. If you went forward that indicates more glute involvement. Landing where you took off from indicates optimal balance. Correcting compensation patterns specific to the VJ and establishing glute dominance will be the topic of a later chapter, but I wanted to give you a general idea of how they can impact optimal movement.

## Gross Motor Skills Vs Fine Motor Skills

Now that I've talked ad nauseum on things that can impact movement efficiency, I want to caution you not to dig too deep into it. Compared to most feats of athleticism the VJ is as simple as it gets. It doesn't require much technique or anything elaborate. Things like technique and compensation patterns I talked about can be an issue for some, but some coaches make them out to be a lot more than they really are. Fortunately, jumping is a gross whole body motor skill, which basically means it doesn't require much conscious effort to perfect. This also means that performance is largely determined by strength qualities and is not as reliant on technical skill.

Gross motor skills are kinda like riding a bike. Once you learn them they don't require much conscious input. Once you learn how to ride a bike you don't have to think about it much do you? Crawling, walking, running, jumping, and throwing a punch or kick can all be put into this gross motor skill category. I also call these *primal* movement patterns because they're highly instinctual. Now, contrast those physical skills to something like threading a needle or executing a double twisting back-flip. These require much more skill, concentration, and focus.

The reason I bring this up is because throughout this manual we're gonna talk quite a bit about a multitude of factors involved in jumping high, including many technical issues, but don't lose sight of the fact that it is predominately a primal gross motor skill. Now, I'd like to spend some more time talking about things that influence HORSEpower.

# *Horsepower & Relative Power…*

Leaping ability is heavily dependent upon lower body ***relative power*** and is in fact a *display* of relative power. Relative power is really just a fancy term for how much force you can generate in a fairly short period of time, relative to your bodyweight. If athlete A weight 100 pounds and produces 200 lbs of power, and athlete B weights 100 lbs and produces 250 lbs of power, athlete B has greater relative power. Simple enough. The amount of power you produce is explained by the equation: force times velocity, or work divided by time.

(Power=Force x Velocity) or work divided by time (w/t)

The *force* component is primarily determined by your pound per pound strength, and the *velocity* component determined by how quickly you apply that strength. Put all that together and you

get ***explosiveness***. So, in essence, explosiveness is really the ability to develop maximal force in minimal time.

***Explosiveness =*** *Maximal Force in minimal time*

Or a little simpler:

***Explosiveness=*** *how strong you are relative to how much you weigh + how quickly you utilize, or express, that strength*

# *What about Speed?...*

I briefly mentioned speed back in the myths section, but now I'd like to give a better real world example to explain the difference between speed and explosiveness: Even the average casual observer realizes that in all the world of sport SPEED IS KING. What the casual observer typically DOESN'T know, however, is that speed as it relates to real world activities such as sprint speed or terms such as speed-strength is not the same thing as speed defined. Huh? Let me explain: The scientific definition for velocity is the speed of movement. However, speed of movement rarely exists in isolation in real world tasks or in speed-strength related endeavors such as the vertical jump.

When a sprinter sprints down the track he's displaying explosive strength and power more than he is pure velocity. Same goes for a jumper coming off the ground. Why is that? It's because during a sprint (or jump) the athletes are working against a significant external resistance in the form of bodyweight and gravity, and that resistance they have to overcome means the speed of movement will never be close to the velocity they could achieve if they didn't have to overcome bodyweight. For a real life example of this simply assess how fast you move your legs during a sprint. Let's say you run a 40 yard dash in 4.5 seconds and you take 18 steps throughout the 40. That means you take an average of 4 steps pure second.

Now, lie on your back with your feet up in the air and see how many mock steps you can take in 4.5 seconds. Guaranteed it'll be a WHOLE lot more than 18. Lying on your back and cycling your legs is a more pure example of velocity and absolute speed. Sprinting down the track or jumping are both examples of explosive power and are primarily determined by how quickly and forcefully you can overcome the weight of your own body, and the ability to do THAT is different than the raw ability to move as quickly in the absence of resistance.

Explosive power, Speed-strength, rate of force development, explosive strength, reactivity, and power are all often used interchangeably and really they're all pretty much demonstrated in the real world as the same thing: Producing maximal force in minimal time. If you only take one scientific term or explanation from this section of the book take this: The entire objective is *generating maximal force in minimal time.*

# *Ground Reaction Force…*

The VJ is really nothing more complicated than pressing hard against the ground. The harder you press against the ground, the higher you jump. It involves a summation of force from the entire body, primarily the calves, quadriceps, and hips, with slight contribution from the upper body (back and shoulders). It involves something called ***triple extension,*** where the joints of 3 major power producing joints (ankles, knees, and hips) extend simultaneously producing a sum greater than the whole. The harder you push, the higher you go. You have limited time to activate your muscles as much as possible and push off the ground as hard as possible, about .2 to .4 seconds. The amount of force you put into the ground is called "***ground reaction force***." The higher you jump, the more ground reaction force you'll create. Let's start off with a more detailed discussion on what factors can influence that ability:

1. ***Strength***- Besides the obvious influence on your ability to create and generate force, strength is also important for *absorbing* force. If you perform a running vertical jump (from both legs or 1), there are tremendous forces involved as you decelerate to gather yourself to jump. You must be able to support anywhere from 3 to 7 times your bodyweight on each leg. That obviously requires a good degree of strength. If an athlete isn't strong enough to absorb the reaction forces he creates, his legs will crumple under his bodyweight and he won't be able to decelerate and absorb force efficiently. If this occurs he obviously won't be able to put out any force either. The ability to withstand force is just as, if not more important, than the ability to put out force. You can see this exemplified anytime you watch a really weak athlete try to do a proper approach jump. They look gangly and uncoordinated simply because they're not strong enough to deal with the weight of their own body.

2. ***"Stiffness" and Plyometric Ability***- When I'm referring to *stiffness* I'm not referring to flexibility, but rather the ability to efficiently stabilize and transfer force in the joints, like a basketball rebounding off the ground. This largely involves the above capacity to withstand high forces without folding under the tension. Watch a good athlete move and they make things look so easy. They stay on the balls of their feet and just kind of "bounce" over the ground with seemingly little effort – like a rock skipping across water. Therefore, stiffness in this sense is a positive thing.

What causes stiffness? Simple. It's a combination of muscular and tendon stiffness. It's determined by how much force the muscles can develop, how fast and proficiently they develop that force, and how proficiently the muscles and tendons work together to transfer force and create movement. Each time you plant to jump or bend down to jump your muscles have to "lock up", or contract, to withstand the oncoming force that occurs as you switch from "down" to "up". The muscles themselves lock up and this allows the tendons to help serve as movement generators. This entire process is also known as ***plyometric ability***. To illustrate this concept for yourself try these 2 drills:

A: First, stand on 2 feet, lock your knees, and simply bounce up and down on the balls of your feet in a rhythmic manner. Each time you hit the ground I want you to concentrate on simultaneously relaxing and LOCKING UP your calf muscles as fast and hard as you can so that your heels drop as little as possible after impact. What happens? First, your calf muscles lock up and absorb the force created from the impact against the ground. Next, your achilles tendon stretches like a rubber band and then

recoils. What happens next? You kinda rebound off the ground effortlessly. The quicker you can lock your muscles up, the less your heels give at impact, and the quicker you can rebound up. That entire sequence is also known as a plyometric movement.

B: Now, try something a bit more advanced: Stand on the ball of only one foot this time and bounce up and down on one leg at about the pace you'd be moving if you were swinging a jump rope. Stay on the ball of your foot and as soon as you hit the ground try to avoid letting your heel descend down any lower. Next, pick up the pace and do the same thing but in a more intense rhythmic fashion. Get a little higher with each hop. What happened? Well, at some point you probably collapsed at the ankle, didn't move worth a darn, and may have even noticed some pain. That's the point where you lost your ability to be *stiff*. Improving stiffness and plyometric proficiency is an important part of jumping higher. You can fail to be plyometrically proficient for one of 3 reasons:

1: Your muscles aren't able to produce enough force when they contract against high resistance, so they give too much at impact. (You lack strength)

2: You aren't able to lock your muscles up quickly enough (or produce force quickly) enough, so your muscles give too much at impact.

3: You are able to lock up and absorb force proficiently, yet are unable to efficiently spring out and recruit and utilize the right muscles, at the right time, in the right order. (You lack movement efficiency and coordination)

A flat basketball can't bounce off the ground because it gives too much. What causes the give? Lack of stiffness (air pressure). The same thing happens with a weak athlete. The lack of strength makes his legs give at ground contact just like the flat basketball. He can't absorb (or produce) force. Now, think of what happens when you throw a softball against a slab of concrete. The softball is strong enough to absorb the force, yet doesn't bounce back really well. Why not? Because it doesn't have a whole lot of rebound to it. In human terms the soft ball would be the guy who is really strong but who lacks spring. Now think of a golf ball. Not only is it stiff and resilient, yet also fairly springy. When it comes to plyometric ability, you want to be more like the golf ball, resilient (stiff) and springy.

**Tendons & Connective tissue adaptations**

Although strength and coordination are the most important factors impacting stiffness, it should also be noted that when it comes to stiffness in the tendons, the stiffer the tendon the more force it can potentially produce. Although not a HUGE factor as far as how much you can influence it, having stiffer tendons can give you a bit of extra oomph in your jump. Tendons are similar to rubber bands - a big, thick tendon flies a lot further when you pull it back. Some people naturally have tendons that are 2 x stiffer than average. Elite high jumpers have been found to have tendons much stiffer than average. Anytime you see an athlete that can naturally jump REALLY high with fairly low levels of strength chances are they were born with very good tendon stiffness.

Development of the tendons could be considered a *connective tissue* adaptation to training. Tendon contribution to activity spares energy contribution from the muscles to make movement more efficient. Not ALL athletic movements involve much in the way of tendinous contribution, but many of them do. Even walking is an activity that involves significant achilles tendon contribution. Tendinous contribution explains why you see lightweights in the gym repping out the stack on the standing calf raise machine - they're not using their muscles, they're using their tendons. As mentioned, tendons are much like rubber bands and the thicker and stronger a rubber band the further it flies when you pull it back. Tendons are the same way. The more collagen they contain the stronger they are and the more force they can potentially exert when stretched.

Although this is mostly a genetic characteristic it is possible to improve tendon stiffness somewhat. The amount of collagen in the tendon is one of the primary factors responsible for increased tendon stiffness. So, what are some things that boost tendon collagen production? Plyometrics and lactic acid will both do it. Ensuring you engage in regular sports movement (jumps) and "being in shape" are the most important things to influence those 2 factors.

*** It should be noted that concentrating solely on tendon stiffness isn't necessarily a positive thing. Lactic acid boosts collagen production but prolonged exposure to lactic acid can also negate positive explosive muscular qualities. A good example of this is looking at sprinters over varying distances: 200 meter and 400 meter sprinters have stiffer tendons than 100 meter sprinters, but they also aren't as fast. The same thing occurs in those who oveuse plyometrics. They might have stiffer tendons but they also suffer from overuse injuries and are often strength deficient. In reality the average person can optimize their tendon stiffness by engaging in jumping related tasks on a regular basis in conjunction with proper strength training.*

3. ***Mobility***- Mobility refers to range of motion. Obviously, before you can generate extreme power and tension in a movement you have to be able to get into an optimal position to carry out the movement to begin with. The vertical jump obviously doesn't require the mobility of a high flying trapeze artist, yet there are certain muscle groups that can become tight, which can cause other movements to become inhibited. This can negatively affect the ultimate power of your vertical jump. This will be covered in great detail in the assessment section of this manual.

4. ***Bodyweight to strength ratio***- For the vertical jump your body-fat level could be considered as important as any other quality. The leaner you are and the stronger you are the higher you're gonna jump – it's that simple. Imagine what would happen if you put a 20,000 pound weight and attached it to a drag racer prior to the beginning of a race? Instead of seeing a drag race you'd be watching a tractor pull! Well, the same thing happens if you're hauling around a 10 to 50 pound tub of lard around your gut or your butt. Being fat simply ain't gonna cut it! If you want to be an explosive athlete, a certain level of leanness is desirable.

Having said that, bodyweight increases in the form of muscle mass increases aren't necessarily a bad thing, as touched on earlier. When a muscle increases in size, it also increases its strength potential. Let's say you take your bodyweight from 140 to 165, while your squat and deadlift go from 200 to 400 pounds. Did your bodyweight to strength ratio go into the crapper? No, it improved! Therefore, one should strive to be lean, yet should not be deathly afraid of bodyweight increases. Each year at the NFL combine some of the biggest vertical jumps are actually by 260 lb defensive ends! They combine long levers (long legs) with lots of muscle mass, lots of strength, and as a result they become ultra explosive.

Instead of focusing so much on bodyweight I believe it's better for an athlete to focus on body-fat. I consider 6 to 10% body-fat ideal for a VJ seeking male and 12-20% ideal for a female. The following internet URL has a handy calculator you can use to identify with quite amazing accuracy what your body-fat level is. Simply take your waist measurement, plug it into the space provided, and figure out where you are at:

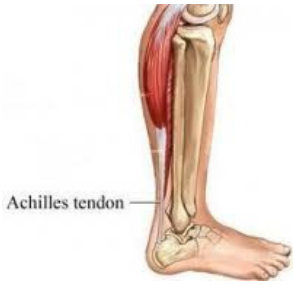
http://home.fuse.net/clymer/bmi/

5. ***Body structure***- Body structure is important because it determines how effective you're able to ***leverage*** a given amount of tension into power. As an example, take a 12-inch bat and hit a baseball with it. Next, take a 32-inch bat and hit the same baseball. Which one goes further? Probably the one hit with a 32-inch bat. This is because the longer bat gives you a longer lever, which gives you more leverage, which means you can generate more power at the moment of impact.

When jumping, think of the length of your bones and tendons as being the same thing as a bat. A longer leg serves as a longer lever and, assuming the amount of force generated by the hips and legs is equal, the longer leg can generate more leverage off the ground. So, with the amount of force generated by the hips, knees, and ankles being equal a person with longer legs will tend to jump higher.

Is there anything a person with shorter legs can do to bridge the gap? Yes. They can produce more force. Let's use a real life example: Imagine if you gave me a 32-inch baseball bat and gave Albert Pujols a 15-inch bat and asked us both to hit a baseball as far as we could. Who do you think would hit the ball further? Would my longer bat and greater leverage make up for their superior strength and power? Hardly. I'd still get blown away. I'd probably even get blown away if he was using a 6 inch bat. He's simply too strong and powerful for me to ever compete, regardless of how much leverage I have with a longer bat. This is largely how a short, squatty, pit bull type sprinter like Ben Johnson was able to beat a weaker Greyhound type sprinter like Carl Lewis or someone like a Nate Robinson can jump so high. ***Disadvantageous limb ratios (leverages) can often be overcome by disproportionate strength.***

The same process I described above with regard to leg length is also true when we refer to variability in the length of the tendons, particularly the length of the Achilles tendon. Take a look at the calf muscles of the average elite level jumper and compare them to the calf muscles of an average person. Many (not all) good jumpers have achilles tendons that tend to be longer than average. It's long been believed that the longer the achilles tendon the greater the potential for jumping, and a cursory real world observation verifies that to an extent.



Why is a longer Achilles tendon advantageous for jumping ability? Well, providing the muscles from the hip down can properly absorb and create force, a person with longer Achilles tendons basically has a longer rubber band in his legs and that can offer an advantage when jumping (or sprinting). Is there anything a person cursed with a short Achilles can do to bridge that gap? Yep. The solution to the "Achilles" curse is the same solution as the "short-legged" curse. Disadvantageous tendon lengths can also be overcome by disproportionate muscular strength or other qualities. **

*** The reverse is also true in that people with naturally good structural and muscular qualities can often perform while being weaker than their opposition. The weak athlete who can jump out of the gym is a perfect example.*

**** Some fairly recent research indicates Achilles length may not be all it's cracked up to be. Many elite high jumpers such as Stefan Holm have "normal" tendon lengths. Research by Earp et. al found that athletes who have shorter Achilles tendons tend to do poorer when jumping from a standing position, but perform better with elastic jumping, such as a depth jump or running jump. Athletes with shorter Achilles tendons will tend to do better respectively with jumps from a running start. On the other hand, athletes who have extremely long Achilles tendons will do better in their standing vertical leaps, but suffer when a running start is involved. This may be because people with short Achilles need not spend as much time "winding up" to allow tendon recoil and this favors running jumps – they don't need to bend their knees as much on an approach jump. On the other hand, people with longer Achilles also tend to have longer lower legs, which favors all types of jumping ability.*

6. ***Neural factors*** – Your nervous system is like the power plant that governs your muscles and in large part determines how much of your existing muscle mass you're capable of using for strength, and how much of your strength you can utilize in a quick movement such as the vertical jump. In the Vert you have between .2 and .4 seconds to activate your muscles as much as possible and get as much power into the ground as possible. The more efficient your nervous system is the better you can do that, the more of your strength you can utilize, and the higher your jump will be. The neural factors we're primarily concerned about include recruitment patterns, motor unit recruitment, and rate coding. I will discuss neural factors extensively later on, but they all refer to how efficiently your nervous system works to coordinate your movements and how strong your neural signaling is. Just like a stronger battery can create a more powerful charge, a supercharged nervous system can create a stronger charge when it comes to your movements.

I will expand upon many of these factors throughout this manual. Let's start at the very beginning with the foundation:

## *Strength = The Backbone*

In essence, for an athlete, maximal strength is like the horsepower of the engine in a vehicle. The more strength we have the higher our other physical attributes can potentially go. A car with a 400-horsepower motor doesn't necessarily go as fast as one with a 200 horsepower motor, but it certainly has the POTENTIAL to be a lot more impressive if all things are equal as far as the tuning of the engines, fuel, and chassis. Just like horsepower is the foundation for how fast or how far any type of vehicle can go, maximal strength is the foundation for our physical attributes, which can be limited by insufficient strength in the contributing musculature. Even a relatively "low force" athletic feat, such as performing as many pushups as possible in 60 seconds, is largely influenced by your maximal strength levels. Someone with a 100 pound bench press typically won't be setting many power endurance records.

The vertical jump is primarily a combination of hip extension, knee extension, and ankle extension. These 3 extensors consist of the calves (ankle extensors), quadriceps (knee extensors), and glutes (hip extensors). All 3 extensors contract relatively simultaneously in any type of jump – synergizing to produce a sum greater than their individual parts. The hip and knee extensors contribute around 30-40% each to the total power in the vertical jump with the remaining 20% taken up by the ankle extensors (calves), and upper body. Thus, in order to jump high you need to be STRONG in the quadriceps and glutes. That doesn't mean that qualities like rate of force development aren't important, but your ability to produce high amounts of raw tension (horsepower) serves as the foundation for those other attributes.

Most people that jump high are quite strong, even if they don't appear to be so thru traditional measures of strength. Some people have a build characterized by long tendons, long legs, and high muscle attachment points that allow them to leverage and transfer force very efficiently. So, for each unit of force they develop they will be able to transfer a lot of that into the ground. That's how someone like a Kevin Durant (and people with similar measurables and body structures) can jump well even though they've never seen a squat rack in their life. If you don't have that great natural body structure and great natural nervous system (and most people don't), you're gonna have to make up for it by significantly increasing your strength. Simple enough.

There are some skinny guys who can jump very well without being strong in the traditional sense, yet you won't find ANYONE with a 35 inch plus vertical jump who doesn't have a lot of "natural" strength. By *natural strength,* what I mean is if you find someone with a naturally high vertical they always have a natural ability to create force, kinda like some people can naturally hit, throw, or kick hard. Even if they don't strength train you can take them in the gym and teach them how to squat and within a week they will generally be squatting over 1.5 times bodyweight. Not all athletes are built the same, and not everyone displays their strength in the same manner, yet I have yet to see a weak individual jump high. If you don't have that strength naturally you're gonna have to train to get it.

For some reason these natural athletes cause people a lot of confusion and often make basic concepts related to the vertical jump difficult for the average person to grasp. Admittedly, the problem

is not NEAR as bad today as it was 5 or 6 yrs ago, yet there are still hoards of people who think the vertical jump is something it really isn't. Think about this: You never see guys with 100 pound bench presses winning any shot-put medals do you? It obviously takes a strong individual to be a good shot-putter. Yet, when planting our feet and throwing our own bodyweight through the air (which is exactly what we do when we jump), people don't seem to comprehend or appreciate the importance of raw horsepower.

Granted, not all good jumpers can effectively demonstrate their strength in the weight room via massive poundages, but the potential to produce a lot of force is still there. It's kind of like boxing - some people can hit extremely hard without ever touching a weight or doing any form of resistance exercise whatsoever. They simply have an innate ability to hit hard and do so the very first day they step in the gym. It's the same way with jumping. If you don't inherently have that innate ability to produce a lot of raw horsepower naturally the only way to get it is to build your strength. This is why many (if not most) Olympic weightlifters and throwing athletes can jump higher from a standstill than some high jumpers. They don't do that from endless jumping drills and plyos, they do it by being very strong and having the ability to utilize that strength very quickly.

## What Can Strength Do For You?

Realize that improvements in vertical jump are related to 2 major factors that can be modified by getting stronger the weight room:

a) Force
b) Rate of force development

For all practical purposes you improve force anytime you increase your strength. You improve rate of force development when you can apply a given amount of force quicker than before. Let me quickly show you the importance of having both good force *and* good rate of force development.

**WHAT FORCE AND HORSEPOWER REALLY LOOK LIKE**

| | **Bodyweight** | **Max force without time constraint (squat)** | **Max force in the Vertical Jump** |
|---|---|---|---|
| **Athlete A** | 175 | 400lbs | 200 lbs |
| **Athlete B** | 175 | 300lbs | 225lbs |

Assuming athlete A and B are both the same size, you can see how they have very different strength patterns. Both of them weigh 175 lbs. Now look at the row that says "maximum force or strength without time constraint". All we're describing here is how much force these athletes can put out regardless of how long it takes them to apply that force. A maximum squat is an example of this, since we have ample time to generate max force in a squat. Power-lifting, arm wrestling, and tug-of-war are some sports that come pretty close to measuring maximum force. In practically every other athletic

event the movements occur so quickly there isn't enough time to allow true maximum force to be developed. In this example you see that athlete A reaches a higher peak force and squats more weight, 400 lbs versus 300 lbs. Yet if you look at the 3rd row, the amount of force he can put out in the vertical jump, athlete A's force output is lower than that of athlete B. Thus, his rate of force development is lower. Therefore, athlete A is going to be able to squat more than athlete B, but athlete B is going to smoke athlete A in a Vert.

So, how much force you can develop in a short period of time can have a big-time effect on performance, which explains why NOT ALL people that are strong in the legs can jump high. Having said that, you still need to have enough raw horsepower (or raw force) to tap into for anything significant to happen. The 200 lb guy with a max squat of 100 lbs is likely not going to be getting very high even if he can apply all that force very rapidly.

Here is an example of what that very weak athlete might look like on paper when we break his strength qualities down like we did above:

| | **Bodyweight** | **Max force in the squat** | **Max force in the Vertical Jump** |
|---|---|---|---|
| **Athlete A** | 150 | 100 | 95 |

Even though this athlete is able to utilize 95% of his force potential (95 lbs) in a jump, he still isn't able to develop enough baseline horsepower to tap into for that awesome rate of force development to do him much good.

Now, here is an example of what an *ideal* athlete's maximal force and rate of force development profile might look like:

| | **Bodyweight** | **Max force (strength) in squat** | **Max force in vertical jump** |
|---|---|---|---|
| **Ideal Athlete** | 175 | 400 lbs | 325 lbs |

This athlete is very strong and is also capable of utilizing a large percentage of his max force in a very short time-span, which is ideal. His max squat is 400 lbs. and he's able to utilize over 75% of that, or 325 lbs., during a jump. *** Keep in mind this example is for illustrative purposes only – It's doubtful you'd ever see any athlete utilizing 75% of their maximal force potential in a vertical jump – typical would be closer to 50% as will be discussed in great detail later in the explosive strength deficit chapter.*

## More on Rate of Force Development….

A better illustration of rate of force development as it relates to an athletic event like the vertical jump is to think of the difference between a big truck or tractor and a high performance race car. A big truck or tractor have a lot of horsepower and torque (strength), but they produce their power at pretty low RPMs. A typical 550 HP 18 wheeler will produce the majority of its power down around 1500 RPM. They are great operating at low rpms and have a ton of strength (horsepower) - they can pull 50 tons or more. But when you get them up higher in the RPM range and step on the throttle you don't get much response - there's just nothing there. If you were driving an 18 wheeler at 65 mph on the freeway and stepped on the accelerator you'd accelerate at a snail's pace – with or without a full load. In contrast, think of a high performance race car. At 1500 RPM it's just barely getting warmed up. Take it on up to 2500 RPM and step on the throttle and you still get an immediate burst. What we want is to have high horsepower and lots of strength but we wanna be able to develop force at low, medium, AND high RPMs - to continue ramping up force even while the speed of movement continues to increase.

So, having good rate of force development is really about being able to produce quality horsepower (force) at higher and higher rates of speed. As a practical illustration try this example: Take your index finger and pound it on your table pretty hard - hard enough that it makes an audible sound and hurts a bit. Pay attention to how hard you hit the table. Now try this: Take that same index finger and, holding it just half an inch or so away from the table, tap it as fast as you can. Next, continue at that same rate of speed but now also try to tap it as HARD as you can. You'll notice you can't hit it near as hard as you could in the first example simply because you're now being forced to try to develop high force at higher RPMS (speed of movement). As speed of movement increases the force potential decreases, BUT, the goal is to be able to develop as much force as you can even at higher speeds. That's what increasing rate of force development (RFD) is all about.

Some people are more like a truck and some people are more like a racecar. The "truck" type individual is the guy that can squat 500 lbs but can't jump 20 inches. He can produce a heckuva lotta horsepower as long as the weight is moving slowly (low RPMs) but when the speed (RPMs) pick up, he's not so great at generating force. We will talk plenty more later on improving the rate of force development, but first I want to explore in a little more detail the concept of building strength for the Vertical Jump:

## What Strength Really Is…

Let's dig a little deeper and look at what really goes on at a physiological level: Strength is really just another name for the ability to produce tension, or force.

**Strength=Tension or Force**

Strength is made up of 2 parts: One aspect is determined by how efficient you and your nervous system are at firing and coordinating the muscles involved in a movement, which is called ***neural***

***efficiency***. The other main aspect is how big the muscles are that are fired, which determines how much force is generated when they fire.

Put those 2 things together and you have ***muscular strength***. So, you can get stronger either by boosting neural efficiency, or by increasing the size of your muscles. First let's talk about improving the neural aspect of strength.

## Neural Aspects of Strength

There are two primary ways the nervous system influences your muscular strength. The first process is called ***motor unit recruitment***. Specifically, I'm referring to your nervous system's ability to turn on and fire more motor units. A motor unit is just a grouping of muscle cells or fibers. A given motor unit may contain a few muscle cells, or it may contain several hundred. A given muscle will consist of hundreds of motor units each consisting of hundreds of muscle fibers. When you decide to contract a muscle a message goes from your brain and down your spinal cord where it eventually reaches and signals individual muscle motor units to fire. When a motor unit fires so do all the muscle cells under its control. The more motor units (muscle fibers) you recruit, the more force you'll produce.

Small force tasks (such as lifting a pencil) recruit few motor units; large force and/or explosive tasks recruit many motor units. Full motor unit recruitment for a given muscle group occurs when maximal force output reaches around 80-85% of your maximum. So, if your 1 repetition maximal leg extension is 100 pounds and you perform a set with 80 pounds (80%) you'll be recruiting, or firing, all of the motor units in your quadriceps.

However, it's also safe to say that under normal circumstances few people are capable of utilizing all of their potential strength in a given movement. In fact, an untrained person may only be able to utilize 50% of their strength potential. Why is that? Because there's another aspect of neural efficiency called rate coding. ***Rate coding*** is what I like to call the *psycho factor* – because it's largely influenced by psychological arousal, the excitability of the central nervous system, and your ability to increase and respond to adrenaline levels.

Rate coding allows your muscles to develop more force by enhancing the frequency at which neural signals get sent to your muscles telling them to contract. At very high efforts a given motor unit will continuously fire and relax and repeat that process at a very high rate of speed. The repetitive firing of all available motor units occurs so quickly that there is a summation of force and the ability to produce tension is magnified beyond what you get from regular recruitment and muscle protein content. In essence, your muscles get supercharged and are able to generate tension above and beyond what they normally would. However, the body normally inhibits the full potential of this process as a protective mechanism to protect you from injuring yourself. If your body didn't have this safeguard in place and you could easily call upon your full strength potential from the get-go you'd probably also stand a good chance of ripping your tendons right off the bone!

A few examples where you see this protective mechanism naturally over-ridden are in extreme life or death type circumstances where the body produces tons of adrenaline. If you've ever heard

stories of people lifting cars up off their children what happens in these situations is the extra adrenaline boosts rate coding and over-rides various mechanisms that normally inhibit the display of full force potential. But these people often end up injuring themselves. Some people have a natural propensity to have elevated adrenal related discharges from the CNS and naturally have better rate coding.

**** Fortunately, with training, one can vastly improve this capacity naturally, which is in large part why elite athletes are able to do what they do.*

Rate coding correlates STRONGLY with rate of force development. This extra motor unit recruitment from adrenaline also explains why people tend to be stronger, more powerful, and faster in competitive situations. For example, a powerlifter will tend to deadlift a lot more weight in a meet than in the gym. A basketball player will tend to jump higher prior to a big game then in training. A sprinter will tend to run faster at a meet than in training etc. Regardless of the endeavor a competitive max can be up to 10% higher than a training max for many athletes.

So, with training, you improve your ability to fire motor units and coordinate motor unit firing (rate coding). That's the major reason why when people first start strength training they gain a whole lot of strength, even in the complete absence of any size increases. Obviously, both motor unit recruitment and rate coding take place when you produce high levels of force, and they are both involved in a vertical jump. Why? Because you need to contract a lot of muscles and do so very quickly. Importantly, the neural gains in motor unit recruitment and rate coding that occur through traditional strength training have a global foundational transference and serve as a foundation for neural gains occurring in speed-strength activities like a vertical jump. In other words, as you increase your strength and learn to powerfully engage your nervous system thru heavy exercises like squats and other assorted lifts, the heavy engagement of your nervous system (recruitment and rate coding) can also carry over into your ability to maximally activate your neuromuscular system when you jump. That's ONE reason why weight training is such an effective stimulus for vertical jump gains – it allows you to *practice* maximally activating your nervous system and in lower level athletes simultaneously improves BOTH strength *and* rate of force development.

## How The Nervous & Muscular System Work Together To Produce Force…

Next, let's talk about how the nervous system and muscular system work together to produce force: Obviously, before a muscle *cell* can contract it has to be recruited, or turned on, by the nervous system. Once it is recruited, it always fires with all of its force. How much force a muscle cell generates when it fires is primarily determined by how much protein is contained in it, or how big it is. Some muscle cells are bigger than others, but how much tension they can generate will be determined by how big they are. How that tension gets expressed will vary based on how you're built and things like bone lengths, tendon strength, and many other factors, but speaking strictly at the muscular level, that's what happens. When you add muscle size, the amount of protein contained in your muscle cells increases and they (the individual muscle cells) get bigger. How much peak tension a given amount of muscle cells can generate will generally be determined by how much space they take up. In other words, a

"slow-twitch" muscle cell might be only half as big as a "fast-twitch" muscle, but 100 slow twitch muscle cells can generate about the same peak force as 50 fast twitch muscle cells if the total amount of space (protein) they take up is roughly the same.

Depending on the inherent speed of a particular movement, factors like the dominant fiber type of a particular muscle play some role into how much tension ends up getting delivered, as fast twitch fibers build up to peak tension quicker. Fiber type plays no role in slower movements like powerlifting, but plays more of a role in very fast movements like throwing, jumping, and sprinting. But to keep things on track, anytime you add muscle size the amount of protein contained in your muscle cells increases and they (the individual muscle cells) get bigger and can potentially produce more force than before.

Thus, the tension generated by a given muscle, such as your biceps, is determined by how many individual bicep muscle cells your nervous system can turn on and coordinate during a movement, along with the total amount of protein (size) contained in those muscle cells being recruited.

***Strength*** = Muscle cell recruitment + Frequency of recruitment (rate coding) + total size of all the muscle cells being recruited

As a practical illustration, let's say you have 2 athletes and you want to measure and compare their strength in the leg extension. Both of them have 1000 total muscle cells in the quadriceps. Athlete B's muscle cells are twice as big as Athlete A's, yet athlete A is twice as efficient at firing and coordinating the muscle cells in his quadriceps:

| | Total Muscle Cells | Size of the Muscle Cells | Total tension if all muscle cells were to fire optimally (potential strength) | Percent of potential strength utilized | Amount of tension generated (weight lifted) |
|---|---|---|---|---|---|
| Athlete A | 1000 | Big | 100 lbs | 100% | 100 |
| Athlete B | 1000 | Twice as big | 200 lbs | 50% | 100 |

You can see that they generate the same amount of tension but through very different means. Athlete A has to take full advantage of his muscular recruitment and rate coding capacity to generate 100 pounds of tension while athlete B, due to his bigger muscles, only has to use half of his neural capacity. Thus, athlete A has twice the neural efficiency of athlete B, but athlete B has twice the muscular size of athlete A. The result is a wash.

Most people are initially a lot like Athlete B in that they're not capable of utilizing all of their muscles in a given task. The more efficient you get at coordinating and firing your muscles, the better your neural efficiency gets. This is how people like gymnasts are able to get so strong for their bodyweight.** They have extreme neural efficiency. Improvements in neural efficiency allow you to bridge the gap between your potential strength and actual strength*** and enable you to utilize more of the muscle you have.

*** From a vertical jump perspective, there is definitely an advantage to having good neural efficiency.*

**** The difference between your potential strength and actual strength is also called the **strength deficit**.*

If athlete B in the above example was capable of utilizing all his strength potential in this task he'd be generating twice the tension of athlete A. In athlete B's case, he could get significantly stronger simply by boosting his ability to coordinate and utilize the muscle he already has. He could do this without any increase in muscle size whatsoever, which would boost his relative strength, or strength per pound of bodyweight. That would then improve his relative power, which would enable him to jump higher. In contrast, the only way athlete A will get stronger (and likely the easiest way for him to jump higher), is by increasing the muscle mass in his quadriceps.

So, the point to take home is that strength can improve either through increased neural efficiency, increased muscle size, or both. When it comes to lifting, performing sets of 3 and below primarily train the neural efficiency aspect. Sets of 6 and more primarily boost the size aspect. Sets of 3-5 do both. There is a lot of crossover and you can't totally restrict gains to either neural or muscular, but that's the basic gist of it. When it comes to other VJ related training methods, plyometrics and explosive weight training train the neural efficiency aspect.

## Building Strength & Exercise Selection….

With that information the foundational role that strength plays in the VJ process should be evident. When it comes to building strength the topic of exercise selection is a topic that really seems to confuse people. You'll find endless opinions and recommendations all touting countless schemes and exercises all supposed to be better than any other. Some people preach only uni-lateral exercises, such as single leg squats and Bulgarian split squats. Some people preach only squats, while others say NEVER do squats. Some people preach either front squats or deadlifts as the cure-all for everything. Some say a person shouldn't lift weights and should instead do something like use bands or push cars. The average person is often left so confused they don't have a clue where to start. I have some opinions and I'm not shy about sharing them, but to be perfectly honest, it really doesn't matter how you go about getting stronger as long as you use movements that engage the correct musculature and get it done somehow. You can build strength with squats. You can build strength with deadlifts. You can do it with front squats. I've seen people do it with bodyweight single leg squats. Others like Bulgarian split squats. I've seen people use banded squats and banded deadlifts.

A stone-age setup would include 2 exercises, the squat and calf raise – and those exercises work just as good today as they did 40 years ago. But at the end of the day all that really matters is that you're improving your ability to bend your knees, extend your hips, extend your ankles, and apply force. You're strengthening the muscles of your hips, quads, calves, hamstrings, and lower back. There are a myriad of ways to do that. The most common and some of the most effective exercises are the aforementioned squats, front squats, lunges, bulgarian squats, and deadlifts. Those movements inherently hit the knee and hip extensors hard and that's where the majority of your power in the VJ comes from. If you want to kick things up an extra notch some form of hip extension exercise like a

Romanian deadlift, barbell hip thrust, or reverse hyper, along with some type of ankle extensor exercise (calf raise).

There is research indicating athletes with a superior vertical jump get more out of their hip extensors (the glutes) than normal athletes and in my experience more glute strength never hurt anyone. Thus, in my programming I prefer some type of squatting based movement (squat, front squat, Bulgarian split squat, lunge), some type of hip extension/glute dominant movement (Romanian deadlift, barbell hip thrust, reverse hyperextension), and depending on the athlete, some type of ankle extension dominant movement (calf raise).

## Give me ONE Exercise?

As I've stated, my attitude when it comes to exercise selection is kinda like a mechanic working on a motor – it doesn't matter how fancy the wrench is as long as it gets the job done. Yet I do have my opinions as to what tools I *prefer* to use. When it comes to exercise choice for building strength specific to the vertical jump, I prefer to keep it simple. **THE** time tested best exercise for an athlete interested in vertical jump development is the basic barbell squat. It's a compound exercise very similar to the vertical jump that strengthens the exact muscles we use when we jump. There are plenty of people that lean away from squats for one reason or another, but if you only had **ONE** exercise to use you could certainly get all you needed to get done with the squat. Having said that, some people have issues such as back problems that prevent them from squatting. For these people exercises like front squats or bulgarian split squats can provide a suitable replacement. Strength exercises should be performed in a controlled manner with an emphasis on technique. Proper technique gives an optimal balance between leverage and safety and should always be used.

## What About Squatting Depth?

The issue of squat depth is another issue that always seems to arise arises anytime we're talking about the vertical jump. Some people say full squats are the only way to go while others say half squats should be used because they're more specific to the vertical jump. After all, when you jump you only squat down about ¼ to ½ way down. There are a few issues here that I'd like to bring up before getting into my recommendations:

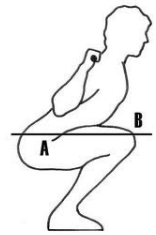
A: **Strengthening muscles vs movement**: Are you trying to make the muscles involved in your vertical jump stronger or are you trying to get better at squats? A deeper squat involves more of the gluteal and hamstring muscles and more fully stimulates the quadriceps. Additionally, the shallower you squat, the less overall muscle mass you stimulate. Keep in mind a loaded barbell squat is a general strength exercise not a specific exercise. General strength exercises build "general" strength in the muscles which can then be applied specifically. If you were trying to turn the squat into a specific exercise for the vertical jump motion you'd be best using exercises that are more specific as far as velocity - like a jump squat. Putting all your focus on 1/4 squats because they're more specific for the vertical jump is kinda like doing isometrics on a leg extension machine to train for a butt kicking contest - it's the wrong velocity and wrong movement.

B: **Safety:** The shallower you squat the more weight you have on the bar and the greater the likelihood for injuries. A 1/4 squat will often be a good 100 lbs heavier than a legal depth squat, if not more. Many people interested in jumping have frail builds to begin with and having them constantly put a super heavy load on their back may create issues. It's for that reason that Yuri Verkoshansky preferred deeper squats for jumpers.

C: **What exactly is your definition of a full squat or half squat?** In my experience there's a big disconnect between what people actually call a squat and what type of squat they actually do. What most people call "full squats" are actually half squats and what most people call half squats are actually quarter squats. Go into a high school locker room and tell everyone to do half squats and film what you see. I can promise you'll be able to count the actual number of good half squats on one hand. By far most athletes will be doing 1/4 squats and 1/8 squats. By telling people to perform squats thru a fairly full range of motion you automatically have a built in buffer against the typical tendency people have to reduce depth on their squats. I can tell 100 people to do legal powerlifting depth squats. Of those 100 about 20 will actually do them properly, 60 of them will do somewhere between a legal and half squat, and the other 20 will find some excuse to do 1/4 squats because deeper squats hurt. I'm not making that up all I've actually tested it!

## So, what's my recommendation?

With all that said, I prefer a legal powerlifting depth squat where the hip joint breaks slightly below the knee joint, or a squat to a depth that the back starts to round, whichever comes first. I see no reason to do full squats UNLESS you prefer doing them. If you can't break parallel without your lower back rounding stop just short of that point. If you can break parallel without your back rounding than do so. Doing squats in this fashion eliminates a lot of guesswork and makes them easy monitor.



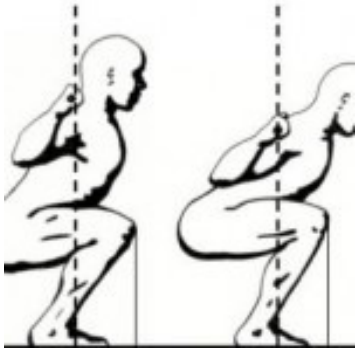

***(note how the crease at the hip (A) temporarily breaks below the top of the knee (B)***

In my experience the average novice trainee will immediately gain a few inches of vertical jump just by performing deeper squats to just below parallel (legal powerlifting depth). That's with no other changes. The extra range of motion stimulates so much more muscle mass they pay off that quick.

## High bar or Low bar Squats?

I'm also not overly anal about whether one performs a high bar squat or low bar squat – use whichever one you prefer. Most people feel more natural with a high bar squat and it does have the advantage of engaging a bit more quadriceps – that can be either an advantage or a disadvantage. If an

individual has overactive quadriceps it can be a disadvantage and a low bar squat might be a better option – but most beginners can use more quadriceps activation.



**High bar vs low bar squat (note the difference in forward lean caused by the different bar positioning)**

## What about stance width?

I do prefer Vertical jump trainees use an athletic stance – about shoulder width apart or a tad wider than shoulder width. Ultra wide stance squats are great for powerlifting, not so great for jumping.

## What about speed of contraction?

Trainees are often instructed to attempt to move the weight as quickly as possible in the lifting phase of a movement, even when lifting heavy weights with the purpose of gaining strength. The idea is you lower the weight under control then explode up. Obviously, for something like a jump squat or a lighter explosive squat with 50% of your max, you WILL move the weight quickly because that's the primary aim of these exercises. Yet, when lifting heavier weights to gain strength, the thought is that even though the weights inherently won't move that fast one can still increase their ability to apply force quickly by *attempting* to do so. There is validity to this and I used to make this same recommendation myself. However, in my experience too many trainees sacrifice form at the expense of attempting to move heavy weights quickly. Additionally, we really don't *need* to attempt to move heavy weights quickly to get what we want out of these movements – their primary stimulus comes from muscular recruitment and the nervous system arousal involved. Thus, my advice is attempt to use good form above all else – use a controlled lowering phase and a controlled lifting phase. If your form is 100% perfect THEN you might look to be a little more explosive with a heavy weight.

## More On Partial Range Squats

Now going back to quarter squats and other partial range squat variations: I know I was pretty hard on them but having said all that, there are 3 times where I feel very shallow squats like quarter squats can be of use. These are:

1. **To peak out strength**: You can occasionally use them at the end of a strength cycle to help peak out your strength. You'll have more weight on the bar which gets you used to lifting a heavier weight.

2. **Nervous system stimulation**: A partial range squat, such as a quarter squat, can be an effective stimulation method because the weight is heavier. This extra load can serve as a decent neural stimulus and this can make them valuable during power and peaking cycles. You engage the nervous system strongly with these overload methods and that stimulation can *carry over* into your specific jumping work.

3. **To unload the legs**: Even though there's more weight on the bar the legs don't have to work as hard in a 1/4 squat. This can make them useful at times when you want to maintain strength but reduce fatigue in your legs, such as during peaking cycles. The workouts in this manual go over how I like to incorporate 1/4 squats.

What about deadlifts? Some people like deadlifts and deadlift variations like trap bar deadlifts. They can be a valuable exercise but there's only one problem: Their value is very form dependent. It's possible to deadlift significant weight with hardly any lower body involvement at all, as people can simply straighten their legs and lift the weight with their spine. In my experience this happens too often for my tastes. The deadlift also doesn't have the same degree of quad activation that a regular squat does. It also takes longer to recover from. For those reasons I shy away from it unless I'm 100% sure I know an athlete is performing it correctly. BUT, if you like it and can perform it correctly there's no reason not to use it, or at least mix it in intermittently.

The verdict: Whether you prefer a basic back squat, front squat, bulgarian split squat, or lunge you can use any of those lifts as a foundational lift for VJ. The deadlift is kinda iffy - take it or leave it.

## What about the calves?

The vertical jump is a combination of knee extension, hip extension, and ankle extension. We place a LOT of emphasis on strengthening the knee extensors (quads) and hip extensors (glutes), but what about the calves? Shouldn't we strengthen them just as much for the same reasons? The answer to that is YES we should, at least to a certain degree, but here's the deal: In the past many vertical jump programs and gimmicks focused solely on the calves to the extent that all the other more important muscles were neglected. From 1990 thru 2005 the majority of vertical jump routines and gimmicks were platform shoes, jumpsoles, and extremely high volume plyometrics. These routines were all calf/ankle dominant, and most of them weren't all that effective.

The key is to train the calves as a SUPPLEMENT to everything else. Having strong, powerful calves will often HELP the vertical jump. That doesn't always mean people with big muscular calves jump the highest, in fact the opposite is usually true. People with big calves often tend to be quite heavy and slow and often carry a build that negates optimal motor recruitment patterns. They often tend towards glute amnesia and extreme quad dominance. However, having stronger calves can only help you. Some research has demonstrated the calves can contribute 50% or more to certain styles of jump, mainly unilateral jumping. Typically they might contribute 15 to 25% to most style of jumps, more or less depending on the individual.

There are also a few arguments away from specific calf training: Any activity you perform on your feet can strengthen your calves to a degree. If you perform a lot of high intensity plyometrics your calves can get VERY STRONG from that alone. In fact, some coaches like Charlie Francis didn't advocate any calf work with athletes because the thought was sprinters get enough of it from all the sprints and jumps they do. Excessive calf hypertrophy can also add excessive weight (in people prone to growing large calves), which can negate optimal movement efficiency. The calves contract thru a fuller range of motion during normal activity than the other prime vertical jump muscles (glutes and quads). Just going out and doing 30 seconds of jump rope, sprints, or an assortment of various plyo drills is enough to give most people a significant calf pump. Where do you think that pump comes from? Muscular activity!

Having said all that, a degree of general strength in the calves is valuable and important and many people can benefit from it. Movements like plyometrics may be high tension, but they don't have the duration of tension that builds good baseline strength. Strengthening your calves with high loads can easily and efficiently help stiffen up your ankles and in this sense being "stiff" is a good thing. My opinion is an athlete should strive for enough baseline calf strength to perform 20 controlled reps of barbell calf raises or exercise equivalent with 1.5 x their bodyweight, or 20 single legged calf raises with bodyweight. It shouldn't take much volume to achieve that, and in my opinion calves don't need to be worked with weights year around. A few sets of 15-25 reps on a calf raise (or equivalent) twice a week in a strength phase should be enough to give anyone the baseline calf strength they need. You can see how I implement baseline calf training throughout the workouts in this manual.

Additionally, the calves are 1 muscle group that respond really well to stretching. Not only does increasing their flexibility increase your movement efficiency, but a recent study verified that daily stretching of the calves also helped slightly with strength increases. Therefore, if you’re interested in improving your vert, I recommend you stretch your calves daily for at least 20-30 seconds. Stretching them multiple times a day is even better.

## Must We Train Fast For Explosiveness?

One debate that often arises between coaches and athletes is whether basic heavy strength training movements like squats and deadlifts with heavy (80% + loads) are superior or inferior to lighter weight, high speed strength training movements (also called power movements) such as olympic lifts, speed squats, jump squats etc. Really, there is no doubt that the heavy strength training movements are far superior when it comes to increasing strength. The only real way to increase baseline levels of strength is to lift a fairly heavy load (70-100% of 1 rep max) which creates high amounts of intramuscular tension. The *duration* of high tension is largely responsible for strength increases. When lifting such a load, the weight does not move very fast, because it is obviously too heavy to move all that fast. However, some say, “Well, since our objective is to move fast on the field, we must move fast when we train!” This leads some to favor using loads with 20-60% of their 1 rep max on basic exercises such as squats, and performing the lifts with great speed.**

This is otherwise known as power, or *explosive strength training*. This type of training can help an athlete learn to express his strength more quickly, yet before an athlete can express strength, he has

to have some strength to express, which is one reason why one who hasn't achieved a minimal level of strength should focus on the heavy basics in the weight room and for the most part seek to minimize the lighter weight high speed variations.

*** As will be discussed in detail in a later chapter, the Olympic lifts such as the power clean and snatch are inherently high-speed power movements so one need not train with lighter percentages on these lifts to be in the "power- training" zone. Even an 80-90% snatch or clean, although heavy, still must be performed fast or it simply won't go up!*

Another way of looking at it is to think of basic strength as the size of an engine and explosive strength, power, rate of force development etc. as the modifications you can make to that engine to make it run faster, or express its horsepower better. You can make a smaller motor run faster by boring out the cylinders, inserting high tech spark plugs, running special fuel, and doing a ton of other high tech things. However, if you don't have a big enough motor to start with in the first place you can do all the modifications you want but it won't do you any good! A weak athlete choosing lots of high speed lighter weight training movements over basic strength movements would be like someone trying to modify a stock issue Honda Civic with the idea of competing against F-1 race cars! No matter how many modifications you make there's simply not enough basic horsepower to compete! Therefore, it's much more economical for an athlete to spend the time laying down a strength foundation before attempting to get overly "cute" in the weight room trying to better express strength that he doesn't even have. That doesn't mean one interested in acquiring strength for the purpose of boosting their vertical jump should ignore or totally eliminate all forms of ballistic and power oriented training – those variations can and should be performed at least at a maintenance volume, but the *focus* should be on acquiring the needed basic horsepower.

What's also debatable is whether or not performing lighter weight exercises such as olympic lifts, speed squats, and jump squats can offer an athlete any extra ability to express strength that an athlete wouldn't get from simply participating in sport. In other words, for someone who wants to jump, why shouldn't he just get strong in the weight room and "jump" for his power training and specific transference? Since sporting movements are already faster and more specific than any explosive movements that can be performed in the weight room, what's the point of trying to work on learning to express strength better in the weight room? Why not just take the straight line approach and build the size and horsepower of the motor in the weight room, and let speedier activities such as jumping, plyometric work etc. take care of the conversion and modifications? Is there any need for specific explosive work in the weight room if one is engaging in these activities? Some say yes and some say no. It's an interesting argument.

Personally, I usually do recommend some explosive weight room training like jump squats, lighter box squats, and Olympic lifts, but it also depends on what other stressors (games, practices etc.) an athlete has to contend with. If I had to choose one or the other I tend to lean more towards the camp that says the weight room should serve as a place to develop strength, while the sport and other activities more closely resembling it (sprinting, jumps, etc.) should be used as the place where the athlete teaches his system how to demonstrate that strength quicker. In other words, if you're strong but have a hard time expressing your strength (you're slow or lack rate of force development), one of

the best things you can do to gain that ability is engage in lots of jumping type activities. If you need to learn to express your strength better in the jump the single best thing you can do is jump.

Having said that, from a loading standpoint there is some value in performing specific explosive variations in the weight room. I'll discuss these advantages in detail later, but one of the advantages is that the explosive variations are inherently less draining than heavier movements, and offer an athlete a chance to stimulate the body without causing excessive drain. Heavy strength training induces a lot of neuromuscular fatigue and can take quite a bit of time to recover from.

## *Why Those Squats Make Temporarily Make You Feel Slow……The impact of Fatigue…*

The issue of fatigue is one reason why people may not always feel as explosive or springy when they're engaging in a lot of strength training. When an athlete is really working to peak, or demonstrate, his explosiveness, he wants to be as fresh as possible. As mentioned, the heavier strength movements can cause a lot of neuromuscular fatigue, and that fatigue can temporarily mask his fitness state. Athletes will often complain they're not jumping well as their strength is going up, yet as soon as they reduce the volume of strength work they remove a lot of that fatigue and suddenly VOILA...they're running on water and jumping out of the gym! They got a lot stronger and were getting more explosive from the strength training, they just weren't able to properly ***display*** that explosiveness until they removed some of the fatigue. Fatigue masks fitness. They didn't get more explosive from eliminating the squats, they got more explosive by removing the fatigue the squats were creating in their legs. Sometimes you have to temporarily take a step back in order to take a step forward.

You MUST consider the impact of fatigue. For the jump trainee fatigue and fatigue management should be looked upon as qualities just as important as your strength, power, reactivity etc. Fatigue can mask fitness and you have to be aware of that and know how fatigue impacts you and know how to manipulate it. A lot of times athletes think they need to do this or that when all they really need to do is manage fatigue better.

When you lift heavy weights with any sort of volume you strongly activate both your nervous and muscular systems. You also create a lot of residual fatigue in the muscles as the activity is prone to create tissue micro-trauma (damage). The junctions between your nervous and muscular system become fatigued, so a given amount of *charge* from your nervous system fails to activate the muscular system to the same degree. Where it's most readily apparent is performance in high speed activities, such as jumping. When your nervous system is fatigued the symptoms will become apparent in faster activities before they become apparent in slower activities. So, your lifts may continue going up even though your jumps go down. So, how do you fix this? There's really only one thing you can do and that's cut down (not necessarily remove completely) the volume of heavy lifting and let your nervous system and muscles freshen up. It doesn't take long - a few days to a few weeks at most. But once you remove the fatigue then you can fully display your fitness (explosiveness). The same thing occurs in athletes that

do a lot of conditioning or cardio work. Those activities create a chronic strain on the neuromuscular system that will negatively impact the display of high speed explosiveness.

When an athlete really wants to temporarily peak out his explosiveness he can simply replace some of the heavier strength movements with lighter more explosive variations so that he can remove some of the fatigue he accumulates. This will allow him to really demonstrate his true explosiveness. If an athlete had an important upcoming testing date then somewhere around 1-4 weeks prior to the test date I'd taper the strength training down to a low maintenance level (a few heavy sets of 1-3 once per week) and replace much of the strength volume with either more explosive weight room work. An athlete who is already strong but who really needs lots of work on rate of force development and movement efficiency work could also benefit from less heavy weight training for the same reason. By not creating excessive fatigue he'd be better able to direct his energy towards improving those qualities.

Tolerance to lifting induced fatigue tends to be variable in my experience. Some people are impacted BIG TIME by heavy or high volume squatting. They don't jump well with any sorta squat volume. On the other hand, other guys will repeatedly hit many of their best jumps right during the middle of a concentrated strength block. In my experience the main variable there is the level of strength and natural levels of speed . Very strong naturally slow people don't jump acutely well when they're in the midst of a high volume of strength training. Weaker, naturally faster people, often THRIVE on it. Additionally, a very strong lifter (2+X bodyweight squat) will generally need to cut back on his strength work more severely than a weaker individual, who probably shouldn't cut back on it at all.

But anyhow, pure explosive training (plyos, explosive lifts) tend not to be all that fatiguing, so you can focus on those anytime you want your nervous system to freshen up. You can usually just cut back on the high fatigue activities but you can go even more extreme and pull it all out. You probably wouldn't want to eliminate ALL strength work for long, but one simple way to tell if fatigue is masking your jump related fitness is to temporarily pull out all strength and conditioning work for 7-10 days or so. That's not enough time to lose much in the way of what you gain from those activities, but you can then assess your situation and state of fatigue.

A more realistic way of de-loading from strength work is to simply use the 1/3 rule: It takes about 1/3 the volume to maintain strength gains as it does to gain them, providing intensity (load) is maintained. So, you can simply reduce your strength work by around 2/3 and look to maintain for a while. This could be as simple as performing 5 sets of 1 rep once per week – an approach I like to use.

You should also design your cycles with this in mind. Unless you’re a beginner, don't expect to consistently hit jump PRs in the middle of a high volume dedicated strength phase. If you CAN jump well during the middle of a concentrated strength phase you're way ahead of the ball game and way ahead of the fatigue management curve. If you’re getting stronger, but your vert is going down, that's a good indication you need to back-off on the lifting volume to fully manifest your explosiveness.

## Basic Strength Training Principles & The Importance of Tension

Now I'd like to talk a little bit about setting up a strength training routine. Honestly, you could start reading everything there is to read about strength training and program design today, and 5 years from today you still might not feel totally confident about what you're doing, simply because there are SO many ways of doing things and none are really right or wrong. Methods are many but principles are few. At the end of the day all that really matters is that you're applying progressive resistance (tension) to your musculature. The body really does not know whether you're doing a Kelly Baggett, Westside, HIT, gymnastics, band, kettlebell, or any other system. It only knows tension!

Most training schemes do provide some stimulation and no routine is perfect. I like to tell people to imagine yourself out on a deserted island without any technology, tools, or anything. Strength stimulation in this situation would consist of dealing with everyday life (chasing prey, running away from predators, lifting rocks to build a hut etc.) You could take an athlete today, put him on a deserted island, and he could stimulate performance improvements without a single modern day tool to work with or any specialized strength training knowledge - his life would depend on it. Having said that, I'd like to give you some general principles or guidelines to follow as far as frequency, volume, intensity, and content of strength work:

1. **Frequency for beginners:** A frequency of 3 times per week with approximately 70-80% of 1 rep max has been shown to work optimally for beginners. I like to have beginners do 3 sets of 5 to 8 reps on a basic exercise like the squat 2-3 days per week. Nothing fancy, just try to add weight or reps most sessions. Most beginners can progress for months on that simple setup.

2. **Frequency for more advanced athletes:** For intermediate or more advanced athletes a frequency of twice a week per muscle group with a bit more volume works well. A typical set/rep scheme would be something like 4 sets of 5-8 reps at 80% or heavier of 1 rep max. Athletes engaged in lots of practice, games, or other work can even progress just fine with an exposure of once per week.

3. **Load:** When it comes to how much weight to use (intensity), strength responds best to loads between 70 and 100% of your 1rm. That generally means you perform anywhere from 1 to 15 reps per set. The more advanced you become, the better you tend to respond to lower reps and heavier weights. If strength is so important shouldn't we just do singles? You certainly can train with singles, but it's not necessary. When engaging in strength training for the VJ we're trying to optimally recruit and strengthen the muscles involved in the movement. Training with extremely low reps can interfere with that recruitment, because the weight is so heavy form can break down. You might be able to lift more weight by loosening up your form a bit, but that's obviously not smart. It's typically easier to use correct form with slightly higher reps, and you also get the benefit of more time under tension. Therefore, most of the time I recommend sets of 5 or 8 rather than sets of 1. Always terminate a set as soon as your form starts to break down. You can stop before that point, as it's not necessary to train to complete failure. I generally recommend stopping a couple of reps shy of failure.

4. **Volume:** When it comes to volume, there really aren't any strict minimal or maximal volume rules, but there are guidelines. Generally speaking, the lower the reps, the more sets you'll want to perform. Five sets of 1 rep is a lot less volume than 5 sets of 10 reps (Five reps vs 50). If you don't feel like counting sets one simple way to monitor volume is by the drop-off method. Work up to a hard maximal effort for a given number of reps. Let's say you work up to 100 pounds for 5 reps on a given exercise. Keep performing sets with the same weight until you can no longer get 5 reps. Simple but effective. This works particularly well for pure neural-related strength gains (relative strength). For neural and muscular (a.k.a. size) related strength gains, which do require a fatigue component, you might work up to a hard effort and stop when your performance drops off by more than a couple of reps. So, using the above example of working up to a hard set of 100 pounds for 5 reps, you'd continue to perform sets until you could only perform 3 reps.

5. **Type of movement:** When it comes to the composition of a strength training workout, compound multi-joint movements such as squats are superior to isolation movements such as leg extensions. One exercise per major muscle group is generally sufficient. Compound movements should be performed prior to isolation movements, and faster movements should be performed prior to slower movements. Thus, a typical VJ workout might consist of depth jumps, jump squats, heavy squats, Romanian deadlifts, and calf raises, in that order.

6. **Percents:** When it comes to percentages, or how much load to use relative to your 1 rep max, I generally recommend basing your loads on effort rather than percentages. In other words, if a scheme calls for you to do sets of 5, simply work with a weight that allows you to complete about 5 reps in good form and increase weight when you can.

7. **Periodization:** This will be covered in detail in a later chapter, but beginners should keep things simple and follow basic linear periodization. Don't make your routines too cute just select a repetition range and number of sets and try to add weight when you can. For example, say your routine calls for you to perform 3 sets of 5 reps on the squat 3 days per week. Simply work up to a fairly hard set of 5 and perform 3 sets with it. Once you get all 3 sets of 5 with a given weight increase the weight by 5-10 lbs the following workout. Continue doing that as long as you can.

People that have been training for a while tend to note slightly better gains by varying the sets and reps on a weekly basis in a step type loading approach. You slightly increase or cycle the load up and down for several weeks, then take a step back to allow recovery to take place. Once every 3 to 6 weeks you'll generally want to have an "easy" or unloading week, where you reduce the volume by about 40 to 50%. I prefer a 4-week cycle for most athletes. Specific set and rep schemes can vary immensely based on what your goals are and what you like to do.

A weekly set and rep scheme for an intermediate might look like this:

Week 1: 3x6
Week 2: 4x5
Week 3: 5x4
Week 4: 3x3 (with week 1 weight)

There are countless ways to set things up based on this principal of step type loading, or undulating periodization, but the general theme is a variance in sets and reps. I prefer to increase the weight and fluctuate the volume on a weekly basis, but there are literally hundreds of ways of approaching it. Just keep in mind regardless of what you do or how you go about doing it, when it comes to building strength you're increasing your ability to exert force.

8. **Rest intervals:** One mistake many beginning trainees make is they try to speed thru their workouts. The shorter you rest between sets, the harder you "feel" like you're working, because your muscles burn and you keep your cardiovascular system activated, but this isn't a positive thing for vert specific gains. When training to boost the VJ, you're not trying to build endurance or power endurance, you're trying to boost your maximum VJ. Thus, in essence you should treat all your exercises like a weightlifter or powerlifter would. Perform a set, rest long enough to recover back to 100%, then perform another set. If in doubt rest a little longer than you think you should. This goes for plyometrics too. For heavy whole body exercises like squats, you might rest as long as 8-10 minutes between sets. Five minutes is typically a minimum for squats, except on warm-ups. For lighter more isolated movements, like simple plyometric drills, you might rest a minute or 2 between sets. Just make sure anytime you train that you're resting long enough between sets that you can give a good quality effort each set.

# *Muscle Mass Increases For a Jumper?...*

While we're on the subject of strength training, I wanted to touch on something that can accompany that, and that is muscle mass. Look at the lower body hamstring and glute development of someone like a Lebron James or Dwayne Wade in comparison to that of a normal person. Quite a difference isn't there? Lots of people are born with lots of muscle cells, good muscular development, and lots of strength in certain areas of their body, such as the glutes. Others are gonna have to work to add muscle in the right areas, so that they can generate more force from key muscle groups. In other words, if you naturally have a butt like a pancake and legs resembling toothpicks you're probably gonna have a hard time generating much force by those muscle groups until you put some muscle on them, regardless of how neurally efficient you are. I'd like to give you a real world example of how muscle mass increases can help a vertical jump, but first let's talk a little more about muscle fiber typing and its importance to the vertical jump:

## What's The Deal With Fiber Type?

The proportion of fast twitch fibers a person carries is often something given quite a bit of attention as far as vertical jump potential. It is true - in men vertical jump performance correlates with fast twitch fiber number. Most people carry around 50% fast twitch and 50% slow twitch fibers in a given muscle, such as the thighs. Yet some people are as much as 80% fast twitch, while others are as much as 80% slow twitch. Both extremes are represented by those of African ancestry: West Africans dominate sprinting events and are typically very fast twitch – around 80% or so. East Africans (Kenyans) dominate endurance events and are typically very slow twitch – around 80% or so. Fast twitch people usually exceed at speed-power events, such as sprinting and jumping, while slow twitch dominant individuals succeed at endurance activities, such as the marathon.

It's long been thought there's not a whole lot you can do to alter this, as you can't change a slow twitch fiber into a fast twitch fiber and vice versa. Thus, your fiber dominance is pretty much genetic, and that is purported to make your potential in events such as jumping and running largely genetic. Well, I don't think it's anywhere near that simple. Before I explain why, let's take a deeper look into muscle fiber typing:

*** Interestingly enough, in women slow twitch fiber number correlates better with power production than does fast twitch. This is due to the fact that exercise usually increases slow twitch expression, whereas sedentary activity increases fast twitch expression. More active females have more slow twitch fiber than average due to being more active, but are also more explosive than their sedentary counterparts.*

Fast twitch muscle cells can produce tension a bit quicker than slow twitch, but the main difference is they're bigger. In other words, if you take 100 grams of slow twitch fiber and 100 grams of fast twitch fiber they will produce the exact same amount of overall maximal tension (strength). This explains why fiber type has no relevance in sports like powerlifting where speed of contraction plays no real relevance. The main difference between the fibers is the fast twitch build up to peak tension quicker, giving them an advantage when rate of force development is more critical, as it is in movements such as sprints and the vertical jump.

However, there are a couple of other things to consider: First of all, one with more fast twitch fibers is more than likely going to carry more overall muscle mass period than one with more slow twitch fibers. More muscle mass means greater natural horsepower. Even a guy like Usain Bolt, a relatively "thin" individual, carries a LOT more muscle mass in his thighs and glutes than the average person - especially an average person of his height and weight. The same is true of most good speed-strength athletes - they may not be large, but their relative development typically favors speedier muscles such as the glutes and hamstrings, which will tend to be naturally well developed compared to the rest of their physique.

Additionally, there is a relationship between testosterone, fast twitch fiber, and body structure. One with more fast twitch fiber is also more than likely to have higher testosterone levels (and more muscle mass) and higher testosterone favors athleticism both on a neural and structural level. Whether thru testosterone, natural selection, or some other undiscovered variable, one with more fast twitch fibers is more than likely to have an "X" frame physique, which is a structure well suited for speed-strength activities. An x-frame physique includes characteristics like broader shoulders, thin hips, low body-fat and higher cut calves (longer achilles). ** As an observational experiment sometime look around at different people and note the correlation you see between different physical characteristics. An x-framed physique also naturally favors other speed-strength correlates, such as enhanced glute dominant movement patterning.

*** Interestingly enough, in women an x-frame physique seems to inversely correlate with calf size, but positively correlates with breast size. In other words, women with big calves are more likely to have narrow shoulders & small breasts, and vice versa.*

Thus, a fast twitch oriented person has 2 things going for him OTHER than the fact that his fibers can produce tension a bit quicker than someone else:

A: He's more likely to carry more muscle in the right places to start with - and bigger muscles can produce more tension regardless of fiber type.

B: He's more likely to have a structure well suited for speed-strength activities - An x framed physique with high calves, thin waist and hips.

How important are these factors? Well, they may not be everything, but they do make a difference. The real question is, “Is there anything you can do about your fiber type if you DON'T naturally have a lot of fast twitch fiber?” Well, yes there is. You can get as strong as you can or add muscle mass. You see, you may not be able to change the ratio of fibers you were born with from a *numbers* perspective, but you can certainly do so from a *volume* perspective, and volume is what matters.

Let's say you're 50/50 fast to slow twitch and you double the size of all your fast twitch fibers. Now, you still might be 50/50 fast to slow twitch from a sheer fiber *numbers* perspective, but since you’ve doubled the size of the FT fibers you DO have, the relative distribution from a volume perspective is now 75% fast twitch and 25% slow. Make sense? Volume is what matters! That means you’ll now function like someone that has 75% FT fibers. The point to take home is you can certainly change the distribution of fibers.

Could an east African (naturally 80% slow twitch) ever transform himself to perform in speed-strength activities somewhat on par with an 80% fast twitch west African? I believe so and here's why:

You see, when you add muscle mass thru resistance training it is always going to be predominately fast twitch in nature. Slow twitch fibers don't grow much no matter what you do. Say you have a 130 pound individual with a ratio of 70% slow twitch fibers and 30% fast. Let's say he has 17 inch thighs. Since slow twitch fibers are very enduring he'll physiologically resemble an elite marathon runner. He'll probably be naturally really good at things like distance running, but not so good at things like jumps and sprints. But let's take that same 130 lb individual, get him in the weight room, and build his muscular bodyweight up to 165 lbs and his thighs around 25 inches. Since the thigh bones don't grow that much we've virtually effectively doubled the muscular size of his legs. All we did was add fast twitch fiber. Now the overall fiber distribution in his thighs will be more like 60% fast twitch and 40% slow and his functionality in power/speed activities will have effectively changed considerably. He'll suffer a bit at the endurance end, but he'll be a different type of athlete at the explosiveness end. THAT's how a person can effectively alter their fiber type!

## Muscle mass and the nervous system

Inefficient natural fiber type (fast to slow ratio) is not really a problem I'd consider giving all that much attention because it's relatively easy to create more FT fiber via strength training. Give me 10 Kenyans any day (80% slow twitch) and I'm confident I can make fairly successful speed strength athletes out of any of them, because they most likely have pretty good nervous systems. But a slow nervous system is a much bigger problem on a physiological level. The nervous system is the engine that powers the ship and it's always been my belief that the engine is the most important factor, regardless of fiber dominance. You can actually take a slow twitch fiber and hook it up to a fast twitch motor unit and the slow twitch fiber will behave in a fast twitch nature.

The ability to ramp up the rate coding aspect of neural efficiency is in my opinion what separates the men from the boys – regardless of fiber dominance. Muscle mass increases are much more viable for people that have excellent nervous systems and good rate coding. They're inherently able to utilize and turn on more of their existing muscle mass.

As a personal example, I'm pretty sure I am about as naturally slow twitch at the fiber level as they come – I am naturally built nearly identical to a Kenyan runner, if not even skinnier. In my freshman year of high school I ran a sub 6 minute mile with only about a week of training. I ALWAYS struggled gaining any sorta muscle mass (and still do). But as slow twitch oriented as I was at the fiber level, I was probably that fast twitch oriented at the nervous system level with regards to neural excitability and rate coding. It's a trait I see in several of my family members. We all tend to get easily adrenalized and have great natural hand quickness and such. ** A highly efficient nervous system can somewhat overcome shortcomings at the fiber level. In a person with a very good nervous system any muscle mass you build is easily converted to power because you have a great nervous system driving it. It's like a governor on a car or truck - allowing your motor to run at higher and higher RPMs before the governor

kicks in is a big key in developing explosiveness, and some people have that to a greater extent than others, independent of natural fiber type variance.

***Natural quickness in unloaded movements is a good indicator of natural nervous system efficiency*

So, in my case over a period of several years I really built up and strengthened my thighs and transformed myself from an endurance oriented athlete into an explosive oriented athlete with a 2.5x bodyweight squat and 40+ vertical jump. However, my physiology also changed. I ran a 6 minute mile in high school with about a week of training. But for the last 7 or 8 years I've done cardio 4 days per week but would be lucky to be able to break an 8 minute mile - those fast twitch fibers I've developed over time simply don't like that high endurance activity.

The problem is not everyone has that level of natural nervous system efficiency. So, when they gain muscle mass it won't always help them to the same extent because they're not able to efficiently use it. That's not the end of the world though all it really means is they'll never be the type that can get away with taking their bodyweight very high and adding a lot of muscle mass. They'll have to pay more attention to *relative strength,* or strength per pound of bodyweight. Additionally, the nervous system can be improved to a decent extent and that's something that will be discussed in later chapters.

# *The Importance of Consistent Progression…*

Now that we've addressed a multitude of factors let's talk about long term progression and show you some actual workouts: One thing people often overlook when it comes to vert training is the importance of making consistent progression over time. For a typical vertical jump seeking athlete I consider a squat weight (or similar exercise equivalent) of around double bodyweight to be a good basic goal for strength. The problem is people want the greatest and best workouts in the short term, but they forget that building the qualities necessary to jump high is a long term process that doesn't always happen overnight. Take the guy who adds 5 lbs of weight to his squat each week...well in a year that's 250 lbs!

That's a TON of weight to add and would most likely not happen! But a lot of trainees get disappointed if they go on a strength phase and don't gain 10 or more pounds each and every week! It is possible to make really good gains like that in the short term - a few weeks or so - but to make really big jumps requires a long term outlook. When training with a long term outlook it is important not to get overly cute in the short term utilizing all sorts of high tech set and repetition schemes and exercise parameters. Those things usually detract from long term gains. To quote a bodybuilding legend Dante Trudel on the topic of steady progression, *"No matter what the method someone uses to make steady strength gains-it's imperative they do so. Again if you put someone out on a deserted island with 135LBS of weights he can superset, giant set, high rep, superslow, and do whatever he wants to his hearts delight...the sad story is his gains will quickly come to a halt because his limiting factor is the amount of strength he will gain. He has only 135LBS to work with. You take that same guy on a deserted island and give him squats deadlifts, and benches and an unlimited weight supply that he constantly pushes, in 5 years I'll show you a big strong Gilligan. If you go in every gym in america 80% of those people are doing laterals. People love to do laterals, they love to look in the mirror at themselves doing laterals. Do they work? Yes for some people. But with all these people doing lateral raises how many do you see with really big and strong delts? One out of 50? One out of 100? Im not seeing many. But how many people front pressing 315 for reps or a whole weight stack with a dumbell chained to it on a selectorized*

*press machine have weak and small delts? This is all about constant progression upward and forward and its very hard to start with side laterals with 35lb dumbells and end up around 135lb dumbells for side laterals over time like you could do with dumbell presses."*

The name of the game is to use basic movements with good form and work to add weight to them over time. With that in mind what follows is an example of how I like to train beginners and using this method it's relatively common for a beginner to add 6 inches to their vertical jump within a month or 2 and a good 100 pounds to their squat within 6 months. I got the basic idea from Mark Rippetoe several yrs back and I modified it specifically for Vertical Jump purposes and found it works wonderful:

The goal is getting your strength up rapidly while simultaneously training rate of force development, plyometric ability, and all the other components of a great vertical jump. You train 3 days per weeks on non-consecutive days (mon, wed, fri or tues, thurs, saturday) and perform squats every workout. Each workout you'll do 3 sets of 5 reps on squats. Start by lifting the bar for 5 repetitions as a warm-up. Rest 1 minute. Add 20 to 40 lbs and do another set of 5 reps. Continue this way until a 5 repetition set feels slightly difficult. Make sure to extend your rest periods out to 3 to 5 minutes as the weight gets heavier. After you find a weight where 5 reps is slightly difficult do 2 more sets of 5 with that same weight. So, you're basically doing 3 sets of 5 with a good working weight. If you get all 3 sets of 5 with PERFECT FORM go up by 10 lbs the following workout and once again try to do 3 sets of 5. Every time you get 3 sets of 5 with a given weight you go up the following workout.

Progression can come from either increased reps or better form. Say you work up to 150 lbs and do 2 sets of 5 with it but you only get 4 reps on your 3rd set. You'd stay with that weight another workout until you get all 3 sets of 5 with 150 lbs. Say you DO get all 3 sets of 5 but your form/depth on the last set is a little shaky. Stick with that weight again until you get 3 sets of 5 PERFECT reps.

Although the workout is based around the squat, as a vertical jump specialist also must get your vert specific work in too. An entire workout might look like this:

**Section A: Plyo/Speed** - do any 2 of the following for 4 sets of 5-8 reps

Multi response tuck jump
Ankle Jump
Low box depth jump
Lateral barrier jump (over midshin level object)

**Section B: Power** - pick 1 of the following and do 3 sets of 5

Jump Squat with 20% of squat max
Hang power clean
Hang power snatch

**Section C: Strength** - Do all of these

Squat: 3 x 5 (regulate as discussed above)

Reverse hyperextension, barbell hip thrust or Romanian deadlift: 2 x 8-10
Barbell calf raise (regulate the same way as squats): 3 x 15-20
Bench press/military press: 3 x 5 (regulate the same way as squats - alternate back and forth between the bench and military press each workout
Mid-grip pullup: 3 sets to failure (if you can get an average of 24 reps over the 3 sets add weight)
Optional Beach work (biceps/triceps) 10 minutes your choice

That's it! Very simple and effective.

If you go a week or more without any progress in weight or reps do a back-cycle. Reduce your weights by 10% and work your way back up. For example, say you were squatting 3 sets of 5 @ 200 lbs and got stuck there. You'd reduce the weight down to 180 lbs the next workout, then go to 190 the next, 200 the next, then continue to progress from there. In my experience most beginners can follow this setup for MONTHS and make consistent and steady gains in strength, power and vert.

This workout is versatile and can be modified. For example, you can set up 2 different workouts and alternate between them. One workout might have squats as the focus exercise and another workout might have bulgarian split squats or deadlifts as the focus exercise. As long as you follow the basic theme it works very well. With this approach most beginners will quickly and consistently gain strength and quickly and consistently add inches to their vert. Although the workout is meant to be performed 3 days per week, it can work just as well for many people at a frequency of just 2 days per week.

# *Strength and Its Relationship To Power, Strength Expression, and Rate of Force Development….*

Now that we've talked ad-nauseum about strength, let's talk about how it relates to other qualities necessary for a great vertical jump: As illustrated earlier, how high you jump is a combination of how much force you can create and how quickly you can develop that force. You gotta have both. You can use terms like strength expression and rate of force development interchangeably. In the big scheme of things they pretty much mean the same thing, which is the ability to quickly demonstrate strength. You can basically think of them as the speed aspect of power and explosiveness. Since explosiveness (power) is a function of strength and speed (strength x speed), and jumping is a display of power, often just increasing the force potential, or strength, of the appropriate muscles will provide a world of improvement.

**Vertical Jump = Power**

**Power = Strength x Speed**

For example, if a ***strength*** score for an athlete was 2, and the athlete's ***speed*** score was also 2, his explosiveness rating would be 4:

2(speed) x 2(strength) = 4 (explosiveness)

Doubling the athlete's strength would double his explosiveness:

2(speed) x 4(strength) = 8 (explosiveness)

Doubling the athlete's speed without altering strength would also double his explosiveness: **

4(speed) x 2(strength) = 8(explosiveness)

***This is really a pretty unrealistic example because the speed (rate of force development) part of the equation is under a lot more genetic control than the strength part. This is why you never see someone double or triple the absolute speed they can move their hands or feet thru the air, yet it's not at all uncommon to see them double their strength on basic movements (bench press, squat etc.).*

BUT, if the same athlete made a 50 percent gain in both speed and strength his explosiveness rating would be:

3(speed) x 3(strength) = 9 (explosiveness)

So, it should be obvious an increase in explosiveness and thus jumping height will result if you either increase the baseline levels of strength, the speed at which you demonstrate strength, or both.

***Relative to this example you increase strength anytime you increase the poundage of key exercises like deadlifts and squats. You increase explosiveness anytime you increase rate of force development, or the ability to **express** that strength.*

So, basically there are 3 ways to improve the VJ. You can:

1. Focus more on the speed side of the equation: Here you're training the nervous system to ultimately produce faster contractions. You're bridging the gap between the amount of strength you have and the amount of that strength you can display at high speeds. Or, simply put, you're teaching your body to develop horsepower at higher RPMs. *** Examples are: plyometric exercises, jump squats, Olympic lifts, and weight training with loads around 50% of your max so you can use lots of acceleration with the lift.*

2. You can also improve explosiveness through focusing on the strength side of the equation. Here you're simply improving the raw horsepower you have. This could take the form of 2 general approaches. They are:

A: Using 80-90% of your max in a given exercise for multiple sets of low repetitions in an effort to improve neural efficiency. (E.g. 3-5 sets of 2-3 reps)

B: Using 60-80% of your max for higher reps in an effort to combine increased neural efficiency with muscle growth. (E.g. 3-4 sets of 6-10 reps)

Or:

3. You can focus on both sides of the equation, focusing on strength and rate of force development.

Which approach would be optimal for you? It's really quite simple: The optimal approach requires either zeroing in on your weak area, whether it's raw strength or the speed at which you display strength, while maintaining the other, or improving them both simultaneously. Obviously, if both factors can be improved with a specific routine it would be more efficient than just improving one aspect.

So, how can you improve the speed at which you display strength while simultaneously getting stronger? Well, to some degree just getting stronger via lifting fairly heavy weights provides a high velocity specific effect and improves neural efficiency. When you lift heavy loads to improve your strength the resistance may move fairly slow, yet the degree of nervous system activation and muscular recruitment can yield results both in maximal strength and rate of force development. Thus you get the best of both worlds. This is particularly true of beginning trainees who yield significant gains in RFD via strength training.

But now let's get a little deeper into things. How much of YOUR strength can YOU actually use in YOUR vertical jump? It will be variable from one individual to the next but there's a name for it and it's called the *explosive strength deficit*.

# *The Explosive Strength Deficit…*

The ***explosive strength deficit*** is defined as the amount of strength that ***ISN'T USED*** in a sports movement. Remember, a movement like a vertical jump occurs too quickly to develop absolute maximum force. The difference between the absolute maximum force you're capable of putting out and the amount of force you actually DO put out is the explosive strength deficit.

So, how do you figure out your explosive strength deficit? It's really quite simple: We can assess how much force you put out in the vertical jump by simply measuring your vertical jump. Simple enough. But how do you assess the absolute maximum force you can put out? One simple way to do it is to test yourself on a movement where there aren't any time constraints. A simple barbell squat will get the job done for that. Thus, we can get a fairly accurate indicator of your explosive strength deficit by simply comparing how your squat relates to your vertical jump. Simple enough!

If your squat goes up but your vertical jump stays the same (or decreases), the difference between your squat and vertical jump increases thus, your explosive strength ***deficit*** increases. If your squat stays the same but your vertical jump goes UP, your explosive strength deficit decreases. Simple enough.

In general, you don't ever want to do anything LONG TERM that *increases* the explosive strength deficit. In other words, over a ***long period*** of time you don't want to continue driving your strength up without seeing any gains in vertical jump. The key phrase there is ***long term,*** as in a few months or better. The gap between the 2 may widen slightly during some particular phases of training where the ***short term*** focus is on acquiring significant strength, but over a longer term period you should always see the explosive strength deficit staying the same or decreasing. The next question is, "What is the typical explosive strength deficit for a vertical jump?" Good question! Former Soviet sports scientist and author Vladimir Zatsiorky gives the following examples in his book "Science and Practice of Strength Training":

*"In movements such as jump takeoffs and delivery phases in throwing the explosive strength deficit is about 50%. Among the best shotputters during throws of 21 meters the peak force applied to the shotput is in the range of 50 to 60 KG. The best results for these athletes in a bench press exercise is about 220 to 240 kg (480-520 lbs), or 110 to 120 KG per arm. Thus, in throwing, the athletes can only use about 50% of Fmax (potential maximum force). "*

Notice how Zatsiorsky compares jump takeoffs with the deliver phases in throwing? That would mean the amount of force that gets put into the ground on a vertical jump would typically be about 50% of the force put into a force plate on a typical maximum squat attempt.

## *The Shotputter Analogy…*

I wrote an article a few years back where I compared vertical jumping to shot-putting. Jumping and shot-putting are VERY similar as far as the motor qualities required, yet the way they're perceived

by the average trainee or coach is ENTIRELY different. The average trainee or coach typically associates training for the vertical jump with jumping. They associate shot-putting with lifting weights or throwing around other heavy objects - typical high tension training. But how similar are shot-putting and throwing? Well, consider this: When you jump you have to "throw" your body into the air. When you throw a shotput you have to "throw" a loaded weight. When you jump into the air you actually have to lift about 80% of your bodyweight. That means if you weigh 150 lbs you have to *throw* 120 lbs into the air when you jump. A shot-put only weighs about 15 lbs. What is a vertical jump? A rapid squat. What is a shotput? An accelerated one arm bench press with a 15 lb weight. Try this illustration: Go to the gym and take a pair of 15 lb dumbells, lie back on a bench, and do 100 bench presses. Next, do 100 bodyweight half squats rising up on your toes at the end of each one. Which one's more difficult? I think MOST people will agree the squats are more difficult. From that exercise it should be obvious that lifting your own bodyweight in the half squat (similar to a vertical jump) requires more strength than throwing a 15 lb weight.

My observation is this: Nobody ever tells an aspiring shot-putter with a 135 lb. bench press that he DOESN'T need to get stronger to succeed in the sport, but there are still TONS of people that will tell an aspiring VJ enthusiast who can't squat 135 lbs that he DOESN'T need to get stronger. If a shot-putter needs a ~500 lb bench press to throw a 16 lb shot-put a good distance is it any wonder that it takes a pretty good barbell squat to throw 80% of your bodyweight a good distance into the air?

Back on topic the next question is, “How much is enough strength to jump high?" Or, “How much is too much?” This will vary depending upon your body structure and the innate speed of your nervous system. In general, the longer your legs are and the faster your nervous system is the more you'll benefit from increased strength, but the weaker you'll naturally be. The shorter your legs are and the slower your nervous system is the less you'll benefit from increased strength, but the stronger you'll naturally be.

In my experience I've yet to see anyone, male or female, that didn't benefit from a squat of at least 1.5 times their bodyweight. That means, providing they were doing regular jump work of some sort, their VJ increased in direct proportion to their squat increases up to the point where they could squat 1.5 times their bodyweight. In my experience the VAST majority of people continue making vertical jump gains as their squat increases to 2 times bodyweight. People that are naturally athletic will often continue gaining vert from pushing their squats to 2.5 x bodyweight. However, I personally don't know anyone who benefitted from getting their squat to 3x bodyweight, although I've heard of a couple of guys.

Now, does all this mean that all anyone needs to do to jump higher is increase their squat? Well, that will certainly work well initially for most people. Remember, you can improve from either increasing the max force or increasing the rate of force development. But the biggest potential for immediate gains will occur as you take your basic leg strength up to the point where you can squat 1.5 to 2.0 times bodyweight because the maximum force is MUCH more trainable than improving the rate of force development. Additionally, initial strength gains will ALSO improve the rate of force development as well. However, at some point the rate of force development and not the maximum

force will become the limiting factor and the focus of your training will need to change. Zatsiorsky gives the following example specific to the vertical jump:

*"DEFINING A TARGET: STRENGTH OR RATE OF FORCE DEVELOPMENT*

*A young athlete begins to exercise with free weights performing squats with a barbell. At first he was able to squat a barbell equivalent to his bodyweight (BW). His performance in a standing vertical jump (VJ) was 40 CM. After 2 years his achievements in the barbell squats was 2x BW and the VJ increased to 60 CM. He continued to train in the same manner and after 2 more years was able to squat with a 3xBW barbell. However, his jump performance was not improved because the short takeoff time (rate of force development) rather than maximal absolute force became the limiting factor. Many coaches and athletes make a similar mistake. They continue to train maximal muscular strength when the real need is to develop rate of force."*

So, initially the athlete made really nice gains from increased squat strength alone and that trend continued as he got his squat up to double bodyweight. But as he continued getting stronger and stronger he eventually hit a point where he was no longer benefitting from increased strength and needed to focus more on RFD.

To summarize this section:

1. For all practical purposes the explosive strength deficit is the difference between your squat and your vertical jump, or the difference between your maximum force and rate of force development.

2. Virtually EVERYONE will initially make quality gains on their VJ by boosting their squat somewhere to 1.5 to 2.0 times their bodyweight. *note: This doesn't mean that the ONLY thing a beginning athlete should do is strength train (squat). It should be a **focus** but a well rounded program incorporating some level of rate of force development work via plyometrics and other methodics will be superior to a program that only includes strength training*

3. Some people will continue making VJ gains from even more additional strength than a 2.0 x bodyweight squat.

4. At SOME POINT the rate that you develop force, and not your maximum force (strength), will become the limiting factor. Thus, at some point you will have to focus on decreasing the explosive strength deficit. For most people this will be somewhere around a 2.0x bodyweight squat. At strength levels beyond this point there simply isn't enough time to use the additional force you're capable of creating.

So, how do you decrease the explosive strength deficit? GOOD QUESTION and that brings me to the next chapter.

## *Decreasing The Explosive Strength Deficit….*

Now that we've identified what the explosive strength deficit is, let's talk about how to improve (decrease) it! Anytime you decrease the explosive strength deficit you increase the rate and speed that you develop force. On a physiological level your nervous system gets better at quickly recruiting and firing your muscles. Rate coding improves. Some of your fast twitch muscle cells convert from a slower contracting IIA subtype into a faster contracting IIX subtype. The end result is the difference between your potential and useable high RPM strength decreases, so you're able to actually "use" more of your strength when you jump. There are 4 ***traditional*** ways to decrease the ESD and they all involve training fast and rapidly accelerating either a load or your bodyweight. They are:

1. The use of ballistic exercises with loads ranging from 0% to 50% of one-repetition maximum (1RM).

2. Weightlifting exercises performed with loads ranging from 75% to 90% of 1RM

3. Plyometric exercises

4. Engaging in the specific movement pattern

Basically, all these methods allow you to "practice" accelerating against load (or your own bodyweight). This creates neural and structural adaptations that can "help" you perform your movement of choice with more power and quicker rate of force delivery. Now let's discuss these methods one by one:

## *Decreasing the ESD Option #1: Ballistic Exercises…*

A ballistic exercise is an exercise like a jump squat where you *throw* either your body or an implement into the air. A medicine ball throw could also be considered a ballistic exercise, yet for vertical jump purposes THE traditional ballistic exercise is a simple loaded jump squat. The jump squat is simply a loaded jump and requires that you extend your ankles, knees, and hips at high velocity against resistance. Research has demonstrated loads of anywhere from 10-30% of maximum squat to be most viable, but probably a better way to manage the weight on jump squats is by using something called ***total system weight***. This is an idea I picked up from Mike Boyle several years ago and it's very effective: Simply take your maximum 1 rep squat, add your bodyweight, then use 40% of that weight.

For example, an athlete weights 150 lbs and squats 300 lbs. You add his bodyweight to his squat and come out with 450. Take 40% of that (180) then subtract his bodyweight (150). That gives you a load of 30 lbs. This method inherently autoregulates load and the great thing about it is athletes that don't achieve minimal strength standards (1.5 x bodyweight squat) won't use any weight. Jump squats can be loaded several ways: You can use a barbell, hold dumbells, wear a weighted vest, or wear a weighted belt. Gimmicks like the "Power-Jumper" and "Vertimax" are also useful for doing jump squats. Don't think you "need" to spend money getting an expensive band setup though, as a simple barbell jump

squat is about the oldest ballistic exercise in existence and works just as well today as it did 50 years ago.

Jump squats can be performed in 3 basic ways: paused, rhythmic, and a mix of the 2. You should perform a jump squat just like you do a regular jump. Don't squat ALL the way down simply squat as low as you do in a normal jump. In the ***paused*** variant you stop and reset yourself each rep. In the ***rhythmic*** variant you execute your reps rhythmically, without any pause between reps. In the ***mixed*** variant you reset yourself, but you do a slight pre-jump right before each of your main jumps, similar to how some people jump rope.

**Volume guidelines:**

I typically prescribe 3 to 5 sets of 5-10 reps per set of jump squats per set, with 2-3 minutes rest between sets.

# *Decreasing the ESD Option #2: Olympic lifts…*

There are quite a few people who think the Olympic lifts, the clean and the snatch, are the answer to everything. Their claim is that the triple extension of the hip, knee, and ankles that occurs with the O-lifts mimicks sprinting and jumping, and because of this the thought is Olympic lifts by themselves will build better athletes. The fact that Olympic lifters are some of the most explosive athletes around definitely helps their claim. People that follow me know I don't try to jam the O-lifts down peoples throat, my thought has always been that Olympic lifts are very good indicators of athletic power, but don’t necessarily build athletic power all by themselves. In other words, as explosiveness increases performance on the O-lifts also improves, but just getting better at the O-lifts might not always make you more explosive.

Now, having said that, if done correctly the hang snatch (or clean) is surely an easy way to self monitor and increase rate of force development and strength speed specific to the VJ over time. They are great “tools” to use to monitor your increases in athletic power. The olympic lifts (snatch and clean) inherently require lots of acceleration and if DONE PROPERLY they require simultaneous extension and contraction of the ankle extensors, knee extensors, and hip extensors. For those reasons they can be a valuable way to decrease the explosive strength deficit and improve rate of force development IF PERFORMED CORRECTLY. A friend of mine did some research on a master’s thesis several years back and the high hang snatch actually came out ahead of the depth jump as far as recruitment pattern similarity to the standing vertical jump.

A study a few years back took 2 groups of athletes: One group performed squats and plyometric exercises. The other group performed squats and olympic lifts. The group performing the olympic lifts improved their vertical jump more than the other group. Impressive, but does that make olympic lifts absolutely necessary? Not necessarily!

Here's the thing: the O-lifts can be difficult for some to perform correctly and an athlete can accomplish the EXACT SAME THINGS with a jump squat as he can with the olympic lifts - triple extension

of the ankles, knees, and hips under load in a movement practically identical to a vertical jump. BUT, the best thing about the o-lifts compared to a jump squat is they're much easier (and I think funner) to monitor than a jump squat because the weight is either getting heavier or it's not - you have something more tangible to shoot for. It's a bit harder to monitor jump squats.

I like the hang versions of the snatch and clean because they are almost EXACTLY like a loaded jump. The hang variations are relatively easy to learn, easy to implement, and also provide a great potentiation exercise. They also help sports specific upper body strength as it relates to rebounding and such. One problem with the O-lifts is, although they do require triple extension of the hips, knees, and ankles, their effect on the ankles is considerably less than the hips and knees. This is more true the better one gets at performing them. In other words, they don't really force you to come up hard off the balls of your feet as you would in a jump. For that reason I often like to have people alternate a set of O lifts with a set of a plyometric variation like depth jumps, tuck jumps, etc. You do the O-lifts primarily for hip and knee extension power, and the plyo for ankle extension power.

As far as load goes, the olympic lifts are inherently fast on their own so you can work with a relatively high percentage of your 1rm and still focus plenty on RFD. Loads of 80-90% of 1rm are about right. You can do them anytime you work your lower body and you'd typically do them early on in the workout before any other heavy lifts. As long as you keep the reps between 1-5 and maintain good form they're hard to screw up and you'll make progress. Typical rep schemes might include 4 x 3, 4 x 2, or 5 x 1. Lots of o lifters really don't even count sets they just work up to a daily single then take a bit of weight off and knock out a few more sets. I know some track coaches like 8-10 x 1. There are lots of effective set and rep schemes and they're all valid as long as you follow the general guidelines of keeping the reps under 5.

How often should you do them? Two days per week per lift is fine but I've seen people make good gains doing the hang snatch one day during the week and hang clean on another.

Now, one problem with the olympic lifts is they DO require some coaching. It is POSSIBLE to teach yourself how to do them but that's not something I'd really recommend. All you need to do is go check out any high school lifting program that tries to incorporate the o-lifts to see the piss poor technique with the o-lifts that typically goes on. A problem you'll often see is people "bump" the weight up off their hips and end up performing a form of reverse curl. On a proper olympic lift the weight should move vertically from the hang position. If you're dead set on performing the lifts and you're in doubt how to do them make sure you have a qualified coach properly instruct you.

## *The formula for becoming a super-freak athlete with the Olympic Lifts….*

If you want a simple formula to shoot for as it relates to gains on the lifts for overall athletic success, add your bench press, mid-stance legal squat, hang snatch, and hang clean together, then divide by your bodyweight. The number to shoot for is 6. If you can hit that chances are you're gonna be one explosive dude (or chick!). I learned that formula from a throwers coach years ago named John Smith. His wife was a bobsled competitor and former college basketball player. According to my notes

he said that in college basketball she ran her butt off for 4 years and was running a 5.1 40 at 208lbs when she left basketball. Two years of weight training later she weighed 212 and ran a 4.7 and improved her vertical 6 inches without running or jumping in training. In 1985 her bodyweight to strength ratio on the 4 lifts was just below 3.00. In 1987 it rose to 5.33 and she became a different athlete.

I had a guy a few years back really dedicate himself to that formula after I discussed it with him. His main focus was getting bigger and more muscular, the explosive gains were secondary. I trained him for a while and got him started and he kinda took things on his own with occasional input from me. Over a span of a couple of years went from 155 to 185 lbs and increased his vert from mid 26 up to 39 inches! All he really did differently was throw in some depth jumps occasionally.

# *Decreasing the ESD Option #3: Plyometrics…*

Plyometrics are traditionally THE way to decrease the explosive strength deficit. I plan to cover plyometric action and plyometric training in great detail in a later chapter, but for now I just want to give you a basic overview: The main purpose of plyometric drills is to enhance the ability to more rapidly express strength (increase rate of force development), develop reactive rebound type strength,** and improve your capacity to use your tendons as movement generators. Plyometric exercises force an imposed stretch on your muscles and their value comes from this stretching action and, in the case of the vertical jump, the fact that they closely resemble the actual vertical jump from a recruitment pattern perspective. Plyometric drills enhance the efficiency of the entire absorption, stabilization, and release of force in a sports specific manner. They enable you to work on expressing your strength in a manner specific to jumping.

***Plyometric strength is also termed elastic strength, reactive strength, reversal strength, and rebound strength. Don't let the terminology confuse you! They all mean the same thing.*

When you engage in a bout of plyometric training it's not necessary to use a ton of various plyo drills in order to have an effective workout. People tend to overcomplicate plyometric work, and I've been guilty of this myself, but all lower body plyometric drills do basically the same thing:

They all involve some type of hopping, bouncing, jumping, or running variation. In my opinion there isn't any magic in any given exercise, the only magic is in the intensity of exercise. There are low intensity movements like double legged line hops, which work best to train movement efficiency and basic coordination on the feet. There are very high intensity movements like depth jumps, which build max power. If possible you should try to choose exercises that most closely hone in on your specific needs, and you should always choose exercises that correspond to your level of development.

You should also choose exercises involving stretch rates and stretch loads that are similar to those encountered in the specific activity you wish to improve. In the case of the vertical jump, you want to improve vertical jumping, so your exercises should obviously come close to duplicating the jump. As far as that goes a simple depth jump is the most specific plyometric exercise.

The original Vertical Jump Bible incorporated a smorgasboard of plyometric exercises. It had about 5 different categories of plyometrics with a good 10 or 15 exercises in each category. As I've grown in knowledge as a coach my use of plyometrics has changed. I now only use a handful of drills. My reasoning will be explained a bit further in a later section, but these are a few of the typical drills I use:

Double legged line hops
Single legged line hops
Ankle jumps
Tuck jumps
Lateral barrier jumps
Depth jumps (various heights and various ground contact times)
Depth jumps between boxes
Skipping
Bounding
Single leg bounding

That's it! Each exercise has a specific purpose. If you wish to get creative it's relatively simple to come up with your own plyometric variants. Here are some ideas:

**Low Intensity Plyo Exercises**

Draw a line on the ground and jump side to side over it with 2 legs

Do the above front to back

Draw 4 imaginary stars on the ground forming a box with each star separated by 12-18 inches. Hop around the box on one leg

Get in a squat position up on the balls of your feet and bounce in place

Put a low box in front of you - jump up on it and step off. Do the same with one leg. From the side…..from the other side.

**Medium Intensity Plyo Exercises**

Get underneath a basketball goal and rhythmically jump up and try to touch as high as you can without any pause between repetitions

Get in a lunge position and jump up as high as possible landing in the same position as you started. Switch legs in midair

Jump side to side over a cone, bench, or other knee-high object

Jump high and bring your knees to your chest

Put a medium to high box (18 to 40 inches) in front of you and jump up on it

Put a low box (6-12 inches high) and bounce rhythmically up and down off and on it with one leg...repeat from the side, repeat from the other side

Skip for max distance

Skip for max height

Stand on a box about 18 to 24 inches high, step off the box, and land softly up on the balls of your feet in a motionless position (depth drops)

Perform a standing broad jump

**High Intensity Plyo Exercises**

Hop forward on one leg

Hop sideways on one leg

Sprint with exaggerated strides trying to get up as high as possible and cover as much ground as possible with each stride (bounding)

Stand on a box, step off, hit the ground, and jump as high as possible...repeat to the left, to the right (depth jump)

Remember to choose exercises based on your developmental level. A depth jump off a high box would be too intense for a beginner or someone who lacks strength. Youngsters should spend the majority of time working with basic low intensity hops, skips, and jumps until they have a modicum of strength in place. The general recommendation is that an athlete should be capable of a 1.5 x BW squat before participating in high intensity plyometrics.

**Volume Guidelines**

A typical workout might have 1 to 3 exercises of plyometrics, for 3-5 sets each, for a total of 50-100 ground contacts per session. Your repetitions per set will typically range from 1 to 10. If in doubt always stop any plyometric workout before, or as soon as, your performance in the movements begins to decline. In my opinion a set should ALWAYS stop prior to 10 seconds unless you're intentionally trying to build up power endurance, which isn't the focus of this book. One way of auto-regulating the repetitions per set is simply start your set and stop at the first sign of performance drop-off in the set. For example, say you're performing tuck jumps and your first rep is ok, your 2nd thru 5th reps are good, and you start to lose it on your 6th rep. That's a good time to stop the set. Exercises should be performed with a mix between effort and efficiency. In other words, you want to go HARD and try to get up high, but you also want to stay smooth and light on your feet, almost like you’re dancing. I find the cue “100% height at 90% effort” works well because it teaches a person to put out high power while still staying relaxed.

Volume of plyometric work will also vary depending on the intensity of the exercise and level of the athlete. A plyometric workout incorporating high intensity exercises like maximal depth jumps (discussed later) can be very intense, but may only have 15-20 total jumps in a session, whereas a lower intensity plyometric workout of various lower level drills like ankle jumps, tuck jumps, etc. might include around 80-100 total jumps (or ground contacts) in a session. There may be some utility for lower level athletes to perform more volumous workouts (more total jumps per session) as it gives them a stronger jump training stimulus and may allow their bodies to better adapt to jumping. However, this is with the caveat that the athlete's body is working efficiently, as high volume jump training on a poorly functioning system (discussed later) is virtually guaranteed to cause overuse aches and pains of some sort. Regardless of the volume, the above mentioned rules still apply: Stop the workout and exercise as soon as any performance drop-off is noted. The gains from plyometric work are largely neurological in nature, and neural related gains require high quality efforts.

***Progress*** in plyometrics can occur 3 ways: Via increased height/distance jumped, via increased efficiency of movement (maintaining the same performance while being smoother and quieter), or by decreased ground contact times between repetitions. Over time one should progress via all 3. In other words, you want to jump higher, while being smoother, while spending less time on the ground between reps.

As far as ***rest intervals*** go, you want to be fresh enough to generate a high intensity effort with every rep of every set. Take your time between sets and don't rush. You might rest 3 to 5 minutes in between sets of high intensity exercises like depth jumps and bounds and a minute or 2 between "easier" exercises like ankle jumps. Go when you're ready and don't rush yourself.

***Frequency:*** Plyometrics can be performed every day depending on the intensity and volume of the exercise, but a typical prescription is 3 days per week for low and moderately intense exercises, and 1-2 days per week for more intense exercises like high volume depth jumps from high boxes.

# *Decreasing the ESD Option #4: Engaging in the specific movement pattern….*

It also so happens that the act of jumping itself is very plyometric in nature. For this reason anyone doing more than a modicum of jumping in their sport typically doesn't need to utilize a ton of plyometric drills, as they can get specific plyometric training AND decrease the ESD simply by going out and engaging in jumps on a regular basis.

A problem many people have is they do lots of plyos but they don't do enough actual jumps. Jumping is so simple many people forget it still can be considered a specific exercise. The end result is they get really good at the plyos in comparison to their actual jumps. This actually happened to be me at one point back when I did a lot of plyometric training. A few months later I was REALLY good at performing all the drills but my maximal jumps suffered because my coordination was off. Plyos can be beneficial, but you have to remember none of them are 100% identical to the vertical jump.

I like to see most athletes do at least 10-20 quality combined running and standing vertical jumps 2 or 3 days per week. Basketball courts are a good place to get your jumps in, and these sessions shouldn't be all that stringent as far as reps, rest intervals, and so forth. Just go out, have some fun, and jump around a bit.

Now I also want to touch on another very valuable way to decrease the explosive strength deficit:

## *Decreasing the ESD Option #5: Heavy Weight Training For RFD Success…*

As I mentioned earlier, building strength using exercises such as squats can be an excellent way to increase the rate of force development for less experienced athletes. A multitude of studies demonstrate beginner to intermediate level trainees performing heavy squats not only get stronger they also increase their rate of force development. One study demonstrated that the "intent" to move heavy loads quickly was effective at increasing rate of force development for more advanced intermediate athletes. But what about advanced athletes?

Traditional American thinking has held that as athletes get stronger their training for speedier qualities such as rate of force development and maximum power should become more specific, as does the need to emphasize RFD in their training. You may initially get gains in RFD from increasing your strength, yet as you become stronger RFD no longer responds to traditional set and rep schemes with typical loads (75-85%) and rep schemes (3-8 reps). The thinking is that as an athlete gets stronger the weights should get lighter with more focus on speed. Instead of focusing on lifting maximal weights, an athlete with a good strength base seeking maximal gains in RFD might perform 6-8 sets of 2 reps at 50-60% of his max on barbell squats in conjunction with jump squats/Olympic lifts and plyometrics twice per week. This type of training CAN and DOES work, but in my experience there might be something that works even better:

Before I get into the method, let me give you the background: In my experience there seems to be a bit of a disconnect between the science and real world when it comes to rate of force development training because in my experience I have found NOTHING increases RFD like PROPERLY implemented low rep barbell training. Many people have heard of the "Max effort method" as espoused by Westside barbell club. Well, it was actually taken from former Soviet strength coach Yuri Verhoshansky, and Verkhoshansky primarily used it not as a way not to gain maximal absolute strength, but as a way to simultaneously gain maximal relative and explosive strength SIMULTANEOUSLY in advanced athletes. The Westside max effort method basically involves working up to a daily max in a particular exercise, but the original max effort method involves doing sets of 1-3 reps in rest pause fashion - usually with a sub-maximal load. What follows is a typical example:

**Option 1:**

2-4 sets of 2-3 reps with 90% 1rm
15-60 seconds between reps
4-6 minutes between sets

In my thinking I think the original max-effort method should be better termed, “The sub-maximal low rep method" because the focus is typically on lifting heavy, but not maximal, weights.

This type of training is effective for decreasing the ESD because the weight is heavy, around 90% of 1rm, but it typically isn't so heavy it has to be “grinded” up. This allows one to focus on accelerating a fairly heavy weight for low reps. Lifting a heavy weight like this strongly engages the central nervous system, which improves the firing frequency of the nervous system. This improves rate coding, and rate coding has a BIG-TIME effect on the nervous system processes controlling rate of force development. The weights are also typically heavy enough that maximum strength can often be improved simultaneously.

The example above is straight from Verkhoshansky himself. However, one thing I'm kind of against is rest pause training on squats. I feel it leads to bad form and it's just not something most people find enjoyable. Rather, I like athletes to perform anywhere from 1 to 3 "traditional" reps with a good rest in between each set.

To maximally influence RFD with heavy low rep weight training the use of a "***buffer***" can be valuable. A "buffer" is simply a difference between the weight you have on the bar and your maximum for a particular rep range. For example, if you're performing sets of 1 rep with 90% of your maximum load you're using a 10% buffer, since you’re capable of doing a set of 1 with 100% of your 1rm (100% - 90% = 10%). If you're doing sets of 3 reps with an 85% load you're using about a 5% buffer, since most people are capable of performing 3 reps with a 90% load. Sets of 1 with 85% are about a 15% buffer.

In my experience the use of a buffer is most valuable for athletes that are already strong (~2 x bodyweight or more squat), or those who are in the midst of a peaking cycle. The buffer allows them to work on accelerating the weight, which is still heavy enough to maintain (or even gain) strength, is heavy enough to stimulate the nervous system, but is light enough to keep both their muscles and nervous system somewhat fresh. What you want to avoid with this method are ultra-slow, *grinding*, repetitions. The weight should move at a fairly consistent speed from bottom to top. In general, the stronger the athlete the bigger buffer you want to use. I will often have really strong athletes perform sets of 1 rep with 85% of 1rm. Rather than get all technical with the percentages I just tell them, "Do sets of 1 rep with a weight you could do 4 or 5 times if you HAD to". You should go more by feel then by absolute percentages on these. Remember, you want to avoid grinding reps and want to be fairly explosive with the weight, but still generate max tension.

As an example of how this might be prescribed in practice, one set and rep scheme I'll often prescribe is 6-8 sets of 1 rep with 85-90% of 1rm. Use a close or athletic stance, control the eccentric, squat down, and explode up. The frequency might be 2 times per week. If an athlete is not yet strong there is no real need for such a large buffer. A low level intermediate athlete might perform sets of 1 rep

with 95-100% of maximum, or sets of 3 with 85-87.5% of maximum, for as many as 20 total reps in a given workout, with a frequency of 2 days per week, or once every 4-5 days.

One scheme I've found that really works well for strength and explosiveness is to have 2 sessions per week, with one of them being more of a traditional strength building set/rep scheme, such as 5 sets of 5 reps (or something similar). The other session would be the sub-max low rep method, such as 6-8 sets of 1 rep at ~90% of maximum. During a peaking phase one might perform 6-10 sets of 1 rep at 90% of maximum one day and explosive speed squats at 60% of maximum another day. Definitely experiment with this sub-maximal low rep method in your own training. I began using it with athletes a few years ago and have had great success with it!

## Where Everything Fits In

Now that we've covered a smorgasboard of training methods, let's talk about how they all fit together. You know you need a baseline of raw horsepower and coordination to jump high, but you also know the importance of how having good rate of force development and plyometric ability positively influence the explosive strength deficit. The following table summarizes the effects different types of training have on various strength qualities:

| **Strength Quality** | **Standard StrengthTraining** | **Ballistic weight training (jump squats) and lighter load explosive weight training with 60% or less of 1rm (speed squats)** | **Plyometrics** | **Olympic lifts** | **Submaximal but heavy low rep weight training (singles with 90% of 1rm)** |
|---|---|---|---|---|---|
| **Max strength** | Good to Excellent | Poor | Poor | Fair | Good to Excellent |
| **Rate of force development (ESD)** | Fair to Good | Excellent | Excellent | Excellent | Good |
| **Plyometric/reactive strength (ESD)** | Poor | Fair to Good | Excellent | Fair | Poor |
| **Coordination and skill** | Poor | Good | Excellent | Fair | Poor |

# *The Mechanisms Behind Maximal Plyometric Power…*

Now I'm going to talk a little bit more why plyometric action (not necessarily plyometric training), is so important to the VJ. In this section I'm going to get into some detail on some modern research that gets into exactly what takes place on a mechanical level during high power plyometric activities and some things you can do to improve it. *Maximal plyometric power* could be considered the greatest combination of force and velocity during any movement performed with the goal of producing maximal velocity at takeoff, release, or impact. This makes it a positive thing during movements such as jumping, sprinting, changing direction, throwing, kicking, striking, and makes it highly relevant to the vast majority of sports - particularly the vertical jump.

Before we get into exactly what happens during a plyometric muscular contraction, it's important to understand the way muscles contract during sports activities, the different types of contractions, and the different anatomical parts involved in contraction. There are 3 different types of muscle contraction: *Concentric*, or positive, where a muscle shortens, *eccentric*, or negative, where a muscle lengthens, and *isometric*, which involves no movement.

**Concentric:** A shortening contraction. Anytime you "flex" a muscle the muscle shortens and exerts force.

**Eccentric:** A negative contraction. The lowering phase of any typical weight training movement is an eccentric contraction, as is the part of a vertical jump when you *squat down* slightly before you jump. The muscles actually lengthen eccentrically.

**Isometric:** No movement takes place – however brief the *motionless* state may be.

A typical sports movement involves all 3 types of contraction. Take the act of throwing: You rear your arm back to throw and this lengthens the muscles on the front of your shoulder. Next, there is a very brief isometric contraction that occurs as your arm switches from going backwards to going forward - for a split second there won't be any movement at all. Finally, there is a concentric contraction that occurs as you whip your arm forward to release the ball. Most other movements and activities use the same sequence. This combination of eccentric, isometric, and concentric actions forms the most common type of muscle function and is termed the stretch-shortening cycle (SSC), plyometric action, reactive strength, or rebound strength - they all mean the same thing.

Now, here's the important thing: When a muscle fiber is activated, stretched, then immediately shortened, the force and power generated during the concentric action is greater than a concentric-only contraction. This is one reason why you can jump higher when you quickly dip down and jump (called a counter-movement) then you can by squatting down in your ready to jump position, pausing, and then jumping.

Before getting into "why" we have to look at what happens when muscles contract: Muscles traditionally contract by cross bridges actin and myosin overlapping and latching onto each other. They pull against each other creating movement. You will hear this type of contraction called several different

things among them, "contraction by the contractual elements", "cross bridge contraction", and simple "muscle contraction." I prefer to keep it simple and refer to it as simple *muscle contraction*. There is an inverse relationship between the force the muscles produce in a movement and the velocity of that movement because it takes time for the muscular cross bridges to develop force. In other words, the faster your muscles have to move the less force they can develop. Slower movements, such as squatting with heavy weights, allow the muscle cross bridges plenty of time to generate maximal force. Faster movements, such as jumping, involve much higher muscle contraction times which do not allow the cross bridges ample time to build up to maximal force.

But anyway, the ability of a skeletal muscle to generate force is also highly dependent on its length. The muscle can produce its greatest force when the length of the muscle (at the time of contraction) provides for optimal overlap between the actin and myosin filaments. At this length, cross-bridge interaction is maximal, which allows for the greatest levels of active tension development. However, research has demonstrated that resting muscle lengths are generally slightly shorter than the optimal length, which means you can increase muscular force in MANY ACTIVITIES with a slight stretch prior to activation.

So, basically, preceding a strong contraction with a slight stretch puts the muscle in better *position* to exert force and this better positioning allows it to develop more force. This is one explanation as to why we can exert more force in a countermovement jump then a paused jump.

But that's not all, a countermovement, or pre-activation stretch, also allows the agonist (the muscle or muscles doing the bulk of the work during the concentric phase) to actually begin building up tension during the stretch, so the cross bridges actually have more *time* to develop force. In other words, as your arm is going backward in a throw the muscles on the front of your shoulder already begin "tightening up" and generating tension. In essence they're *winding up* as you're rearing back. When you quickly squat down during a jump you're basically putting a quick stretch on the muscles of your calves, quadriceps, glutes, and hamstrings. But as you're squatting down these muscles are already *priming* for the jump.

So, you're more powerful during a plyometric movement because you get more time to fully activate your muscles, you allow yourself to position the muscle in its sweet spot (put it at optimal resting length), and you allow yourself to start activating the muscle beforehand.

But that's STILL not all! Plyometric movements also seem to increase the quality of the cross bridge attachments that occur during the subsequent positive phase. Strained cross-bridges are detached in a state that permits them to re-attach more rapidly than cross-bridges not exposed to a prestretch, and they deliver more force per cross bridge.

If all that isn't enough there are quite a few MORE reasons why you're more powerful during a plyometric movement. We talked about traditional muscle contraction now let's talk about something known as *non-contractile aspects* of the muscle:

Muscles are attached end to end by tendons. When an active muscle-tendon unit (MTU) is stretched mechanical work is absorbed by the muscle tendon unit and some of this work can be stored as potential energy in the series elastic component (SEC includes fibre cross-bridges, aponeurosis and tendon). It is believed that some of this potential energy can then be used to increase the energy released during the following concentric contraction. This recoil of the tendon-muscle complex is thought to contribute to the increased force at the beginning of the concentric phase in plyometric movements and ultimately to enhanced maximal power production. Or simply: The stretch increases tension contribution by the tendon, which acts like a rubber band. This has long been the view of traditional "plyometric" action. Although more recent research tends to demonstrate the actual increased power we get from tendons may have been over-rated in the past, it's still relevant, and it can significantly enhance the concentric rate of force development an estimated ~20% according to some research.

The combination of muscular contraction and tendon contribution during a plyometric type movement allows the best of both worlds: The muscles can contract at relatively slow velocities - nearly isometric in nature. This allows the muscle to create more cross bridge attachments and develop more FORCE than if it were shortening faster. While the muscle is contracting slowly the tendon is contracting rapidly. With the muscle generating more tension via slow contraction, and the tendon contracting rapidly, you get the best of both worlds. High force and high velocity. Put those 2 things together and you get more power, which is another reason why you produce more tension in plyometric activities .

BUT, those STILL aren't all the reasons why you're more powerful in a plyometric movement. A deep stretch may also increase passive contribution by non-contractile aspects of the muscle - namely titin. *Titin* is a protein found in muscle that increases muscular tension WITHOUT cross bridge attachments and WITHOUT tendon recoil. Titin becomes relevant at high degrees of stretch. For example, when you get really deep into a squat and fire out, or when you lie on your back and grab your knee and stretch your hamstring. It you take a deep bodyweight bulgarian split squat position and get low enough you'll eventually get low enough that you get an INVOLUNTARY contraction. This type of deep stretch contraction is thought to be largely influenced by titin. So, titin helps you generate tension DEEP in the stretch position and is independent of muscle cross bridge or tendon activation. The research on titin as it relates to real world activities is still in its infancy.

My buddy Brent Contreras and I were having an email discussion about titin and he brought up an interesting point. He was watching a lifter do standing push presses. The lifter was so tight that he couldn't bring the bar all the way down to touch his upper pecs, but that seemed to provide him with an advantage during the initial blast off. Are increased titin levels something we can train for? It's hard to say, but one study demonstrated greater titin levels in trained powerlifters and weight lifters, so it could offer an additional explanation as to why and how gaining strength and muscle mass may help athletes in power oriented activities. But now that you understand how exactly plyometric power occurs let’s talk about how to improve it:

## *Improving Plyometric Power and The Power Development Cycle*

The simple thing about improving plyometric power is you already know how if you've read this far. Anything that improves your vertical jump also improves plyometric power because the vertical jump IS inherently a plyometric movement. All the aforementioned methods we talked about in the "Decreasing the Explosive Strength Deficit" chapter (and before) will also influence plyometric power. As you gain muscular cross sectional size you increase the potential amount of potential horsepower you can generate. When you gain strength (horsepower), you then increase the amount of tension your muscles can both absorb and generate. The more force your muscles can absorb and generate, the better you can potentially perform in plyometric activities. You then follow this up with methods that stimulate the nervous system and closely duplicate the vertical jump involving similar recruit patterns, velocity, and stretch rates to really hone in on the power conversion process and learn to transfer and express ALL the strength you've gained. These methods include ballistics, plyometrics, and methods such as the sub-maximal loaded squats I discussed.

This whole process is known as the POWER DEVELOPMENT CYCLE. You get bigger so you can get stronger, so you can get more powerful. Once you're as powerful as you're gonna be at a given level of strength you simply repeat the cycle again. Simple enough!

Now let's take a closer look at the best plyometric exercises, particularly depth jumps:

# *A Closer Look At The Best Plyometric Exercises….*

According to the rules of sports specific training, in order to be an effective sports specific exercise, an exercise must overload a similar portion of a sporting movement, and do so with the same amplitude and direction of movement. In other words, if you want to increase your jump, you should use sports specific exercises which are similar to jumping, but provide overload to it. The reality is all plyometrics involve some type of jumping, so they fit the bill here. Plyometrics can be broken down into several different categories – bounding, skipping, jumping, and hopping. They can also be broken down into single response, multi-response, and multi-directional. Here is a brief description of each:

**Bounding**: Jumps where you take off one foot and land on the other foot

**Skipping:** pretty self explanatory if you know how to skip

**Jumping:** Jumps where you take off both feet and land on both feet

**Hops:** Jumps from one leg where you take off one leg and land on the same leg.

**Single response**: Single response plyo drills are characterized by a pause between one rep and the next. You jump, land, pause, jump again, land, and pause. Single response plyos are best for those who haven't built up the necessary landing efficiency/eccentric strength to perform consecutive repetitions efficiently.

**Multi- response**: Multi -response drills are characterized by their rhythmic nature – there isn't any pause between the end of 1 rep and the beginning of the next. Virtually all plyo drills can be performed in mult-response fashion as well. They can be performed with 2 legs or one. Some of my favorites include knees to chest tuck jumps, ankle hops, and rhythmic squat jumps.

**Multidirectional single or multi response**: Multi-directional multi-response drills involve various changes of direction, sometimes over hurdles, cones, or other obstacles. Examples include lateral barrier jumps over a low cone or 3 side hops into a sprint.

**Shock :** Now we're getting to the good stuff. There are nearly as many plyometric exercises as there are foods to eat, the problem with most plyometric exercises is they really don't provide overload above and beyond what you'd get from a regular jump. In fact, in my opinion MOST plyometric exercises don't even provide the same degree of overload as a simple running jump for height. Pick any plyometric drill other than depth jumps and do 50 reps of it. Next, go out and do 50 reps of running jumps for max height. Which one was harder? Probably the running jumps. This helps explain why people can do a ton of plyometrics but not always increase their jumps, unless they also do a lot of actual jumps in the process. The ONE exception are the shock plyos and that's what I want to talk about now:

# *The Shock Methods - Depth Jumps and Drops…*

The depth jump (and drop) are *the* original *plyometric* exercises. They were invented by Russian sports scientists as a way to cause involuntary recruitment of dormant muscle fibers . The athlete would stand on a box up to 3 ft. high, drop off, and upon hitting the ground immediately jump up as high as possible. The belief was the action of imposing a significant involuntary stretch on the muscles at impact was enough to SHOCK the muscles into protecting the body from injury, and in doing that the muscles would learn to contract more forcefully than normal voluntarily. Over time the body would learn to except this increased motor unit recruitment as normal and a new heightened athletic state could be reached. The soviets used only 2 exercises, depth jumps and depth drops, and called them plyometrics. Word quickly spread and soon the whole world was on to what they were doing. The problem in the United States is coaches began throwing every jumping drill into the category of plyometrics so the importance of the *original* plyometric exercises was kinda lost.

In my opinion, providing they do so in a smart, safe, and progressive manner, one *could* conceivably skip ALL other forms of plyos and focus on shock, because the shock variations have the greatest training effect by far. That doesn't mean everyone should do true shock depth jumps from high boxes, but a depth jump from an 8 or 12 inch box is still a very good exercise. There is value in performing other low level plyometric drills – various skips, tuck jumps, ankle jumps, etc. They do offer SOME training effect and help athletes avoid boredom, but are they really necessary? Good question!

If I only had to choose 1 plyo drill I would choose a depth jump variation where you step off one box, hit the ground, and immediately rebound back up to another box. Here is why I love depth jump variations: When you perform a depth jump you're really just performing a loaded vertical jump. The movement is a vertical jump. The difference is dropping off a box inherently creates overload. Whereas loaded jump squats work primarily via concentrically overloading the quadriceps, the depth jumps work via eccentrically overloading the calves and quadriceps. In the depth jump you get a movement efficiency stimulus from being able to practice your actual jump, and you get an overload stimulus from dropping off the box. That's all there is to it really. The one weakness of depth jumps is the overload occurs primarily in the plantar flexors and to a lesser extent the quads. You still need to strengthen the hips. But, if your routine consisted of nothing but depth jumps and a high quality posterior chain exercise like RDLs, hip thrusts, or hang snatches, you could conceivably have a pretty effective VJ routine just from that.

Depth jumps are also very versatile. They can be performed in a variety of manners, with varying box heights and ground contact times. Probably the simplest, and perhaps the most effective way to perform them, is just perform them with the aim of jumping as high as you can when you hit the ground. Theoretically, in order to be most applicable to a given movement, your performance in plyometrics (and choice of exercises) should mirror the ground contact times in your sport. This is something I will discuss in a later chapter, but depth jumps can be tailored to a wide variety of jumps and goals just by manipulating how long you stay on the ground and the height of the box you use. For a simple bilateral

jump (2-leg vertical jump) probably the best way to perform them is don't worry about your ground contact times too much, simply focus on getting up as high as possible when you jump.

## *Finding Your Optimal Box Height For Depth Jumps..*

One issue that sometimes arises with depth jumps is people have different abilities to tolerate overload, so a box height that works perfectly well for one person might be too much or too little for another. You want a degree of overload, yes, but you don't want TOO much overload. The soviets routinely used box heights of up to 36 inches. In my opinion that's too high for most – some recent research indicates no real tangible difference between 12 inch depth jumps and those from higher boxes. Probably the best way is to customize your ideal box height to you. One simple way of monitoring the box height is to always make sure the box is no higher than your maximal standing vertical jump. Thus, if your standing vertical jump is 24 inches you should NEVER use anything higher than a 24 inch box for depth jumps. That's a decent general recommendation, but it's not perfect. Here's a really simple way to dig significantly deeper and determine the optimal box height that you should use:

1. Start off with about a 12 inch box, step off of it, and as soon as you hit the ground immediately rebound up as high as possible, and try to measure the height you jump either thru an object touch, jump pad, or something else.

2. Go up in 6 inch increments to an 18 inch box (the height of a standard bench press) and do the same thing. If the jump from the 18 inch box is about the same (within a couple of inches) or higher than your jump from the 12 inch box, you know the 18 inch box is more optimal for you than the 12 inch box.

3. Keep going up in 6 inch increments until you find a point where your jump really "drops off" and you fail to come anywhere close to the 6 inch lower box. For example, let's say I jump 30 inches from a 12 inch box, 29 inches from an 18 inch box, but only 24 inches from a 24 inch box. That constitutes a pretty severe drop-off, so I know 24 inches is too high for me. Thus, I'd stick with 18. Some people will find their optimal height will be 30 inches or higher, but for MOST people I find a box of around 18 inches works quite well.

4. It's ok to exceed your "optimal height box" by a small margin – 20% or so. But you probably shouldn't exceed it by much more than that. Say you determine your optimal height box is 18 inches. Well, 18 x 20 percent is 3.6 inches. That means you probably shouldn't ever use a box higher than 21.6 inches for any type of depth jumps. You CAN, but your risk of injury will be greater. If you decide to exceed your optimal height it probably should only be done for short periods of time – a few weeks during the middle of a plyometric focused phase. To give you an idea, back in my training heyday I spent 90% of my time doing depth jumps from an 18 inch box and only rarely would perform them from a 24 inch box for short periods of time.

## *Performing A Depth Jump….*

Now I want to talk a little about performing a depth jump: The first thing you should keep in mind is the landing is key. You have to prepare your body for the landing while it's in the air as you approach the foot touchdown. Thus, your body should be preparing for landing and take-off while in the air, sorta like a plane extending its landing gear. As you step off the box you should be "winding up" for an efficient touchdown. Pre-tense your plantar flexors by slightly dorsiflexing your foot (pulling your toes back). This helps increase the rate of force development on your initial blast off the ground. The rest of your body should be in the same optimal position as if you were going to jump. This pre-activation phase actively pre-stretches the muscles and tendons to store more energy before the foot ever comes in contact. You should ALWAYS land on the front half of your feet – never land on your heels. Upon landing you should absorb the force and leap out and up without any unnecessary hesitation or break in the action. Don't speed things up TOO much – you'll likely be on the ground nearly a half second (.400 ms), but if you have to pause and reset yourself, if you stumble and fall forward, OR if your heels hit the ground, the box is probably too high and/or you need to practice your technique. Here are a few more tips for depth jumps:

1. Land about as far away from the box as the box is high. In other words, if you're using an 18 inch box you should land about 18 inches away from it.

2. Consider experimenting with different landing styles. The traditional way of performing a depth jump is landing with both feet simultaneously, yet when performing an actual jump most people use a 1,2 style plant - absorbing force with one foot first then lightly touching down with the other foot before blasting off full force on the concentric action. There may be some utility in performing your depth jumps in this manner. It might be worth experimenting with anyhow.

3. Never perform your depth jumps in a fatigued state. Always perform them when you're relatively fresh. Rest long enough between reps & sets to generate a high quality effort between reps. Typically 10-30 seconds between reps and 2-3 minutes between sets is about right, but if in doubt simply go by feel and performance. I can't say I've ever actually monitored rest intervals of the athletes I train with a stopwatch – we just kinda go by feel and that's fine. A typical set/rep scheme would include around 20 total jumps per session broken down into something like 4 x 5, 2 x 10, or even 10 x 2. I typically prefer to keep it simple and stick to 4 x 5 or 6 x 3. If in doubt let your performance be a guide as far as when to stop, or how many reps to do. Stop a session as soon as it's obvious your performance on the depth jumps has begun to decline, or just before that point. Typically, your first few reps (or sets) will be a little stale then you'll seem to improve every rep for several ground contacts. Then you'll find you hit a plateau and your performance will begin to immediately tail off. That's when it's time to shut it down.

4. If you're fairly advanced and have a significant plyometric training base you might consider adding bands and/or light weight to your depth jumps. Use a light weight of no more than 10% of your bodyweight via either a vest, belt, or banded resistance. Holding weights in your hands is kinda awkward and not something I recommend. Although the research is sketchy, in my experience few things are as effective for improving ankle stiffness as banded depth jumps, or depth jumps with a light weight. Be careful and never do these from a high box, but if you have a good strength base and have some jump training experience under your belt (a year or more) they might be worth experimenting with. One particular potent combo is a form of contrast training. Do a couple of reps of weighted or banded depth jumps followed by a couple of reps of un-weighted depth jumps or running jumps for height.

5. Keep in mind depth jumps are seemingly more trouble than they're worth for some people. The depth jumps are significantly more ankle dominant than a regular jump and not everyone is built in a way they can effectively and efficiently generate much power out of their ankle extensors. Fairly large individuals with wide hips and big legs are often in this group. They just don't get much out of depth jumps (and most other plyos) because they're not built right for them. This will be discussed more in the chapter on joint by joint training for the vert, but the problem is these people are built in a way that they get a disproportionate percentage of their power out of their hips and thighs. They also tend to be heavier, so when they hit the ground on a depth jump they're largely taking their larger more powerful hips OUT of the movement while simultaneously asking their ankle extensors to take on a TREMENDOUS load (due to their higher bodyweight). These people tend not to perform well on any type of plyometric drill, even if they have tremendous measured vertical jumps. Not only do these people not respond well to depth jumps, but they often end up injured from them. If you're the type of guy that has a pretty good standing VJ, but your performance on depth jumps has always been significantly less, and you don't seem to respond to them, you might be one of these individuals, and it might be a better idea avoiding them altogether.

## *What About Depth Drops?*

A depth drop is just the landing phase of a depth jump. You land in a ready to jump position as smoothly as possible, but you don't actually jump. I used to use them quite a bit but don't really make depth drops a big focus anymore because I found they just didn't seem to be as effective as I wanted them to be. Having said that, they do offer *some* utility and are certainly a viable training option. They allow you to focus on proper landing technique and can help prepare your muscles, tendons, and ligaments for depth jumps. I like to use them during a max strength or hypertrophy cycle and then transition into depth jumps. You can use a fairly high box for depth drops – a height at least equivalent to the height of your best running vertical jump. Just make sure you land softly and can control the landing 100%. You should be able to land on the front half of your foot up on the balls of your feet without your heels ever touching the ground and without your knees coming forward way over your

toes. If you can't do that the box is probably too high. A typical set and rep scheme I use for depth drops is 4 sets of 5 reps 2 or 3 days per week.

## *A Forgotten Shock Method - Bounding*

Bounding variations are certainly not new but their relative value may have been overlooked in years past. Next to depth jumps I'd have to consider bounding variations the next best plyometric variation, and some would argue they're even more effective. I personally consider bounds a shock method just like depth jumps. High intensity bounds are perhaps even more stressful than high intensity depth jumps. Bounding is nothing more than a sprint with exaggerated strides, hopping down the track from one foot to the next. They can be easy or difficult based on the length of your strides and the height of your jumps.

Bounding is particularly helpful for single leg jumps due to the stress they put on the ankle joint – that stress also can make them rather dangerous if you're not ready for them. I suggest you start off with very easy bounds, not much more than a sprint stride with a jump. See the videos on the site for an example. From there gradually increase the intensity as you get used to them. It's best to perform bounds on a relatively soft surface such as grass or a good track. Bounds can be performed 2 ways: You can do them for max height or for max distance and you can do them with 2 legs or with one leg. The distance variation is more helpful for sprinters and the height variation is better for jumpers. Personally, I prefer a mix of height and distance.

I would never suggest anyone perform intense bounds on a hard surface such as concrete. You can get away with that on any other plyometric drill, including depth jumps, but performing single leg bounds on concrete is just asking for injury. When performing bounds distances will typically cover anywhere from 15-30 yards with anywhere from 3-6 sets. The combination of bounds and depth jumps is a potent and formidable combo for practically any vertical jump seeking athlete.

## *A Special Type of Depth Jump…*

As noted earlier, the regular depth jump is basically a loaded vertical jump with more calf and quad involvement than a regular standing jump. There is one other way to perform depth jumps that can increase ankle/calf involvement even further: Joel Smith had an interesting study published in January '11. He had noticed that people performing depth jumps that have an overhead object to try and touch usually performed better and got up higher off the ground. The study was meant to determine the difference between a regular depth jump from a 18 inch box, a depth jump from an 18 inch box with an overhead goal, and a depth jump from an 18 inch box over a hurdle. The authors used Division III male athletes and after a standardized warm-up had them perform four of each type of depth jump.

The hurdle height was individualized for each athlete. Prior to the test the athletes determined the highest hurdle they could jump over using a counter-movement jump. For the testing conditions, the hurdles were set to be 5cm lower than that height. The overhead goal variant used a Vertec and when the subjects landed from the depth jump they were instructed to jump as high as possible with that height being recorded by the Vertec.

The results showed the following:

• The depth jumps over a hurdle had almost 25% shorter ground contact time than the other two types of jumps.

• The jumps over the hurdle resulted in less knee and hip bend than the other types of jumps.

• Ground reaction forces were greater for the hurdle group by 16-17%.

• Power generation and power absorption were greater in the ankle and knee extensors for the hurdle group. This may be important as the ankle is the largest power absorber and generator in unilateral power production.

So, what does all this mean? Well, basically the jumps over the hurdle involve more of the ankle extensors and also favor shorter ground contact times. The natural tendency when jumping over a hurdle is to lift the legs up off the ground quickly and that may have some positive impacts for the development of reactive strength in the ankle extensors (calves, achilles etc.). The less knee bend involved in a jump the more elastic energy will be involved, and the greater the knee bend the more likely some of that elastic energy gets dissipated. Thus, the depth jumps over hurdles may be valuable for athletes that are "heavy-footed" and really need to work on having better footwork and quickness. The typical strong powerlifter type often fits this description. They may also be really valuable for athletes in sports that require a shorter ground contact time, such as the high jump (or sprint).

That doesn't mean you should immediately replace all your depth jumps with depth jumps over hurdles, but they may be worth including, particularly for single leg jumping. The one drawback to jumps over the hurdle is there's a pretty significant skill component and injury can result if you're not

careful. It's also worth noting that one can likely get the exact same benefits by performing regular depth jumps with a short ground contact time, which will be discussed in the next section.

## *Short SSC vs Long SSC…*

Have you ever wondered why some people can be very successful in one type of jump but not another? For example, some people are really good at jumping off of one leg, but are terrible when jumping off of 2. Additionally, some people can jump really high, but when it comes to sprint speed (a form of jumping) they don't do so hot. Well, some researchers have observed that jumps using extremely short ground contact intervals of .2 seconds or less could be considered an entirely different motor skill than jumps that utilize longer ground contact times of .4 seconds. Ground contact time is just a name for how long your feet are on the ground during the final plant phase of a jump, or how long they're on the ground in a sprint. If an activity uses contacts of around .2 seconds or less it's considered ***short response reactivity***. Sprinting is a perfect example of this. Top sprinters will often have ground contact times of .1 second or less. Track and field jumping events have a ground contact time during the takeoff phase between 0.11 to 0.12 seconds for the long jump and 0.17 and 0.18 seconds for the high jump.

In contrast, a regular 2-foot jump could be categorized as ***long response reactivity***. Ground contact times during a bilateral jump, or regular depth jump for height, will typically be around .35 to .4 seconds, so quite a bit longer than something like a high jump. Here's the thing: It's very possible for an athlete to be proficient in short or long response activity but not the other. There are other factors involved, but this helps to explain why some people can jump VERY well bilaterally but have a horrible running unilateral jump, and vice versa.

One factor that may partially explain this is inherent muscle fiber type. Years ago a sports science researcher by the name of Carmelo Bosco determined that individuals who jumped well in a countermovement jump with a very slight knee bend, or jumped well in a depth jump with an intentionally short ground contact time, inherently had more fast twitch fibers than normal. They were also faster in a sprint. People with less fast twitch fibers may have jumped just as high, but they tended to use larger knee bends or stayed on the ground longer in a depth jump.

Anyhow, there are quite a few interesting correlations between performance in various jumping drills and various jumps. For example, one study found that 61% of performance in a single leg running jump for height is can be predicted by performance in a depth jump for maximal height where the ground contact time during the depth jump is kept to a very minimal level. In the same study, it was found that only 38% of running single-leg jump performance was explained by performance in a depth jump for maximal height with no regard for ground contact times. In other words, people that could jump VERY HIGH on a depth jump didn't get much carryover into their running single leg jump unless they were also able to perform a depth jump very well using short ground contact times. Performance in depth jumps using very short ground contact times has also predicted sprinting success. In other words, performance in depth jumps where you spend as little time on the ground as possible better correlate with the ability to jump high off 1-leg or run fast.

Young and colleagues demonstrated in a study done in 1999 that a 6-week training program involving depth jumping while simultaneously striving for low ground contact times can increase a running single-leg jump for height more so than a training program involving depth jumping for maximal height with no regard for time spent on the ground. In jumping for an optimal combination of height and ground contact time, they found a ground contact time of .2 seconds to be optimal for single-leg jump training improvement.

So, what does all this mean? Basically all it means is that if you want to jump high off one leg and/or run fast you're better off performing your depth jumps with a short ground contact time, or performing variants such as a hurdle depth jump previously discussed. Instead of performing your depth jumps with an aim to simply jump as high as possible, do them with an intent of jumping high while coming off the ground as fast as possible with little knee bend. This could help your single leg jumping and it could also help your sprint speed. If you're mainly interested in improving your bilateral (2-foot) jump, just try to jump as high as possible when you hit the ground and don't worry how long you're on the ground, or how much you bend your knees.

# *Common Myths About Plyometrics….*

Now that I've given you a bunch of reasons why you SHOULD do plyometrics, and given you a ton of science on the subject, I'm gonna back-track a little and tell you why plyos are in many ways over-rated or misunderstood:

**Myth #1 – Isolated plyometric ability Is a HIGHLY trainable quality**

The truth is, how much additional force you get from a *plyometric* movement is mostly unique to you as an individual. True isolated plyometric ability isn't something you can really impact through training to a large extent. In fact, it's likely to get less as you grow as an athlete. ** Let me explain:

Let's say your standing stationary vertical jump is currently 20 inches and your 3-step countermovement vert is 26 inches . Since the approach jump is more plyometric than the standing jump, that basically means you're getting 6 additional inches from *plyometric* action. What I'm saying is you're unlikely to see the difference between your standing and running vert grow much. It can (and often does) get smaller, but it rarely grows much, even as your overall performance gets better. The general trend in any developing athlete is to get more proficient at exerting force under smaller windows of opportunity, so over time you don't need as much wind-up to generate the same amount of power. In sticking with this same hypothetical example, let's say you improve your stationary jump from 20 up to 35 inches. You're still unlikely to find much more than a 6-inch difference in your running countermovement jump (41 inches), regardless of what type of training you did to get there. Generally speaking, the longer your limbs and legs, the greater you'll benefit from plyometric wind-up. The shorter your legs, the more likely you are to jump almost as high from a virtual standstill.

***There are some exceptions, such as seen in single leg jump specialists, yet that's basically an entirely different event than a 2-foot (bilateral) jump.*

That brings us on to our next myth:

**Myth # 2 Plyos work because they isolate and train "reactive" (plyometric) ability**

It's not that plyos don't train reactive ability, it's that the average person is confused on what reactive ability really is. It's largely an observation of movement and, rather than existing as an isolated strength quality, it's largely a combination of strength qualities also present in other common training methods (starting strength/RFD). In other words, it is possible to significantly impact reactive ability even without performing exercises specifically geared towards improving it. For example, it's definitely possible for barbell exercises alone to positively impact it. Going back to the 2 basic qualities that all training methods influence they are:

A: Movement Efficiency
B: Horsepower

Plyos can affect both qualities: They directly enhance the movement efficiency aspect because they allow you specific practice and rehearsal carrying out a variety of jumps, bounds, and hops - which closely rival jumps for height or sporting movements. With regard to the horsepower aspect, power in most sporting movements is a result of how much force you exert at the hips, knees, and ankles. The ankles are most important for absorbing forces, allowing your quads and hips to drive your movements. Most plyo variations provide a low degree of overload to the muscles acting on the hip, a moderate degree to the muscles acting on the knee, and a **HIGH** degree of overload to the muscles acting on the ankles, due to the constant eccentric forces involved in landing. Many athletes are less than stellar because they lack "stiff ankles". Practicing plyo drills on their feet allows people to improve their coordination and ability to generate high-velocity force thru the ankles.

So, in a nutshell, plyos work largely because they allow you to practice your jumps and increase high velocity jump specific recruitment, inter-muscular coordination, and intra-muscular coordination between your muscles and tendons - largely neural factors which I consider movement efficiency stimuli. They also provide a great deal of eccentric forces and make the muscles acting on your ankle (plantar flexors/calves) stronger, more powerful, and help you become "stiffer."

**Myth #3 – Shock plyometrics (depth jumps and depth drops) give supra-maximal muscle recruitment**

***Muscle recruitment*** refers to how well you use, or *turn-on,* your existing muscles. The age old explanation is that great athletes are great because they can turn on more of their existing muscle in a given task compared to regular folk. The age old thought was that normal people could only call upon around 50% of their muscles in a sports movement, while superstar athletes could utilize up to 90% or more. However, because of the shock method all hope was not lost for the average Joe athlete. The Russians believed imposing a high amplitude involuntary forceful stretch on a muscle, such as what happens when you drop off a box, hit the ground, and jump back up (the shock method), causes the muscles to involuntarily stretch. In an effort to protect the joint this caused the involuntary recruitment of **ALL** existing muscle cells. Over time this would allow the average Joe to become like a superstar. Sounds great eh?

Unfortunately, we now know this is partly a myth. It is true that great athletes get more out of their muscles, but not because of muscle recruitment. Even beginners can recruit all their muscles. Anything equivalent to around 80-85% of max voluntary tension will do it. The reality is great athletes can really ramp up something called rate coding and this is largely due to the inherent excitability of their central nervous systems. What the depth jumps really are is:

A: A form of loaded jump training
B: Specific strength training for the plantar flexors, and to a lesser extent, the knee extensors (quadriceps).

As mentioned earlier, depth jumps are basically a loaded Vertical Jump. They provide a substantial eccentric overload to the plantar flexors (ankles/calves) and the quadriceps. The depth

jumps CAN be somewhat effective as a lone training method, but they have a weakness in that they are quad and calf dominant and don't affect the glutes and hamstrings strongly. They're still effective, just know what they're capable of and what they're not. You need good baseline strength at the hips and knees (a good squat) before you can really benefit much from them. Think about it - If shock methods were all they were cracked up to be all these guys doing parkour would be jumping out of gyms.

**Myth #4- Plyos/Depth jumps don't improve standing vert**

This is one I hear a lot. The reality is anything you do to improve one form of vert will likely improve the others. Providing you can benefit from them, in my experience depth jumps do a good job of improving standing vert. Providing you work at it, squats will also improve running vert. The reality is it's about impossible to improve your running vert without improving your standing and vice versa. The exception is increased strength in squats won't always impact your running single leg jump.

**Myth #5- One MUST perform plyos**

It is certainly possible to jump high without performing plyometric drills as long as perform plenty of actual jumps and get stronger. You can certainly add in some plyos for enhanced effectiveness, BUT, in my observations more people screw up by doing too many plyos than those that don't do enough. Many athletes already border on overusing their ankles and knee extensors thru sport, and plyos surely don't do anything to help that. If you play a sport requiring jumping it's best to put plyos on the backburner when you have lots of fatiguing practices to contend with. The majority of athletes I've worked with, consulted with, or heard from that reported spectacular VJ gains did fairly little in the way of plyo training. That's not to say they wouldn't have benefitted from plyos, but with frequent practices and games they weren't really an option. If you're 100% structurally sound and you're in offseason mode you might be the type that can get away with considerably more plyo work.

**Myth #6- Plyo workouts should be fatiguing and exhausting**

Most young guys think a plyometric workout should be as fatiguing as an Army Ranger obstacle course. The reality is completely opposite. In general you should finish any plyo workout just as fresh, if not fresher, than you were when you started. You should keep the reps low enough so that each rep is a quality rep. I typically prefer to use plyos as sort of a warm-up to strength training, as you'll see in the workouts section of this manual. Part of the problem is even high intensity plyometrics typically don't *feel* all that exhausting, so athletes are prone to do more than necessary. Remember, the effectiveness of plyos is largely due to their ability to hone your mind to muscle connections for jumps, which takes *quality* rather than *quantity*.

All this isn't to say that plyos don't have a solid place in your routine or shouldn't be emphasized. For best results you want to make use of **all** available tools at your disposal. I personally prefer a routine that has strength training, ballistic training, and plyometric training.

# *Rate coding and The Stimulation Method…*

Remember when I was talking about the depth jumps and how they were purported to work due to the superhuman recruitment they would allow one to achieve? Well, they fell short in that regard, but in my opinion exercises designed to boost rate coding are better at doing what the shock training was supposed to do. So, let's talk a little more about rate coding and how to improve it: As discussed earlier, ***rate coding*** refers to the frequency that your brain sends messages telling a muscle to contract. The faster the frequency, the greater the intensity of any given muscular contraction. Rate coding is similar to beating the heck out of a drum with your hands. The faster and harder you move your hands, the stronger the beat. Rate coding appears to be highly related to the excitability of the CNS. Activities that cause a great psychological arousal fit the bill here. This explains why we're capable of exerting more force in our movements when we're excited or slightly nervous. A highly excitable nervous system can produce greater force at a faster rate, due to the positive impact on rate coding.

## *The Importance of Adrenaline…*

We all know that when we're excited, really pissed off, or nervous, we seem to be able to function at a different level. The extra rate coding resulting from adrenaline explains why people tend to be stronger, more powerful, and faster in competitive situations. Even trained athletes can have as much as a 10% difference between what they're able to do in a regular non-stimulated setting compared to what they're able to do in a competition effort. It also explains why the more advanced you become the less likely you are to be able to match your personal best every single day. A beginner athlete can run, lift, or jump their personal best any day of the week. A highly advanced athlete might have 6 months to a year or more in between their truly top efforts.

The greater the extent to which a person can tap into rate coding, the more stressful each effort tends to become. This makes sense as it's largely an adaptation built on the stress response pathways. Things like testosterone sensitivity in the CNS are thought to have a significant influence on rate coding. The best look into natural rate coding capacity in my experience is looking at a person's natural quickness. How fast are your hands? How quickly can you move your feet? Although you can't influence rate coding as much as you can a quality like strength, it can still be improved to a significant degree.

# *How to improve rate coding*:

**A: Practice and hone your "skill" of choice**

Imagine taking someone who's never driven a car before out on the track and handing them the keys to a souped up F-1 race car. They might as well be driving a Honda civic, as their lack of driving skills would leave them unable to benefit from the supercharged motor. In much the same way, you have to

be coordinated and fluid enough in your movement patterns of choice to benefit from additional supercharging.

**B: Increase your strength**

The neural activation that takes place via maximally engaging your muscular system thru strength training trains your nervous system to function more efficiently and powerfully in all activities. Boost your strength and you automatically tend to boost your rate coding capacity in other activities involving like muscle groups. This is the concept of "organism strength" that sprint coach Charlie Francis used to preach. The thought is that being "strong" all over would positively affect any activity requiring maximum force or power.

**B: The Stimulation Method**

This is the most direct way for the average person to improve the rate coding aspect of neural efficiency, and one I want to focus on: Here you perform one higher tension exercise to really activate or excite the CNS. You then follow it up with an exercise to take advantage of the CNS excitability, which temporarily boosts rate coding. In essence you're "tricking" your nervous system. Think of a baseball player swinging a bat loaded up with weights before he steps in the batter's box. The idea is the loaded bat allows him to swing a regular bat with more speed and power. The stimulation methods are also known as ***post-activation potentiation methods,*** *or PAP for short*. With potentiation methods not only do you get a temporary boost in performance, but over time the thought is your body becomes more sensitive to the neural discharges from your CNS and learns to accept a new level of force as being normal for a particular movement, even in the absence of psychological excitement. Here are a few examples:

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**Stimulation Methods for Jumps**
==========

Squat x 5 with 85% load - rest 5 minutes
followed by:
Jump squat- x 6-8 with 20% load
---
Jump with weighted vest or barbell with 10-20% of bodyweight x 5 rest 3 minutes
followed by:
Unloaded jump x 3
---
Depth jump x 5 from 24 inch box rest 3 minutes
followed by:
Running jump for max height x 3
---

¼ Squat 2 x 2 at 90-95 % - rest 4-6 min
followed by:
Depth jump- 2 x 5 from 24-30 inch box.
---
Lightly loaded depth drops (~10% of bodyweight) x 2 rest 3 minutes
followed by:
Bodyweight depth jump for max height x 2
---

Hang snatch x 1-3 reps
Followed by:
Jump or plyometric variation of your choice
---

45 degree back extension iso hold by 10 seconds rest 2 minutes followed by: jump variation

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## *Stimulation Methods for Strength*…….

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Squat walkout x 110% (or more) of 1rm walkout and hold x 10 seconds rest 3-5 minutes
followed by:
1rm Squat
==========
1/4 Squat x 110% of 1rm x 2 rest 3-5 minutes
followed by:
1rm squat
==========
Snatch grip partial deadlift from high blocks with shrug - x 3 reps rest 5 minutes
followed by:
Hang Power snatch x 2-3
==========

Potentiation/stimulation methods such as these can be effective at any time, but they're most effective during specific peaking phases, as overusing them can burn you out and make them less effective. They're also most effective for intermediate level trainees. Beginners don't benefit as much because their nervous systems are unable to clearly ramp up force production from one movement and carry it over into another. Their nervous systems are unable to really differentiate the load between the 2 movements effectively. Advanced athletes can benefit, but not as much as intermediates because they don't need them - their nervous systems are psycho enough already without any external tricks. That leaves these methods most effective for intermediate level athletes. How do you tell which group you're in? If in doubt play around with some of them and see.

## *What Not To Do..*

The stimulation methods like those above are great for promoting immediate personal bests and are great for peaking. However, they also have their drawbacks: Anytime you set a significant PR (personal record) in a high state of arousal you run the risk of burning yourself out. This is particularly true when someone sets a PR and gets so amped up they continue the same type of training for days or weeks on end trying to match or improve upon the PR. Some of these methods can be used year around in manageable volumes, but it's generally advisable to restrict their use to short 2-3 week cycles when you want to peak.

# *Assessments: Determining Your Ideal Training Focus…*

Now that I've covered in detail a multitude of training methods and background a logical question is, "How do you know what YOU as an individual should focus on?" Should you focus on strength training or rate of force development? Ballistics, O-lifts, plyos, or all of them? It's a great question and what this section is designed to answer. First of all, you gotta look at the body like a mechanic and ask yourself what needs tuning and what needs fixing? Rarely do 2 cars that come into a mechanics shop have the exact same problem, and before you can identify the solution you gotta identify the problem. That's why when people ask me, "If I do this how much will I increase my vert?" I honestly couldn't tell you . For one thing I haven't even looked at your motor and determined what's wrong with you and what TOOLS and tool combinations you might require. The process of training is not a perfect science -everybody responds differently.

Having said that, a well designed program will be 80-90% applicable to all athletes involved. Individualization is about the 10-20% of modifications you have to make to correct individual weaknesses or imbalances. Several years ago when I released the Vertical Jump Bible I helped introduced the concept of reactive deficit testing to the mainstream world. The concept had been around a while but mainly thru obscure foreign literature. But anyhow, the idea with reactive deficit testing is to individualize workouts for each particular individual: The basic idea is that a vertical jump is part strength and part plyometric and each individual usually relies more on one than the other. Thus, people are either strength dominant or plyometric dominant and should assess using a simple bounce depth jump test to see where their dominance is. They should then seek to emphasize the weak part to "balance out" their weaknesses. You'd compare a regular standing vertical jump to a vertical jump done while dropping off a 12-18 inch box and bouncing off the ground. A person who jumped higher in a normal countermovement jump from the ground (compared to a bounce jump from an 18 inch box), relies more on strength (or is plyometrically weak) and should emphasize plyometrics in their training. A person that tested better bouncing from the box is plyometrically efficient and should emphasize strength training. It works decent for most people most of the time, but there are several issues with this that need to be expounded on, as I feel after I started talking about individualizing programs some people took that concept a bit too far.

## *A Few Issues…….*

Coordination issues and lack of a sufficient training base can significantly influence the results of these types of tests. Without a low level base of strength tests such as the reactive deficit test are relatively useless. Too many people overlooked the first part of the assessments section in the VJ Bible where I recommend assessing leg strength as an initial qualifying assessment. Lack of strength will cause one to test plyometrically weak because they're not strong enough to absorb the force of the landing in a bounce jump. Imagine a 90 year old woman doing a depth jump. What's gonna happen? She's gonna crumble as soon as she hits the ground because she's not strong enough to absorb the force. Before you can "spring" out of a landing you have to be able to absorb the force from the landing. The science

doesn't lie: The best predictor of eccentric power (such as landing from a depth jump) is maximal eccentric strength, which is highly correlated with any traditional measure of strength (1 rm testing etc.)

Thus, beginners without a solid strength base should almost ALWAYS focus on basic leg strengthening exercises and movement efficiency exercises (low level plyometrics) to emphasize coordination in the jump. Once they have some semblance of coordination and strength, assessments like the bounce test then become more valid. The problem is too many people think they're different and can skip the beginner stages. I would consider a 1.5 x bodyweight squat the minimum amount of strength needed before worrying about anything else than basic strength and low level plyometrics. Now please go back and read the first sentence in this paragraph again: "Beginners without a solid strength base should ALWAYS focus on basic leg strengthening exercises and low level plyometrics to emphasize jumping coordination!" With that said there are a couple of other problems with attempting to always emphasize weaknesses:

## *Strengths Tend To Remain Strengths and Weaknesses Remain Weaknesses...*

Strengths tend to remain strengths and weaknesses remain weaknesses: In the bounce jump test there is a significant percentage of the population that will NEVER get up higher off of a bounce than they will off a standstill from any height of box. They're simply not wired for it structurally and neurologically. Tests like the bounce jump test really do 2 things:

A: They assess movement efficiency (coordination) and power in the jump and the relative explosive efficiency of one's inner neurological makeup

B: They help measure where a person is powerful and where they are weak with regards to muscular contribution and power in the Vertical Jump, as I'll go into much more detail on later.

A person can be a hip dominant jumper, knee dominant jumper, ankle dominant jumper, or any combination. One who is extremely knee and hip dominant with a thicker build who is not genetically wired for speed will not tend to perform well with any sort of plyometric assessment. They are strength dominant people and will typically make immediate and long term gains emphasizing strength qualities. They will always tend to be "slower" jumpers. They will always tend to use a naturally deeper knee bend when they jump, and they will RARELY test superior plyometrically regardless of what they do.

A better approach in my opinion is to train to keep an eye on your weaknesses and know what they are, but still train and plan to address all qualities long term. Minimize your weaknesses while building your strengths. Yes, you can and should work on weaknesses, but within the context of a properly designed longer term program that "hits" on all qualities. If all you ever do is work your weakness it won't always get you where you want to go. Now I'd also like to talk a little more about assessments:

## Assessments

When it comes to plyometric assessments like the bounce jump test, virtually all of them try to characterize the display of power into strength dominant or speed dominant. Rather than getting extremely technical with the assessments, I prefer to look at several of them and make the following holistic observation:

A: Either you're stronger then you are fast
Or:
B: You're faster than you are strong

For all practical purposes being faster than you are strong is the same thing as being *plyometric dominant,* and being stronger then you are fast is the same thing as being *strength dominant*.

Those who are ***stronger than they are fast*** will tend to have many of the following characteristics:

A: You will tend to be much faster at the start than the end of a sprint race (40 yd or 60 meter sprint)

B: You will tend to have a shorter, thicker build, with larger ankles and calves, thicker wrists, shorter legs, and longer torso.

C: Your running 2 foot vertical jump will be nearly the same as your standing vertical jump (around 4 inches or less difference). You will tend to perform a standing 2 foot jump with plenty of knee bend and bend over quite far before you jump.

D: Your running bilateral vertical jump will be higher than your running single leg vertical jump

E: The bounce depth jump (off any height box 12 inches and up) will tend to be about the same or lower than your stationary countermovemet jump

F: Your strength will be ahead of your speed and movement efficiency (You have naturally good weight room numbers but not so naturally impressive speed and vertical jump numbers. See the vertical jump calculator at www.higher-faster-sports.com/verticaljumpcalculator.html for a reference point)

G: Naturally quad and calf dominant - (Calves and quads are naturally better developed and stronger than glutes and hamstrings)

H: Naturally less ripped and defined

I: More strong and powerful than “quick.” Naturally less quick in rapid fire low tension movements (slower hands and feet) - even if explosive & powerful

Look for general trends and characteristics. You probably won't answer yes to ALL of these but look for a majority. Those are all indicators that your body is more genetically wired for strength than speed.

Those who are ***faster than they are strong*** will tend to have many of the following characteristics:

A: You tend to be faster at the end of a sprint than the start (Naturally good top speed)

B: Your build is lean and muscular and it's not difficult for you to stay that way

C: Your strength has never been dramatically ahead of your speed in any event

D: Your best bilateral (2-legs) vertical jump from either a run-up or a depth jump is about 20% or more higher than your best jump from a standstill.

E: Your bounce depth jump (from a box height of 12-18 inches or more) will be higher than your best standing vertical jump.

F: Naturally glute dominant - (Naturally less muscular calves and quads and better developed glutes).

G: Naturally longer Achilles with higher calf insertion points

H: Naturally more quick - even if not explosive and powerful

Again look for general trends and characteristics. These are all indicators that your body is more genetically wired for speed than strength. If nothing jumps out at you then you likely don't have a lean in any direction.

## *Which Way Are You Wired?*....

Those who are wired more for strength (stronger than fast), often need to pay close attention to movement efficiency work to optimize their coordination. They often have body structures, postures, and physiques that are not naturally 100% conducive towards being athletic. They need to pay close attention to muscle recruitment issues and mobility impairments. They tend to not respond particularly well to muscle growth or weight gain. They do tend to respond well to explosive training, such as jump squats, olympic lifts, speed squats, and plyos, at least initially, but they shouldn't necessarily neglect strength work. They rely on their strength to get things done and often will continue to get better and better as they get stronger and stronger, at least up to a certain point. **

Those who are more wired for speed than strength (faster than strong) are relatively easy to train and do not typically benefit much from pure speed/plyo/and explosive work. They're naturally good at expressing their strength well, so the way for them to progress is by building more strength. If

they're not getting stronger in some fashion on a consistent basis they're often not progressing. They respond well to muscle growth and often continue getting faster and more explosive as they gain bodyweight thru muscle growth. In large part the main thing they need to focus on is getting stronger.

Those who have no obvious lean one way or the other could be considered balanced. Balanced individuals can train like those that are faster than strong, they just can't benefit from the same things to the same extent. In other words, someone that is truly faster than they are strong primarily benefits from more raw horsepower – muscular strength – but they're a little extreme as far as how much of it they can truly benefit from. They're the type that can perform a routine of nothing but squats and benefit immensely from it. The balanced individual can benefit from strength, but needs more balance I his routine. Those highly wired for speed may continue gaining inches on their vertical jump all the way to the point where their squat is 2.5 or 3 x their bodyweight and may jump higher and run faster after gaining 30-50 lbs of muscle. A balanced individual may only benefit from a 2 x bodyweight squat and might only be able to gain 10 or so pounds of muscle.

*** It should also be noted that those who are naturally stronger than fast will have a naturally lower potential for improvement. There's only so much you can improve CNS processes and leverages and those tend to become the ultimate limiting factors.*

All groups should seek to train properly long term with a lean towards maintaining or building their strengths and addressing their weaknesses. See the chapter on periodization for more detail on how to do exactly that.

## *Just Give Me The Simple Formula….*

Now I realize all that stuff above may be confusing for some people, so I'm going to make it a lot easier for you: Here is a very general and simple outline to test to determine where your focus should be for a vertical jump program:

**A: Look at your general leg strength:** If you can't squat at least 1.5 times your bodyweight you should ignore the results of any other test and simply focus on improving your strength. If your strength is up to par you can move on to the next assessment:

**B: Reactive Resource Assessment:** Measure your standing vert. Next start off with a 12-inch box and perform a depth jump. If you got higher on the standing vert AND your strength is up to par you likely would do well initially with more of a focus on plyometrics, rate of force development training, and other explosive oriented training – basic explosive oriented templates such as those you'll find in the programs section of this manual. Once you've spent a couple of months working on your explosiveness you can follow more of a balanced protocol. If you got higher on the bounce jump test you will likely do better continuing to drive up your strength, maintaining plyometric efficiency, and perhaps even benefit from some hypertrophy oriented training, so there's a good chance you'll do better on the strength oriented or balanced templates you'll find in this manual. That's all there is to it!

# *Assessing and Developing the VJ – Joint by Joint…*

To expand a bit more on athletic assessments specific to the vertical jump, as noted, the traditional view was that the VJ was a combination of strength and plyometric ability. Those who used their "strength" to jump were considered very different than those who used their "bounce" to get up. That is, those who could jump a lot higher with a bounce or a run-up were said to be plyometric dominant because they relied more on the stretch reflex. The belief is this type of individual relied more on non-contractual muscular elements to get up, such as the tendons. In contrast, one who could jump higher from a virtual standstill was said to be more strength dominant, because he relied less on the stretch reflex and more on raw strength.

With the *modern* approach we know the concept of plyometric performance via non-contractual tendinous contribution has been overstated in years past. Some people will test relatively better plyometrically than others, but not necessarily so much because of tendons or any other fancy physiological characteristics. In the practical sense there is only strength, but that strength can exist to varying degrees, in various body-parts, in each individual. In addition to that, body structure, bodyweight, and neural processes (such as CNS processing speeds) are also highly variable. Those things, in combination with varying degrees of strength, can cause people to demonstrate their jump in different ways, due to either strength or leverage.

## *Your Strength and Where It's At…*

The vertical jump (any type of jump) is a combination of ankle extension, knee extension, and hip extension. As noted earlier, the muscles primarily contributing are the calves (ankle extensors), quadriceps (knee extensors), and glutes (hip extensors). Science has determined that during a vertical jump no single extensor muscle operates at 100% peak power (or force), but they combine to produce a sum greater than their parts. On average, the contribution to the VJ consists of about 35% hip, 35% knee, 15-20% ankle, and the rest upper body. BUT, it is possible for there to be individual differences in the amount of particular extensor/muscle contribution people primarily use to get off the ground, and it does change a bit based on the style of jump.

In a nutshell, some people are more quad dominant, others are more glute dominant, and others are more ankle dominant. The dominance affects the jump, affects performance on different types of jumps (running vs standing, 1 leg vs 2 etc.), and also affects the optimal training for the jump! For years I have had a hunch this was the case and last year a really neat study kinda verified it. It analyzed people completing different styles of jumps and found there could be a LARGE variance among various extensor contribution between different individuals in different types of jumps. *(See "The Effects of Three Modified Plyometric Depth Jumps and Periodized Weight Training on Lower Extremity Power" United States Sports Academy)*

One who has exceptionally powerful and strong calves (ankle extensors) might be able to step from a box to the ground and rebound very high with very little knee bend and a very short ground contact time. They might also be able to perform consecutive straight legged ankle jumps in place and rebound very high, as these are inherently ankle (calf) dominant exercises. One with exceptionally

strong and powerful quadriceps and glutes (respective to the rest of his body), can probably jump nearly as high from a standstill as they can with any sort of a bounce or run-up. So called "back-jumpers" usually fall in this group. They bend their legs a lot when they jump and really bend their back over to really get a lot of wind-up out of their hips and quads. One with balanced development will typically have a running jump around 15% higher than a standing jump. One with strong calves and hips, long legs, and light bodyweight, will tend to excel in the unilateral jump. One with wide hips and heavy bodyweight probably won't be jumping really well off 1 leg no matter what he does – nor will he perform well on depth jumps or ankle jumps. This is simply because his structure is naturally heavy and he's unable to get enough oomph out of his ankle extensors (calves) in those movements to overcome his higher bodyweight and poor leverages.

It stands to reason your body will tend to emphasize the stronger muscles, which isn't always a good thing. For example, if you're TOO much of a quad dominant jumper you'll inherently tend to put a lot of stress on the knees.

We can't control limb lengths and body structures, but we can control or influence your strength and where it is at. With the joint by joint approach to VJ we're simply taking the 3 joints involved in the VJ, taking the muscle groups associated with those joints, and hitting them with various exercise along the speed and strength continuum. First, let's take a deeper look at what joints/muscles are dominant in you as an individual:

## *What's Your Best Jump?..*

The standing vert is mostly all hip and knee with relatively little ankle contribution. If someone has a standing vert really close to their other jumps (less than 15% difference), and they bend down and lean over a lot when they jump, they're almost assuredly a hip/knee extensor dominant jumper. (This also explains why squats are so notorious for the impact they have on the standing jump – the squat is knee extensor dominant). The depth jump is a lot more ankle dominant than the standing vert and also a bit more knee – it largely takes the hips OUT of the movement. The running bi-lateral jump is pretty much a combination of all 3 extensors. Thus, if ones best depth and running jumps are well ahead of their standing jump (around ~20% more) they will tend to be more ankle dominant. Lastly, the running uni-lateral jump is VERY ankle dominant and more hip dominant than the running bilateral jump. So, someone who does well (or terrible) on it will tend to fit the profile of hip *and* ankle dominant.

Looking at it test by test:

**Standing vert** = more hip and knee (hip and knee extensor contribution about equal)
**Running vert** = equal hip, knee, and ankle (roughly)
**Depth jump** = more ankle and knee, less hip
**Running 1 leg jump** =more ankle and hip, less knee

Everyone is a combination, but most people will have one that stands out a bit. Let's go thru a few typical scenarios: Roy has a really good standing jump, a terrible depth jump, and a terrible running

1-leg jump. Where do you think his dominance is? Well, we know it's not in the ankles right? Might it be his quadriceps? Maybe so - let's assume he squats double bodyweight. That means his quads and hips are likely very strong. That leaves ankles and inherent body structure as the only possible weaknesses.

Wayne has a good 1-leg jump, a good running bilateral jump, but his standing vert and depth jump are both terrible. He jumps a good 20% higher off a run-up than he does standing. He weighs 150 lbs and squats 185. Where is his weakness? Well, it's obviously his quadriceps. He needs to get his squat strength up.

It's important not to overanalyze this too much because individual body structures can also have a significant influence on your performance in various jumps. I'll use myself as an example:

My standing vert is (or was) very good and there's never been a very large difference between my standing vert and running vert (maybe 4 inches). Based off that I could be considered a hip/knee dominant jumper. However, my depth jumps are even better relatively than my standing vert – I could step off a 12 or 18-inch box and rebound higher than my standing jump, so that indicates my ankle extensors are very powerful too. BUT, my running 1 leg jump sucks and always has.

So, let's analyze it joint by joint and jump by jump: I know my ankles aren't weak because I do so well on the depth/bounce jumps, which we know are ankle dominant. I do good on standing jumps too, so that indicates either my knee extensors or hip extensors are really good, maybe both. The standing jump is more quad and hip and the running single leg jump is more ankle and hip. I have a really good standing jump *and* a good squat so that seems to indicate my quads are good to go. We know my ankles are good to go because of the way I perform on depth jumps, but might my hips be weak? Maybe so, but it could also indicate that I just have a crappy body structure or poor coordination for doing a 1 leg jump.

One thing I could do is a lot of hip/glute specific work and lots of 1-leg jumps and see how my running 1 leg jump responds. Let's say I do and my 1-leg jump still doesn't respond favorably. Well, in that case the only thing left to say is my body structure isn't well suited for the 1-leg jump. That's why you don't want to read into this TOO MUCH. Things like body structure and weight can and will influence things to a very large degree.

Now, let's take a closer look at the muscles involved with each particular joint:

Muscles acting on the hip: Glutes (primarily), hamstrings (secondarily)
Muscles acting on the knee: quadriceps
Muscles acting on the ankle: Calves (gastrocnemius and soleus)

As noted earlier, if you look into the research on muscular contribution to the VJ the majority of the power comes from the muscles acting on the hip and knee, followed by the ankle. Relative contributions vary somewhat for each individual but a general breakdown would be:

30-35% hip

30-35% knee
15-20% ankle
10% upper body (arm swing from shoulders, lats, upper back)

However, some individuals might be 50% knee, 30% hip, and 20% ankle. Other individuals are likely 35% ankle, 35% hip and 30% knee. You can get a general idea of where you're weak and where you're strong looking at your performance in various exercises or athletic feats:

**Ankle extensor dominant feats and exercises**

Calf raise, depth jump, bounce in place with straight legs, jump rope, agility ladder

**Knee extensor dominant feats and exercises**

Standing vertical jump, squat, front squat

**Hip extensor dominant feats and exercises**

Standing broad jump, barbell hip thrust, reverse hyperextension, flying 20 yard dash, hang clean and snatch

Chances are you're relatively better at some of those than others. If nothing jumps out at you then you're likely pretty balanced. Next, we'll lay out exercise targets for each particular joint. For each muscle group we have a strength exercise, a loaded speed-strength exercise, and an unloaded speed-strength exercise. You could also say we'll have a strength exercise, a rate of force development exercise, and plyometric exercise, if you prefer that lingo.

**Ankle dominant exercises**

Strength exercise: calf raise

Loaded RFD exercise: Rhythmic jump squat or barbell skip

Speed-strength exercise: short response depth jump, alternate leg bounding (note: most common plyometric drills are very ankle dominant but the short response depth jump and alternate leg bound are VERY ankle dominant)

**Knee dominant exercises**

Strength exercise: squat

Loaded RFD exercise: jump squat

Speed-strength exercise: depth jump for max height

**Hip dominant exercises**

Strength exercise: barbell hip thrust, reverse hyper, Romanian deadlift (or PROPERLY DONE deadlift)

Loaded RFD exercise: hang snatch, hang clean, dumbbell 2-hand swing

Speed-strength exercise: single leg bounding, standing broad jump, flying 20 yard dash

It should be noted that almost all garden variety plyometric variations are ankle dominant, with the exception of horizontal jumps or bounds for distance, like a broad jump or single leg triple jump. In most plyometric variations the hips are involved in the concentric action, but the majority of the training effect comes from the landing phase, where the ankles contribute heavily to brake the body and reverse direction.

If someone has a terrible running vert relative to their standing vert (only a few inch difference), the likely reason would be due to relatively inefficient ankles (bad feet), or poor coordination, so they might experiment with more ankle dominant exercises in their routine – calf raises and plyometrics.

If someone has a really good depth jump, particularly a depth jump performed with very short ground contact times, but poor 1 leg jump, the hips would tend to be the weak link. So they could emphasize exercises like snatches/cleans, bounding, and barbell hip thrusts/reverse hypers.

If someone has a really good running 1 leg jump or running bilateral jump, but their standing jumps don't measure up, the quads would tend to be the weak link, so they could really emphasize the quads thru exercises like squats and jump squats.

If no obvious weakness stands out, one could simply train all 3 joints about equally.

Next, we'll write out a complete routine based upon these principles. For each muscle group we'll use a strength exercise and at least one speedier strength-speed or speed-strength exercise. The exception is the hips, which we'll just hit with a strength exercise.

**General Workout:**

Depth jump: (or any other medium intensity plyo drill) 4 x 5 (ankle speed-strength)
Jump Squat: 4 x 5 with 15% of max squat (stop and reset before each rep) (knee speed-strength)
Squat 3 x 5: (knee/hip strength)
Reverse hyper or barbell hip thrust: 2 x 10-15 (hip strength)
Calf raise: 2 x 10-15 (ankle strength)
Frequency 2-3 x per week.

The above workout fits the criteria perfectly, which is why this setup has become by far my FAVORITE setup for beginners. It works great because hits all 3 joints in the right ratios in equal proportion. However, we can also set up more strength-oriented or speed-oriented workouts based on the same joint by joint principle:

**Speed-Strength Workout**

Depth jump 4 x 5 (or any other plyo variation) (ankle)

Clean/snatch 4 x 3 (hip)
Jump squat 4 x 5 (knee)

**Strength Workout**

Squat 3 x 5 (knee)
Romanian deadlift, barbell hip thrust, or reverse hyper 3 x 6-10 (hip)
Calf raise (ankle)

With this knowledge you can kinda custom design a split based on where you know you're weak or strong. If all this sounds a little confusing you might wanna read it again, but I caution you don't read into this TOO MUCH. If nothing in this section jumps out at you immediately I wouldn't delve into it too much. Also, make sure you DON'T neglect the other baseline assessments such as your squat strength. That should always take precedence over detailed over-analysis such as that described in this section.

# Workouts

Note: Go to www.higher-faster-sports.com/assessmentsforvert.html password: vertical to see videos of exercises

## Beginner/Novice Workout **(Do this workout if you have little experience training, if you squat less than 1.5 x your bodyweight, or if you're in doubt of which workout to follow)**

**Section A: Plyometric** (choose 2 of the following exercises from section A each workout) (each ground contact equals 1 rep)

1. Low box depth jump: 4 x 5 (use a box around 12-18 inches high)
2. Ankle Jump: 4 x 8
3. Tuck jump: 4 x 8
4. Lateral barrier jump: 4 x 8
5. Low box depth jump onto another box 4 x 5 (stand on one box, drop off and rebound up onto another box, turn around and come back the other way)

**Section B: Strength-Speed (Perform *one* of the following from section A each workout)**

Hang Power Snatch (3 x 3), Hang Power clean (3 x 3), Jump squat with 45 lb bar* (3 x 5) or 2 hand dumbbell swing (3 x 10)

**Pause and reset each rep if you choose the jump squats*

**Section C: Strength (Do all of these)**

Medium stance squat 3 x 5

Start with the bar and perform a set of 5 reps as a warm-up set. Do a couple of more warm up sets of 5 reps continuing to add 25-40 lbs to the bar each set. Continue doing sets of 5 and adding weight until you have enough weight on the bar that your form towards the end of a set starts to get a bit shaky. Then do 2 more sets with the same weight. As soon as you can get 3 sets of 5 perfect reps with a given weight you increase the weight by 5-10 lbs the next workout. Here is an example of how it'll be implemented in practice.

45 x 5 (warm-up)
95 x 5 (warm-up)
135 x 5 (warm-up)
185 x 5 (form gets a bit shaky towards the end so you count that as 1st "real" set)

185 x 5 (work set #2)
185 x 4 (oops - missed a rep. No big deal. The next workout you try for that extra rep then move up by 10 lbs the following workout)

Calf raise 2-3 x 15-20 (use either a standing calf machine, leg press, or you can also do them with a barbell, or one leg at a time holding a dumbell – use a nice smooth 2 second lowering phase with a full range of motion)

Barbell hip thrust, reverse hyper, or Romanian deadlift: 2 x 6-8 (choose 1)

Bodyweight bulgarian split squat iso hold: 1 x 60 seconds per side (get in the bottom of a Bulgarian split squat and hold for 60 seconds per side, making an attempt to emphasize your glutes. Perform your weak leg first. If you don't get 60 seconds initially no big deal just hold as long as you can and gradually work up to 60 seconds)

(Optional upper body work)

Bench press/Incline press (alternate between them workout to workout): 3 x 5 (regulate these the same weight as squats – Work up to a heavy set of 5 that you can do on your own and stick with that weight for a total of 3 sets at that weight).

Medium grip overhand pullup: 3 x as many reps as possible (if you get a total of 24 reps over the 3 sets start to add enough weight to bring your average down to 6-8 per set)

Front and side ab bridge hold for time: 2 x 30 seconds front, 1 x 30 seconds each side

**Scheduling and Frequency:** Try to hit this workout 2-3 x per week on alternating days. In a perfect world you'd hit it every Monday, Wednesday, and Friday (or Tuesday, Thursday, and Saturday) but if you can only get 2 workouts in per week you'll still make good gains from it.

**Rest intervals:** Don't rush yourself in between sets. You might take 5 minutes between sets on heavy exercises like the squat, 1-2 minutes for less intensive exercises like calf raises or the plyos. If in doubt rest a little longer than you think you should.

**Manipulating the Workout**: If you prefer you CAN substitute Bulgarian split squats or front squats in place of the regular squats. You can also split the workout up into 2 sessions and do deadlifts every other workout like this

**Session A:** Squats 3 x 5, RDL, reverse hyper or hip thrust 2 x 6-10

**Session B:** Deadlifts 3 x 5, Bulgarian split squats (or front squats) 2 x 6-10

On that format you'd simply alternate back and forth between session A & B on an every other day basis. On exercises where you have options you can either stick with one each workout or alternate between them.

**Deloading weeks:** If you go 3 workouts in a row without any gains on the squat reduce the weights by 10% and build back up to where you were over the course of a few workouts. Do the same for any other exercise. Once you stall out 3 times over the course of several months it's probably time to switch to one of the intermediate templates or go on an explosive phase.

**Progression:** Can come from either increased reps or better form. Say you work up to 150 lbs and do 2 sets of 5 with it but you only get 4 reps on your 3rd set. You'd stay with that weight another workout until you get all 3 sets of 5 with 150 lbs. Say you DO get all 3 sets of 5 but your form/depth on the last set is a little shaky. Stick with that weight again until you get 3 sets of 5 PERFECT reps.

# The Break-In Workout

This workout is for someone who has some decent training experience in the past but who is coming back after a lay-off. Rather than jump into everything full bore and experience a ton of soreness, gradually work yourself back into things over the course of a week or so. Here is what I recommend.

Pick 2 low to medium intensity plyo exercises such as bilateral line hops, ankle jumps, tuck jumps, or low box depth jumps, and perform 2 sets of 8-10 reps each.

Pick one strength speed exercise such as jump squats, cleans, or snatches, and perform 2 sets of 5 reps - If you choose cleans or snatches don't push the weight. Use a weight you could do 10 reps with if you had to. If you choose jump squats just use a standard 45 lb bar.

Do one work set of squats for 6 reps your first workout but don't push the weights - stop 4 or 5 reps shy of failure. Add one set per workout until you hit 3 sets the 3rd workout.

Do one set of barbell hip thrusts, romanian deadlifts, or reverse hyperextensions for 6-8 reps, stopping well shy of failure, and add one set per workout just like squats.

Perform a workout every other day, or every 3rd day if you're sore, and do 3 consecutive workouts. You'll have built up to 3 sets of squats by then and should be good to go on a "real" routine.

# Intermediate Strength-Focused workout

This workout is for the intermediate level trainee who needs to work on building strength up. An intermediate will squat at least 1.5x bodyweight or more, and have already performed something similar to the novice template for at least 3-6 months prior – Generally speaking an intermediate will have at least a year of combined training experience behind him/her.

| | Mon | Tue | Wed | Thu | Fri | Sat | Notes |
|---|---|---|---|---|---|---|---|
| **Week 1-3** | **Upper body #1**<br><br>**Section A: Speed/Plyo**<br><br>Standing and Running Vertical Jumps: 10-15 total reps (these can be done on your off days if you prefer)<br><br>Depth jumps (from optimal box height) or bounds: 2 x 5 (DJ) or 2 x 25 yds bounds (add 1 set per week until you reach 4 sets in week 3)<br><br>**Section B: Strength**<br><br>YTWL Drill x 10 reps each x 2 sets<br><br>Bench Press | **Lower body #1**<br><br>**Section A: Speed/Plyo**<br><br>Choose 2 of the following and perform 4 sets of 8 reps each:<br><br>Ankle jump, tuck jump, lateral barrier jump<br><br>**Section B: strength**<br><br>Hang clean/Hang snatch or jump squat: For hang clean or snatch follow set and rep scheme under notes. For jump squat do 4 x 5 reps with 15-20% of max squat. Pause and reset each rep.<br><br>Squat<br><br>See set and rep scheme | **Off/ Optional core conditioning**<br><br>A)Hanging Leg/knee Raises 3xas many reps as possible (maximal pelvic rotation at top)<br><br>B) Swiss ball weighted situp 2 x 15-20<br><br>C) Front & side planks 1 set each x 1 minute | **Upper body #2**<br><br>**Section A: Speed/Plyo**<br><br>Standing and Running Vertical Jumps: 10-15 total reps (these can be done on off days if you prefer)<br><br>Depth jumps (from optimal box height) or bounds: 2 x 5 (DJ) or 2 x 25 yds bounds (add 1 set per week until you reach 4 sets in week 3)<br><br>**Section B: Strength**<br><br>YTWL Drill x 10 reps each x 2 sets<br><br>Standing military Press:<br><br>See Set and rep scheme under notes<br><br>Wide Grip pullups 3 sets as many reps as possible | **Off/ Optional core conditioning:**<br><br>A)Hanging Leg/knee Raises 3xas many reps as possible (maximal pelvic rotation at top)<br><br>B) Swiss ball weighted sit-up 2 x 15-20<br><br>C) Front & side planks 1 set each x 1 minute | **Lower body #2**<br><br>**Section A: Speed/Plyo**<br><br>Choose 2 of the following and perform 4 sets of 8 reps each:<br><br>ankle jump, tuck jump, lateral barrier jump<br><br>**Section B: Strength**<br><br>Hang clean/Hang snatch or jump squat: For hang clean or snatch follow set and rep scheme under notes. For jump squat do 4 x 5 reps with 15-20% of max squat. Pause and reset each rep.<br><br>Deadlift or | For squat, deadlift, bench, military, hang clean, and hang snatch follow this set and rep scheme:<br><br>**Week 1**: 3 x 5, 1 x max reps @ 65, 75, 85, 85% (1 set of 5 @ 65%, 1 set of 5 @ 75%, 1 set of 5@ 85%, then 1 set of as many reps as possible with 85%)<br><br>**Week 2**: 3 x 3, 1 x max reps @ 70, 80, 90, 90%<br><br>**Week 3**: 1 x 5, 2 x 3, 1 x max reps @ 75, 85, 95, 95%<br><br>**Rest Intervals**: Are self monitored. Go when you're |

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| | See set and rep scheme under notes<br><br>Incline DB Bench Press 2 x 6-10(NTF: Not To Failure – stop a couple reps shy of failure)<br><br>Seated Cable or Chest Supported Row 3 x 6-10 NTF<br><br>Bicep exercise (your choice) 2 x 8-15<br><br>Tricep exercise (your choice of 10 inch grip decline bench, lying triceps extension, or vertical dip) 2 x 6-10 | under notes<br><br>RDL, barbell hip thrust, or manual reverse hyper: 3 x 6-8<br><br>Barbell or machine calf raise 2 x 20 | | w/bodyweight<br><br>DB flat bench 3 x 8-10 NTF<br><br>Bicep exercise of your choice 2 x 6-12<br><br>Tricep exercise of your choice 2 x 6-10 | | Front squat<br><br>See set and rep scheme under notes<br><br>Bulgarian split squat 2 x 6-8/side<br><br>RDL, barbell hip thrust, or manual reverse hyper: 3 x 6-8<br><br>Barbell or machine calf raise 2 x 20 | ready and don't rush yourself.<br><br>ESTIMATE your starting percentages if you're not sure. In week 1 you should be doing somewhere between 5-8 reps on your last set. In week 2 you should be doing 3-5 reps. And in week three 1-3 reps.<br><br>Use **STRAPS** on deadlift if you like.<br><br>**Separate** workouts into AM and PM if you prefer<br><br>**Tempo:** Use a controlled negative for each exercise other than the snatches, cleans, and deadlifts. Raise the weight with |

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| | | | | | | | some acceleration but still controlled. The exception is deadlifts. On those stay controlled throughout) |
| **Week 4** | **Upper body #1**<br><br>**Section A: Speed/Plyo**<br><br>Standing and Running Vertical Jumps: 10-15 total reps (these can be done on your off days if you prefer)<br><br>**Section B: Strength**<br><br>YTWL Drill x 10 reps each x 2 sets<br><br>Bench Press: 3 x 3 with 5rm weight<br><br>Incline DB Bench Press 2 x 6-10 (reduce | **Lower body #1**<br><br>**Section A: Speed/Plyo**<br><br>Choose 2 of the following and perform 2 sets of 8 reps each:<br><br>Ankle jump, tuck jump, lateral barrier jump<br><br>**Section B: strength**<br><br>Hang clean/Hang snatch or jump squat: 2 x 3 with 10% under 3 rm weight. On jump squats keep the reps at 5 and do 2 sets.<br><br>Squat | Same | **Upper body #2**<br><br>**Section A: Speed/Plyo**<br><br>Standing and Running Vertical Jumps: 10-15 total reps (these can be done on off days if you prefer)<br><br>**Section B: Strength**<br><br>YTWL Drill x 10 reps each x 2 sets<br><br>Standing military Press:<br><br>3 x 3 with 5rm weight<br><br>Wide Grip pullups 2 sets x 2 reps shy of failure<br><br>DB flat bench 2 x 6-10 with | Same | Eliminate this workout and do VJ testing on this day | |

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| | the weight 5-10% below week 3 weights)<br><br>Seated Cable or Chest Supported Row 2 x 6-10 (reduce the weight 5-10% below week 3 weights)<br><br>Bicep exercise (your choice) 2 x 8-15 (reduce the weight 5-10% below week 3 weights) | 3 x 3 (with 5 rm weight)<br><br>RDL, barbell hip thrust, or manual reverse hyper: 2 x 6-8 (with 10% less weight than week 3) | | 10% less weight than normal<br><br>Bicep exercise of your choice 2 x 6-10 with 10% less weight than normal<br><br>Tricep exercise of your choice 2 x 6-10 with 5-10% less weight than normal | | | |

Alternatively, one can follow this schedule with a bit more recovery utilizing this every other day template:

Mon: upper body #1
Wed: lower body #1
Fri: upper body #2
Sun: lower body #2
The 4th time you hit each workout take a deload, using the parameters laid out in week 4.

## Intermediate Squat Focused Protocol

For people that do better with more squatting frequency this is a workout I like a lot. You do squats twice a week, an intensity oriented workout along with a volume oriented workout. This workout will definitely get your squat up in a hurry. A 4 week setup would look like this:

| Week | Monday | Tuesday | Wednesday | Thursday | Friday | Saturday | Notes: |
|---|---|---|---|---|---|---|---|
| 1-3 | **Lower body #1**<br><br>**Section A: Speed/Plyo**<br><br>Choose 2 of the following and perform 4 sets of 8 reps each:<br><br>Ankle jump, tuck jump, lateral barrier jump, low-box depth jump (12-18 inches)<br><br>**Section B: strength**<br><br>Hang clean<br><br>Week 1: 5 x 5 (same weight each set)<br><br>Week 2: 5 x 3 (same weight each set)<br><br>Week 3: 5 x | | Choose 2 of the following and perform 4 sets of 8 reps each:<br><br>ankle jump, tuck jump, lateral barrier jump, low box depth jump<br><br>**Upper body:**<br><br>Horizontal press variation (bench or incline):<br><br>Week 1: 5 x 5 (same weight each set)<br><br>Week 2: 5 x 3 (same weight each set)<br><br>Week 3: 5 x 1 (same weight each set)<br><br>Week 4: 3 x | | **Lower body #2**<br><br>**Section A: Speed/Plyo**<br><br>Choose 2 of the following and perform 4 sets of 8 reps each:<br><br>ankle jump, tuck jump, lateral barrier jump, low box depth jump (12-18 inches)<br><br>**Section B: Strength**<br><br>Hang Snatch<br><br>Week 1: 5 x 5 (same weight each set)<br><br>Week 2: 5 x 3 (same weight each set)<br><br>Week 3: 5 x | | The cleans and snatches can be replaced by jump squats. If you do jump squats just do 4 x 5 at 15% of max squat.<br><br>The weights on Monday's workout should be slightly submaximal. You'll be using the same weight for all 5 sets so start off a little light.<br><br>On Friday's workout when I say "building up to max" that doesn't mean you only do one set it means you slowly build up to at least one heavy set for the listed rep range. |

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| | 1 (same weight each set)<br><br>Week 4: 3 x 3 (with week 1 weight)<br><br>Squat<br><br>Week 1: 5 x 5 (same weight each set)<br><br>Week 2: 5 x 3 (same weight each set)<br><br>Week 3: 5 x 1 (same weight each set)<br><br>Week 4: 3 x 3 (with week 1 weight)<br><br>RDL, barbell hip thrust, or manual reverse hyper: 3 x 6-8 (all weeks)<br><br>Barbell or machine calf raise 2 x 20 | | 3 (with week 1 weight)<br><br>Pullup variation:<br><br>4 sets to failure (all weeks)<br><br>Dumbell press variation: 2 x 6-10 (all weeks)<br><br>Bicep exercise of your choice 3 x 6-10<br><br>Tricep exercise of your choice 3 x 6-10<br><br>Leg raise or leg raise: 2 x as many reps as possible<br><br>High to low cable woodchop 2 x 15-20 | | 1 (same weight each set)<br><br>Week 4: 3 x 3 (with week 1 weight)<br><br>Squat<br><br>Week 1: 1 x 5 (building to max 5 rep set)<br><br>Week 2: 1 x 3 (building to max triple)<br><br>Week 3: 1 x 1 (building up to max single)<br><br>Week 4: 1 x 3 (with week 1 weight)<br><br>RDL, barbell hip thrust, or manual reverse hyper: 3 x 6-8 (all weeks)<br><br>Bulgarian split squat:<br><br>Week 1: 2 x 6-8/side<br><br>Week 2-3: 3 x 6-8/side | | For example, a typical progression for a max set of 5 at 300 lbs might look like this:<br><br>45 x 5, 95, x 5, 135 x 5, 205, x 3, 255 x 3, 275 x 3, 300 x 5<br><br>Find a couple of days during the week and go out and perform 10-20 combined standing and running verts. The days you do them don't matter as long as you get them in at some point.<br><br>After 4 weeks on this protocol start back over at week 1, or move on to something else.<br><br>Upper body is optional but if you prefer to do more or less upper body work it can be |

| | | | | | | | |
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| | | | | | Week 4: 2 x 6-8/side | | incorporated on Tuesday, Wednesday, Thursday, or saturday |

Now for those athletes who know you need to focus on explosiveness, here are a couple of explosive setups.

# Intermediate Explosive focus

**Session 1: Monday**

Tuck jump, ankle jump, or lateral barrier jump: 4 x 8

Hang clean or jump squat: 4 x 3-5 (use 15-20% of max squat for jump squats)

Complex with:

Depth jump onto box (or over hurdles) 4 x 5 (use short ground times)

Squat: 8 x 1@ 85-90% (use a weight you *could* do 4-5 reps with if you had to)

RDL, barbell hip thrust or reverse hyper 2 x 6-10

**Session 2: Wednesday**

Tuck jump, ankle jump, lateral barrier jump: 4 x 8

Depth jump onto box or over hurdle:  4 x 5

**Session 3: Friday**

Tuck jump, ankle jump, or lateral barrier jump: 4 x 8

Hang snatch or jump squat: 4 x 3-5 (use 15-20% of max squat for jump squats)

Complex with:

Depth jump onto box (or over hurdles) 4 x 5 (use short ground times)

Explosive squat: (descend under control and explode up): 6 x 2@ 60, 65, 70, 70, 65, 60% of max squat

RDL, barbell hip thrust or reverse hyper 2 x 6-10

Follow that for 3-4 weeks and you should see significant differences in your levels of explosiveness.

Here's another variation incorporating ¼ squats once per week.

# Explosive Focus Option II

**Session 1: Monday**

Tuck jump, ankle jump, or lateral barrier jump: 4 x 8

Hang clean or jump squat: 4 x 3-5 (use 15-20% of max squat for jump squats)

Complex with:

Depth jump onto box (or over hurdles) 4 x 5 (use short ground times)

Squat: 8 x 1@ ~85% (use a weight you *could* do 5 reps with if you had to)

RDL, barbell hip thrust or manual reverse hyper 2 x 6-10

**Session 2: Wednesday**

Tuck jump, ankle jump, lateral barrier jump: 4 x 8

Depth jump onto box or over hurdles: 4 x 5

**Session 3: Friday**

Tuck jump, ankle jump, or lateral barrier jump: 4 x 8

Hang snatch or jump squat: 4 x 3-5 (use 15-20% of max squat for jump squats)

Complex with:

Depth jump onto box (or over hurdles) 4 x 5 (use short ground times)

¼ squat: 4 x 3

RDL, barbell hip thrust or manual reverse hyper 2 x 6-10

Follow that for 3-4 weeks and you should see significant differences in your levels of explosiveness.

Those are some templates designed with a primary focus in mind (strength or explosiveness), but what about everybody else? Well if you're fairly balanced you should be hitting strength, explosiveness, and peaking out your ability over the course of a training phase. Here is an example of a 6 week workout hitting on all major qualities:

# Intermediate General Template

This workout is for people who know they are fairly balanced. This workout is also for people who have a good strength base (1.5 x bw squat) and aren't sure exactly what they're deficient in.

**Phase 1**: Do 2 consecutive Max strength workouts, alternating back and forth between session A & B, while performing a workout about every 4th day, followed by an explosive workout (shown further below), then 2 more consecutive max strength workouts, then move onto phase II.

**Max Strength Session A:**

Depth drop 4 x 5 (from box approximately equivalent to the height of your running vertical jump)

Jump squat with 45 lb bar 4 x 5 (pause and reset after each rep. Go down only as far as you would in a normal jump) or hang clean/snatch 4 x 3

Squat: 8 x 1 @ 90% (add weight for at least the first 3-4 sets until you hit a close to maximal squat single, then reduce the weight by about 10% and continue doing singles until you've done 8 total sets

Bulgarian split squat or lunge 2 x 8-10

Romanian deadlift, Barbell hip thrust, or manual reverse hyper 2 x 6-10

Standing Calf raise or single leg calf raise 3 x 20

**Max Strength Session B:**

Depth drop 4 x 5 (from box approximately equivalent to the height of your running vertical jump)

Jump squat with 45 lb bar 4 x 5 or hang clean/snatch 4 x 3

Squat 4x6-8 (with same weight each set)

Romanian deadlift, Barbell hip thrust, or manual reverse hyper 3 x 6-10

Bulgarian split squat 2 x 6-8/side

Standing Calf raise or single leg calf raise 3x20

**EXPLOSIVE WORKOUT**

Depth jump (from fairly low box): 4x5

Rhythmic Jump squat 4 x 5 @ 15-20% of max squat

Paused box squat 6 x 2 @ 60, 65, 70, 70, 65, 60% of 1rm squat - Use a box slightly below parallel. Sit slowly back on the box, pause momentarily, and explode up.

Barbell hip thrust, RDL or manual reverse hyper 2 x 8-10

Standing Calves : 3 x 15-20

So Phase 1 will go something like this:

Mon: Strength workout Session A
Fri: Strength workout Session B
Wed: EXPLOSIVE WORKOUT
Mon: Session A
Fri: Session B

Then move onto PHASE II.

## Phase II

Do 4 consecutive explosive workouts alternating between session A & B.

**Explosive workout A:**

Depth jump (from optimized box height): 4x5

Rhythmic Jump squat 4 x 5 @ 15-20% of max squat or hang clean 4 x 3

Paused box squat 6 x 2 @ 60, 65, 70, 70, 65, 60% of 1rm squat - Use a box slightly below parallel. Sit slowly back on the box, pause momentarily, and explode up.

RDL, Barbell hip thrust or manual reverse hyper 2 x 8-10

Standing Calves : 3 x 15-20

**Explosive Workout B:**

Depth jump (from optimized box height): 4x5

Rhythmic Jump squat 4 x 5 @ 15-20% of max squat or hang snatch: 4 x 3

Bulgarian split squat 3 x 6-8/side

RDL, Barbell hip thrust or manual reverse hyper 2 x 8-10

Standing Calves 3 x 15-20

So **Phase 2** will go something like this:

Tues: Explosive workout session A
Sat: Explosive workout session B
Wed: Explosive workout session A
Sun: Explosive workout session B

Then move onto PHASE III.

## Phase III

Potentiation/Peak Do 2 consecutive potentiation workouts (one every ~4 days), followed by 1 explosive workout (shown above), then repeat 2 consecutive potentiation workouts.

**Potentiation/peaking Workout A:**

3x5 depth jump (from box height that allows you to jump the highest at ground contact) Each set alternated with running vertical jumps x 3 reps

Squat 6 x 1 @ 85% (heavy not maximal – Use a weight you could do 5 reps with I you had to. Descend smooth under control then use good acceleration on the concentric)

Rhythmic Jump squat 6 x 5 with 45 lb bar (alternate back and forth between squat and jump squats) 1 set of squats, rest , 1 set of jump squats, rest, etc.

Hip thrust/Reverse hyper/RDL 2 x 8-10

**Potentiation/peaking workout B:**

3x5 depth jump (from box height that allows you to jump the highest at ground contact) Each set alternated with standing vertical jumps x 3 reps

Half Squat 5 x 3 (heavy not maximal – use a weight you could do ~5-6 or so reps with if you had to – Descend smooth under control then use good acceleration on the concentric. You can use a box on the squats if you prefer)

Rhythmic Jump squat 5 x 5 with 45 lb bar (alternate back and forth between squat and jump squats) 1 set of squats, rest , 1 set of jump squats, rest, etc.

Hip thrust/Reverse hyper/RDL 2 x 8-10

So **Phase III** will go something like this:

Mon: Potentiation workout A
Fri: Potentiation workout B
Wed: EXPLOSIVE workout A (listed in previous phase)
Sun: Potentiation workout A
Fri: Potentiation workout B

**Entire Workout Over**

Take 4-14 days of active rest after this workout before repeating it or starting something else.

What about upper body training? Upper body training can easily be mixed in with the above workouts. Here are a couple of versatile upper body workouts I like to use. After the upper body work I've listed how to mix it in conjunction with the above phase.

# Upper body training:

Throughout the duration of the workouts you can use the following versatile upper body workouts. I recommend you train your upper body on the same schedule as lower body. Here are the workouts:

## Upper body workout 1 (heavy)

**Heavy horizontal press** – work up to a max set of 3-5 reps in one of the following exercises:

Barbell bench press, Incline barbell bench press, Close-grip bench press

**Supplemental Exercise** – 2 x 15-20 (choose one)

Flat DB bench press, Incline DB bench press, decline DB bench press

**Horizontal pulling / Rear delt superset** - 3 x 8-12. Superset one exercise from "Group 1" with one exercise from Group 2

**Group 1**

DB rows, Barbell rows, Seated cable rows (various bars), T-bar rows, Chest supported rows

**Group 2**

Rear delt flyes, Face pulls, Seated DB “power cleans”, Band pull-aparts

**Traps** – Perform 2-3 sets of 30-50 reps of one of the following exercises:

DB shrugs, Barbell shrugs, Behind the back barbell shrugs

**Bicep exercise** - Perform 3 sets of 8-15 reps of one of the following exercises:

Barbell curls, DB curls (standing), Seated Incline DB curls, Hammer curls

## Upper Body Workout 2 (Light):

Repetition Dumbell Pressing – 3 sets of 12-15 reps (choose one)

Flat DB bench press, Incline DB bench press, Decline DB bench press

**Vertical pulling / Rear delt superset** – 3 x 8-12 Superset one exercise from “Group 1” with one exercise from “Group 2.”

**Group 1**

Lat pulldowns or Chin-ups

**Group 2**

Rear delt flyes, Face pulls, Band pull-aparts

**Delts / Arms superset** - Superset one exercise from “Group 1” with one exercise from “Group 2.” Perform 3 supersets.

**Group 1** (Perform 8-12 reps)

Vertical db press, laterals, cable laterals

**Group 2** (8-10 reps/set)

Barbell curls, DB curls, Seated Incline DB curls, Hammer curls, Skullcrushers, DB triceps extensions, Triceps pushdowns

**Grip / Forearms** – choose one of the following exercises:

Wrist roller (2 sets of 45-60 seconds), Wrist curl ( 2 sets of 30-50), Thick bar or heavy DB holds (2 sets of 40-60 seconds), Plate pinch gripping (2 sets of 10-20 seconds)

**Overall Scheduling:**

Putting it altogether your overall schedule might look something like this:

Mon: strength lower body #1 (Start phase 1)
Wed: upper body #1
Fri: strength lower body #2
Sun: upper body #2
Tues: lower body power workout
Thurs: upper body #1
Sat: strength lower body #1
Mon: upper body #2
Tues: strength lower body #2
Thurs: upper body #1
Sat: START PHASE 2 (power phase) – Lower body power workout
Mon: upper body #2
Wed: lower body power workout
Fri: upper body #1
Sun: lower body power workout
Tues: upper body #2
Thurs: lower body power workout
Sat: upper body #1
Mon: START PHASE 3: Lower body potentiation
Tues: upper body #2
Thurs: Lower body peaking workout
Sat: upper body #1
Mon: Lower body POWER workout
Wed: upper body #2
Fri: Lower body peaking workout
Sun: upper body #1
Tues: Lower body peaking workout
OVERALL PHASE OVER

If you need an extra day or 2 of rest each week or miss a workout here or there it's no big deal just get back on the schedule the following day.

# Very Basic Intermediate/Advanced Workout

It should be noted a workout does not have to be complicated at all to be effective. In fact the best gains I ever made in VJ as a more advanced trainee were following this very basic and simple template.

It involves only a few exercise - an olympic lift from the hang (or jump squat), a squat variation (I used a front squat), and a posterior chain variation. Perform the workout every 4-5 days (I did a workout every 5 days) and follow this set and rep scheme for the o-lift and squat variation:

Session 1:4 x 5
Session 2: 5 x 4
Session 3: 5 x 5, 4, 3, 2, 1
Session 4: 3 x 3 with 5rm load
Session 5: repeat set/rep scheme from session 1
Session 6: repeat set/rep scheme from session 2
Session 7: repeat set/rep scheme from session 3
Session 8: repeat set/rep scheme from session 4

Then each session add on:
RDL/hip thrust/or reverse hyper: 3 x 6-10. Also pick 1 or 2 plyo movements and do them either before the workout or in between sets of the o-lift or jump squat. You might do 4 x 8 tuck jumps and 4 x 5 depth jumps. You can replace the o-lift with jump squats if you prefer. Also pick a day in between your workouts and either do a similar plyo workout or go out and do ~20 or so combined actual jumps for height.

For squatting variations you can choose back squats, front squats, or some type of lunge variation like a step-back lunge from deficit or a bulgarian split squat. This is a really effective way to train in my experience.

# The Tempo Variance Workout

Here's another really effective workout based on similar principles. The only difference here is we use a variance in tempo of movement to provoke gains. We start off using slow tempos on the squat, which really helps amp up hypertrophy, then we build upon that. Each workout will consist of the following:

Rhythmic Jump squat with 15-20% of max squat: 4 x 5
Complexed with:
Depth jump: 4 x 5

Squat (see below)

Romanian Deadlift, hip thrust, or reverse hyper: 3-4 x 6-8

Perform a workout once every 4-5 days. For the squat your reps and tempo will vary every 3rd workout as follows:

**Workouts 1-2**: Squat 5 x 5 at 5-0-5 tempo (5 seconds down, 5 seconds up)
**Workouts 3-4**: Squat 5 x 5, 4, 3, 2, 1 with 2 second pause at bottom of each rep, explode up
**Workouts 5-6:** 1/2 squat 4 x 5 at normal tempo
**Workout 7-8:** Speed squat 6 x 2 @ 60, 65, 70, 70, 65, 60% control down explode up

Give it a shot!

# *The JackM Split*

This is a post I made on a forum a few years back that became immensely popular. Someone had asked a fairly complicated question about VJ training and I replied with a summary of what I thought he should do to avoid wasting time and get to point A to B as quickly as possible. Many people have followed this workout with great success and I received feedback from one fella who told me he'd wasted years overanalyzing things getting nowhere but when he saw this post he decided to simplify everything and follow it. Well, he ended up putting 7.5 inches on his vert in a few months when he hadn't made any gains in years prior. Sometimes simplifying things can be a huge advantage and this post/workout is the definition of simplicity. Here it is:

> *Knowing that you're probably gonna ditch what I'm fixing to say in favor of a bunch of complicated BS that will take you the next 6 months of your training time to figure out, here's a very simple answer as to what I'd recommend you do:*
>
> *1. Get stronger. Until you're going all the way down with 300 lbs on your back you simply don't have the raw horsepower necessary to blast 192 lbs (your bodyweight) up in the air like you'd want.*
>
> *2. After you accomplish #1, shift into more explosive work.*
>
> *3. There is an easy (fast) way and a hard (long) way to accomplish the above.*
>
> *A: The easy way consists of going to the gym every monday and friday or Monday and Thursday. On one day knock out sets of 5 in the squat followed by some Glute Hams for sets of 6-8. On the other day knock out sets of 6-8 in the bulgarian split squat followed by some more glute hams. (Do around 4 sets of each exercise). Try to put more weight on the bar everytime you hit Mondays workout. Do this until you can throw around 300 for reps.*
>
> *Prior to your workouts on Monday and Friday, as well as on Wednesday, do a low volume of some garden variety plyometric drills such as a few sets of lateral jumps, low box depth jumps or any other exercises you like.*
>
> *4. Once you have accomplished #3 you will now have a bigger motor in your car and can then focus on modifying that motor to get the most out of it.*

*4A. Keep the same basic schedule in place. (Monday and Thursday or Monday and Friday) On one day do some depth jumps followed by some jump squats for sets of 5 for a total of 4-5 sets of depth jumps and 8 sets of squats. The squat weights will vary between 20-50% of your max.*

*On jump squats do one set with more weight followed by one set with lighter weight and alternate back and forth until you've done all 8 sets. At the very end of your workout you might do a few sets of 3 reps with 90% of your squat, as well as a few glutehams, just to maintain your strength.*

*4B. On the other weekly workout make it a workout based around jumping and sprinting for PRs. Simply go out and get warmed up and try to jump as high as possible from a variety of starts (running, single leg, standing, bouncing off a box). Maybe add in some sprints if you want.*

*Thus, your workouts will now look like this:*

***Monday:***
*Depth jumps 6 x 3*
*Jump squat with 40% 4 x 5*
*Jump squat with 20% 4 x 5*
*(alternate back and forth between them)*
*Regular squat 3 x 3 @90%*

***Friday:*** *PR Day - running jumps, standing jumps, sprints, - whatever you want to do.*

*5. When you no longer are getting consistent weekly results from 4A and 4B (which might take 3 weeks and might take a couple of months), then it's time to start over with #1.*

*6. Keep repeating Steps 1 through 5 for as many years as you like until you either get to where you want to be or until you get old, gray and worn out*

MOST people could take that simple template there, follow the concepts, and achieve 90% of their potential. It's not complicated at all but it's got the 90% that matters.

# High Frequency Option for Intermediates

Here is a great intermediate/Advanced workout that is REALLY effective for peaking out unilateral 1-leg jumping. It will work for any type of jump but if you really want to drive up your unilateral jump this workout is EXCELLENT.

| Week | Monday | Tues | Wed | Thurs | Fri | Sat | Notes |
|---|---|---|---|---|---|---|---|
| Week 1 | Hang Clean: 4 x 3 (stop each set a couple of reps shy of failure)<br><br>Jump Squats with minimal knee bend and ground time: 2x10 (45 | Jumps for height x 8-10 reps<br><br>Alternate leg bounding 5 x 30 yds<br><br>Depth jumps onto box (moderate box height ~18 | | Hang snatch: 4 x 3 (stop each set a couple of reps shy of failure)<br><br>Deep Squat: 3x5<br><br>Barbell calf raise or standing calf | Jumps for height x 8-10 reps<br><br>Alternate leg bounding 4 x 30 yards<br><br>Depth jumps onto box (moderate box height | | Cleans and snatches can be replaced with 2 hand dumbbell swings for sets of 8-10 if you don't know how to do them<br><br>Take a full recovery between all |

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| | lb bar)<br><br>10-12" Barbell Box Step-Up: 3x6<br><br>Glute ham raise or leg curl 3 x 6-8<br><br>Barbell Calf Raise: 3x15 | inches) 4 x 5 | | raise: 3 x 15<br><br>Glute ham raise or leg curl 3 x 6-8<br><br>Jump Squats: 2x10 with 45lb bar | ~18 inches) 5 x 4 | | exercises/sets/reps<br><br>On half squats you can use a box if you like<br><br>On the jumps for height you can use any style of jump you prefer but it's best if you focus on the style you're most wanting to improve (unilateral, bilateral, etc.)<br><br>On off days feel free to do a dynamic warm-up, core, or upper body work |
| Week 2 | Clean 5,4, 3,2,1 building up to heavy single<br><br>Jump Squats with minimal knee bend and ground time 2x10 with 10 lbs more than previous week<br><br>10-12" Barbell Box Step-Up, 4 x 5<br><br>Glute ham raise or leg curl 3 x 6-8<br><br>Barbell Calf Raise 2x10 | Jumps for height x 8-10 reps<br><br>Alternate leg Bounding: 4 x 30 yards<br><br>Depth jumps onto box (moderate box height) 6x4 | | Hang snatch: 5, 4,3,2,1 building to heavy single<br><br>Jump Squats with minimal knee bend and ground time 2x10 with 10 lbs more than previous week<br><br>Squat: 3x5 heavy weight<br><br>Glute ham raise or leg curl 3 x 6-8<br><br>Barbell calf raise: 2 x 10 | Jumps for height x 8-10 reps<br><br>Alternate leg Bounding<br><br>4 x 30 yards<br><br>Depth jumps onto box (moderate box height) 6x4 | | |
| Week 3 | Dynamic warmup<br><br>60 yard buildup sprints x 80% of max speed | | Dynamic warm-up<br><br>2 x 5 jump squat<br><br>2 x 8 regular squat (very | | **Test Jump** | | |

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| | x 4 reps<br><br>Skipping 3 x 30 yards | | light about 80% of 8rm) | | | | |
| Week 4 | Hang Clean: 5 x 5, 4, 3, 2, 1<br><br>6-8" Barbell Box Step-Up: 3x6 (use a lower box then you did the last phase)<br><br>Glute ham raise or leg curl 3 x 6-8<br><br>Barbell skips: 2x10 (45 lb bar) | Jumps for height x 6-8 reps<br><br>Bounding (with optional 8-15 lb weighted vest):<br><br>LRLRLR x 30m (left, right, left right, left right)<br><br>LLRR x 30m<br><br>LLLRRR x 30m<br><br>LLLL x 30m<br><br>RRRR x 30m<br><br>Depth jumps over hurdle: 5 x 4 (use a box height that "challenges" you – it should be fairly difficult to get up over the hurdle) | | Hang snatch: 5 x 5, 4, 3, 2, 1<br><br>1/2 Squat: 6 x 1 @87.5-90% (use a weight you could do 3-4 reps with if you had to)<br><br>Glute ham raise or leg curl 3 x 6-8<br><br>Barbell skips 2 x 10 with 45lb bar | Jumps for height x 6-8 reps<br><br>Bounding (with optional 8-15lb weight vest)LRLRLR x 25m<br><br>LLRR x 25m<br><br>LLLRRR x 25m<br><br>LLLL x 25m<br><br>RRRR x 25m<br><br>Depth Jump over hurdle 5 x 4 (use a fairly challenging box height) | | |
| Week 5 | Hang Clean: 5 x 5, 4, 3, 2, 1<br><br>6-8" Barbell Box Step-Up: 4x6<br><br>Glute ham raise or leg curl 3 x 6-8<br><br>Barbell skips: 2x10 (45 lb bar | Jumps for height x 6-8 reps<br><br>Bounding: (with optional weighted vest)LRLRLR x 30m<br><br>LLRR x 30m<br><br>LLLRRR x 30m<br><br>LLLL x 30m | | Hang snatch: 5 x 5, 4, 3, 2, 1<br><br>1/2 Squat: 6 x 1 @92.5% (use a weight you could do 2-3 reps with if you had to)<br><br>Glute ham raise or leg curl 3 x 6-8<br><br>Barbell skips | Jumps for height x 6-8 reps<br><br>Bounding (with optional 8-15lb weight vest)<br><br>LRLRLR x 25m<br><br>LLRR x 25m<br><br>LLLRRR x 25m | | |

| | | | | | | | |
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| | | RRRR x 30m<br><br>Depth jumps over hurdle: 5 x 4 | | 2 x 10 with 45lb bar | LLLL x 25m<br><br>RRRR x 25m<br><br>Depth Jump over hurdle 5 x 4 | | |
| Week 6 | Dynamic warmup<br><br>60 yard buildup sprints x 80% of max speed x 4 reps<br><br>Skipping 3 x 30 yards | | Dynamic warm-up<br><br>2 x 5 jump squat<br><br>2 x 8 regular squat (very light about 80% of 8rm) | | **Re-Test Jump** | | |

# *High intermediate/advanced workout*

This is a balanced workout for an advanced or high intermediate level trainee. Normal intermediates could do this routine, but I wouldn't recommend doing it without going thru some of the other intermediate templates first.

| | Mon | Wed | Fri | Notes |
|---|---|---|---|---|
| *Phase I (4 weeks)* | Pick 2 medium intensity plyos and do 4 x 8 (ankle jump, tuck jump, lateral barrier jump etc.)<br><br>Hang Clean or paused jump squat: 3 x 5 (all weeks)<br><br>Squat<br>Week 1-3: 4 x 6-8 (same weight each set)<br>Week 4: 2 x 6-8<br><br>Bulgarian split squat: 2 x 6-8 (all weeks)<br><br>RDL, hip thrust, reverse hyper: 3 x 8-12 (all weeks) | Pick 2 medium intensity plyos and do 4 x 8 (ankle jump, tuck jump, lateral barrier jump etc.)<br><br>Dumbell 2 hand swing: 3 x 10 (all weeks)<br><br>Speed Squat 6 x 2 @ 60, 65, 70, 70, 65, 60% of max squat (all weeks) (descend under control & explode up – use a box if you like) | Pick 2 medium intensity plyos and do 4 x 8 (ankle jump, tuck jump, lateral barrier jump etc.)<br><br>Hang snatch or paused jump squat: 3 x 5 (all weeks)<br><br>Deadlift<br>Week 1-3: 4 x 5<br>Week 4: 1 x 4 (with week 1 top weight)<br><br>Front squat: 3 x 5 (all weeks) | |
| *Phase II (3 weeks)* | Depth Drop (from box height approximate to best running vert)<br><br>Week 1: 2 x 5<br>Week 2: 3 x 5<br>Week 3: 4 x 5 | Jump squat (pause and reset each rep) 3 x 5 with 25% of max squat<br><br>Dumbell 2-hand swing 3 x 10 | Depth Drop (from box height approximate to best running vert)<br><br>Week 1: 2 x 5<br>Week 2: 3 x 5<br>Week 3: 4 x 5 | |

| | | | | |
|---|---|---|---|---|
| | Hang clean (use same weight for all sets each given week – so the first couple of sets will be submaximal)<br><br>Week 1: 5 x 5<br>Week 2: 5 x 3<br>Week 3: 5 x 1<br><br>Or<br><br>Jump squat (pause and reset each rep) 5 x 5 @ 30% of max squat<br><br>Squat (use same weight for all sets each given week – so the first couple of sets will be submaximal)<br>Week 1: 5 x 5<br>Week 2: 5 x 3<br>Week 3: 5 x 1<br><br>Rdl, hip thrust, or reverse hyper: 3 x 6-8 | Speed squat 6 x 2 @ 50, 55, 60, 60, 55, 50% of max squat | Hang snatch (use same weight for all sets each given week – so the first couple of sets will be submaximal)<br><br>Week 1: 5 x 5<br>Week 2: 5 x 3<br>Week 3: 5 x 1<br><br>Or<br><br>Jump squat (pause and reset each rep) 5 x 5 @ 30% of max squat<br><br>Squat<br>Week 1: 1 x 5 (building to max set of 5)<br>Week 2: 1 x 3 (building to max triple)<br>Week 3: 1 x 1 (building to max single)<br><br>Rdl, hip thrust, or reverse hyper: 3 x 6-8 | |
| *Phase III (3 weeks)* | Depth Jump (from box height allowing you to jump the highest at ground contact)<br>Week 1: 2 x 5<br>Week 2: 3 x 5<br>Week 3: 4 x 5<br><br>Rhythmic jump squat 4 x 10 (with 45 lb bar)<br><br>Hang clean: 8 x 1 with 3 rep max (rest about 45 sec between reps)<br><br>Or<br><br>Dumbell 2 hand swing: 5 x 5 | ½ squat: 4 x 5 (stop each set a couple of reps shy of failure – use a box if you prefer)<br><br>RDL/Hip thrust/Reverse hyper: 3 x 6-8 | Depth Jump (from box height allowing you to jump the highest at ground contact)<br>Week 1: 2 x 5<br>Week 2: 3 x 5<br>Week 3: 4 x 5<br><br>Rhythmic jump squat 4 x 10 (with 45 lb bar)<br><br>Hang snatch: 8 x 1 with 3 rep max (rest about 45 sec between reps)<br><br>Or<br><br>Dumbell 2 hand swing: 5 x 5 | After phase III take a week off – do nothing more intensive than dynamic stretching, a handful of sub-maximal jumps or plyos (less than 50 ground contacts 3 days per week) and light lifting. Re-test jump the following week. |

# *High Frequency Concentrated Loading*

Here is a higher frequency option that works really well to get strength and reactivity up in a hurry. On this template you'll be squatting 4 days per week. It's structured as a weekly concentrated loading cycle where you ramp up volume and frequency for several days then have several days to recover.

Here's what it looks like:

**Mon:** Test vert, then do 6-8 sets 5-8 reps of plyometrics (ankle jumps, low to medium box depth jumps, tuck jumps, lateral barrier jumps, bounds) followed by:

**Squat:** 1 x 1 @ 100% (max out) followed by 5 x 1 @ 85% of that weight.

**Tues:** 4-6 sets of 5-8 reps of plyometrics (same as monday) followed by: squat: 5 x 1 @90% of Monday's top weight

**Wed:** 4-6 sets of 5-8 reps of plyometrics (same as mon. and tuesday) followed by: Squat 5 x 1@ 95% of Monday's top weight

**Thurs:** off

**Fri:** vert measure followed by moderate volume/high intensity depth jumps ex: 20 reps (5 x 4 or similar set/rep scheme) from ~24 inch box (or box at least as high as one that you can jump the highest on rebound from) followed by: squat 5 x 1 @ 80% (very light)

**Sat:** off/dynamic warmup/skill work

**Sun:** off/dynamic warmup (low volume dunks/jumps optional)

**Mon:** Start over

Every 4th week cut your frequency and loading down by switching to a template similar to this:

**Monday:** 4-6 sets of 5-8 reps of plyos followed by Squat: 3 x 3 @ 85%

**Friday:** 4-6 sets of 5-8 reps of plyos followed by Squat: 3 x 3 @ 85%

You can follow this as long as it works but after 8-12 weeks would generally be a good time to switch to something else. Basically, you're ramping up strength Monday-Wednesday, a time when you'll also accumulate significant fatigue. Wednesday's squat workout will likely be quite difficult, but then you have 4 days to recover from that so you'll be nice, fresh, and strong by the following monday. Then you do a fair amount of depth jumps on friday which (hopefully) gives a little bit of delayed super-compensation so by monday (possibly sunday if you have good recovery) you'll be seeing the results of that. So, basically at the start of monday your fresh and both your strength and explosiveness are at high levels.

Plyos can consist of any drills that you like, just don't kill yourself. Jump squats and/or o-lifts can also be added in a couple of days if you like, but they're not 100% necessary on this type of template. For the squats use a fairly narrow stance with good form. Go conservative on the weights at first.

# *High Frequency Step Type Loading*

With this high frequency approach you'll train 4 or 5 days per week with 2 lifts and take 3 steps forward and one step back approach using singles. This approach can make you explosive in a hurry and doesn't take much time at all per workout. I had one athlete put 40 lbs on his clean, 75 lbs on his squat, and 6.5 inches on his vert in about 2.5 months following this approach. You take 2 exercises, a hang power clean (or hang power snatch), and a squat variation. Take your current hang clean and hang squat 1 rep max and subtract 20 lbs on the squat and 10 lbs on the clean. Let's say your hang clean max is 135 and your squat max is 250. You'll start off with 125 on the clean and 230 on the squat. Then you increase 10 lbs per workout on the squat and 5 lbs on the clean. Increase the weight for 3 workouts and go back in weight for one. So your weights will look like this:

Workout 1: 125 x 1 clean, 230 x 1 squat
Workout 2: 130 x 1 clean, 240 x 1 squat
Workout 3: 135 x 1 clean, 250 x 1 squat
Workout 4: 130 x 1 clean, 240 x 1 squat
Workout 5: 135 x 1 clean, 250 x 1 squat
Workout 6: 140 x 1 clean, 260 x 1 squat
Workout 7: 135x 1 clean, 250 x 1 squat
Workout 8: 140 x 1 clean, 260 x 1 squat
Workout 9: 145 x 1 clean, 265 x 1 squat
etc.

As your weights get heavier lessen the amount of increase on your squats to 5 lbs instead of 10. In between sets of your cleans (including warm-ups) perform a plyometric variation such as depth jumps or tuck jumps for sets of 5-8. Add a couple of sets of auxiliary exercises like calf raises, romanian deadlifts, barbell hip thrusts, or reverse hypers for sets of 6-10 and sets of 15-25 on calf raises. Make sure on your squats and cleans that you do a thorough warmup and "work-up" to your heavy single. A typical warm-up for a 135 lb clean would look like this:

45 x 5
95 x 5
115 x 3
125 x 1
135 x 1

This approach can also work well with doubles or triples. The basic tenet is you use few lifts each workout and follow the step type loading approach. You can train as often as you want but I suggest you take at least one day off each week. If you feel energetic and ready to train then train. If you feel like taking a day off feel free to do so. I've had good success with athletes using this approach 4 days per week. If you stagnate on a lift 2 workouts in a row subtract 10 or 20 lbs and work your way back up. This workout can be modified a multitude of ways as long as you follow the basic principles. Also,

remember if you're following this or a similar approach make sure to go out and get some actual jumps in on a semi-regular basis - 2 x per week or so. ***Note: See the high frequency chapter for a more periodized variation of this workout.***

# Bodyweight Workout

Some people don't have access to weights for one reason or another but still want to train. Although I don't recommend it if you have a weight room available, sometimes bodyweight exercises are the ONLY answer. In my original vertical jump bible had a long, lengthy bodyweight workout that many people got success from. However, I felt it was unnecessarily overcomplicated and many of the exercises were fairly odd. Anytime you're working with bodyweight exercises your primary objective should be to gain strength - and to that end I now only use a couple of primary exercises - the single leg squat and single leg deadlift. There are progressions for each one shown on the videos in the video section. Simply start off with whatever progression you need and work your way up. This workout is very simple but effective. I personally still use it myself when I'm on the road and don't have access to a gym, so don't be fooled by how easy it may appear: Here is the workout:

**Section A:** Plyo Pick any 2 of the following exercises each workout:

Depth jump onto box: 3 x 5 (use 2 equal height boxes about 12 inches high - rebound onto one turn around and come back the other way)

Ankle Jump: 3 x 8

Knees to chest tuck jump: 3 x 8

Lateral barrier jump: 3 x 8

Jump rope: 3 x 30 seconds

**Section B: Strength** (do all of these)

Bulgarian split squat with bodyweight: 2 x 15-20/side

King deadlift progression: 3 x 10/side (start with whatever progression you can comfortable get ~10 reps with and work towards doing them full range) Work towards performing a set of 10 full range without any assistance on the last progression. When you can do that you'll be quite strong.

Single leg squat progression: 3 x 8-10/side (start with a high box and progress down thru the progressions as strength improves) Work towards performing a set of 10 full range without any assistance on the last progression. When you can do that you'll be quite strong.

Single leg calf raise: 3 x as many reps as possible per side

**Rest intervals:** self monitored- go when you're ready and don't rush yourself. 2-3 minutes is about right for most exercises

**Frequency:** Ideally 3x per week on nonconsecutive days (Mon, Wed, Fri or Tues, Thurs, Sat) If you can only get to it 2 x per week that's fine, but 3 x is ideal.

# The 40 Inch Blueprint

Now with all these workouts you might be wondering how would an average athlete follow all these over a period of a year or more. Just to give you an idea I'm going to lay out a typical gameplan for an "average" athlete. In the past few years I've had several athletes follow this type of format over the course of a year and hit 40 or close to 40 inch VJ. We'll assume this athlete doesn't have much in the way of training experience, so he'll start off with the novice workout.

**January-April:** Novice workout

That's 4 months on the novice workout - After that he might go 8 weeks on the general intermediate routine

**April-May** - General intermediate

After that you'd re-evaluate - Read the assessment section in this book. Are you stronger than you are fast or faster than you are strong? How do you test in the reactive resource test? Based on the results of that then you'd either focus on strength or explosiveness so we might go:

**June thru mid-July**: 6 weeks on an explosive based template

or

**June thru mid-July**: 6 weeks on an intermediate strength based template

From there it might be a good time to experiment with one of the higher frequency templates. Let's say he decides to experiment with 4 weeks of each higher frequency template:

**Mid-July thru mid September**: 8 weeks high frequency training

Next, he decides to back off on the frequency and follow 1 cycle each of the very basic intermediate/advanced workout and the tempo variance workout

**Mid-September thru mid November**: basic intermediate/advanced workout and tempo variance workout

**Mid-November thru January 1st:** Active Rest – show off new dunks

That's it! That's pretty much an entire years worth of training, starting from the beginner level and working up thru high intermediate. Chances are this prototypical athlete would have advanced by leaps and bounds working thru these templates.

# *Plyo Only Routines*

Many times people already have a weight training routine they just want to add some plyometrics onto it and that's what these workouts are for. I know many people will ignore my advice here, but unless you have a phenomenal strength base already I DON'T recommend you follow these workouts in the absence of strength training. You might get SOME gains but your success will be limited because the Vertical Jump is largely about strength. Having said that, here are some plyo workouts – these workouts can either be performed in isolation or they can be added on to any existing strength workout you're currently performing.

## Beginner

Bi-lateral low line hop: 2-3 x 10 seconds

Single response* squat jump: 2-3 x 6-8 reps

Single response* tuck jump: 2-3 x 6-10 reps

Depth drop: 2-3 x 5 reps

Frequency: 2-3 x per week

Length: 3-4 weeks

*Single response means you pause and reset yourself each rep.

## Novice

Single leg lateral low line hop: 2-3 x 10 seconds

Multi-response** ankle jump: 2-3 x 8-12 reps

Multi response tuck jump: 2-3 x 6-8 reps

Depth jump: 4 x 5

Frequency: 2-3 x per week

Length: 3-4 weeks

** Multi-response means you perform your reps rhythmically with no pause or hesitation between reps - true "reactive" training.

## Intermediate

Single leg lateral low line hop: 2-3 x 10 seconds

Lateral barrier jump: 2-3 x 6-10 reps

Alternate leg bounding: 2-3 x 25-30 yards

Depth jumps onto box or over a hurdle: 4 x 5

## Advanced

Depth jumps (with optional 10-15 lb weighted vest): 4 x 5

Bounding: 2-3x 25 yds per side

Single leg bounding: 2-3 x 25 yds/side

That's it! Progression comes via increased intensity of exercise and progresses right into true "shock" training - depth jumps and higher intensity bounding. See videos on the main site for exercise examples/descriptions

# *SPECIAL TOPICS*

## *How Much Can Vert Be Improved?...*

How much can vert be increased? This is the age old question everyone wants to know. More specifically, how much can YOU increase your vert? Unfortunately, there isn't any hard and fast answer. I increased my vert over 20 inches. Should I expect everyone to do the same? Hardly. Unfortunately, VJ marketers tend to broach realistic expectations and there's always another VJ program promising more gains in less time. But reality is reality. All you can do is train properly and let things fall where they may. I can remember even back when I was in high school there were an assortment of gimmicks promising 14+ inches within a matter of weeks.

The ultimate limiting factor for vertical jump is how fast you can apply force, which is largely controlled by things you have no control over, such as CNS sensitivity to androgens and catecholamines, density of motor neurons, and other neural characteristics. You can ALWAYS increase the "force" part of the power equation and get stronger, but as discussed in the strength deficit chapter, there comes a point in time where your speed of force development will be what holds you back. There was a heated debate on Mark Rippetoe's forum a while back when he said it was impossible to increase vert more than 30%. People that train for vert and had nearly double those type of improvements were all over him. But I can see his point of view. Before I get into answering the question, "How much can you increase your vert?" I think we really have to differentiate what type of vert increases we're talking about here. In my mind there are 3 relatively distinct types of improvements and they are:

**A: Gains that come about thru maturation**

Gains that come about thru the maturation process can be very significant. If an individual participates in a jumping oriented sport from their early teens thru their early 20's and stays healthy and keeps their body composition in check (they don't get fat), they will likely gain a good 5 or 6 inches of vert just due to the extra strength and power they get from maturing. One major factor that needs to be considered is that most people that train for VJ start at an early enough age where they still do a lot of growing and physical maturity over time and, assuming an active enough lifestyle, the general maturation process alone tends to contribute to gains up until about the mid 20's. A 25 year old athlete may have very little in common with his own physical self at 15 yrs of age. He'll be more muscular, stronger, and have the potential to be more powerful even WITHOUT training. In contrast, one who begins training at 25 yrs of age doesn't have the same potential for improvement as a 15 yr old, because he's already physically developed. Having said that, I have one 37 year old client who didn't start training until he was in his 30's and still managed to put 10 + inches on his SVJ. The guy has some very unique physical traits though and his results would definitely be outlier in nature. Simple enough.

Now to the 2nd way gains can occur:

**B: Gains that come about thru increased movement efficiency**:

These are gains that come about as one "practices" the vertical jump and becomes efficient at it. Specifically, I'm referring to grease the groove concepts like synaptic facilitation. MANY people use the vertical jump only as a barometer and don't really "get into" it. They may lift, do some plyos, and *assess* the VJ, but they don't spend hours and days on end at the basketball court trying to throw down dunks. They don't jump up and touch every ceiling they walk under. They may train for the jump, but they don't train for the jumps, if you catch my drift. Those who really "train for the jumps" will get significant improvements from simply mastering the activity. My best guess is the average athlete can get a good 3 inches of standing vert improvement and 4 or 5 inches of running vert improvement simply by getting better at doing it over time. It doesn't take long to get these gains, but a TON of trainees never really do enough jump specific work to get them.

Now for the 3rd way gains can occur:

**C: Gains that come about thru increased motor qualities:**

These are the gains most people refer to when they talk about "vertical jump gains". Increase your power, strength, reactive ability, etc. and you'll jump higher. The typical powerlifter or football player who goes thru a protocol will typically ONLY deal with gains in this department.

Now put all that together: Let's say we have a 14 year old athlete who gets interested in vert. He's kinda uncoordinated and never really been interested in jumps. Let's say he begins heavy participation in a jump oriented sport, does a lot of growing, and pays his dues with proper strength and power training. Let's say he starts off with a 20 inch vert and sticks with jump training every offseason until he's 20 years old. He gets 3 inches of vert improvement via technique/movement efficiency, 5 inches from physical maturity, and another 6 inches from increased motor qualities (strength and power). Now he's reached physical maturity and he will have put a good 14 inches on his jump. That's a 70% increase from the 20 inch vert he started out with. I've seen this happen countless times! It's important to note he didn't get ALL of those gains from doing any particular program or training - a good 40% came thru his own physical maturity.

Where the potential for gains is most limited is when referring to mature athletes who've already been involved in plenty of jump oriented activity prior to ever training for vert. Take a 25-yr old volleyball player whose played volleyball all his life but never lifted. He's already achieved any improvements he's going to make thru physical maturity, he's already likely achieved any improvements he's going to make thru increased movement efficiency. ALL he can do is get the gains thru increased motor qualities. Chances are his potential improvements will be somewhere around 30%.

I'll use myself as an example of how things like the maturation process can kinda obfuscate reality: The first time I ever measured my vert it was at 23 inches with a 3 step lead in. Ten years later at one time I hit 45 inches on the same type of jump and could hit 42 inches from a standstill. So that's 22 inches. Realistic? Well, let's take a look at what happened: First of all I was kinda gangly and uncoordinated when I first got interested in jump training. I was 15 yrs old and had never really been interested in jumping and had spent little time doing it. I got interested in jumps on a whim at school when I got into a little competition with a friend of mine to see who could touch an air conditioning

vent in one of the classrooms. Despite being a couple of inches shorter than me, he could out-jump me, and he let me know about it. That kinda pissed me off so I made it a goal to increase my VJ. I measured it the next day and it was 23 inches. For the next year all I did was jump every chance I got. I didn't do any particular training, I just did lots of jumps. Within about a year my jump was up to 27 inches. My vert stayed around there until I finally began training properly in my early 20's. I quickly got another 8 inches once I started proper strength training. That brought my vert up to 35. Then something odd happened. Between the ages of 23 and 25 I grew an inch and a half. I also continued gaining a lot of additional strength and power. That brought my vert up to the 40 inch mark and eventually a PR of 42 standing and 45 running. How much of it was due to physical maturity and how much of it was due to training? I really don't know. All I know is I did what I could do and got the results I did. That's all YOU can really do.

It's also worth noting that I have a structure fairly well suited for jumping: Extremely thin hips and long femur (thigh) bones. I also have a very good nervous system. As noted earlier, speed and rate of force development have never been weaknesses for me. My only weakness was creating force. Generating maximal force wasn't something I was born with. I had to work at it. YOU might (and probably do) have an entirely different set of physiological circumstances to work with.

But to get back to the point: How much can you improve your vert? If people ask I'll tell them a 30% increase on a relatively untrained jump should be attainable over time for just about everyone. That's 6 inches on a 20 inch jump and 9 inches on a 30. There will be plenty who get more than that, but I think the vast majority can reasonably expect that with proper training over a period of time. Your potential to increase your vertical jump is largely based on natural neural efficiency, lever length (leg length), and strength levels.

Those with the greatest potential for improvement will have long levers, good neural efficiency, and poor strength levels. Those with the least potential for improvement will have short levers, poor neural efficiency, and good strength levels. Where you are depends on YOU! If you're young and long-legged with a thin structure, a good nervous system, and poor natural strength levels, you can probably expect significantly more than 30% improvements, as you'll have a lot of unfilled potential to work with. If you're older and short legged with a slow nervous system and good natural strength levels, your potential gains might be towards the lower end. I also tell people my increases were way outside the norm and should not be expected unless you have the same genetics and/or circumstances I did.

In summary, you have to identify what "type" of athlete you are and just do what you can do with what you have to work with. It should also be noted that consistency over time will be the most important variable. Not many people have the tenacity to stay dedicated for months and years on end. Those that do are often rewarded handsomely.

## Why the Best Dunkers Are Born and Not Made…

Along these same lines, most can't help but notice that most of the top high flying dunkers don't do any sorta training other than dunking. Most of the best jumpers are just naturally that way - there's

an inefficiency in training for vertical jump and it's difficult to duplicate natural talent. I liken it to a sport like dog fighting or horse racing. All the dogs are pit bull, or closely related to pit bull, and all the horses are million dollar creations of focused breeding. They don't really do anything special to fight or run, they're born that way. You could get a regular old farm horse and try to turn him into a race horse and you could improve his ability, but it still wouldn't match the animals that are truly born for it. The VJ can be significantly improved, yes, but the things you do to improve it do so mostly indirectly and the people born with better potential will always have an edge. What I mean by "indirect" is that during something like a VJ under 50% of your muscle fibers are actually active - perhaps as few as 30%. So the idea is you wanna get more force and power from those few fibers, or try to activate more. How do you do that? Well either thru increased neural factors or increased force (strength and hypertrophy) of the recruited fibers.

The problem is the neural aspects are limited in how much you can improve them. Even if we go out on a wild limb and say you can improve those characteristics a whopping 30%, you're still pretty much left with strength and hypertrophy as the main things you can change. The only way to improve strength and hypertrophy is to make those fibers that are active in the VJ bigger and stronger, and in order to do that you have to recruit and overload them. However, in order to do that effectively you also have to hit the other 50% of dormant fibers too. Because of that, eventually you'll reach a situation where the inactive fibers get in the way. So, it's not a perfect process. When you increase your vert you're indirectly trying to "manufacture" something that wasn't there to begin with.

That's not to say if you take an untrained race horse that he wouldn't get his butt whipped by a non-genetically gifted horse that DOES train for it. If the high flying dunkers you see on youtube sat on their butt all the time they probably wouldn't be able to jump very well. But just going out and dunking consistently also has a pretty significant training effect. The only thing these athletes are really lacking is overload, but most of them don't need it. They're naturally born with very good nervous systems and MOST of them have enough muscle fiber in the right places to generate the force they need – just like the pit bull and race horse were born with the somatotype (muscular build) to succeed in their endeavors.

If someone already has the right nervous system and the right amount of muscle in the right places, all that's really left to do is improve movement efficiency. That explains why most of the youtube dunkers you see don't necessarily need additional training. They don't need to manufacture anything because they already have what they need - just like the pit bull and race horse are born with what they need. What might be different about YOU is you might need to tear the hood of the car and completely rebuild your engine from the ground up. You gotta modify the nervous system a bit, gotta build the muscular system, gotta add muscle in the right places, and still gotta boost the movement efficiency. It won't be perfect, but it CAN work. Do all that and MAYBE then you can even surpass the high flying dunkers who didn't have to do anything for it.

## Potential, Fiber type, and Neurological Characteristics…

Now I'd like to talk a little more about potential and how it relates to your innate muscle fiber characteristics and neural characteristics: This is something I touched on a bit in the brief section on muscle fiber typing, but now I'd like to dig a little deeper. There are different combinations of fiber and neurological characteristics and they go a long way in determining your potential. You have to look at things at both a muscular level and a neurological level. For the most part the muscular characteristics are an indicator of your fast twitch dominance - how much muscle mass you carry naturally, how strong you are, and how easily you gain muscle. People with more fast twitch oriented muscular systems naturally carry more muscle size and strength, and gain strength and size at an accelerated rate.

The neurological characteristics are basically an indicator of the "excitability" of your nervous system and your ability to ramp up rate coding. The best indicator of that in my opinion is natural quickness - how fast are your hands and feet in unloaded movements? There's not really a test you can do that says, "Oh you're quick or you're slow", so don't read into this TOO much. Just ponder it a bit and think of how you relate to other people in the natural quickness department. Ideally, you'd have both a fast twitch physique at the fiber level, and a great nervous system. Athletes with these characteristics make easy gains, but they also tend to be explosive even without any training, because they naturally have things at the muscular level already in place. Unfortunately, they don't tend to train consistent and hard. But here are the various combinations of muscular/nervous system and the traits of each:

**Fast Twitch Muscles and Slow Twitch Nervous System**

One subset of athletes I run across have a fast twitch oriented muscular system and a slow twitch oriented nervous system. These people will be strong but slow. Lots of powerlifter types fit into this group. They're naturally big, thick, strong, and gain muscle extremely easy, but are not typically all that quick. He may hate me for saying this, but the first guy that comes to mind is a guy like Mark Rippetoe - big, strong, and thick. In the NBA a guy like Nick Collison comes to mind. These individuals likely will have a very easy time gaining the required strength they need, but will need to spend more time working on various speed-strength and plyometric drills to improve RFD. Oftentimes a big challenge for this group is getting their body-fat down. They tend to be thick and often struggle staying lean.

**Fast Twitch Muscles and Fast Twitch Nervous System**

One can also have a fast twitch oriented muscular system and a fast twitch oriented nervous system. These are the true freaks of the athletic world. They respond to weight training EXTREMELY quickly and easily transfer any gains they make into increased power production. The majority of the NFL fits into this group as do "highlight reel" dunkers like Golden Child, T-dub, and many others. In the NBA Lebron James and Dwayne Wade would both fit into this group. These people don't need to do anything special to make gains and will often respond to ANYTHING - particularly strength training. They can pretty much look at a squat rack and put 5 inches on their vert. They typically don't need or respond

all that well to plyometric training because they already naturally have everything they'd get from plyos (increased RFD, movement efficiency etc.) - they simply don't need much of it.

**Slow Twitch Muscles and Fast Twitch Nervous System**

One can also have a slow twitch oriented muscular system and a fast twitch oriented nervous system. I'd put myself in this group. Kenyan distance runners would fall here. I'd also put a guy like Chris "Birdman" Andersen in this group. The characteristic of this group is a naturally frail build coupled with good natural quickness - skinny and weak, but potentially explosive. They may not actually be *all that* slow twitch at the fiber level, they just don't have much natural muscle fiber to work with. The challenge with these people is getting them strong. Even though they may be naturally good at endurance activities, they have good nervous systems, so tend to respond really well to increased strength and size. However, they tend not to have the greatest endocrine profiles (testosterone and stress hormone profiles), and may have recovery issues that tend to make consistent progress difficult. I'd put a high jumper like Steffan Holm in this group as well. Interestingly enough, in my observations a LOT of top high jumpers fall into this group and it's a good example of how fiber type is over-rated. A good nervous system & structure wins out. These people can respond very well to VJ programs, but they need to be consistent and patient over a good period of time - particularly with their strength training.

**Slow Twitch Muscles and Slow Twitch Nervous System**

Lastly, one can have a slow twitch oriented muscular system with a slow twitch oriented nervous system. Despite shortcomings at both ends (fiber and neural), these types can make good VJ gains because one thing they usually have going for them is a relatively easy time staying lean. Having a preponderance of slow twitch muscle tends to make staying lean relatively easy - and being lean is a big part of having good relative strength. These folks respond well to a general all around vert program including all components of proper training: Strength, explosive, and reactive work. You could put most distance athletes in this group. It's also possible to have a lot of slow twitch fiber characteristics but still be relatively "big". In my observation a lot of NFL quarterbacks and NBA centers would also fit into this group - guys like Brad Miller, Peyton Manning, Philip Rivers, Matt Schaub - long but still pretty good sized.

It should be noted the majority of athletes won't be an all out "pure" type - you'll most likely be a mix. For that reason I wouldn't read into this TOO much, but do make a note of it. If you have to think too hard about what group you fit in, you're giving it too much consideration. On the neural end go back to the assessment chapter: If most of the assessments have you as being "stronger than fast" you're most likely more slow twitch at the neural end, and vice versa. At the fiber end just ask yourself a few questions:

1. How much muscle do you carry naturally?
2. How strong are you naturally?
3. How quickly do you make strength and size gains relative to others?

If you'd like to dig a little deeper, one thing you can do is pick a handful of basic strength training exercises (squat, bench press, etc.), put 85% of your max on the bar, and do as any reps as possible. If you get more than 8 reps you're more likely to be more slow twitch oriented at the fiber level. If you get less than 5 you're more likely to be more fast twitch **.

*** Keep in mind on this type of assessment the more training experience you have under your belt the more you will tend to test as being fast twitch dominant.*

To summarize, regardless of your genes you CAN make good gains, but everyone has a different innate potential. All you can do is work with what you have!

## *How Long Does It Take To Get Gains?…*

Just as frequently as people ask how *much* they can gain, the next question is, "How many inches can I gain in xx weeks?" What I always tell people when it comes to how long it takes gains to manifest is they need to look at things realistically: People that make really big gains typically make those gains over a period of YEARS, not months or weeks. It took me many years of training to make the kind of gains I did. Most others who make really big gains are the same. You might be fortunate enough to gain 6-8 inches your first couple of months of training, and that's not terribly uncommon for someone that hasn't been training regularly. The Flyin in 4 routine I released with Alex Maroko a couple of years ago promised 4 inches in 4 weeks. Those numbers weren't exaggerated. I tested the protocol on a basketball team prior to the launch and the results were right around 4 inches. I expect most beginners following my routines to get at least 4 inches their first month. I know a few other trainers who can take a group of athletes and consistently get an average ~5 or 6 inch improvement from a 6-8 week offseason camp, but once you exhaust these newbie gains the gains typically take much longer to get.

Let’s say you're fortunate enough to get a 6 inch improvement your first 2 months of training. That's an average of 3 inches per month. Doesn't sound like a lot does it? But let's say you continue that rate of progression for an entire year. That's a total of 36 inches! Hardly realistic. The reality is once you get those newbie games progress won't come linearly. The VJ is a finicky animal.

All you can really do is turn your body into an efficient vertical jump power generating monster by boosting your strength to elite levels with at least a 2 x bodyweight squat, (or 2.5 x bodyweight half squat), jump on a frequent enough basis to maintain jumping efficiency, keep your bodyweight and body-fat in check, pay attention to your mobility and muscle recruitment patterns, and incorporate proper de-loading protocols. The rate you progress from doing those things will be variable. The routines in this manual will help you take care of the training part. All you really need to do is follow them and be consistent with your training. That's really all you CAN do.

I think a better method of forecasting the rate of potential improvement is to start with a raw number for how much you think you can improve overall, and work backwards from there. Let's use 30%. Basically what we're saying is you're capable of putting 30% on your starting vertical jump. So, if you have a 25 inch vertical jump today, it's possible for you to have a 32.5 inch VJ at some point in the future. We can then do some assessments of things that correspond to the VJ and see what you need to work on and figure out how long it'll take you to make those improvements. The most basic measure we can look at is your base relative strength. Let's say you weigh 200 lbs., have a 25 inch vert, and squat 250 lbs. Well, we already know that most people will progress on their vert as they increase their squat to double bodyweight. We can logically assume you're not immune to that rule.

So, the real question is how long will it take you to get your squat from 250 lbs to 400? That's also variable, but it's a WHOLE LOT less variable than the rate of progress on your vertical jump. Many newbie athletes can add 5 lbs to the bar in a squat exercise week in and week out for 6 months or more on end. A more advanced (but not super-strong) athlete might get half that. Let's be conservative and

figure you can increase your squat 10 lbs per month. That means in 15 months you should be able to take your squat from 250 to 400 lbs IF your training is high quality and you're a normal responder.

Would getting your squat to 400 lbs automatically put 7.5 inches on your VJ? It might or it might not. But it gives us more of a CONCRETE goal to work with. In my experience there's a GOOD CHANCE going from a 250 to 400 lbs squat would put 7.5 inches on your VJ, IF you perform jump related drills on a frequent enough basis to maintain jumping efficiency, keep your bodyweight and body-fat in check, and get enough recovery in. One thing that has been beneficial for many athletes is the Vertical Jump calculator on my website. What this calculator does is use your squat to forecast how much you SHOULD be jumping in comparison to where you're currently jumping. It's not perfect, but it does a good job telling you if you're on the right track. You can check it out here:

www.higher-faster-sports.com/verticaljumpcalculator.html

## Do You LOVE To Train?

It also goes without saying that if you're gonna make REALLY BIG gains, at some point you're gonna have to fall in love with training and get to the point where you train because you like to train, not because you have a workout sheet you must complete to achieve XX. A problem many VJ trainees have is they have a very short term mindset and don't really like to train - they look at each workout as a means to an end and don't enjoy the process. The workouts to them are something they have to "get thru" and not something they do because they like doing it.

It's kinda like I used to do in Jr. High – After several years of begging I got a weight set for christmas because I'd always wanted to be bigger and stronger. I'd decide that I was going to dedicate 6 weeks (or however long) towards a particular routine. I'd keep a sheet of paper and I'd draw a bunch of empty boxes on the paper. Each box would signify a workout that I'd vowed to do over the next 6 weeks. Each time I'd complete a workout I'd shade in one of the boxes. My goal was simply to get to the end. I didn't enjoy the process (training), I just wanted the end result. The end result was at least 4 or 5 different times over the next couple of years I'd start a workout and quit after a week or 2. I don't think I ever did get all those boxes filled in.

I don't know what changed, but at some point I began to train because it was something I HAD TO DO, and not something I was just doing to accomplish something. There hasn't been one week in the last 14 years that I haven't got at least a couple of workouts in, and the only time that will change is if I'm physically unable to train. I simply love to train and everyone I know who makes great long term gains does as well. Granted, you can make gains without loving training, but your chances of sticking with it for long are slim. If you don't currently love to train, the best thing you can do is get a good workout partner and dedicate yourself to a month of consistent training. After a month it tends to become a habit and having a workout partner makes those initial few weeks much easier.

# *The Importance of Recovery…*

I mentioned earlier in this manual the automatic assumption people carry into training is that if they just "outwork" everyone else they'll get where they want to go. Training hard on a consistent basis is important, but it's really only half the battle. You can have the greatest work ethic and willpower imaginable, yet you also need to train smart and pay attention to your body. A trainee could go out and run 5 miles and do 2 hours of weight training and an hour of plyometrics every single day. It would definitely be tough, but would likely do absolutely nothing for the vertical jump. It's simply too much volume and too much stress. Another individual could go out and train efficiently a couple of hours TOTAL per week, and might achieve earth shattering results.

It is fairly rare to find a young individual who does just the right amount of training and not too much or too little, and most people inherently want to do TOO much. The body only has so much energy available to perform, and only so much adaptive energy to recover from the demands you place on it. Progress is basically just adaptive energy being utilized to enhance performance. When you train you're placing a stress on your body. When you rest your body it then recovers and adapts to this stress, so that it can better handle the stress you're placing on it. Progress occurs just as much when you rest as it does when you train.

When most people reach a sticking point in their progress the first thing they look to is their training. If a given plan isn't working the natural instinct is to do more. So, when most people hit a sticking point they inherently want to add to their existing protocol. So, they add this, add that, start doing this, start doing that. Many times this is the worst thing they can do. When I talk to athletes that are stuck and haven't been making progress I'll have them show me on paper what they're doing. About 80% of the time I can get them progressing again by just reducing their overall training volume. Sometimes these changes come through a reduction in training frequency, and other times by a reduction in exercises, sets, and reps, but the important thing is all I did was improve their recovery. It's a 50/50 battle. **

***Note, don't overdramatize under-recovery and change your program just because you're not at your strongest and most explosive each day of the week. What I notice is many athletes freak out when they see their performance fluctuate on a daily or weekly basis. If you did a heavy squat workout yesterday your probably not going to feel all that springy tomorrow. If you're not as explosive or as strong this week as you were last week or 2 weeks ago do you really think you lost your explosiveness and strength that quickly? No way. You're not as explosive because your body is fatigued. So, pay attention to any consistent regressions in performance over time. Even in the absence of any training at all it takes a fairly lengthy time for strength or performance parameters to diminish to any appreciable degree. A powerlifter will often lay off a lift a full 10 days prior to a meet. Their strength doesn't suddenly dwindle away during that time span it actually improves as they allow their bodies to rest.*

The right amount of training and recovery is of utmost importance. The workouts are designed to take care of both. Just realize that it takes time for your body to adapt to the stresses put upon it and you can only train so much and make progress so fast. I can say that in most cases if you're training on a consistent basis and not making progress it is because you're doing too much, or a combination of too much work and not enough recovery. The over-work doesn't happen so much from the programs per se, but rather all the other activities that you're doing. Don't expect to make much progress following a vertical jump specialization program while you're also playing full court basketball for 2 hours per day 7

days per week. In that case you'd be lucky to progress at all. In fact, in that situation, most people actually regress! What follows are some tips and some guidelines you can use to make sure you're getting enough recovery. Ask yourself these questions to help assess your state of recovery and thus your state of adaptation:

1) Are you making consistent strength gains on a weekly basis? If not that's definitely a sign you're doing too much.

2) Do you remain excited about training every week?

3) Are you sleeping well?

4) Do you wake up feeling refreshed each morning or do you feel stressed out upon wakening?

5) Are you completing each workout feeling as if you could go back and do more?

6) Is your vert and strength at least staying stable? A reduction in vert combined with a reduction in strength is a surefire sign that you're not recovering from your workouts. Vert by itself may go down a bit during a dedicated strength phase, and strength itself might go down a bit during a dedicated explosive oriented phase, but if BOTH strength and vert are declining at the same time you most definitely have some sorta recovery problem.

If you answered "yes" to all of those questions you're on the right track. If you answer "no" to a couple of them it's time to take a closer look at your other activities, the amount of rest days you have between workouts, and your nutrition. Are you spending too much time playing other sports or spending too many hours on the court? In my experience anymore than a few total hours of basketball (or similar activity) per week is bordering on the point of being counterproductive for vertical jump gains. Also consider your nutrition – are you losing weight or is your weight at least stable? Are you getting enough sleep each night? Weight loss in someone who isn't overweight and lack of sleep will often interfere with VJ gains. The first thing I do with a recovery challenged athlete is manipulate their frequency. Most intermediate vertical jump routines are set up on a weekly split such as this:

Mon: lower body workout A
Thurs: lower body workout B

There's nothing wrong with reducing the frequency of training to one workout every 5 days or so like this:

Mon: lower body workout A
Sat: lower body workout B
Thurs: lower body workout A
Tues: lower body workout B
etc.

A beginning athletes routine might be set up like this to start:

Mon: lower body workout
Wed: lower body workout
Fri: lower body workout

They could insert an extra rest day between workouts like this:

Mon: lower body workout
Thurs: lower body workout
Sun: lower body workout
Wed: lower body workout
Sat: lower body workout
etc.

Often those minor frequency changes can make a BIG TIME difference in your results. When you're well recovered you stay fresh, and fresh (not stale) muscles are powerful muscles. If you've ever taken a long 10 hour drive in a car and tried to get out and walk around you experienced stale muscles. It doesn't take much, but muscles function best with some regular light activity – even a short walk. If you're worried about getting too stale in between workouts you can always throw in a dynamic warm-up, engage in some 10-20 actual jumps, do 6-8 sets of lower intensity plyometrics, or take part in some other light non-strenuous activity on your off days. These activities won't wear you out but will enable you to stay fresh while getting some activity in. It generally takes about 48 hours after an intense training session for full systemic (neural) recovery to occur.

In general, if an activity makes you sore or has the potential of inducing lots of soreness, it has the potential to impact systemic recovery. If it doesn't it's probably not something to worry about. That's why low intensity activities like jump rope, cardio, low intensity plyometric drills, and lower intensity conditioning work can be done every day without any detriment in performance, unless they're performed at insanely high volumes. Higher intensity activities like intense weight lifting, sprinting, intense agility work, and high impact plyometric activities often require a day or more of rest in between workouts for optimum performance.

If you see you have major problems with recovery and are not making strength gains, aren't making vertical jump gains, and you aren't motivated to train, you should probably take a week off and resume training with lower volume, lower frequency, and reduce the volume and frequency of your extraneous work, such as basketball. Remember, long-term progress is the most important thing and there is nothing at all wrong with taking a planned rest period every 4-8 weeks. Many coaches have their athletes take a mandatory 1/2 to full week of active rest after every 3-8 weeks of training. Athletes get better results and they come back to train with renewed energy and excitement and typically quickly surpass their previous performances. This is also the reason that de-loading weeks and active rest is already built into many of the programs I've designed. This can take the form of a few days to a week of lower intensity activity. You can and should stay semi-active - do some dynamic stretching, outdoor activities, work on your sport, but cut back on intense activities during this time.

The training programs I've put together are designed to be "stimulatory" in nature. That is, they are designed to stimulate your body to increase strength, power, speed, and explosiveness. All these factors combined lead into increased performance. The workouts are not designed to kill you! Stimulate is not the same thing as annihilate! Generating fatigue is not the same thing as generating results. You should finish each workout feeling slightly refreshed, not totally worn out. You should feel as if you could go back and complete 50% of the workout again no problem. If you're feeling totally drained you're probably doing too much and need to cut down on volume.

Now for some practical tidbits and some answers to common questions:

## How do you know if you're recovered or not?

If, over the course of a week or so, you can't at least match what you did last time, you're not recovered. For example, on Monday you measured your vertical jump and it was 32 inches. You lifted weights and played full court basketball on Tuesday. On Wednesday your vertical jump was 30 inches. On Thursday you did some skill work and played some half court ball. You measured your VJ again on Friday and it was 32 inches. On Saturday it was 33 inches. Would you want to plan another weekly training session on Thursday or Friday? Probably not. It's already taking you until Saturday to fully recover from your Tuesday training. If you were going to plan another workout Saturday would probably be the day to do it.

## What about "recovery" drinks?

Recovery drinks are something promoted heavily in the bodybuilding and performance world. Let's look at some of their benefits. After an intense workout:

1. Glycogen stores are low
2. Protein breakdown is increased

As a result, there are potential consequences:

1. Poor subsequent performances in the gym – particularly if you plan on exercising the same muscles again within the next 24-48 hours.
2. Potential loss of muscle mass

So, what we want to do after a workout is quickly:

1. Replenish low glycogen stores in muscle
2. Decrease muscle protein breakdown that occurs with exercise

Under normal conditions the body restores itself and recovers (gets into a positive protein balance) in anywhere from 8 to 24 hours. However, proper post-workout nutrition can shift your muscles to a positive protein balance within an hour. Over the relatively short term this leads to increased performance and muscle growth.

To accomplish this you want 3 things post-workout:

1. A proper amount of carbohydrates
2. A proper amount of protein with a good ratio of Branched Chain Amino Acids (BCAAs) and essential amino acids
3. High blood levels of insulin

High insulin levels along with high levels of amino acids can increase protein synthesis over 4 times that of normal post-workout amino acid and insulin levels. So, to create these conditions, you ideally want a protein that is high in BCAA and is also fast acting. For this purpose whey protein is perfect. You also need something that can stimulate insulin, like dextrose or maltodextrin. A good dosage for a postworkout recovery drink will generally consist of 0.8 g/kg (2.2 pounds) of carbohydrate and 0.4 g/kg of protein.

However, even though a post-workout recovery drink can obviously be beneficial, a home made post workout meal or standard drink can provide identical benefits to a manufactured recovery drink. A carbohydrate meal consisting of oatmeal, rice, cereal, pasta, juice, or other carbohydrate source, along with a scoop of whey protein or a few glasses of low-fat milk will work just as well as any recovery drink, but may not be as convenient. A few glasses of chocolate milk also work well.

Additionally, unless you're training a muscle group hard and heavy every single day, you will have plenty of time to replete muscle glycogen through your standard diet. For one thing it takes quite a bit of training to become fully glycogen depleted. For example, with it takes 15-20 sets of 15-20 reps to fully deplete a muscle of glycogen. Most athletes, particularly in speed, power, and strength dominated sports, don't train with nearly that much volume.

Additionally, even though your muscles will be somewhat depleted of glycogen after a workout, if you don't train the same muscle group until 48 hours (or more) later is it really going to be that difficult for you to replenish that glycogen? No, not really. Therefore my recommendation is as follows: If you're trying to gain muscle a post workout recovery drink or meal consisting of 20-40 grams protein and 50-100 grams of carbohydrate will definitely provide muscle gaining benefits. If muscle gain is not your goal it's optional. You can get away without it, just make sure you eat enough to maintain your weight and energy and don't worry about it.

## How important are nutrition and supplementation for recovery?

For the recovery of the energy used up in training, if you don't eat enough calories you're probably going to find things difficult. From a systemic perspective, lack of nutrition or a diet containing inadequate calories can negatively influence systemic fatigue by creating an additional stress. That's why if you need to lose fat you shouldn't attempt to lose it TOO fast – no more than 2 lbs per week. The positive effects of nutrition on systemic recovery are fairly minor - sleep and rest are much more important. Having said that, eating more of an alkaline diet can relax the nervous system and can be of

some benefit. Green veggies are highly alkaline. Other supplements that can help systemic recovery are magnesium and fish oil. I recommend 300 mg of magnesium per day and 6 grams of fish oil.

## Does the type of diet a person eats have that much of an impact on recovery?

Not near as much as many people believe. The important thing is that a person gets enough calories, protein, and carbs. Where the protein and carbs come from isn't so important. With the volumes that most athletes eat this really isn't an issue. Believe it or not, most high level athletes aren't consuming spartan or fancy diets. In fact, it's not uncommon for athletes that try and eat perfect to end up doing more harm than good, simply because they don't eat enough to meet their needs. For an active person with a good metabolism it can be hard to get enough to eat if your diet is ultra clean and consists of nothing but egg whites, boiled chicken breasts, steamed veggies etc. Those foods are good, but can be hard to live on unless your whole life revolves around buying, preparing, and eating food, like a lot of bodybuilders.

## What about recovery workouts?

Recovery workouts are often used to get blood into sore and fatigued muscles with the thought being this helps speed recovery. Sprinters often use "tempo" workouts - they'll sprint intensely one day and go out the next day and sprint at lower speed over increased distances. A powerlifter might bench heavy one day and might come back the next day and do several light sets of 50 to 100 reps on exercises like pushups and triceps pushdowns, just to get some blood flowing to the muscles. Or they'll squat heavy one day and come back the next and do some sled dragging. A baseball pitcher will pitch one day and go for a jog the next couple of days in hopes the jogging will increase blood flow to his sore arm and speed recovery.

My personal feeling is that many athletes who use recovery workouts often do more harm than good and actually end up interfering with the recovery of local muscles. This is particularly true if the muscles used in the recovery workout are the same muscles used in the main workout. The intentions are good, and recovery workouts can provide some psychological benefits - they make you "feel" better and more recovered, but at the cellular level they really don't truly "speed" recovery. Additionally, people tend to perform these workouts at too high of an intensity with too much volume. My preference is that if athletes are using the same main muscles in the recovery workout that they are in the main workout, they should limit these workouts to activities that are unlikely to cause much fatigue. For example, instead of performing tempo workouts, which involve sprinting, a jumper or sprinter might perform a long dynamic warm-up. Another option is to engage in activities that involve different muscles. For example, a jumper might swim some laps in a pool, hit a heavy bag, or swing a sledgehammer as a recovery workout. If you get creative you'll find there are dozens of ways to stay active without interfering with recovery. I don't think one should feel that recovery workouts are absolutely necessary.

## What about recovery methods like sauna, contrast showers, massage, foam rollers etc.?

I recommend athletes try all of these. None of them are world beaters but they all provide some good benefits. A sauna provides many positive hormonal benefits, burns calories, and also has the benefit of increasing whole body blood flow. If nothing else this can improve your perception of how recovered you feel. Contrast showers can increase blood flow. Massage is great for adhesions, loosening up tight muscles, and plenty of other things. You can also do a form of self-massage yourself with a foam roller or even a basketball.

## What is the most important thing an athlete can do for recovery?

Besides consuming a diet that contains adequate protein and carbohydrates and paying attention to the content and timing of workouts, one word – SLEEP! A chronic lack of sleep (as in more than a few days) will kill even the best designed training program. An occasional stretch of bad sleep probably won't have any ill effects, but don't expect to make many gains if you're chronically sleep deprived. Most athletes need anywhere from 7 to 8 hours of sleep per night. Some can get by with only 6, but I don't know many people who can only sleep 5 hours a night for long without suffering a detriment in performance.

# *Breaking Out Of a Training Rut…*

Sometimes no matter what you do as a VJ seeking athlete, you get in an inevitable rut. It's really hard to make consistent and steady gains in VJ. If you increase your vert just 1 inch per month, in 3 years that'd be 36 inches. Obviously not gonna happen! The reality is vertical jump gains tend to come in spurts, followed by weeks and month of stagnation. Sometimes all you can do is train and do what you know you should do and hope for the best. You get stronger, so you can get more powerful, but sometimes while you're building strength your jump doesn't respond. Only when you dedicate yourself to explosiveness do you see the fruits of your labor. So, you go on a strength phase and gain some strength and your vert goes nowhere....but you follow that up by working on your explosiveness and all the sudden your vert shoots up 3 inches. But SOMETIMES even that doesn't work.

Your body operates on circadian rhythms which can be affected by a ton of different things: the time of year, heat/cold, stress levels, sleep/wake schedules, nutrition, your moods, and so many other things impossible to predict that perfectly forecasting gains is impossible. So, really all you can do is all you can do. Having said that, here is a checklist that might be able to help you bust out of a rut:

1. What type of training have you been doing lately? Is your routine strength dominant, balanced, or plyometric dominant? If in doubt do whatever you HAVEN'T been doing. If you've followed a strength dominant routine cut back on the strength work and work on more of a pure explosiveness type format such as this:

**Monday or Tuesday**

tuck jump, lateral barrier jump, or ankle jump: 4 x 8
depth jump 4 x 5
jump squat, hang clean or hang snatch: 4 x 3
squat: 8 x 1 @ 85-90%
RDL , manual reverse hyper, or barbell hipthrust: 3 x 6-8

**Friday or Saturday**

tuck jump, lateral barrier jump, or ankle jump: 4 x 8
depth jump 4 x 5
jump squat, hang clean or hang snatch: 4 x 3
speed squat: 6 x 2 @ 60-70% of max
RDL , manual reverse hyper, or barbell hipthrust: 3 x 6-8

Oftentimes a routine like that will get you moving in the right direction, particularly if you've been overblowing the strength aspects for a good while. What if you've been performing too much plyometric work for too long? Then take the opposite approach. Decrease frequency and go on more of a strength dominant routine.

**Session A:**

tuck jump, lateral barrier jump, or ankle jump: 4 x 8
jump squat, hang clean or hang snatch: 4 x 3
squat: 5 x 5
RDL , manual reverse hyper, or barbell hipthrust: 3 x 6-8
calf raise 3 x 20

**Session B: (5 days later)**

tuck jump, lateral barrier jump, or ankle jump: 4 x 8
depth jump 2 x 10
jump squat, hang clean or hang snatch: 4 x 3
Bulgarian split squat: 3 x 6-8
RDL , manual reverse hyper, or barbell hipthrust: 3 x 6-8
calf raise: 3 x 20
Alternate between A & B every 5 days

Next, look at your outside activities, how much basketball and conditioning work do you do per week. If more than 2-3 hours total then CUT BACK on it and/or reduce your training frequency and see what happens. If you routinely train for vert twice a week or more, AND you're engaged in more than a couple of hours of basketball, practice, games, etc. it'll probably be tough to make gains. If you have to maintain a busy practice/competitive/conditioning schedule you might be better off with just one workout a week. I've had MANY athletes make good gains with just one workout per week. A sample one time per week workout might look something like this:

Tuck jump, lateral barrier jump, or ankle jump: 4 x 8
depth jump 2 x 10 from low box
jump squat, hang clean or hang snatch: 4 x 3
squat: 5 x 5
bulgarian split squat: 2 x 6-8
RDL , manual reverse hyper, or barbell hipthrust: 3 x 6-8
calf raise: 2 x 20-30

If your outside activities aren't an issue just take a simple look at your training frequency. Do you ALWAYS train on the same weekly schedule? If so, try changing it up. If you normally train twice a week, switch to one workout every 5 days. You might also give high frequency training a shot. There is more than enough info. in this manual to give you plenty of things to experiment with. Also, take a look at how long it's been since you've had a week off. If you've trained 3 months or more without a de-loading week go ahead and take 4-5 days off. You can stay semi-active and do light conditioning and stretching on a daily basis, but give your body a break. Sometimes that's all that's needed to get things moving in the right direction. The bottom line is you WILL run into training ruts. In fact, the longer you train the less frequently you'll make gains, and that should be expected. You just gotta do what you can,

train properly, tweak your setup, and eventually you'll get thru it! If you still can't figure things out contact me thru my website www.higher-faster-sports.com and I'll evaluate you & your program.

## Competitive Maxes vs Training Maxes

One thing that often causes people confusion when it comes to jump progression is differentiating between highly stimulated "competitive" maxes vs "training" maxes. A competitive max is an effort that typically occurs in a high state of arousal and is typically only reserved for competitions. Amongst olympic athletes these type of performances would be reserved for competitions like the olympic games - they don't occur often and won't occur often. For example, Usain Bolt couldn't go out any day of the week and match his world record 100-meter performance. Does that mean he's not improving as an athlete? Hardly. But if you looked at it from the point of view that failure to exceed your best effort is regression it certainly would look that way.

For the typical VJ trainee a competitive max is a PR type effort you hit every once in a while when you just feel really good. You typically have something external motivating you - whether a crowd, competitive environment, testing environment, or something else. These types of true max efforts can occur in the VJ, or they can occur in various other exercises like squats. I've mentioned elsewhere in this manual there can be a 10% difference between a competitive max and training max. That means you might see a 2-4 inch difference in your jumps just based on how you feel. Every once in a while you'll REALLY be feeling good and have one of those true competitive max type days (even if you're not competing). DON'T expect to see those efforts all the time and DON'T expect to be able to match them all the time.

The important thing about competitive maxes is they're basically a true "110%" effort and not something you can do every day of the week. If you TRY to hit them every day of the week you'll get discouraged and burn yourself out in no time flat. The trouble comes when an athlete hits a rare super motivated competitive max and expects to be able to duplicate that effort day in and day out - ain't gonna happen. For example, just a few weeks ago I had an athlete hit a 10'7 inch single leg jump in his training - a PR for him. It was a rare competitive max type effort for him - a very high quality effort. But in the following couple of weeks he failed to equal that effort and began to get discouraged. He was STILL progressing with his training, he just didn't think he was. Finally, about 6 weeks later he had another quality PR type day and was able to surpass his previous best. Imagine if he would have gotten discouraged when he couldn't match his previous best and quit? It happens to many people.

Progress in jumps always tends to come and go in spurts. You go weeks (or sometimes months) without any progress, then BAM, all the sudden you're up 3 inches seemingly overnight. That's just how it goes. Don't expect to duplicate PR performances every time you jump!

Instead of always basing your progress on your PRs, I recommend you pay attention to your fairly relaxed, unmotivated jumps. How do they feel compared to how they used to feel? Are you jumping higher with less effort? That's progress. Back when I was playing basketball in high school I KNEW my best efforts would be reserved for game conditions. That's just the way it worked for me. No

matter how hard I tried, or how many jumps I did, outside of games I would never equal or approach my best “game” jumps. But one thing I'd do is go to school in the morning and play a couple of games of half-court basketball . During or after these games I'd take several 90% effort jumps. They were quality jumps, but I wasn't super motivated like I would be in a game.

I began using THESE unmotivated relaxed jumps as a barometer on how my training was going. I knew when I was touching a given height with less effort a true PR type performance was just around the corner. You can do the same sorta thing in your training. Look at your personal bests but also look at your standard unmotivated easy effort jumps. What you wanna see is consistent and gradual increases in your unmotivated jumps.

# *Training For the Unilateral (1-leg) Jump*

Over the years I've had a lot of athletes interested in increasing their 1-leg jumps and I used to be one of them myself. Unfortunately, as much as any other demonstration of athleticism, the 1 leg jump is something you either have or you don't. People are for the most part either naturally good at it or they're not - usually from an early age. Take a group of 15 yr. olds out to a basketball court and ask them to jump as high as possible at the rim with no limits on how many steps they take or what style of jump they use. What you'll find is some people do better going off both feet, some do better going off 1, and a handful (like Michael Jordan), excel either way. Those that inherently excel in the 1 foot jump will tend to have a greater gap between the 2 styles.

In my experience, it's relatively uncommon that those who jump significantly higher off both feet ever "convert" to being excellent unilateral jumpers, although the reverse is not terribly uncommon. In other words, if you've been training for a while and you jump higher off of 2 feet right now, you'll probably ALWAYS jump higher off 2 feet. I've seen relatively few exceptions to that observation. It is certainly possible to improve the bilateral (2-leg) vertical jump substantially. These days gains of 15 inches or more over a period of time are relatively common. Physical growth and physical maturity in conjunction with consistent training will improve either style. However, most of the *training* that improves jumping ability tends to improve bilateral jumping to a greater extent than unilateral jumping.

Fortunately, although not everyone can be an elite high jumper, I do believe most people can significantly improve their unilateral jump.

## *What Are The Differences?*

First, let's look at some of the differences between the bilateral and unilateral jump: There are an assortment of muscular and structural factors which contribute positively or negatively to unilateral jumping. Let's start off by taking a look at what happens in the unilateral jump and the differences between it and the bilateral jump:

You run up to a predetermined take off point, plant your foot, and basically take off one leg like you're doing a lay-up. If you think of the action of a pole vaulter, and imagine the pole vault is the leg of an athlete, you have a somewhat accurate picture of what takes place in a unilateral jump. The foot plants and the body levers over the plant leg and rebounds off the plant foot up into the air. After the plant, and as you begin to leave the ground, the action of the plant leg is fairly similar to a bull pawing the ground, pushing down and back, which involves a good degree of hip extension, much like a cross between a single leg squat and single leg reverse hyper. Thus, it should be no surprise that science has found that a unilateral jump favors greater ankle and hip extensor activation than a bilateral jump. Thus, strong calves and glutes are a must.

What other differences exist? Well, based on my research and observations, the main difference is the muscles acting on the knee and ankles (quads and calves) are unable to contribute as much

positive energy to the actual jump. They do act eccentrically during your plant phase to help you absorb the force and avoid collapsing when you plant, yet when you perform a bilateral jump you're able to get more wind-up and generate significantly more "oomph" from your quads and calves, particularly your quads. You're able to explode thru the positive motion like you're doing a ¼ squat and pop off the balls of your feet – and you're able to do it with both legs instead of just 1. In the unilateral jump all you can really do is plant 1 foot, absorb the force, and generate a quick push. You have to be able to generate enough additional leverage to make up for the fact that you're only using one leg to generate power, and that's where structural factors come into play.

## *Structural Factors*

Now let's discuss the structural factors responsible for that leverage: By structural I'm referring to the length of your bones, muscle attachment points, length and makeup of your tendons, and your posture. Some of these you can do nothing about, while some you can influence to a decent degree.

Go out and do a few unilateral jumps and you'll probably find if you go fast enough and hard enough in your approach you can induce a good degree of pain in your ankles or knees. This is due to the impact forces involved, which throw a lot of force onto the knees, ankles, and hips. In fact, these forces can be up to 10 x bodyweight! One of the keys to a good unilateral jump is being able to keep a somewhat stiff leg at impact and not allow the knee to buckle too much. This requires the ability to absorb forces in the ankles and knees, of which strength in the ankles and quadriceps is a primary variable and a consistent focus amongst high jumpers.

However, this is also influenced by your body structure, since the structure in large part determines how efficiently your body deals with force. The structure of your hips, limbs, and feet determine how your feet impact the ground, how the joints are aligned at impact, how much stress is absorbed into the joints, and in large part how much leverage you're able to generate from a given amount of muscular force. People that unilaterally jump well tend to inherently have a structure and posture that lessens impact stress in the ankle, knee, and hip when they plant and optimizes their leverage when they jump. In other words, if you've ever skipped rocks on a pond of water good uni-lateral jumpers function like a longer ultra thin rock. They inherently deflect and leverage force well. Let's take a look at some of these structural factors:

**Leg length:** It would obviously be fairly difficult to effectively pole vault with a pole that is only 2-feet long. You simple don't have a long enough lever to swing up and over the bar. By the same token, individuals with longer legs have better leverage in the unilateral jump. The best high jumpers typically have long legs and disproportionately long lower leg bones. The shorter your legs are the less likely you are to be a good unilateral jumper, although there are exceptions.

**Hip width:** The wider the hips are the greater the natural deviation the center of gravity is at the plant. When a person with thin hips plants their foot their center of gravity is more easily directed over their plant foot. Ideally you want thin hips, but again, there are exceptions. This does help explain why you

don’t see many women who jump off 1 foot as well as they do 2, as they inherently have a wider relative hip structure.

**Tendon length and stiffness:** As noted earlier, the tendons sorta act like rubber bands. However, plenty of excellent high jumpers don’t appear to have relatively lengthy Achilles tendons compared to the rest of their body. A more important factor is tendon stiffness. In this sense the word stiffness does not mean lack of flexibility. To grasp the concept of tendon stiffness, think of the difference between an ultra thin rubber band and a thick rubber band. The thick rubber band generates more force when pulled back, and is more difficult to deform. Top high jumpers have been found to have Achilles and patellar tendons much stiffer than normal. This allows them to jump using faster approach speeds and more forceful take offs, as they inherently deal with and generate force better, and can handle and deflect a given amount of force with less knee bend. Tendon stiffness improves with regular training, but the extreme deviation from the norm seen in elite high jumpers in this area is largely genetic.

Unfortunately, those factors are fairly difficult for you to influence to a great degree, short of choosing the right parents. Now let’s touch on some factors you can control:

**Bodyweight:** With the exception of rare specimens like a young Charles Barkley, most unilateral jumpers are relatively thin and lean. Bilateral jumpers can benefit from increased muscle mass to a greater extent because increased thigh and hip size allows them to generate more power during the positive phase of the jump, and they’re able to use 2 legs to generate power instead of just 1. Increased upper body size and strength throughout the spinal erectors, shoulders, and traps also can help them (or at least doesn't hurt them) drive off the ground. The increased whole body power that comes with increased muscle mass makes up for any increased weight they gain, which explains why many of the best bilateral 2-foot jumpers often push 200 lbs bodyweight or better. Unfortunately, since it's more about leverage and less about strength, the unilateral jump doesn’t tend to respond well to weight gain. That doesn’t mean you need to starve yourself like an anorexic or be overtly paranoid of any muscle gain. Keep your body-fat down and restrict your muscle mass increases primarily to your glutes, hams, and quads and you should be good to go.

**Posture and muscle balance:** By controlling your posture you can optimize impact stress in the ankles and knees when you plant and optimize muscular contributions to get as much leverage as possible when you jump. Unilateral jumping favors a forefoot dominant posture, which is primarily controlled by the muscles acting on the pelvis, the glutes and hip flexors. A sure sign that your posture and muscle balance are off is if you feel pain or lack of coordination when you jump. Unilateral jumping is a lot like skipping walks across the water. It's not a feat that should feel like you’re exerting a ton of effort. It should feel relatively smooth and effortless, even if your jumps aren’t extremely high.

Theoretically, the superior impact short ground contact depth jumps have on unilateral jump would tend to support the notion that having more of a fast twitch dominant profile would lead to success in the unilateral jump, much like the sprints, but observationally this doesn't seem to be the case. A disproportionate number of elite high jumpers are built more like distance runners - white with long frames with rail thin upper bodies - typical characteristics of more of a slow twitch or mixed

muscular fiber type. This lends credence to the notion that body structure and tendon structure/stiffness are more important and, much like the regular VJ, fiber type may not be all that important.

## *5 Keys For Increasing The Unilateral Jump*

Now let's go over the 5 keys for increasing the unilateral jump:

A: **Keep your body-fat in check**

This should be self-explanatory. Anything more than 10-12% body-fat and I'd start to worry, and 8% body-fat or less would be optimal.

B: **Establish glute/hip extension dominance and strength**

Here you need to get your posture and muscle balance correct, so you execute the movement with optimal recruitment patterns. You want to be glute dominant. See the mobility and assessment section later for information on getting this accomplished.

C: **Strengthen the quadriceps and ankle extensors**

Remember what I said about the ankles and quads helping you absorb force in the plant? You need enough strength in the relevant muscles to dampen that impact stress. The knee and ankle extensors (quads and calves) are most important, so squats, lunges, calf raises, various low level plyometrics, and other loaded quadriceps strengthening movements are your friend here. To give you an idea, 150 lb champion high jumper Steffan Holm routinely does barbell step-ups with 400 + lbs!

D: **Use shorter response plyometric variations**

The running 1 leg jump or high jump take place a lot faster than a regular jump and could be considered a "short response" plyo movement with an amortization phase of around .150 milliseconds. Thus, ankle extensor dominant plyometrics like depth jumps over hurdles and depth jumps with intentionally short ground contact times should be favored over normal longer response plyometrics, such as depth jumps for height. Bounds are another good plyo exercise for single leg jumps. Research indicates the calves can contribute up to 60% in the unilateral jump, so having powerful ankle extensors is a must.

## *What about single leg movements?*

One of the questions I'm frequently asked is if a person wants to jump off 1-leg should they exclusively favor uni-lateral exercises like lunges, stepups, and bulgarian split squats in their training? Unilateral exercises are good for a lot of things and can certainly be incorporated, yet given the fact that weight training has very general affects, I personally don't think there's any need to emphasize these lifts much more than you usually would. I'd still consider squats the foundational strength lift, as do most high jumpers. In other words, if you took 2 twin brothers competing in the high jump and one of

them could only train with 1-leg lifts and the other with only 2 leg lifts I wouldn't expect to see much of any difference in their performance.

One thing that is true of the unilateral jump with is it does seem to respond exceptionally well to partial range movements. Exercises like half squats, lunges, and low box step-ups carry over particularly well. Next, there's one other important thing you must do:

E: **Hone/practice jumping technique to optimize specific leaping patterns, recruitment patterns, muscular, and tendon strength**

This you accomplish by actually practicing your jumps. This does 2 things: First, it obviously enables you to practice and optimize your technique. Secondly, moreso than most athletic events, unilateral jumping is intense plyometric activity. Jumpers in the field events tend to take several days of rest between sessions of jumping because the speed of the approach and unilateral plant sends a lot of force into the muscle tendon complex. Thus, the best plyometric exercise for a good unilateral jump IS a unilateral jump. That doesn't mean there aren't some effective auxiliary plyometric exercises, like depth jumps, bounding, and skips. But keep in mind these are secondary to the actual event of jumping.

## *A Sample Routine*

Sound complicated? Well to make it a bit easier here is a sample routine designed for an intermediate level athlete. Before I get into it one caveat: Even at the more advanced levels the differences in training for the unilateral and bilateral jump are relatively miniscule. For relative beginners the training is EXACTLY the same. So, make sure you've established a strength base and have 6 months to a year of solid training under your belt before worrying about most of the stuff in this article. Having said that, here is a sample 3 week routine:

| Week | Mon | Tues | Wed | Thurs | Fri | Sat | Sun | Notes |
|---|---|---|---|---|---|---|---|---|
| 1 | Barbell skips with 15% of max squat– 2 x 8-10/side<br><br>¼ rhythmic Jump squat with 45 lbs – 2 x 8-10<br><br>Barbell hip thrust or manual reverse hyper 3 x 8-10<br><br>Bulgarian split squat- 3 x 8/side | | short response depth jumps or hurdle jumps 4 x 5<br><br>Alternate leg bounding: 4 x 30 yards<br><br>Jumps for height x ~10 jumps | | Barbell skips 2 x 8-10/side<br><br>¼ rhythmic Jump squat with 45 lbs – 2 x 8-10<br><br>Romanian deadlift or Barbell hip thrust 3 x 8-12<br><br>Barbell Squat- 4 x 6/side<br><br>Single leg calf raise- 2 x 20 | | short response depth jumps or hurdle jumps 4 x 5<br><br>Alternate leg bounding: 4 x 30 yards<br><br>Jumps for height x ~10 jumps | Use an "optimized" box height for your depth jumps, but when you perform them come off the ground as quickly as possible |

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | Standing calf raise- 2 x 20 | | | | | | | |
| 2 | | Barbell skips 2 x 8-10/side<br><br>¼ rhythmic Jump squat with 45 lbs – 2 x 8-10<br><br>Manual reverse hyper or Barbell hip thrust 3 x 8-12<br><br>Barbell bulgarian split squat- 4 x 5/side<br><br>Standing calf raise- 2 x 20 | | short response depth jumps or hurdle jumps 4 x 5<br><br>Alternate leg bounding: 4 x 30 yards<br><br>Jumps for height x 10 jumps | | Barbell skips 2 x 8-10/side<br><br>¼ rhythmic Jump squat with 45 lbs – 2 x 8-10<br><br>Romanian deadlift or Barbell hip thrust 3 x 8-12<br><br>Barbell Squat- 2 x 5<br><br>Half squat- 2 x 5<br><br>Single leg calf raise- 2 x 20 | | |
| 3 | short response depth jumps or hurdle jumps 4 x 5<br><br>Alternate leg bounding: 4 x 30 yards<br><br>Jumps for height x 10 jumps | | Barbell skips 2 x 8-10/side<br><br>¼ rhythmic Jump squat with 45 lbs – 2 x 8-10<br><br>Depth jump- 4 x 5<br><br>Half squat 2 x 5 | | short response depth jumps or hurdle jumps 4 x 5<br><br>Alternate leg bounding: 4 x 30 yards | | Barbell skips 2 x 8-10/side<br><br>1/4 rhythmic Jump squat with 45 lbs – 2 x 8-10<br><br>Depth jump- 4 x 5 (18-24 inch box) | Take half the next week off then test between thurs and sunday |

Although that routine is fairly generic and won't serve everyone well it should serve the average intermediate trainee quite effectively. There is also a higher frequency program in the Programs section that works really well for uni-lateral jump.

# *Periodization Made Simple…*

I regularly get a ton of questions about different types of periodization: linear, undulating, concurrent, conjugate and others. It's a topic that confuses a lot of people, and I've seen many with good intentions screw up a good training plan trying to get overly cute and technical with their periodization. So, let's take a look at periodization.

First of all, *periodization* is defined as an organized approach to training that involves progressive cycling of various aspects of a training program during a specific period of time. It was originally conceived as a method of organizing training into periods of time, each devoted to a particular goal or emphasis, so that a competitive athlete would "peak" at a pre-determined time (such as the Olympic games). One premise of periodization is that certain qualities are foundational to others. For example, for the vertical jump your "target" motor quality is explosive strength. This being the case, you'd want to identify the qualities which are foundational to explosive strength and start your cycle with training to address those foundational qualities, then progress into qualities that are more specific to your primary event/goal.

Specific to the vertical jump, you know that strength is important and you know that qualities like rate of force development are important as well. Thus, your foundation would progress from strength (heavy barbell work), to power (RFD), and speed-strength/plyometric (also called reactivity or speed-strength). Thus, we'd progress from exercises like heavy squats, to exercises like jump squats, to exercises like depth jumps. A typical periodized 22 week training phase might look like this:

**Adaptation Phase (4 Weeks)**

**Hypertrophy Phase (4 Weeks)**

**Maximal Strength Phase (6 Weeks)**

**Speed Strength Phase (6 Weeks) ****

**Active Rest (2 weeks)**

*** Speed-strength, rate of force development, explosive strength, reactivity, and power are all often used interchangeably and really they're all pretty much demonstrated in the real world the same way: Producing maximal force in minimal time. Thus, a speed-strength phase would include typical explosive work like plyometrics, jump squats, and other "quick" variants.*

The first phase, ***adaptation***, is an introductory period of time where the athlete kind of shakes off the rust, develops a foundational training base, and takes some time to get back into the swing of things. The next phase is devoted to increasing lean body mass, which is thought to be foundational to the goals of the next phase (maximal strength). Finally, the athlete progresses into a 6-week phase devoted to speed strength development, which is the target motor quality for the vertical jump. During this phase the athlete would perform lots of methods designed to decrease the explosive strength deficit: plyometrics, ballistics (jump squats), and o-lifts. The idea is that phases earlier in the cycle set the foundation for the qualities achieved in the latter phases and towards the end of the 20 week period (or however long) the athlete would be at a peak, as he would have successively developed all the

motor qualities allowing him to peak successfully. He'd then take a couple of easy weeks and go thru the entire thing again. Let's take one exercise, the squat, and look at what a typical set/rep scheme would look like thru the duration of the above 20 week cycle:

**Adaptation phase:** 3 sets of 12-15 reps twice per week

**Hypertrophy phase:** 3 sets of 10-12 reps progressing to 4 sets of 6-8 reps twice per week

**Maximal strength phase:** 5 sets of 5 reps progressing to 6 sets of 1 rep twice per week

**Speed strength phase:** 5 sets of 5 rep jump squats at 30% of maximal squat twice per week

See how thru the max strength phase the volume starts off high and decreases while the intensity starts off low and increases? That's a hallmark of pure ***linear periodization***. Another hallmark of linear periodization is you start with general exercises (squat) and progress towards more specific exercises (jump squat and depth jump). This approach CAN work well. In fact, variations of it are my preferred method for lower level athletes. One problem is when this approach is applied too literally the abilities you gain in one phase can be lost in the next. Thus, if applied very strictly most athletes would lose a great deal of their strength/hypertrophy during the speed-strength phase.

Most athletes are probably better off using a form of ***concurrent periodization***. With concurrent periodization, instead of having separate phases for each general motor ability, you work on everything at the same time. In the vertical jump that means you'd be concurrently focusing on strength and speed-strength together – you'd be focusing on strength exercises like squats as well as speed-strength exercises like jump squats and plyos. Take a look at my beginning routine. You perform plyos to start your workout. You do o-lifts or jump squats next. Then you finish up with strength work. You do all that in the same workout. Then you just keep progressing as long as you can. THAT's a GREAT example of a linear concurrent model. It works absolutely BEAUTIFUL for beginners and other low level athletes. MOST athletes should follow this model for as long as it works. Why change something if it ain't broken?

Now, having said that, there's also a problem with pure concurrent periodization: When you always work on EVERYTHING (all motor abilities) at the same time, eventually you reach a point where you need more emphasis on a given quality to continue bringing it up, and you really can't FOCUS on ANY of them because you can only tolerate so much volume. So, how do you get around that? Well, you can use what's called ***conjugate sequencing periodization***. The basic tenet of conjugate sequencing is that as an athlete advances he needs more FOCUS on a given quality to improve that quality. The idea is similar to concurrent periodization. You always incorporate some degree of volume for every given motor ability necessary in your primary athletic endeavor, most of the time. But the basic idea is the general FOCUS shifts from one ability to the next thru different phases (strength to speed strength to reactive ability etc.) The major difference from pure linear periodization is the qualities not being focused on in a given phase are maintained with lighter volume, so each phase builds upon the next and you're able to maintain the previously developed qualities. Thus, at the end the sum is greater than its parts.

Although I've never really heard it described as a "conjugate system" so to speak, in my opinion a great example of it is the sprint training system designed by the late Charlie Francis. He called his model "Vertical integration" but if you take a deep look it fits nearly perfectly. A conjugate system also often has one more aspect unique to it and that is the incorporation of ***concentrated loading***.

The idea behind concentrated loading is discussed in an article I wrote a while back on "Planned Overtraining". The idea is, instead of recovering fully between each workout, you load the athlete up with high volume over a period of weeks, then let him recover over a period of weeks. An athlete trains with very high volumes during a particular phase with the idea that the gains from that training won't be realized until several weeks after as he recovers. This is also known as the ***long term delayed training effect***, or ***delayed transformation***. Many of the popular 5 x 5 approaches to strength training are good examples of concentrated loading. An athlete might train the squat in some form 3 x per week for 4 weeks straight, then only 2 x per week with less overall volume per workout during the next 4 weeks. The idea behind the long term delayed effect is the gains from the high volume phase will be delayed, and the athlete will continue to make gains for several weeks after the conclusion of that phase, as he fully recovers. Thus, if you had a 4 week loading phase, you should see gains from that phase continue to manifest over the next 4 weeks of training.

Let's consider an athlete that uses a concentrated strength phase followed by a speed-strength phase with the idea of reaping the benefits of delayed transformation:

**Strength Phase**

Squat 3 x per week for 5 x 5 for 4 weeks

Low volume speed-strength work (plyos, jump squats etc.)

**Speed-Strength phase**

Squat 1 x per week for 3 x 5 for 3 weeks

Depth jumps and jump squats 3 x 5 three times per week for 3 weeks

Even though his squatting volume drops by a HUGE margin, there's a good chance he'll CONTINUE making strength gains during his speed-strength phase, as he fully recovers from the previous high volume phase. THAT, combined with his fresh emphasis on speed-strength work, allows him to peak his overall abilities. The volume of depth jumps is concentrated in the speed-strength phase, so the long term delayed training effect from them will be realized a couple of weeks after the end of THAT phase. So, you get delayed gains from the strength work and those persist into the delayed gains from the shock work, allowing an athlete at the end of the phase to simultaneously peak in strength, reactivity, and thus vert.

A true "shock" plyometric depth jump cycle as written about by Verkhoshansky could also be considered a form of concentrated loading. Here an athlete might perform 40 high depth jumps 3 x per week for 2 weeks, followed by 2 weeks of relatively light loading (20 depth jumps once every 3-4 days

from lower boxes). Explosiveness should peak towards the end of the lighter 2 week period as the athlete continues to fully recover from the shock cycle.

*Concentrated loading* can really be applied to any endeavor or any skill, and often is, even if the athletes and coaches are unaware of it. Think of a sport like football. The idea is players go into training camp for 4 weeks, run themselves into the ground, and beat the hell out of each other. They also lift and train daily. Towards the end they're so tired and beat up they can barely move. A couple of weeks before the season starts the coaches cut back on all the drills, hitting, running, and the athlete recovers. If they're lucky the athletes bounce back even better than before.

Another great aspect of conjugate sequencing is it allows YOU to focus on qualities unique to YOU. No matter who you are your weakest link is always foundational to everything else. You should prioritize motor qualities which are poorly developed in you as an individual. By motor qualities I'm referring to hypertrophy, maximal strength, and speed-strength. If you really wanted to get practical, you could also include body composition as a quality to be addressed specifically in a periodization cycle, and you could effectively do this in conjunction with a conditioning dedicated phase. You should focus on the motor qualities that you need the most. For example, if you're already as big as a house, you probably have little need to focus on hypertrophy in your training. If you're already strong as an ox, you have little need for maximal strength. But you should prioritize qualities which, when improved, also cause improvement in other qualities. For most people, in most situations, prioritizing maximal strength will go a long way.

Next, I'm gonna show you a sample conjugate cycle using only 3 exercises, the squat, jump squat, and depth jump. Note how we incorporate all 3 exercises at all times during each phase, but the FOCUS and volume of each changes from phase to phase. This example fits an athlete looking to boost vertical jump. We know we need a base of general strength, good RFD, and good reactivity. Thus, an athlete seeking to improve his vert would perform general strength exercises like squats for strength, lighter explosive exercises like light jump squats for explosiveness, and exercises like depth jumps for plyometric ability.

## Phase I - Focus: Strength

Perform each session once per week for 3 weeks

**Session A:**

Depth jump: 2 x 10 from low box (use a box height you can easily rebound off and on, almost like a jump rope)

Jump squat: 2 x 8 @15% of max or 45 lb bar

Squat: 5 x 6 (add weight for first 3 sets, maintain it the last 2)

**Session B:**

Depth jump: 2 x 10 from low box

Jump squat: 2 x 8@15% of max or 45 lb bar

Squat: 8 x 1 @ 90% (add weight the 1st 4-5 sets working up to heavy but solid form single. Reduce the weight by 10% and perform 3-4 more sets of 1)

## Phase II - Focus: General explosiveness (2 weeks)

**Session A:** Depth jump 2 x 10 from medium box

Rhythmic 1/4 Jump squat 6 x 5 with 15, 20, 25, 30, 20, 15% of max.

Squat: 6 x 1@90% of 1rm

**Session B:**

Depth jump 2 x 10 from medium box

Rhythmic 1/4 jump squat 6 x 5 with 15, 20, 25, 30, 20 & 15% of max.

Explosive squat 6 x 2 @ 60% of max squat

## Phase III – Focus: Reactive/plyometric ability (2 weeks)

**Session A:** Depth jump 4 x 5 from high box

Jump Squat 2 x 5 with 15%

Half Squat: 6 x 1-3 @ 85%

**Session B:** Depth jump 4 x 5 from medium box

**Session C:** Depth jump 4 x 5 from high box

Jump squat 2 x 5 with 15%

Squat 3 x 3 with 5 rm

## Phase IV - Peak (2 weeks)

**Session A:** depth jump 2 x 10 low box

Half Squat: 2 x 5 complexed with jump squat:

Jump squat 2 x 5 with 15%

**Session B:** Jumps for max height

See how the volume and emphasis given to each exercise changes in each phase? That's how conjugate periodization works. That's just one example of potentially hundreds one could come up with. Don't take that example too literal, but think of it from a conceptual point of view - If you remember nothing else just remember this:

1. You want to maintain MOST of your reactive ability and explosiveness during your strength or hypertrophy phases, without hindering the acquisition of strength.

2. You want to maintain MOST of your strength during your more explosive or reactive oriented phases, without hindering the development of those qualities.

You can maintain a motor ability with about 1/3 the volume it takes to build it. Throughout a complete cycle of training you should find your general strength exercises (squats) getting heavier and/or faster, your explosive exercises like jump squats and/or cleans and snatches getting higher or heavier, and your specific reactive exercises like depth jumps getting higher and/or smoother. Keep those concepts in mind and you'll be just fine!

The number and length of phases can be unique to you and your needs. The most basic periodization schedule you might set up would have you alternating between 2 phases, strength and speed-strength (explosive). People that don't participate in a sport and just want to train for vertical jump can do this indefinitely. You could focus on strength for 4-6 weeks and explosiveness for 3-4 weeks and simply alternate back and forth between those cycles indefinitely. (*See the JackM split in the programs section or an example of that)*

Generally speaking, your explosive/reactive oriented phases will be a bit shorter than your strength oriented phases – typically 2/3 to half as long. This is because strength is a quality that takes longer to develop, while in large part the explosive oriented work is just how well you *express* your strength, so the gains are neural and kinda like learning a skill. One can also add general physical preparation and conditioning phases. For example, here is what a longer term schedule might look for a basketball player:

Adaptation Phase (2 Weeks)
Hypertrophy Phase (4 Weeks)
Maximal Strength Phase (6 Weeks)
Speed Strength Phase (6 Weeks)
Conditioning (4 weeks)

Here we simply added a phase of focused conditioning to the end where the goal would be to maintain strength and overall explosiveness while bringing conditioning to a peak. There are plenty of more examples of that in the conditioning bonus. The key point with periodization is don't be a slave to it but fit it to YOU and your unique needs. A plan gives you a general roadmap – a framework to work from, but in practice most good coaches and/or athletes will adjust on the fly. So don't feel you always have to be a slave to the schedule.

# *How to Maintain Your Gains Thru a Grueling Season…*

I routinely get questions from athletes in the middle of their season who write me and say something like, “I have basketball practice 2 hours per day. I have 2 games per week. Can you set me up on a routine to put 10 inches on my jump within the next 2 months?” Unfortunately, that’s an impossible goal and it’s just not gonna happen! Beginner athletes may be able to muster up some decent gains during a season, but most people involved in sports like basketball will see their VJ performance decline throughout a season, for reasons which are fully explained in the conditioning guide. Unfortunately, this is largely an inevitable occurrence. You're practicing a lot, you're running a lot, you're under significant stress, and you get worn down physically and emotionally. Your legs simply get worn out. Virtually ALL athletes experience this to a certain extent, even professionals. The ONLY time professional football and basketball players are truly at 100% is on day 1 of training camp. After that it's all downhill. There are some relative beginners and really out of shape athletes who play their way into shape over the course of a season and can experience significant explosive gains in-season, but unless you’re overweight and drop a lot of fat, or very weak and manage to gain a lot of strength, the best MOST people can hope for is to maintain the majority of their gains during a season. Having said that, here is a list of keys that’ll give you the best chance of making gains during a grueling season, or at least maintaining what you have:

## 1. Seek to maintain your size and strength

Here's what typically happens: You grow increasingly fatigued, you lose muscle, you lose strength, and that causes you to lose power. It's not uncommon for players to drop 15-20 pounds over the course of a season without really even trying to. A lot of that weight may be fat, but inevitably some of it will be muscle. It does seem rather contrary that a decrease in bodyweight and body-fat would lead to big time decreases in relative power, but all of the running and conditioning that typically takes place in-season has a lot to do with this. All the conditioning causes extra fatigue and eventually causes you to begin to lose fast twitch muscle. So, even though you're dropping body-fat your relative power decreases. There's also a nervous system aspect as continued daily pounding causes your nervous system to become inhibited and high velocity neural rate coding suffers.

Since it's primarily an increase in fatigue, along with a loss of muscle and strength, that causes us to lose power in-season, we want to focus on the opposite of that. That means we want to try and do the following:

1. Decrease fatigue/increase recovery
2. Maintain muscle
3. Maintain strength
4. Be kind to the nervous system

## Keep Tabs On Your Bodyweight

Unless you're significantly overweight to begin with, one of the most important things you can do to maintain your performance in-season is keep your bodyweight up. You can expect to lose a few pounds regardless of what you do, but if you're going down in weight more than about 4-5%, you're more than likely to run into problems. That means if you started day 1 of practice at 200 lbs you probably shouldn't let your weight drop much under 190. Some people will need to pay particular attention to the quantity and frequency of their meals. You need calories and lots of them. Now isn't the time to be on some esoteric restrictive fat loss diet. If in doubt make a concerted effort to eat more, even if it means you have to eat some things you know aren't that great for you.

Your training should be focused on maintaining strength while limiting fatigue. That means you'll want to keep your training at maintenance levels. One big mistake many people make in-season is they try to continue training at the same volume they did during the offseason. This just contributes to the problem of fatigue and they end up running into the ground even further. The in-season is definitely not the time to focus on qualities like reactive strength and explosive power. You get enough actual jump training stimulus from practices and games - focus on maintaining your foundation. A few hard sets of any given exercise once per week will allow most people to maintain 90% or more of their strength levels. Here is a sample template I recommend:

## Sample In Season Template

Hang power clean or jump squat: 3-4 sets x 3-5 reps

Squat, front squat or bulgarian split squat: 3 x 5

Reverse hyper, glute ham, or romanian deadlift: 3 x 6-10

Horizontal press variation (barbell or dumbell bench press or incline bench press): 3 x 5

Pullup or row variation: 3 x 5-8

That's the basic idea. You might add a few other auxiliary exercises if time allows, but the basic tenet is keep things short and sweet. Perform each exercise with good form and don't go overboard seeking to push the weights. You don't want to injure yourself or burn yourself out - your goal should be to stay close to 10% of your offseason bests over a given rep range. That means if your best set of squats is 300 x 5, you should seek to hit 270 x 5. Anything better is just gravy. Do this (or a similar) workout once every 4-7 days as your schedule allows. You can even do this type of workout on game days, just make sure you have a 4 hour break in between your workout and game.

## 2. Supplement appropriately

Most products that help promote size and strength are also good for maintaining size and strength. In my opinion, creatine is great to use in-season because it increases anaerobic endurance and helps maintain strength and size gains. It's an osmolyte which promotes intracellular hydration, and that increases strength leverage. There have been reports of increased cramping with creatine use, but those haven't been verified thru research. If in doubt just make sure you stay well hydrated. I recommend 5-10 grams of creatine per day. Supplemental protein and carbohydrate powders and drink mixes can also be valuable for nutritional convenience.

I also recommend a good mutivitamin (like Adam from NOW nutritionals) along with at least 1000 mg extra vitamin C per day for immune support. The last thing in the world you want to do is get sick.

## 3. **Pay extra attention to sleep**

If you get a chance to sleep in take it. Get to bed on time and on a consistent schedule. If you have trouble sleeping try taking 300 mg of magnesium a couple of hours before bed. Choose a magnesium form that ends in "ate" like citrate, gluconate, or orotate. If you're on a tight budget you can go to pretty much any grocery store in the world and pick up a 12 gram bottle of liquid magnesium citrate for a couple of bucks. It's sold as a laxative and will work for that purpose if you drink the whole bottle. You don't need the whole bottle for daily magnesium supplementation just take a couple of capfuls once per day.

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Although a bit more on the pricey side, some of the ZMA products are also valuable because they combine magnesium with extra zinc, which will help maintain your testosterone levels.

## 4. **Take advantage of rest days**

If you get a chance to take a day of rest take it. Ideally take at least one day per week and do absolutely nothing. Your nervous system needs time to recover and one whole day of rest each week during the season can be invaluable.

# *The Genetic Relationship Between Metabolism and Power and The Genetics of Rate Coding…*

I will warn you in advance this section is fairly technical and will be over the heads of many, but I decided to include it for those analytical types who really like to get into the physiological processes behind power. One can look at the various neurochemicals involved in metabolism and fat burning and note how they also influence muscular activation. Neurochemicals like adrenaline, nor-adrenaline, thyroid, testosterone, leptin, insulin, and SENSITIVITY to them all play key roles. Given the role of adrenaline and nor-adrenaline in both the ability to strongly activate the muscular system and the ability to burn fat, it would appear there are strong relationships between metabolism and power. That is, the stronger your metabolism and easier you burn fat the more natural relative power you tend to have. Upon cursory observation this holds up quite well. Powerful athletes are typically lean and even those that are rather robust (Charles Barkley) typically lose weight extremely quickly and easily once they dedicate themselves. Science has verified this to an extent - at least one study demonstrated relative body-fat to be the key variable explaining speed differences among adolescents.

One of the key variables responsible for great rate coding appears to be the ability of the nervous system to respond to stimulatory neurochemicals like dopamine, adrenaline, testosterone, and thyroid. Testosterone increases the density of beta-adrenoreceptor (adrenaline receptors) in muscular tissue, and thyroid increases their expression via migration to the cell membrane. Better adrenaline receptor expression (combination of number of receptors and sensitivity of those receptors), allows a given amount of adrenaline to give a greater muscular activation response. Testosterone also has DIRECT effects on CNS discharges in the central nervous system.

According to soviet scientists, individual CNS sensitivity to testosterone could vary as much as 7-fold between individuals. On that note there is plenty of research out there demonstrating testosterone when given to healthy young men really doesn't influence mood and aggressiveness all that much. However, one has to wonder if these studies are flawed due to the type of people recruited in the studies. Would the same experiment carried out on high level athletes or violent criminals induce more of an aggressive response? My hunch is it would simply due to the fact that these people are more than likely to have greater CNS sensitivity to testosterone. It would be an experiment worth seeing anyhow.

To get to the point, what all this means is that someone that has a great internal hormonal milieu with regard to the above mentioned neurochemicals is more likely to be fast, explosive, and have a great metabolism. That partly explains why the best sprinters and jumpers in the world are typically very lean - and typically don't have to work hard to achieve or maintain that leanness. The exception to this appears to be naturally thin SLOW TWITCH dominant athletes - also known as hardgainers. They are lean and have fast metabolisms but there's appears to be an adrenal dominant metabolism with often low testosterone. Their fiber type is usually slow twitch oriented. A sprinter and typical anglo marathon runner are both lean, but that's pretty much where the similarities end. How their metabolisms are

expressed are worlds apart. For more info. on that check out my "Skinny-Fat Ectomorph" articles on my website www.higher-faster-sports.com

## *Why Powerful Athletes Don't Always Live Long....*

One semi-interesting observation I've noticed is the inverse relationship between relative power and longevity. Great speed/power athletes often die fairly young - and often from cardiac related ailments. Approximately 10 years ago I began noticing there was some type of relationship between metabolism and the ability to express power. That is, for a given level of body-fat powerful people expressed a *stronger* metabolism. Blood pressure can be a relatively simple and crude measure of metabolism because things that increase blood pressure can also elevate the metabolism. It's a rather crude marker but a useful one.

Typically, as body-fat increases blood pressure and metabolism increase, and as body-fat decreases blood pressure and metabolism decrease. What I had noticed over a period of years is many good athletes tended to run a little higher blood pressure than you might expect given their body-fat levels. Not all, but enough for me to take notice. No one is surprised when a 350 lb obese individual runs a high blood pressure, but when a healthy 150 lb 20 year old does you take notice. But anyhow, I noticed many of the best power/speed athletes I was around tended to run towards the high end with regards to blood pressure, even though they were often extremely lean and seemingly extremely healthy.

My observations were semi-verified fairly recently in a study on NFL players: As a general rule NFL players (including skill players) run slightly high normal blood pressure. These are highly trained highly conditioned athletes at low body-fat. So what's going on? Well, it may back to strength of metabolism - these athletes likely have enhanced metabolic expression, which includes increased dopamine, adrenaline and various catecholamine expression (the same things which stimulate the nervous system). Thus, even at low body-fat levels their metabolisms run strong and that’s evident in a multitude of ways – including the ease they maintain a lean state and likely their ability to express power on the field.

Most diet related health problems (heart disease, diabetes, etc.) could be considered a result of too much metabolic expression. As body-fat increases so does metabolic stimulation - metabolic signaling in the hypothalamus increases, which increases dopamine, cortisol, epinephrine, thyroid etc. The rate of calories we burn may not increase *that* much, but these could all be considered things the body does to "fight back" against further weight gain – they tend to reduce appetite making fat loss relatively easily.

Unfortunately, in modern society high calorie food usually beats a good metabolism and sedentary lifestyle. MOST people continue eating excessively and EVENTUALLY as a person continues to gain weight things like insulin and leptin resistance set in and a cascade of health problems often results: high blood pressure, elevated blood fats, cholesterol perturbations, polycystic ovarian syndrome (in females) - all things associated with common cardiac disease. Naturally explosive athletes will often

inherently have a higher set-point for metabolic signaling. So, for a given level of body-fat they will have an easier time staying lean, because their bodies tend to resist fat gain and don't resist fat loss. But, they ALSO develop health problems at lower thresholds – particularly lower thresholds of body-fat. We've all known very obese people who weigh 400 lbs with perfectly normal health markers (blood pressure, triglycerides, etc.) An individual like this will naturally have very low metabolic signaling, so even though they gain a lot of weight, which further increases the metabolic signaling, they're still normal. People like this have a VERY STRONG propensity to gain weight from an early age. Their metabolisms don't really "fight back" when they gain weight, so they keep piling it on. Contrast this to the multitude of fairly lean black men who suffer from high blood pressure or stroke. They have an easier time getting and staying lean, but also a greater susceptibility towards health problems with weight gain.

Keep in mind these aren't absolute just general observations and trends. It also turns out good athletes may have something else going on:

***The ACE Gene***

Genetic variants in the angiotensin/ACE gene seem to have a lot to do with individual differences in body-fat distribution patterns, insulin sensitivity, muscle fiber type, and health. ACE is short for "angiotensin converting enzyme." It's a circulating enzyme that participates in the body's renin-angiotensin system (RAS). ACE mainly seems to be involved in vasoconstriction and extracellular volume (i.e. that of the blood plasma, lymph and interstitial fluid). Variants in the ACE gene cause some people to produce more or less ACE than others and that leads to more or less restriction of blood vessels. ACE catalyses the conversion of angiotensin I to angiotensin II, a potent vasoconstrictor. ACE also breaks down bradykinin, a potent vasodilator, There are 3 types of ACE gene. These are:

A: I/I
B: I/D
C: D/D

The D/D types have the MOST ACE and the I/I types have the least. The important thing is research has demonstrated a link between the high ACE variant, strength, and fast twitch fiber. They tend to respond to weight training with great strength increases and have greater fast twitch fiber. On the other hand, it also tends to make those same people more susceptible to high blood pressure, depression, visceral fat accumulation, and heart disease. Having the lower ACE genotype makes some people have a propensity to be scrawny and better at endurance activities, with soft looking muscles and subcutaneous fat storage patterns. The low ace gene is also associated with longevity and lower risk of cardiac related ailments. One study demonstrated African Americans are 50% more likely to have one of the "D" genes than whites. This might not explain much, but maybe it does. The combination of metabolic signaling and genetic factors such as ACE likely contributes to the large variance we see in neural expressions like rate coding and physical ability. For more on things related to this topic check out the section on genetics in the *How Much Can Vert Be Improved* section of this book.

# *How Often Should You Train and How Much?...*

How much jump training and lifting is enough per session or per week? How much is too much? There is quite a bit of variance in the recommendations you'll find in this area. I've seen routines calling for enough volume to kill an elephant, and I've also seen routines calling for 40 total minutes of training per week. To understand my volume recommendations it helps to understand how improvements in motor qualities like strength and explosiveness occur. The 2 primary ways are:

A: **Gains in inter-muscular coordination**: This is otherwise known as simple *coordination* and is highly relevant to improvements in movement efficiency. With gains in inter-muscular coordination, the various muscle groups involved in the jump cycle become more proficient at carrying out the movement. Think of a youngster first learning to walk or run. Initially their arms and legs are flailing all over the place and they have a hard time coordinating their movements. With practice they become more proficient until one day they just *get it*.

If the coordination is lacking someone will often increase their vert just from the improvements in coordination they gain as they practice jumping. But there's only so much you can improve from increased coordination. This is particularly true of gross primal movement patterns like jumping and sprinting, which really don't require a whole lot of technique in the first place. Once a person develops a certain level of coordination in a movement they don't need to focus near as much on it - kinda like riding a bike. Once you learn how to do it you don't forget. I haven't been on a bike in over 6 years but I'm sure I could get on one tomorrow and be just fine. Muscle memory is very real.

B: **Gains in intra-muscular coordination:** The 2nd way gains in strength and power can improve is through gains in *intra-muscular coordination*. This is what I often refer to as ***horsepower***. With gains in intra-muscular coordination each muscle group involved in the movement becomes more proficient at generating force in the movement. Activities like weight training, jump squats, and Olympic lifts work by boosting this aspect.

So, like I've said many times, you can jump higher by improving your ability to carry out and coordinate a movement pattern, or you can get more powerful jumps by putting more force behind that movement pattern. You get the best of both worlds when you put inter and intra-muscular coordination together. It should be noted that gains in coordination generally always occur prior to gains in horsepower. One first learns to carry out the movement pattern effectively, then learns to put more force behind the same movement pattern.

## Frequency

As far as frequency goes, gains in *inter*-muscular coordination respond better to increased frequency. This is why when a baby is first learning to walk, he or she doesn't get up and try to do it just once every few days. No - he practices constantly. Learning to jump is the same way, and so is any other type of movement or skill. A frequency of 3-7 times per week is optimal for initial gains in coordination .

The more quality exposures you get when learning a movement pattern, the faster you pick up the technique required to efficiently carry out the movement pattern.

In contrast, gains in horsepower (intra-muscular coordination) often respond well to lower frequency with much more intensity and volume per session. In this respect gains in intramuscular coordination for a jumper are much like gains in strength for an advanced lifter. Think about that. A very strong powerlifter will often only train a given lift once every 7 to 10 days. His technique for the lift is well developed and he doesn't need endless repetition practicing the various lifts. Rather, his time is spent stimulating and strengthening the muscles involved in the powerlift. After a hardcore workout it'll often take his muscles and nervous system a week or more to recover. Because he's already spent years perfecting his technique, he need not worry about losing any technical prowess not hitting his lifts every other day. He can simply focus on getting stronger in the various muscle groups overall, then apply that increased strength to his powerlifts.

The same sorta thing can often be observed with short sprint training like the 40-yd dash. You can take an active group of young athletes off the track and throw them in the weight room for 3 months and get them really strong. Providing they maintain their mobility and leanness, you can take them back out on the track and within 3 sessions most of them will be setting PRs in the sprints. They didn't lose much technique from not sprinting because it's something they've probably been doing since they were kids. However, they did gain a lot of strength, which they were then able to transfer to the sprints.

You see the same thing in jumping related events, though not as often, because people tend to jump year around, where in many places it's too cold for athletes to get outside and sprint year around. But this is one good reason why one need not spend endless hours all year around performing high volume plyometrics, providing they've reached a baseline level of proficiency in their ability to jump efficiently. So, basically, you use frequency to learn. You use intensity to enhance what's learned.

## Maintaining Movement Proficiency vs Improving Movement Proficiency

It takes a lot less volume to maintain a skill, movement pattern, or strength quality then it does to improve a quality. In general, it only takes 1/3 the volume to maintain a given quality. Thus, you might become more coordinated by performing jump related training 3 times per week, but once you've made those improvements you can maintain the majority of them jumping one time per week. The same is true of most other qualities as well. You can maintain strength with 1/3 the volume it takes to build it.

When training to improve maximal power, as a general rule of thumb you should terminate a jump session prior to, or as soon as, your performance starts do decline during a workout. The same goes for any other movement skill or explosive movement you're trying to improve. Wanna improve your agility? Your sprint speed? Your footwork? Your martial arts kicks? Your gymnastics ability? Then treat all those just the same way. Perform a lot of quality reps with good recovery and stop a session as soon as fatigue begins to interfere with performance. It will work for the acquisition and improvement

of any movement or skill you're trying to improve. It's the simplest way to monitor volume you'll ever hear and is also one of the most effective. The reason it's so effective is because pure movement related gains (inter-muscular coordination) are mostly neurological in nature and making neural improvements requires fresh exposures. That's why when you're training for increased coordination, movement proficiency, and skill, you can and should train more often. A frequency of 3-5 days per week or even every day works very well for coordination and movement acquisition. The more fresh exposures you give yourself to a given skill or movement, the more proficient you're gonna get at carrying out that movement. As long as fatigue is not accumulating on a day-to-day basis, and as long as your performance is not deteriorating on a day-to-day basis, you can train as often as possible.

## Frequency When Training For Horsepower

When I refer to *horsepower,* keep in mind I'm referring to things like weight training, explosive lifts, and high intensity forms of plyometric training – basically anything other than very basic low intensity movement efficiency drills like jump rope. In these higher intensity tasks you're trying to improve the amount of "oomph" that you put behind your movements. To give you an example, squats don't improve the coordination of your jump, they improve the amount of force you put into the ground. Prescribing frequency when training for horsepower is a little more difficult. Why is this? Well, the main reason is because most of the things associated with boosting horsepower tend to be highly intense activities that tend to causes a significant amount of whole body central nervous system fatigue, or muscular micro-trauma. Either of these will require recovery time. How do you tell if something is causing whole body fatigue and/or micro-trauma? Very simple: If you perform a task today and can't come back the very next day and repeat the task at the same level, your body is not recovered.

If I went in the gym yesterday and did a fairly heavy and volumous squat session I probably couldn't come back and repeat that exact same session tomorrow with the same weights. In fact, I might have to wait 4-5 days before I could repeat that same squat session. If I were to do some depth jumps tomorrow those depth jumps would induce enough fatigue that I probably wouldn't be able to come back the next day and jump as well either. I wouldn't have to rest 4-5 days like I would with the squats, but I probably would have to rest 48 hours or so. The more micro-trauma (muscular damage) that you induce in any given workout the longer it's going to take to recover. This is why weight training tends to require the most recovery time. However, nervous system fatigue can be induced even without microtrauma.

The depth jumps may not induce much micro-trauma, yet still drain the system enough that they require some recovery time. This is due to the neural arousal that takes place and the effect that has on the neuromuscular junctions that join the muscular and nervous systems together. When the neuromuscular junctions (NMJs) become fatigued a given amount of "charge" from the nervous system will fail to illicit the same degree of muscular activation.

One other very important point: Things like basic sprinting and jumping, when performed at a high level, can also require significant recovery time. An elite level sprinter or jumper may only be able to sprint or jump maximally 2-3 days per week. If they try to go at full speed every day they'll most likely

find their times from one day to the next would gradually go down as they build up fatigue. But allow a day of rest between sprints or jumps and their fine. However, a grade schooler can jump or sprint every day with no day to day deterioration in performance. Why is this? Since the elite athletes are much more advanced shouldn't they be able to go hard more often? You would think so. So what's going on?

Well, the body can double, triple, or quadruple the capacity to generate stress (increase performance), yet the capacity to recover from that stress does not improve nearly as much from baseline. The elite level athlete is putting out a TON of force when he goes all out and, in comparison to the lower level athlete, puts a lot more stress on his body and nervous system. Have you ever wondered why pro athletes or sprinters tear and strain hamstrings left and right yet you never see a single person in a class full of grade schoolers have any such problems? They're simply not able to create enough stress to challenge their bodies. It's like the difference between a thoroughbred and camel or a funny car and a Honda civic. The thoroughbred and funny car are so strong and powerful they'll blow out if you try to run them full bore all the time. The camel and Honda Civic don't operate such a high intensity and thus can be ridden further and more often.

The point to take home is this: If you perform a task today and can't come back the very next day and repeat the exact same task you need some recovery time. The exception is if you're intentionally trying to ramp up volume and fatigue via either concentrated loading or high frequency training. Activities such as heavy weight training may require 2-7 days for full recovery while activities such as intense plyometric and speed will typically require 48 hours recovery. It's generally best to allow 48-hours rest between any highly intense activities for a given muscle group. So, if I did a heavy squat session today I wouldn't wanna come back tomorrow and perform depth jumps or sprints. I'd wanna give my legs and my nervous system at least 48 hours rest. For that reason, for those athletes who have advanced past the coordination stage, it's often best to put any activities that require significant recovery time all on the same training day. So, instead of performing weight training one day and high intensity plyometrics the next day, you'd do them all on the same day with 48 hours rest in between workouts. On the days in between your higher intensity workouts you could either rest or engage in lower intensity activities.

Activities that could generally be considered ***high-Intensity*** activities would be the following:

1. Strength work (anything above 80% of 1rm for lower body and "whole body"movements such as squats, deadlifts, cleans etc.)

2. Maximum effort bodybuilding work (8-12 reps to failure)

3. Maximum effort speed work with full recovery between reps

4. Maximum effort shock plyometric work (depth jumps)

5. Maximum effort agility and deceleration work will full recovery between reps

6. Maximum effort conditioning work (ie. Timed max effort intervals)

7. MMA or boxing sparring and heavy bag

9. Any activity performed with heightened and competitive emotional intensity (competitions)

10. Any activity performed under the influence of artificial stimulants (ephedrine, various energizing supplements)

11. Any activity involving PR type performances (moreso for advanced athletes)

Activities that could generally be considered ***low-Intensity*** activities would be the following:

1. Aerobic work

2. Sub-maximal conditioning work

3. Dynamic warm-ups and form running drills

4. Sub-maximal bodybuilding or upper body isolation bodybuilding work

5. Sub-maximal speed work (runs less than 80% top speed)

6. Easy plyometric work (basic low level uni-lateral and bi-lateral hops etc.)

7. Footwork drills (agility ladders and dot drills)

8. Jump rope

9. Martial arts kata, mitt work, or shadowboxing

These activities that don't tend to induce a ton of fatigue or strongly engage the nervous system – at least when done in moderation – They can typically be performed on a daily basis if desired.

# *High Frequency Training…*

The more often we train the better results we get right? It sounds good but that belief has been the downfall of many a VJ trainee. I get questions all the time from people asking me, "Kelly if I do these workouts twice as often won't I get results twice as fast? Don't we wish it was that simple. A better phrase is, "The more often we "can" train OPTIMALLY the better." Most people can't train as often as they want because they don't recover fast enough to benefit from frequent workouts. People are typically limited by their ability to recover from training, not by their ability to put the hours in. If that weren't the case all anybody would need to do to become great is train 24/7.

Having said that, if all things are equal the more often we *can* train in an optimal physiological state, the better gains we *can* potentially make. Think about it: If you train once every 5 days over the course of a year that's 70 workouts. If you train 5 days a week that's 260 workouts over the course of a year. Lots more stimulation over the course of a year and lots more *potential* gains. For that reason you might look into dedicated high frequency training. The basic tenet behind high frequency training is, "Train as often as possible while staying as fresh as possible." It IS possible to train every day (or almost every day), as long as you limit the volume and stay fresh.

The problem many people run into is they try to perform workouts designed for lower frequency, while doing volume that requires lower frequency. You can train long or you can train frequently, but you can't do both at the same time. Anytime you train all the frequently you have to make sure your workouts are short and don't create excess fatigue, or you'll burn yourself out. Another PROBLEM with many VJ trainees is they already have inherent problems with recovery for 2 major reasons:

1. They're involved in sports which inherently contribute lots of fatigue with lots of running.

2. A disproportionate number of them are long limbed naturally thin ectomorphs, and many ectomorphs don't tolerate stress well as is. They have a hard time recovering from life itself, much less life itself in addition to hours and hours of training per week.

Having said that, IF you recover well and don't have a whole lot on your plate as far as extraneous activity (practices, games, running, stress levels, lack of sleep etc.) it is possible to train more often, you just have to take precautions to keep yourself fresh.

Here are some guidelines to consider:

1. **Keep your workouts short:** A high frequency workout shouldn't take long to complete - 45 minutes tops.

2. **Keep the exercise number low:** You should have no more than 1 or 2 exercises for each component (speed-strength, strength-speed, strength) and no more than 5 total exercises per workout. You might have one plyometric exercise, one strength-speed exercise like an olympic lift, and one strength training exercise like a squat. Throw in an auxiliary exercise and that's it. A good maxim to follow is the 3 to 5

rule: Three to 5 sets of 3 to 5 reps with 3 to 5 exercises 3 to 5 days per week. For exercises like plyos and speed-strength movements like jump squats, you can increase the repetitions above that, but follow the basic 3 to 5 exercise rule.

3. **Keep your sets and reps low:** The strength component of the workout on a heavy exercise like a squat should typically have no more than 10 total reps per workout if you're going to hit it more than 3 days per week. That's PER WORKOUT. Typically your reps would be in the 1-3 rep range. This limits volume accumulation so you can hit your nervous system without trashing your muscles.

4. **Keep your adrenaline low:** If you train very frequently you can't also train very intensively. *Intensiveness* refers to how much psychological effort you put into your workout. If you're the type that likes to get all jacked up and yell a lot when you're training, you'll have to put that on hold. High intensiveness workouts are inherently quite draining and can't be repeated all that frequently. If you're going to train frequently you CAN train hard and put a lot into your workouts, you just have to learn where to draw the line as far as workout motivation is concerned. If you get too jacked up every time you train you won't be able to train as often, because that extra arousal prolongs nervous system recovery time. If in doubt lay off stimulants like caffeine, as they can artificially increase intensiveness.

5. **Use back-cycles:** Don't try to set PRs each and every workout. Use buffers and back-cycles. A back-cycle is where you intentionally reduce the weight for a workout or 2. Here is an example of what your top weights on an exercise like a squat might look like:

Workout 1: 100 x 1
Workout 2: 105 x 1
Workout 3: 110 x 1
**Workout 4: 105 x 1**
Workout 5: 110 x 1
Workout 6: 115 x 1
**Workout 7: 110 x 1**

See how you took 3 steps forward and 1 step back? That's a backcycle. You can use 1 rep, 2 reps, 3 reps or whatever, but keep that basic idea in mind.

6. **Get your recovery in:** If you have problems sleeping, high frequency training might not be the best choice for you. If you have problems eating enough to keep weight on, it might not be for you. If you're gonna train often you need to make sure your sleep and calorie intake are in order. You don't need to be on any cutting edge esoteric diet, but if you're the type of guy that looks at a workout and loses 5 lbs you'll probably have a hard time making gains training frequently. Also, if you have a full plate of conditioning/running on your plate, via practices and such, high frequency training might not be a good option for you. There are examples of high frequency workouts in the workout section of this manual but here's a quick example of a very general one:

Depth jump for height from 18 inch box
Hang Power clean: work up to max triple

Squat variation: work up to heavy double (no more than 4 sets total)
Calf raise: 2 x 15-20

You'd repeat that workout up to 5x per week. Here is an example of a periodized mini-cycle:

**Week 1:** Work up to heavy triple on power clean and squat, use an 18 inch box for 10 reps of depth jumps

**Week 2:** Work up to heavy double on power clean and squat, use a 24 inch box for 10 reps of Depth jumps

**Week 3:** Work up to heavy singles on power clean and squat, use a 30 inch box for 10 reps of depth jumps

**Week 4:** Cut your frequency in half, perform singles on power clean and squat, but with 10% less weight than previous week. Eliminate depth jumps and replace with running jumps for height.

**Week 5:** start over with week 1

There are many options available, for example, you could follow the exact format but do your strength work one day and jumping/plyometric work the next. High frequency training can make you very strong very quickly, and very explosive very quickly, but there are also a couple of additional things to consider:

**A: High frequency training makes you really good at the particular exercises you're doing and a big portion of your gains can be attributed to that:** When you hit an exercise really frequently you really learn to push your inter and intra-muscular coordination to the max and truly "figure out" and *grease the groove* of a particular exercise. Thus, a substantial percentage of the gains you make can be due to increased coordination and specialization on the particular exercise. That's not a negative thing. In fact, it's one reason you WANT to jump fairly frequently if you're a beginner - the more often you jump the more coordinated you get at it. But, keep in mind if you put 50 pounds on your squat on a high frequency protocol a substantial percentage of that increase may be because you simply got much better at squatting, and not necessarily because your muscles are really that much stronger.

When we strengthen our muscles for the purpose of jumping we really shouldn't care about whether we're doing better on a particular exercise as long as we get stronger generally on *most* exercises. Baseline *general* strength is what matters for athletic applications. It's the "meat" strength that can be applied elsewhere versus the "grooved" strength specific to the lift. As an example, say an athlete eliminates squats from his routine and focuses on bulgarian split squats and front squats for 2 months. After 2 months his split squats are up, his front squats are up, AND his VJ is up, but his squat is down. Does that mean he got weaker? No, it just means he lost coordination on the squat. Should he care? I wouldn't. Strength exercises are just tools and barometers we use to gain and monitor strength improvements. It is possible to get stronger overall but weaker on a particular exercise. It's also possible

to get stronger on a particular exercise but not much stronger overall. This isn't something to be overly concerned about just something to keep in mind.

**B: High frequency training can amplify movement pathologies:** If your technique on a particular exercise is poor, or if you have movement and mobility imbalances that show up on a particular exercise, hitting that exercise at a frequent rate can make those pathologies worse. So, make sure your form is GOOD if you're going to hit an exercise at high frequency. One of the benefits of high frequency training is it gives you more opportunities to work on your technique.

That about sums it up. Check out the programs section for some more high frequency templates!

# *Mobility, Flexibility, and Movement Assessments…*

I've already talked about the importance of proper movement efficiency when it comes to increasing performance. You might be able to make a better athlete by getting stronger without addressing factors such as muscle activation patterns and range of motion, but if an athlete has issues, the stronger he becomes, the greater the injury potential. Eventually injury WILL result. The answer is balance. Thru what you learn in this section you essentially want to accomplish 3 things:

**Establish optimal femur control**

You want the muscles that sit up higher on your body, the glutes and hip flexors, to be in the "drivers seat." They should control your thigh bone and lower body muscles, rather than the muscles residing lower on your legs. It's akin to a baseball player swinging a bat. A good hitter gets his bat power primarily from his hips, torso, and lower body. The shoulders, arms, and wrists are just "hooks" that help transfer this power and add to it a bit. When you jump you will inherently get a lot of power from your knees and ankles, but you want your hips (butt muscles) to be in the drivers seat. Not only does this help you perform better, but many cases of knee pain, IT band pain, plantar fascitis, and other common issues are a direct result of this imbalance and strengthening the glutes, hip flexors, and abdominals, while increasing mobility in the quads, hamstrings, and ankles will help keep you moving efficiently and help you avoid common overuse injuries.

**Optimize the way you drive off your feet**

Many athletes move in what is called a "rearfoot" dominant posture. They sit back on their heels and don't naturally drive off the balls of their feet with efficiency, or what's called "forefoot dominance". In the world of professional scouting one of the first things scouts look at is an athlete's feet. Regardless of what sport the athlete plays, or whether he weighs 100 or 350 pounds, the phrase, "He moves like a ballerina" is generally a positive thing. On the other hand the phrase, "He moves like his feet are stuck in cement", is negative. Although the logical thing to do to boost this capacity would seemingly involve lots of footwork drills, in reality this trait is largely influenced by the lower body posture and the primarily thing controlling the lower body posture is how "in control" of your lower body your glutes and hip flexors are. One thing you should notice after a week or so performing mobility and proper activation work is more lightness on your feet.

**Move Freely and Easily**

Excelling in sports is largely about having the mobility to get into the proper positions needed in your sport and having the strength to maintain or move from those positions. Before you can have the strength, you have to have the mobility to get in the proper positions, so you can activate the correct muscles. One of the best ways to fix a crappy movement is to do your basic barbell movements correctly with good form. This can be accomplished by concentrating really hard and getting feedback -

ideally from a coach. Yet many people have tight ankles, quadriceps and hip flexors which make it difficult for them to perform certain movements correctly, which we'll identify.

## The Cure

What follows are several mobility and strength tests for various movements and muscles I've deemed important for jump training. Honestly, there are quite possibly about as many assessments as there are stars in the sky. My objective isn't to put you thru every test under the sun and try to identify every single little thing that might be wrong with you, I just want to hit on a few of the major problems I see in a lot of trainees, problems that may impact your ability to properly benefit from VJ training. Go thru the videos and see how you do. If you have problems with any of them you'll want to address them thru the prescribed corrections until you can do the assessment correctly. How you do on these tests determines how much additional mobility/flexibility/activation work you need to do in addition to the regular warm-ups, which I'll detail next. If you pass all of these you can pretty much get by with just a regular warm-up and moderate amounts of additional mobility/flexibility work.

To view the assessments go to: www.higher-faster-sports.com/assessmentsforvert.html
Your password is vertical.

I've actually had some athletes gain as much as 4 inches on their VJ doing nothing but mobility and activation exercises in just a couple of weeks. Your results will probably not be THAT dramatic, but don't be surprised if you gain an inch or 2.

# *More on Mobility and Flexibility…*

Many people confuse mobility and flexibility. ***Mobility*** is defined as range of motion of a joint. ***Flexibility*** is defined as the length of a given muscle. Typically, groups of individual muscles are responsible for the range of motion of a particular joint, and thus have a large impact on mobility, BUT it is possible to have joint or neurological restrictions that prevent range of motion of a joint, even if the muscles are flexible. It's also possible to have excellent joint mobility, even if individual muscles acting on a joint are tight.

***Flexibility*** is the ability to take an individual muscle through a range of motion with external assistance, also known as passive flexibility. For example, if you lie on your back, grab your knee, and stretch your hamstring, it is a test of hamstring flexibility. The typical "hold for 20-30 seconds" type of stretching is passive flexibility work.

***Mobility*** is similar to flexibility in that it involves range of motion at a joint, however it involves active movement. For example, performing an overhead squat. You have to take the joints and muscles that impact the overhead squat thru their range of motion actively. Most ballistic type stretching, such as rhythmically touching your toes 10 times, could also be considered mobility work, as could any type of

typical dynamic warm-up. In a dynamic stretch you're simply carrying out some of the movements that occur in sport over a greater range of motion. You're integrating mobility into the actual movements that occur when you move.

I like to think of mobility and flexibility training as lubrication for the moving parts of an athlete - similar to the oil in the motor of a car, or the grease that lubricates various working parts. Ever seen what happens when you take a strong so called muscle-bound bodybuilder type and throw him out in a flag football game? Strained hip flexors and a host of other injuries are the norm. What most people call a muscle-bound state isn't caused from excessive muscle, it's caused by lack of mobility. If you run a car without oil, or run a wheel without grease, you get all sorts of clanking, squeaking, and other annoying problems. It's the same way for an athlete without mobility. Lack of mobility causes movement to be restricted, and you get all sorts of squeaks, clanks and other assorted problems, often involving injury. Mobility training is like grease or oil that helps parts run smoother.

Mobility and flexibility will typically complement each other and if you have tightness you SHOULD do both, but many people make mistakes as far as how often and when they do one and not the other. When you passively stretch your muscles you are relaxing them and your nervous system. For this reason it's not always a good idea to do passive flexibility work BEFORE a workout. You can relax a muscle too much and can this can negatively impact strength and power. The only time you should engage in passive stretching before a workout is when you have an overly tight muscle restricting you from achieving proper movement or muscular activation. For example, if your hip flexors are tight you'll have trouble properly engaging your glutes when you squat. If your calves are tight you might have problems with your heels raising up when you squat. Performing a static stretch on these muscles before your workout can help you achieve proper activation and range of motion in your workout.

After a workout is the time when you should really focus on passive flexibility work. This is the time when you want your muscles to relax. Your muscles will also be warm and warm muscles more easily respond to stretching at this time. Full body mobility type movements are great for warm-ups. They enhance neural drive and warm your muscles up efficiently, and actually enhance strength and power. Ideally you'd perform mobility work prior to doing any type of jump or plyometric work.

To summarize: Do full body, rhythmic, mobility type movements before exercise, and do passive stretching type movements after exercise. What about if you really want to make gains in your mobility and flexibility? Good question!

I've often said it's not how ***much*** flexibility and mobility work you do DURING a particular workout, but how ***often*** you do it throughout the day. Most mobility & flexibility impairments are caused by the nervous system trying to keep stress off of tissues out of fear of injury. For example, say you have problems with your heels coming off the ground when you squat. This would typically be assigned as a problem with ankle and hamstring mobility. But the problem isn't really the muscles, it's your nervous system inhibiting the muscles, because for some reason your body is "afraid" you're going to get injured.

So, what you need to do is allow your nervous system to "reset" itself, and the way to do that is to teach it the new skill. For any neural related task *frequency of exposure* is king. Anytime you're learning a new skill you'll progress quicker on it the more often you do it. Think of flexibility and mobility as skills - the more often you work on them the better off you'll be. So, IF you have issues with tightness, you'll make the most rapid gains doing a combination of mobility/flexibility work very often throughout the day. The worse the issue is, the more often you'll want to do it.

I have some people do particular movements and stretches every hour of the day, but for progressive improvements you'd pick a mobility and flexibility movement and hit them once in the morning and again at night for one set each. Say you're having problems with tight hip flexors: You might get up in the morning and choose a mobility exercise like an alternating reverse lunge for 10 reps per side, followed by a static hip flexor hold for 20 seconds per side. Do this again during your workout and again before bed. With the high frequency approach you don't spend gobs of time stretching, but you do get lots of exposure. Using myself as an example I have problems with mobility in my lats - they're tighter than piano strings. So I start every morning off with a set of 10 overhead squats followed by a static lat stretch for 20 seconds per side. Mobility + flexibility.

## *More On Warming Up For a Great Workout….*

Now I'd like to touch on how to warm-up for a great workout: How you warm-up really depends on what type of activity you plan to do. If you're planning to lift weights all you really need to do is take your first exercise of the day, start with an empty bar, do a few reps, and continue adding weight. Basically all you're telling your body is, "This is the movement I want you to do and the weight is going to get heavier." Let's say you plan to do a squat workout and you plan to build up to a max set of 225 lbs for 5 reps. Your warm-up might look like this:

45 x 5, 95 x 5, 135 x 5, 185 x 3, 205 x 3, 225 x 5

A general rule of thumb is one warm-up set for each 45 pound plate on the bar. So, if your first work set is 135 lbs (2 plates) you'd wanna do at least 2 warm-up sets. If your first work set is with 315 pound (6 plates) you'd wanna take around 6 warm-up sets. That's it! Now, for a weight training workout you can add in some dynamic stretches if you want, but it's not absolutely necessary.

If you're planning to warm-up for jumps or plyometrics you want to get a little more blood flowing and get a good sweat going. You don't necessarily have to turn your warm-up into a physical therapy session, but a dynamic mobility warm-up helps get the blood flowing while lengthening tight muscles, which is beneficial to explosive movements. Having said that, in my experience, 15-20 minutes of recreational basketball or something similar will typically produce a superior warm-up to any cutting edge dynamic warm-up you can come up with. That's not saying I don't recommend dynamic warm-ups – ideally you'd do both a dynamic warm-up and a few minutes of rhythmic activity, but the reality is left to their own devices most people hate specific warm-ups. Regardless, you want to get your sweat going at least a little bit and lengthen the appropriate muscles. What I typically recommend is a couple of

minutes on a bike or a few minutes jump rope followed by a simple dynamic warm-up. Or a series of basic track drills - high knees, skips, buildup sprints followed by a short dynamic warm-up. Here is an example of a quick dynamic warm-up:

Skip: 25 yds x 3
High knee run: 25 yds x 3
Straight leg kick moving forward: x 10 reps per leg
Side lunge: x 10 reps/side
Deep lunge moving forward: x 10 reps per leg
Sumo squat: x 10 reps
Pullback butt kick walking forward: x 10 reps
Walking forward elbow to foot: x 10 reps/side
Go through each movement for 1-2 sets per move.

(See videos at www.higher-faster-sports.com/assessmentsforvert.html (password vertical)

If you want to get into a bit more of an extensive warm-up, an example of a longer dynamic warm-up on the assessments page. The purpose of the warm-up is to make gains in active range of motion, provide a proper neural stimulus, and obviously get your muscles warmed up and ready to go for the more intense work to come.

If you want to take things a bit further you can add on a few loaded/resisted jumps to the end of the aforementioned warm-up to stimulate your muscles a bit. One study found that performing a set of 5 jump squats with 10% of bodyweight immediately following a dynamic warm-up led to temporary improvements in jumping height.

## *Static stretches*

You saw several static stretches in the mobility/flexibility assessments, but here are a few more I recommend for just about everyone, every day. For best results perform each stretch twice a day, one after your workout and another one either first thing in the morning or before bed:

### Kneeling rectus femoris/quad 2 x 20 second hold

### Calf 2 x 20 second hold

**Hamstring 2 x 20 second hold**

**Groin  2 x 20  second hold**

**Hip Flexor 2 x 20 second hold**

## *Overvaluing Warm-ups*

I know warm-ups and stretches aren't typically something most young athletes really look forward to, and I can empathize.  If in doubt just try to make sure you do a little something to get your blood flowing, but don't read into it or overanalyze it too much. The vertical jump doesn't require much in the way of flexibility to perform, and I've known hundreds of successful athletes and coaches and I don't think I've ever seen the exact same warm-up twice.  Here the past several years there's been a gradual trend towards coaches and researchers really over-thinking warm-ups, stretching, muscle activation, and things related to it.  This is my opinion, but a cursory search of the phrase "vertical jump" on the NSCA website brings up hundreds of fairly recent research papers, a disproportionate number of them having to do with warm-ups and stretching.

As stated, I'd be surprised if your best warm-ups don't occur by accident and involve you actually playing games or doing something besides going thru pre-rehearsed motions.  *Optimally*, you want to get your muscles and joints warm before throwing significant stress on them, but don't let your warm-ups become a "chore" that you hate doing.  Although it may not be safest from an injury perspective, if in doubt remember your body IS designed to respond and react to physical situations without any preparation beforehand.  A caveman getting attached by a saber tooth tiger surely wouldn't have time to thru a 15 step dynamic warm-up beforehand, and there are still plenty of athletes that exemplify this.  They show up, do a couple of toe touches and arm swings and BAM! – They're ready to go.  Not ideal in the eyes of many, but hey, if it works it works – your mileage may vary.  There is actually one school of thought that says the less we normally warm-up the less we NEED to warm-up, and vice versa.  Not saying I necessarily subscribe to that theory, but there MAY be some truth to it – at least enough that I don't get my panties in a bunch if someone doesn't normally do what others would call a 100% perfect "optimal" warm-up.

# *Vertical Jump Q&A…*

**Q: I'm not sure what workout I should do? I'm not really sure if I'm lacking explosiveness or strength. Can you help?**

Ok, if you're not sure what to do just look at your basic strength first. Can you easily squat 1.5x your bodyweight with good form? If not, do the beginner workout. If you can, I suggest you start off with the general intermediate workout.

**Q: You say I need to do jumps regularly, but how many days and how many jumps are enough?**

It really depends on how advanced you are as a "jumper", how often you train and perform plyometrics, and how often you're involved in sports that require jumping. There aren't any hard and fast numbers on how many jumps you do, or when you do them. Pay attention to how you feel - if you feel good on an off day and feel like getting some jumps in, don't hesitate to do so. If you feel a little run down, it's probably best to recover. Jumps aren't something you really wanna do if you don't feel energetic. The important thing is you always want to stop a jumping session at the first sign of drop-off, and if your jumps are going down, instead of getting better, that's a sign you need to cut back.

Generally speaking, if all you do is lift weights and don't perform plyometrics, or play any jumping oriented sports, you can do jumps 5 days per week. Try to get 15-20 jumps in 5 days per week. These jumps don't ALL have to be 100% high intensity . You might have 3 days where you jump as high as possible, and a couple of days where you just jump easily and stay smooth - maybe 50-60% max effort. Don't overanalyze your jump sessions too much. A good way to get jumps in is to practice dunking of some sort. My friends and I used to dunk on lowered rims all the time when we were still too ground bound to even touch a regulation goal.

If you perform a couple of plyometric sessions per week, you can do regular jumps 2-3 days per week. Again, these don't ALL have to be 100% high intensity.

If you play basketball a couple of days per week you might do jumps a couple other days during the week. However, if you play a sport like basketball or volleyball 4 or more days per week, I wouldn't really recommend you do any extra jumps, aside from what you get in your games and warm-ups.

Lastly, if you lift, do plyometrics regularly, AND play basketball/volleyball more than a day or 2 per week, I wouldn't recommend any extra jumps, aside from what you get in your games and warm-ups.

**Q: Can I play basketball while I'm training or do I need to give it up?**

Sure, you can play basketball just don't go overboard with it. A couple of days of moderate full court basketball is fine, but if you play hours per day it'll wear you out quick. Basically, if you're not making progress in the gym as far as your training weights go, and aren't getting more explosive, you

might need to cut back. Also, if you're playing much basketball at all I wouldn't recommend you do more than minimal plyometric work.

**Q: What should I do on my off days? Just relax or do some type of activity?**

It really depends on how much low level moving around you do. If you're the type that just sits around in the house all day you might get out and get your blood flowing a bit - go for a walk and do a dynamic warm-up and a few sub-maximal jumps. However, if you normally walk around a lot, or work standing up, you can get away without doing anything. You want to recover, but you don't want to get stale. Here's an example of what I mean by stale: Have you ever ridden in a car all day and get out and almost feel like you have trouble walking? Well, that's a little bit too stale. If that's how you feel on your off days definitely do something to get your blood flowing a bit. In my experience, any level of staleness can be overcome with a 10 minute dynamic warm-up. If you're unsure what to do see the conditioning manual for some ideas on implementing "tempo" work.

**Q: Aren't sprints really good for vert? Why don't you have them in the programs?**

Sprints are a decent plyometric drill and can help potentiate the nervous system for jumps in someone that doesn't normally do them. I don't require them because not everyone has a convenient place to sprint, it's hard in many climates to sprint year around, and they really aren't a requirement. Having said that, they do some positive things and optimally I recommend vert trainees do them every other month. Do them in place of a plyometric drill. For example, if your workout calls for you to do ankle jumps and tuck jumps for 4 sets of 8 reps each, you'd do 4 sets of 40 yard sprints in place of one of your plyos. Keep them in your program 1-2 days per week for a month, then pull them out for a month. I've noticed sprints, when done chronically, tend to create a low grade state of fatigue in the legs. However, when done intermittently, they do a nice job potentiating the jump. Thus, I think it's best to cycle them in and out. Additionally, if you're going to do sprints keep the following parameters in mind:

1. If you're not used to sprinting ease back into them slowly. Go at 80% top speed your first workout, 90% your 2nd, then you'll be ready to go. Failure to do this can lead to painful quad and hip flexor strains.

2. Keep the distances relatively short - less than 50 yards/meters. The perfect distance is probably around 40 yards.

3. Always stop a workout before or when your times start to decline.

If you want to use sprints for conditioning that's also another option. I recommend you take a look at my Conditioning For Vert manual for some options on that.

**Q: Do you have any tips on jump technique?**

Yes, I wrote a rather lengthy article on jump technique on my website you can read it here: www.higher-faster-sports.com/lastlittlebit.html For 1-leg jump technique go to Joel Smiths website at www.just-fly-sports.com and read his free e-book on unilateral jumping. I must caution you don't try to

delve TOO much into technique. I've noticed too many people worry too much about it and it's not something that's gonna give you all that much anyway. If you practice your jump enough you'll inherently take on the technique optimal for you as an individual. Like I said earlier, the VJ is a gross movement pattern and not that difficult. However, things like recruitment patterns can impact your jump moreso than sub-optimal technique. Make sure you pay attention to the assessment section if your jump feels "off".

**Q: What if any relationship is there between jumps and sprinting speed? I was watching some NFL combine footage and anyone that ran a good 40 also jumped in the mid 30's. One of the guys was 5'11 200 pounds and still ran a 4.4 and jumped 33 inches.**

The 2 events are very similar from a motor quality perspective. They both involve relative power of the lower body and both rely on triple extension of the ankles, knees, and hips. The primary difference is the sprints involve the hips and hamstrings to a much greater degree than the VJ (particularly the hamstrings), which involves the knee extensors (quads) to a greater degree. The sprints also have shorter ground contact times, which favor more fast twitch muscle fiber. Having said that, most people that can run fast can also jump high because it NEVER hurts to be more glute dominant in a vertical jump and people that can run fast are almost universally glute dominant. Fewer people that can jump high are glute dominant ENOUGH to be extremely fast, although there is some correlation.

For the most part, anytime you increase your vertical jump you increase your potential for a great 40 yard dash, and anytime you improve your 40 yard dash, you increase your potential for a great vertical jump. You need to do enough specific training in either event to fully transfer those gains though. In other words, if all you do is jump you can't expect to go out and set the world on fire in the 40 if you never run sprints. You'll need to take some time actually running sprints to fully transfer your gains. The same is true of the sprinter who wishes to jump high. I trained one guy who ran a 4.3 40 but couldn't vertical jump 30 inches simply because he'd never cared about jumping and had absolutely no coordination/efficiency in the movement.

**Q: Should female athletes train any differently than men? Can they benefit from the same things men benefit from?**

There is no inherent reason to train a female athlete any differently than men. They respond to the same things, but females actually tend to respond a little better to strength increases because they're naturally weaker. The one major difference is females have less strength potential, so whereas a man should initially seek to establish around a 1.5 x bodyweight squat, I'd set a goal for a female around 1.25 x bodyweight.

**Q: The big squatting debate: Which squat is best? Back squats vs front squats vs box squats vs athletic stance squats vs wide stance squats - which squatting style is best?**

Well, let's take it from the beginning: Regardless of what style squat you're talking about (with one exception) I like an athletic stance squat, feet about shoulder width apart, because it engages more muscle over a fuller ROM. A close stance squat will also get the job done. Wider stance squats decrease

muscular involvement shortening the ROM. They're great for lifting more weight, but not so great for creating strong legs. As far as a back squat vs front squat, the front squat forces you to remain upright and due to that it lessens hamstring involvement and puts more stress on the quadriceps. That isn't necessarily a bad thing. Since you can't lean forward in the front squat it forces you to keep stress on your thighs and off your back.

Take 2 groups of people and have one group do back squats and jumps and one do front squats and jumps. It wouldn't surprise me if the front squat group made better vert specific gains due to the inherent increased lower body loading involved, since you can't lean forward and take stress off the legs. They do have some advantages, they're harder to cheat, there's less lower back involvement, and more lower quad involvement. However, they're also kinda a pain in the butt. It can be difficult to keep the bar in place in the groove along the front delts, they can be hard on the wrists, and me personally I just have never looked forward to doing them. I realize that's a personal thing - I recognize how effective they can be but if you don't "like" and exercise that exercise is going to be less effective for you. Because of the increased skill involved in a front squat they're also more difficult to load effectively. A beginning athlete will make better gains per session on a back squat because it doesn't require as much skill to keep the bar in place. In case you're curious what a good front squat is, you should be able to front squat 80% of what you can back squat.

As far as the bulgarian split squat goes, it's almost the perfect exercise. You take the lower back out of the movement, you get an increased range of motion, AND you can adjust your stance width to target more glute and/or quad. A closer stance targets more quadriceps, while a wider stance takes stress off the quads (and knees) and targets more glutes. I generally recommend a stance where the front of the knee stays about even with the toes on the descent. The bad thing about the bulgarian split squat is you have to do twice as much work to get the same amount of volume as a squat, since you have to train each leg individually.** There's also a balance aspect involved, which can make adding weight difficult for lower level trainees. It's relatively easy to perform 4 sets of 6-8 rep regular squats from a mental and physical perspective. But try doing 4 sets of heavy 6-8 rep bulgarian split squats. It sounds easy until you actually do it. By the time you've done you've really performed EIGHT total sets - which is REALLY tough. When was the last time you did 8 sets of 6-8 reps on squats? It's a TOUGH workout. For that reason I prefer to use bulgarian split squats as assistance exercises, most of the time.

As for the box squats, I'd consider them a separate exercise altogether. They're more of a "specialist" movement. The wide stance deep box squat is more of a specific glute/hip exercise. I'd put it in the same assistance exercise category as exercises like romanian deadlifts, barbell hip thrusts, etc. The parallel box squat is more of an overload exercise. It does have a purpose but I wouldn't really consider it a foundational lift for most people.

With all that said, *theoretically* I'd rank the exercises this order: front squat/bulgarian split squat (tie), back squat, box squat . However, we also have to ask ourselves if we really need to complicate things so much? A back squat works fine and there are plenty of standards for it, why make things more complicated? So, from a *PRACTICAL* perspective I'd rank the exercises in this order: Back squat, bulgarian squat, front squat, box squat. (***When performing Bulgarian split squats always do your weak leg first)*

**Q: What about wearing a belt for squats? Some say a belt makes you weak, others say the provide benefits.**

The main benefit of a belt is it thickens your waist which gives you additional leverage and it also gives you something to push against to increase intra-abdominal pressure. Both these things will enable you to lift more weight. You don't have to worry about a belt weakening your core as long as you don't use one ALL the time. I would save a belt for your heavy attempts – 5 reps or less, and even then I'd consider it optional.

**Q: How important is core strength for vertical jump? I notice you don't talk much about the core. Don't I need a strong core to jump high?**

Core strength and its relationship to most athletic tasks, including the vertical jump, is one of those things that sounds great theoretically. The core (abdomen, obliques, lower back) joins the upper and lower body together. The thought is that if the "middle" is weak you can't transfer force efficiently. It sounds like a great concept but the problem is in the real world it doesn't pan out. There have been a ton of studies done on the topic of core strength and how it relates to performance and to date increases in core strength haven't been found to contribute significantly to strength and power measures, such as the VJ. Even in an endeavor that involves a TON of direct core activation, a tennis serve, increased core strength wasn't found to be predictive of increased serving power.

In a vertical jump the front of your core (abs and obliques) aren't prime movers. Your lower back strength could be considered somewhat important since it's a prime mover, but if you're training hard and consistent you should be able to get all the lower back strength you need from exercises like squats. That's not to say you should neglect your core - it may be somewhat valuable from an injury perspective, but you shouldn't expect performance gains out of it and it need not be a primary focus in your training. I like people to have enough core strength to control their pelvis optimally during movement, (see the leg lowering drill video in the assessment section), and hold a front and side plank in good alignment without the hips dropping.

In my observations many people confuse core strength with "general organism strength." When you're weak throughout your body you appear to lack stability and move like jello. So, when you take a weak athlete and watch him move around he'll appear to flangle around a lot. The opposite is true of someone that's generally overall strong. This is one argument in favor of folks that say no direct core training is necessary in athletes - they can get all of it they need from basic hard training and their sport, which is perfectly exemplified in sports such as gymnastics. In my opinion, mobility and postural imbalances are more important than core strength. For example, if someone has forward head posture, anterior rotated shoulders, tight hip flexors etc. their upper body will be chronically flexed over, and they'll likely have a hard time reaching full extension in something like a vertical jump.

**Q: Do you have any tips for palming a basketball? I can jump high enough to dunk I just can't palm the ball.**

Sure thing. Open hand grip strength (the type required when palming a ball) is a relatively specific form of strength. That is, you can do regular grip work like captain of crush grippers, barbell holds, plate pinch gripping, and forearm work, and it won't necessarily help your ability to palm a ball. You want to work your grip in ways fairly specific to palming a ball and the best way to do that in my opinion is to use isometric progressions actually PALMING A BALL. There are 4 different progressions to palming a leather men's basketball. These are:

1. Palming a rubber women's basketball
2. Palming a leather women's basketball
3. Palming a rubber men's basketball
4. Palming a leather men's basketball.

Rubber is easier to grip than leather and a women's basketball is obviously smaller than a men's basketball, so that's a good starting point. What you'll wanna do is start off with a women's rubber basketball and see how long you can palm it (if at all). If you can't palm it at all I really can't help you, you'll have to wait until you do a bit more growing. But, assuming you can palm a women's rubber ball, see how long you can hold it - you want to work up to 30 seconds. Do 2 sets of 30 second palms with both hands a few times a week. Once you hit 30 seconds move up to a leather women's basketball. At first you might only be able to palm it a few seconds, but gradually work your way up to 2 sets of 30 seconds. It helps if you lick your fingers a bit before you attempt to palm it - or use chalk. You'll progress relatively quickly using these 30 second progressions as long as you work at it consistently. Then do the same thing with a rubber men's basketball, then finally a leather men's basketball. That's all there is to it! If you have a good selection of medicine balls to work with they can also be used the same way - work your way up in 2 sets of 30 seconds in bigger and heavier med balls. That's the basic idea. Good luck with it!

**Q: How important is the tibialis anterior (the front part of the lower leg) for jumping and athleticism?**

The tibialis anterior is the muscle that runs up the front of your leg on the opposite side of your calves. Its job is to raise the front of your feet up like a reverse calf raise. It's not really important from a performance perspective, but it is relatively important from an injury perspective. It can become inflamed and can cause pain known as shin splints, particularly if you do a lot of running on hard surfaces like concrete. Shin splints are most common in people that do lots of jogging. In my experience, the most problematic activities as far as shin splints are concerned are activities that involve a lot of heel pounding the pavement. Fortunately, low volume plyometrics like I recommend tend to strengthen the tibialis anterior because the volume is low enough that you can use good technique and stay mostly on the balls of your feet, but if you ever start to feel pain, or if you just want to throw in some remedial work, you can perform bodyweight reverse calf raises for sets of 25-50 reps a couple of sets a couple of days per week. If you have stairs let the front half of your foot hang off the stair and just perform a reverse calf raise from that position.

**Q: Should I actively seek to gain bodyweight or muscle while I'm on a hypertrophy or strength phase? Also what supplements should I take?**

This really depends on you as an individual. If you're trying to gain strength it's typically VERY hard to do if you aren't at least maintaining your bodyweight, unless you're overweight (over 15% body-fat). People with more body-fat can get away with losing weight and getting stronger at the same time. But, if you're already lean it's a good idea to try to at least maintain your weight. If you're really skinny you'll tend to have an easier time making strength gains if you intentionally try to gain some weight. Look at your progress on exercises like squats - if you don't seem to be gaining much strength you might consider going up a few pounds in bodyweight. On the other hand, people that are performing explosive oriented workouts where the focus isn't on gaining strength can typically get away with losing bodyweight while training.

If you're skinny and lean (within shouting distance of 10% body-fat), and test out faster than you are strong on most of the general assessments, gaining bodyweight will definitely help you gain strength, and will most likely help your VJ. If you're already up around 15% body-fat, and/or test out stronger than you are fast on most of the assessments, I'd be more hesitant to recommend intentional weight gain. People that are inherently stronger than they are fast don't often do well with bodyweight increases.

For people interested in weight/muscle gain I recommend they shoot for an average of a steady ½ to 1 pound per week. You'll have to increase your calorie intake when gaining muscle and typically I recommend a starting point of about a 300 to 500 calorie increase, with about 30% of that increase coming from protein, 40% coming from carbohydrate, and 30% coming from fat. You'll typically gain 3-5 pounds your first week (which is mostly extra water and glycogen storage), but after that you should see an average of a pound gained per week, which typically will be 2 or 3 pounds gained 1 week, zero the next, .5 pounds the next, then 2 or 3 lbs the next, or something similar. Weight and muscle gain won't ever be completely steady, but you should see a general trend of about 2-4 lbs per month. It is possible to gain at a faster rate, but doing so increases the risk of additional fat gain. You will likely gain SOME fat anytime you're intentionally pushing your bodyweight and muscularity up, so don't be deathly afraid of it - you can always get rid of it later

As for supplements, I generally recommend a good multivitamin, extra vitamin C (1000 mg per day), extra zinc at 30-50 mg per day, and extra magnesium at 300 mg per day. Fish oil is a great general health supplement and I recommend taking 6 grams per day. Creatine monohydrate is good for strength and usually impacts the VJ favorably – I recommend the plain creatine powder and take 5 grams per day. If you’re older and have joint problems you might try glycine for joint health, 5 grams twice a day. Protein shakes or meal replacement drinks (whey, milk & egg, or casein) are great for convenience if you have trouble eating enough, but are by no means necessary.

**Q: When it comes to strength training how specific does the lifting need to be in order to help the VJ? Can I use a leg press instead of a squat since the leg press also recruits the thighs? Also, what about using accommodating resistance such as chains or bands on exercises like squats?**

Good question! You can get your legs strong using a leg press but the recruitment pattern is not as similar to the VJ. In other words, you sit on your butt so your lower back does not have to work to help stabilize you. The great thing about squats is they've very similar to a vertical jump from a movement and recruitment standpoint. Having said that, you certainly can get results with a leg press and it CAN help your jumps but I don't typically recommend it unless you don't have free weight equipment. In my experience squats, front squats, bulgarian split squats (with barbells or dumbells), and regular lunges are all superior to the leg press for vertical jump. In these exercises you're not locked into one plane of movement and you do them standing up, which is how you jump. Personally, I think most people who like doing leg press do so because they're not tough enough to do squats - so don't be a wuss! If you do decide to use a leg press make sure you use a full range of motion - I rarely see trainees use a full range of motion on it when they train. Also, keep in mind you'll be MUCH stronger on a leg press than a squat. Two plates of each side on a squat is approximately equivalent to 5 or 6 plates (on each side) with most leg press machines.

The main problem with machines such as the leg press, smith machine, hack squat machine, etc. is they lock you into one plane of motion, so your body can't move 3 dimensionally. For example, if you watch someone do a barbell squat (or any type of free weight movement) the bar won't travel a completely straight line up and down, it'll kinda curve. The machines keep you locked into one plane. This forces you to recruit muscles you normally wouldn't recruit and takes stress off muscles that should be active. For example, exercises like the smith machine squat take the hamstrings out of the movement, since people put their feet more forward then they normally would. This increases stress on the knees (but also the quadriceps). You can use machines and make some gains off of them - they're probably not as bad as some people say, but better to use free weights.

As for accommodating resistance with bands and chains, they can be beneficial but I don't find them necessary. Remember, in the VJ you only need to strengthen the muscles a certain degree - it's not like you're training for an 800 lb squat. You're using weights to develop general strength. Will bands and chains help get you there quicker? Well, they inherently promote more RFD while lifting fairly heavy loads, so they may be somewhat valuable, but 100% necessary? No. Remember, you can train RFD just fine via jump squats, speed squats, o-lifts, etc. Same goes for concepts such as overspeed eccentrics. Useful? Yeah, but why not just perform a hang snatch, clean, or dumbbell swing? What we do in the weight room does not have to be 100% specific – the effects are general. Just make sure your overall program and periodization is solid before you mess around with making things more complicated than they already are.

**Q: What is the better exercise to increase vertical jump the power clean or power snatch?**

They’re both very good, but if you had to choose one or the other I’d go with the snatch. It’s more similar to a vertical jump from both a recruitment pattern and movement speed standpoint.

**Q: What's the best thing to eat before a workout to get the best workout (most energized) aside from caffeine? Like an actual meal, and how long before my actual workout? Thanks!**

You don't get your workout energy from what you eat before a workout you get it more from what you ate the day before. You just need to make sure your glycogen levels are replenished, and really the only time they wouldn't be is if you were actively dieting. So, if you're maintaining your weight you should have more than enough "energy" to do a typical VJ improvement workout. There really aren't any foods that can "stimulate" you. A LOT of people actually feel better and more energetic eating nothing, or at least no carbs before a workout. If you try to eat a lot of high carb foods (or drinks) before your workout you're more apt to feel lethargic, as all that glucose boosts insulin, which tends to make you tired and lower your blood sugar. If you want to drink a carb drink wait until you start your workout and get into the flow before you consume it. Many people feel best eating a high protein meal such as steak and eggs or chicken and veggies about an hour or 2 before a workout. Most people will feel best with a source of protein and moderate carbs. One example would be a protein drink and banana, or something similar like a tuna or chicken sandwich. People that are overweight generally do best with very little carbs in the hour or 2 before their workout begins. On the other hand, some people can eat a heavy meal before a workout, like an entire pizza, and still do fine. So, basically eat whatever you want, unless it puts you to sleep. If you're getting tired as your workout starts look at how many carbs you're taking in within the hour or 2 before your workout - most likely you're eating too many and you're crashing your blood sugar.

**Q: How important is 'Compensatory Acceleration', ie moving the weight as fast as possible, in training for Vertical Jump? On that train of thought, when should lighter weights be used, to train speed-strength and strength-speed?**

It's really not important when referring to strength exercises. If your main goal with an exercise is to gain strength, the recruitment plus time under tension takes care of it, so no need to intentionally try to accelerate the weight. It's not a bad thing to do, but when you have a heavy weight on the bar and try to explode with the weight, form often breaks down. Thus, I put form over speed when it comes to exercises done with the intention of gaining or maintaining strength. In fact, you can gain plenty of strength performing intentionally slow repetitions. But when trying to "express" strength better and develop RFD you inherently want to try to accelerate the weight, so any strength-speed and speed-strength exercise will inherently use compensatory acceleration. The one instance where a heavy strength movement should be done with compensatory acceleration is when performing the sub-maximal low rep method, where you lift a heavy weight for low reps using a buffer. For example, sets of 1 @ 90% of max in the squat. So, if you're primarily using an exercise to push your strength up higher don't worry about using compensatory acceleration. You CAN, but you don't NEED to.

**Q: Can you talk about some of the different VJ testing methods (mat, vertec, stand & reach) and their relationship?**

There are 3 traditional ways of testing the vertical jump and one non-traditional way:

1: **Traditional stand and reach:** Here you measure your maximum extended reach by reaching up and making a mark or placing a piece of tape on a wall, then you jump up as high as you can and make a mark or put a piece of tape on a wall. Next, you simply calculate the distance between the 2 marks and

whatever that difference is your vertical jump. The disadvantage of the stand and reach method is it's possible to cheat the result and give yourself a higher jump by not reaching up as high as you can when you measure your standing reach. The average person can easily cheat himself up an additional 3 inches by locking the shoulder blade down to give the illusion of an extended reach. For this reason stand and reach assessments are often over-inflated. They can be effective, but their accuracy depends on whether a person tries to cheat or not.

2. **Vertec:** The vertec is a specialized stand and reach device but is able to get a more precise measurement because the target is not on a wall. However, the vertec has the same drawbacks as the regular stand and reach method - it's possible to cheat and give yourself an additional few inches by not reaching up as high as you can when you calculate your standing reach.

3. **Jump pads:** Jump pads are specialized electronic mats you stand on. When you jump up in the air they calculate the length of time you are in the air then factor the flight time into an equation to figure out your vertical jump. The flight time and equations are very accurate, but there are a couple of major drawbacks:

A: It is possible to cheat a jump pad by bending your knees just before touchdown to artificially inflate the vertical jump. The longer you're in the air, the higher your calculated jump will be. I know people who've consistently cheated jump pads for an additional 5 inches or so of vert.

B: Jump pads calculate your vert from the toes extended position instead of the flat footed position. This would be like measuring your standing reach up on your toes. This measurement will inherently decrease your vert measurement by at least a few inches. There's nothing wrong with that unless you get into comparing different vertical jumps assessed by other methods.

There is one other form of VJ testing that is rarely talked about that is actually my favorite way to assess the VJ:

4. **Head height testing**

In my opinion head height testing is the most reliable vertical jump assessment and one that you should do if possible. It's fairly impractical to use it with big groups of people, or when testing teams, but individuals can easily figure out a way to get it done. Here you simply measure your standing height then measure how high you can touch with your head. I've seen people use everything from strings hanging down from objects, basketball nets, basketball goals, basketball backboards (just make sure they're padded) and ceiling tiles. You will need a good tape measure and a little bit of creativity. Just make sure you don't hit your head on anything hard or you might knock yourself out! As a teenager I was obsessed with head height testing. My friends and I used to walk around trying to pop up the tile ceiling panels. It was simple and effective. I knew if I could hit my head on an 8 foot ceiling tile standing 5'6 tall I was jumping 30 inches. Then I knew when I could pop a 9 foot ceiling tile at 5'9 tall I was jumping 39 inches. Then I knew when I hit a 9.5 ft string hanging down from a ceiling I was jumping 45 inches (running vert). In my opinion you should use stand and reach testing if you want to brag, but use head height testing for

what you want to really KNOW. If you're 6 foot tall and can graze your head on a 10 foot regulation goal you're jumping 48 inches...no ifs, ands, or butts.

**Q: In regards to altitude Drops or shock Drops, what is the correlation between the height of the box an athlete drops from (assuming the athlete lands appropriately) and how high the athlete can or should be able to jump? If you think about the amount of force from a drop...and how much energy it takes to absorb it...is it reasonable to think that the higher the box an athlete drops from and lands appropriately the higher the athlete should be able to jump? Is there a box height at which this is no longer the case? Could you almost solely rely on altitude drops, depth jumps and variations to improve your vertical performance and not to mention speed and reactive capabilities?**

There isn't necessarily a direct correlation between the box height one can land from and how high they can jump, but there are subtle relationships. Generally, the more efficient you are stabilizing and absorbing force (the higher the box you can efficiently land from), the higher you're going to jump, but there are exceptions. Some people are really good at absorbing force but not so good at putting it out. In my opinion, there is no need to do any form of plyometric variation using a box height greater than your best running vertical jump. You CAN use higher boxes but the results probably wouldn't be any greater than using more moderate box heights, but the risk of injury would be greater. As for the last part of your question, YES it definitely is possible to rely exclusively on depth jump variations to improve reactive ability, although the effects for most people would be significantly less than depth jump variations in conjunction with weight training, as it takes a certain duration of tension to optimally strengthen the muscles involved. Depth jumps inherently require large amplitudes of tension which is better at increasing the rate of force development specific to the vertical jump. They COULD be used to improve max force potential as well (strength gain), although the volume one would have to use to optimally train this quality could create an assortment of overuse injuries.

**Q: I am interested in the difference between training hard getting sore and pushing yourself versus overtraining and how to know when to back off b/c of overtraining versus being a wuss and slowing training too much b/c one is afraid of overtraining. I think sometimes I'm too worried about overtraining and don't train hard enough**

It's ok to be sore, and ok to push yourself, but you don't want to run yourself into the ground. If in doubt of whether you're pushing too much look at your progress in the gym, particularly with strength lifts. Are your lifts getting consistently heavier? Are your explosive lifts getting faster? Are your plyometrics getting smoother? Even if your jump isn't improving you can look at other things.

It is possible to set personal records when you're feeling mentally at your worst - feeling sore, tired, and overtrained. However, any consistent decline in performance is a sign you're doing too much and need to get back. Think about it. If you did nothing at all you'd at least maintain your abilities, but if things are getting worse, or you go several workouts in a row without any progress, you likely need a de-load of sorts.

**Q: My question is how should older guys approach when working on improving the vj and what kind of expectations? I'm turning 40, still quite athletic @ 5'8", no history of injury, etc.**

In my experience older guys can respond to "training" induced gains just as well as younger folks, but their overall increases will be less dramatic because they've already gone thru physical maturity, which can account for a significant portion of your overall results. I've seen plenty of older guys put a good 6 inches or more on their vertical jump though and they will benefit from the same qualities any other athlete will benefit from. There are a few things older athletes should pay special attention to though:

1. Mobility & muscle recruitment patterns: As we age we acquire scar tissue, injuries, and an assortment of other things that increase the risk of overuse injuries. Older athletes need to pay special attention to warm-ups and flexibility work.

2. Be careful with plyometrics: Due to the above mentioned mobility issues, older athletes are at greater risk of injury during high amplitude movements like plyometrics. Always start with a low volume of plyos and use manageable progressions.

3. Overall recovery/volume: Recovery for older athletes tends to be a little less as testosterone, growth hormone, and other important neurochemicals begin to decline. Older athletes can find it advantageous to insert an extra day or 2 of rest into the typical training split. For example, instead of performing a strength workout twice a week, an older athlete may perform the workout once every 5-6 days.

As long as you pay attention to those things there's no reason you can't make great gains at 40!

**Q. When it feels like it's time for a deload, do you feel it's better to use a higher rep scheme, dynamic effort method, less weight as in take it easy, or just plain recovery.**

There is a time and place for all of them, but generally speaking, the approach where you use less weight for the same movements is the one I recommend the most. For example, here is a 4 week set and rep scheme with week 4 being the deload:

week 1: 3 x 8
week 2: 4 x 6
week 3: 4 x 3-5
week 4: 3 x 3 with week 2 weight

Or if you're on a scheme where you don't vary sets and reps here is an example of a deload:

week 1-3: 3 x 5 @ 100% max effort
week 4: 2 x 5 @ 90% of week 1 to 3 weight.

You still hit the movement fairly hard, but keep a couple of reps in the hole. Deloads can come about via reductions in intensity, load, volume, intensiveness, frequency, or all of them.

**Q: I have often seen athletes make big gains with regard to one or two specific lifts, but less so with regard to their jumping and sprinting ability. For example, I have a couple friends who have doubled both their deadlift and squat in the past year. At first glance it would appear that they are twice as strong. However, they do not look twice as muscular, and, more importantly, they have neither doubled their vj nor significantly improved their 100 meter times. Why has the progress of their physiques and verticals not matched that of their lifting numbers?**

In a nutshell it's because they're not good at applying force quickly. Their rate of force development and not maximum force is the limiting factor for them. See the section on the explosive strength deficit in the VJ bible 2.0 where I really get into this.

**Q: What, if any potential benefit toward their vertical jump could an athlete get from training with unstable surfaces such as an IndoBoard or SlackLine?**

None really. If someone had a propensity towards sprained ankles those devices are semi useful to re-establish ankle stability, but that's really all they're good for. By making the surface unstable you actually take away from the target motor qualities you're after, like maximum strength and power, because you can't fire the prime movers as strongly.

**Q: I've noticed an inconsistency in my jumps throughout the week. On some days my vertical jump is 31 inches, other days it's 34 inches. Does this mean something is wrong or is that common?**

That's actually pretty typical. A beginner typically won't see much day to day variance, but the more advanced you become (the higher your VJ gets), the more day to day variance you'll see, because the more advanced you become the greater the degree of neuromuscular activation you're capable of engaging, and that tends to vary based on arousal, what you did the day before, how good you're feeling, and a lot of other things. When my VJ was 42-45 inches I'd have as much as a ~5 inch variance one day to the next and would really only be able to hit somewhere close to a top effort maybe 1 day per week. Additionally, anytime you hit a big vert PR don't be surprised when you don't equal that PR for the next week or 2, and you really shouldn't try to. It takes time for gains to stabilize.

**Q: How important really is upper body in the vertical jump? I've noticed most good dunkers are quite muscular throughout their upper bodies too. Do I need to be?**

This is a real good question. Realize traits such as having a large % of fast twitch muscle fiber and excellent endocrine production correlate with success in power/speed dominant sports. Having those traits also tends to promote a muscular physique. Thus, good dunkers (and sprinters) will tend to be more muscular naturally ALL OVER. It's kinda like pit bulls - they're highly athletic and muscular, but they didn't get that way via training, they were born with it. The upper body contribution to a vertical jump is really quite nil, and in my opinion you can gain all the upper body strength you need via correct training thru hard lower body movements. If you incorporate cleans or snatches they're a nice added bonus. Having said that, training your upper body definitely won't hurt anything, might help a tad (although it's arguable whether it'll help enough to be noticeable), and it's about impossible to become

overly heavy via too much upper body muscle mass. So, if you want to train your upper body, go ahead and do so.

I also get a lot of questions from people asking me how much they can gain from a strong arm swing. Arm swing does not actually lift you up in the vertical jump, but what it does is make you a bit lighter, which makes the job of your legs a bit easier. Here's an example that illustrates that: Stand on a scale and see what you weigh. Next, without bending your legs at all swing your arms up as if to jump. Watch what happens to the weight on the scale as you lift "up". It decreases! The more it decreases the easier it is for your legs to "lift" you off the ground. Having said that, much is overblown by many coaches when it comes to how much strength is required and technical areas related to the arm swing. The lift of the arms must be timed to the release point when you leave the ground, but the swing is not maximal. Getting more muscles involved in the unloading of the upper body (traps, upper back) does seem to be beneficial. I've tried a lot of things and the only thing I've ever seen that can improve arm swing are hang cleans and snatches. In my experience, most people that do hang or block cleans or snatches for a while will notice a difference in the 2 hand jump – particularly a 2 handed jump with the ball in your hands. I believe it's because it teaches you how to activate your upper back and traps to *help* your jump and arm swing. It's worth experimenting with, but as stated, you're not going to get a big vertical jump by getting your upper body as strong and muscular as possible.

# *Vertical Jump Training Review…*

I'm now going to go thru and give you a quick summary of all the information covered thus far. Keep these things in mind and review them often if you feel yourself slipping up:

**A: Make sure you jump often enough to gain or maintain efficiency in the act of jumping**: You get your strength and power from extraneous training methods - lifting, plyometrics, etc. But in order to fully transfer those gains you need to be efficient in the act of jumping itself. That means you need to regularly get some quality effort jumps in.

**B: Make sure your jump is efficient:** You want to make sure you are able to jump efficiently. This occurs via a combination of addressing any movement pathologies or muscle imbalances (see the mobility assessment section) and working on your approach and takeoff (for approach jumps). If you're not sure whether you're jumping efficiently, contact me thru my website higher-faster-sports.com for a video analysis. The most common sign you have movement pathologies or muscle imbalances is chronic knee pain.

**C: Take a focused, yet correct, mentality to strength increases**: Remember that strength forms the base for your jump, but not to the exclusion of everything else. The squat is very important in regards to how high you jump, but you shouldn't obsess about your squat number in relation to your vertical. Remember that RFD and reactivity training should also be included in your program. In other words, boosting your squat in a jump specific manner is important, but you shouldn't perform powerlifting routines or super wide stance squats just because you might put a few extra pounds on your squat. Use an "athletic" stance squat with feet slightly wider than shoulder width. Descend to slightly below parallel

(legal powerlifting depth) or go down until your back starts to round, whichever comes first. Seek progressive increases on your squat while you engage in other jump specific work.

**D: Strengthen ALL the contributing joints to the VJ**: Remember the VJ is a combination of hip extension, knee extension, and ankle extension, so make sure your strength program addresses all 3. Squats will hit the hip extensors and knee extensors pretty good, but don't neglect focused calf work either (calf raises). Extra glute work never hurt ANY athlete, so make special efforts to train the glutes via such exercises as barbell hip thrusts, romanian deadlifts, and reverse hyperextensions.

**E: Know your plyos:** Know the difference between low intensity and high intensity plyometrics (depth jumps) and realize when to implement each type in your program. Remember, always stop a plyometrics session as soon as any performance drop-off set to set is noticed. If in doubt about volume, pay attention to total ground contacts - more than 100 per session is too much. Always attempt to be smooth and quiet while performing plyometric drills and make use of overhead targets, jumping heights, and distance to make your training more goal oriented. Attempt to either beat your mark each time you train, or progress via improvements in efficiency (quietness and smoothness while maintaining the same height). Remember, a good cue for plyometric drills is "100% performance at 90% effort." Seek to consistently reduce your ground contact times in your plyos.

**F: Use a well rounded program:** Have a well rounded program incorporating strength, strength-speed, speed-strength, mobility, and skill. Also realize that while some conditioning/stamina work can have benefits, too much can kill your progress.

**G: Properly periodize your training**: Beginners really don't need periodization but if you're an intermediate level athlete or higher remember you can't expect to move every motor ability up simultaneously. Seek to maintain some qualities while maintaining others via a reduction in volume. How do you know if you're an intermediate level athlete or higher? Simple. As soon as the smorgasboard approach stops working.

**H: Manage your recovery:** If you're not progressing on a fairly regular basis it's likely not from doing too little, but doing too much. If you're going backwards it's DEFINITELY from doing too much. If in doubt insert an extra day or 2 of rest between workouts and adopt the mantra "improve or go home." Beware of the impact various extraneous activities and stress can have on your recovery. Consume a proper diet, get enough sleep, and manage the stress in your life. Remember to take periodic recovery/deload weeks and realize you can't train at 100% intensity all the time.

**I: Set goals and find daily inspiration to train:** Write your goals down on paper, make them realistic, and place the paper at a place where you can see your goals on a daily basis. Keep a training log and record your progress session to session.

# *Personalized Coaching and Consultation Service…*

It is my hope that the information in this book has solved all your training problems and will be all you ever need to develop a sky-high vert. However, I find some people desire more of a personal touch or have special problems/situations they need help figuring out. For these people I do offer a satellite consultation and program design service. My rates are quite affordable and I also have an advantage in that I've "been there and done that." Thru my own experience and almost 20 years coaching I can safely say there probably aren't any situations with regard to vert training that I haven't encountered. I offer several different services. For athletes who just need a little help I am available for phone consultations, or I can do a simple video analysis to identify what I see and what I think you should do to make quality gains. I can evaluate your current program and suggest changes, or help you choose which direction you as an athlete should head as far as training. For those who desire more interaction, I can do a full video mobility, power, and strength assessment on you, custom design a workout and diet for you, and work with you 1 on 1 via email/phone. For more information on my services and current rates send me an email to ***kelly@higher-faster-sports.com***

Good luck!

-Kelly

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